Supplementary Materials for

Comorbidities in Juvenile-onset Rheumatic Diseases: a systematic review and meta-analysis

Shall be considered in conjunction with the

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Supplementary Table S4.1: Prevalence and incidence rates of comorbidities in JIA, jSLE and JDM patients presenting in childhood or apparent in adulthood

Childhood / JIA

Study	Gender	Age in yrs	Onset age, yrs	Age diagnosis, yrs	Disease duration	Follow up period for the study	Comorbidity	Sample size, N (patients/ controls)	Prevalence	Incidence (per 1000 person-yrs)	Ethnicity /Race
Demirkaya, et al, 2011, Turkey[1]	F: 354	Mean 11.84±4.66 yrs			Mean 4.35 ±3.25 yrs	Mean 7.6±4.4 yrs		634			
							Macrophage Activation Syndrome	14	0.022082	2.905528806	
							Infections	14	0.022082	2.905528806	
							Uveitis	62	0.0977918	12.86734186	
Jain, et al, 2001, India[2]	F: 10	Mean 8.7± 3.2 yrs			Mean 16.4 ± 5.23 months			50			
							Keratoconjunctivitis sicca	7	0.14		
Rangel, et al, 2003, UK	F:20	Mean 15.2±2.4 yrs	Mean 5.5±3.6 yrs		Mean 9.8 ± 3.8 yrs			30			Caucasian 27; Asian (SE) 3
							Major depressive episode	2/27	0.0740741		
							Over-anxious disorder/generalized anxiety disorder	3/27	0.1111111		
							Dysthymic disorder	2/27	0.0740741		
							Night terrors	1	0.0333333		
							Oppositional defiant disorder	1	0.0333333		

Saurenman, et al, 2007, Canada[26]	F: 711	Mean 13.8 yrs (1.2 to 34.1)	Mean 6.9 yrs (0.5-17.8 yrs)	Mean 6.9 yrs (Range 0.08 to 25.7)	Mean 6.9 yrs (Range 0.08 to 25.7)		1081			
						Uveitis	142	0.1313599	19.03765971	
						Cataracts	33	0.0305273	4.424244862	
						Synechiae	31	0.0286772	4.15610881	
						Glaucoma	22	0.0203515	2.949496575	
						Band keratopathy	20	0.0185014	2.681360522	
						Macular oedema	7	0.0064755	0.938476183	
						Legal blindness	10	0.0092507	1.340680261	
						Impaired vision	6	0.0055504	0.804408157	
Chabra, et al, 2020, Canada[27]	F: 142	Median 16.9 yrs (IQR 12.9 - 18.0)			Median 5.6 yrs (IQR 3.3-8.3)		247			
						Uveitis	24	0.097166	17.35106998	
						Band keratopathy	1	0.0040486	0.722961249	
						Decreased visual acuity	1	0.0040486	0.722961249	
						Cataract	1	0.0040486	0.722961249	
						Micrognathia or facial asymmetry	11	0.0445344	7.952573742	
						Leg-length discrepancy	12	0.048583	8.675534991	
Beukelman, et al, 2012, USA[28]	F: 5,421	Mean 9.7±4.4 yrs			Median 1.1 yrs (IQR 0.5-2.2)		8479			White 4402; Black 1448; Latino 1721; other/unknown 932
						Upper respiratory tract	110	0.0129732	11.79384361	
						Pneumonia	87	0.0102606	9.327858131	
						Bacteraemia/septicaemia	67	0.0079019	7.183522928	
						Urinary tract/pyelonephritis	65	0.007666	6.969089408	
						Skin and soft tissue	44	0.0051893	4.717537445	
						Abdominal abscess	20	0.0023588	2.144335202	
						Gastroenteritis	30	0.0035382	3.216502804	
						Diabetes mellitus	56	0.0066046	6.004138567	
						Asthma	669	0.0789008	71.72801252	

van Rossum, et al, 2007, Netherlands[29]	F: 41	Median 18 yrs (range 10-25)		Median 10.7 yrs (range 8-23)	Median 9 yrs (IQR 8-9 yrs)		61		
						Uveitis	12	0.1967213	21.8579235
Gerloni, et al, 2008, Italy[30]	F:124	Mean 17.1 yrs SD 9.2 yrs (range 1.9-49.8)	Mean 6.4 yrs SD 4.8 (Range 0.5-16.8)	Mean 10.7 yrs SD 8.6 (range 1.8-49.8)			163		
						Flare or new onset of chronic iridocyclitis	7	0.0429448	
						Gonadal effects	6	0.0368098	
						Hypertension	8	0.0490798	
						Infective events	41 (split below)		
						Recurrent URI or UTI or mycotic vaginitis	21	0.1288344	
						Viral thyroiditis	1	0.006135	
						Varicella	4	0.0245399	
						EBV infections	3	0.0184049	
						Single URI	3	0.0184049	
						Rotavirus gastro-enteritis	3	0.0184049	
						Helicobacter gastritis	1	0.006135	
						Scarlet fever	1	0.006135	
						Hip prosthetic infection	1	0.006135	
						CMV pneumonia complicated by MAS	1	0.006135	
						Pneumonia	1	0.006135	
						Impetigo	1	0.006135	
						Neuropsychiatric manifestations	53 (split below)		
						Headache	10	0.0613497	
						Severe headache (dose-dependent)	4	0.0245399	
						Psychoses	2	0.0122699	
						Depression	3	0.0184049	
						Panic attacks	2	0.0122699	

					Anxiety	4	0.0245399
					Anorexia nervosa	1	0.006135
					Nervousness/hyperactivity	7	0.0429448
					Memory and concentration deficit	1	0.006135
					Aggressiveness	3	0.0184049
					Fatigue	5	0.0306748
					Hyperalgesia	1	0.006135
					Pain amplification syndrome	1	0.006135
					Sleepiness	2	0.0122699
					Insomnia	1	0.006135
					Vertigo	3	0.0184049
					Paraesthesia	1	0.006135
					Hypoglossal paralysis	1	0.006135
					Neuroalgodistrofy	1	0.006135
Klotsche, et al, 2016, Germany[31]	F: 1,537	Mean 11.3 yrs (SD 4.6)	Mean 7.8 yrs (SD 3.7)	Mean 3.5 yrs (SD 3.6)		2263	
					Malignancies	8 (split below)	
					Acute lymphoblastic leukaemia	2	0.0008838
					Thyroid gland carcinoma	1	0.0004419
					Yolk sac carcinoma	1	0.0004419
					Non-Hodgkin’s lymphoma	1	0.0004419
					Hodgkin’s lymphoma	2	0.0008838
					Oligodendroglioma	1	0.0004419
					Uveitis	361	0.1595228
					inflammatory bowel diseases (IBD)	14 (split below)	
					Colitis	4	0.0017676
					Crohn’s disease	10	0.0044189
					Infections*	471 (split below)	
					Bacterial	29	0.0128148
					Viral	107 (split below)	

				Epstein Barr infection	8	0.0035351		
				Varicella zoster infection	34	0.0150243		
				Chickenpox	16	0.0070703		
				Herpes Zoster	18	0.007954		
				HPV	2	0.0008838		
Beukelman, et al, 2013, USA[32]	F: 5,421	Median 10.1 yrs (IQR 5.9-13.3)	Median 1.2 yrs (IQR 0.6-2.4)		8503			White 4,402; Black 1,448; Latino 1,721; Other/unknown 932
				Infections			(300/100,000 [95% CI 216-406] yrs split below)	
				Herpes zoster			225 /100,000 yrs (95% CI 154–318)	
				Salmonella			86/100,000 yrs (95% CI 44–150)	
				Coccidioides			35/100,000 yrs (95% CI 11–81)	
				Nontuberculous Mycobacterium			21/100,000 yrs (95% CI 4–61)	
				Nocardia			7/100,000 yrs (95% CI 0.2-39)	
				Pneumocystis			7/100,000 yrs (95% CI 0.2-39)	
				Toxoplasma			7/100,000 yrs (95% CI 0.2-39)	
				Blastomyces			0/100,000 yrs (95% CI 0.2-39)	
				Histoplasma			0/100,000 yrs (95% CI 0-36)	

						JC virus			0/100,000 yrs (95% CI 0-36)
						Cryptococcus			0/100,000 yrs (95% CI 0-36)
						Tuberculosis			0/100,000 yrs (95% CI 0-36)
						Legionella			0/100,000 yrs (95% CI 0-36)
						Listeria			0/100,000 yrs (95% CI 0-36)
						Aspergillus			0/100,000 yrs (95% CI 0-36)
Schenck, et al, 2018, Germany[33]	F: 8,043	Mean 11.6 yrs		Mean 4.2 yrs			12,269			
						Type 1 Diabetes	58	0.0047274		
						Type 2 Diabetes	2	0.000163		
						Mature onset diabetes of the young	2	0.000163		
Angeles-Han, et al, 2015, USA[34]	F: 208		Median 6.4 (2.9-11.7) yrs	Median 3.4, range (1.8 – 5.9) yrs	Mean 15.1± 12 months		287			Caucasian 221, African American 36, Others 30
						Uveitis	52	0.1811847	11.9989847	
						Cataracts	14	0.0487805	3.230495881	
						Glaucoma	7	0.0243902	1.615247941	
						Synechiae	12	0.0418118	2.76899647	
						Band keratopathy	11	0.0383275	2.538246764	
						Cystoid macular oedema	7	0.0243902	1.615247941	

					Cataract extraction	3	0.010453	0.692249117	
Thiele, et al, 2021, Germany[35]	F: 2,490	Median 12.3, IQR (8.6-15.2)				3258			Caucasian 95%, African and Asian 5%
					Pneumonia	30	0.0092081		
					Herpes zoster	34	0.0104359		
					Varicella	23	0.0070595		
					SAE infection	108	0.0331492		
					Infections	813 (split below)			
					Upper respiratory tract infection	412	0.1264579		
					Otitis media	28	0.0085942		
					Urinary tract infection	32	0.009822		
					Tonsillitis	67	0.0205648		
					Gastroenteritis	63	0.019337		
					Streptococcal infection	20	0.0061387		
Kearsley-Fleet, et al, 2022, multiple countries[36]	F: 2,014	Mean, 6.5 (SD 4.6) Median 14, IQR (10–17) yrs	Median 1, IQR (0, 4) yrs, N=2894	Median 5, IQR (3, 9) yrs, N=2926		2693			
					MAS: macrophage activation syndrome (systemic JIA only)	37/250	0.148	29.6	
					Varicella	822/2238	0.3672922	73.45844504	
					Tuberculosis	17	0.0063127	1.262532492	
					Uveitis	556	0.2064612	41.29223914	
	F: 1,046	Mean 13.2 (SD 6.1) Median 22, IQR (19–25) yrs	Median 3, IQR (1, 7) yrs, N=1531	Median 14, IQR (7, 18) yrs, N=1514		1541			
					Uveitis	238	0.1544452	11.03179753	
					MAS: macrophage activation syndrome (systemic JIA only)	5/82	0.0609756	4.355400697	
					Varicella	127/871	0.1458094	10.41495818	

					Tuberculosis	2/1541	0.0012979	0.092704181	
	F : 2,476	Mean 6.5 SD (4.5) Median 13, IQR (9–17) yrs	Median 3, IQR (1–6) yrs	Median 6, IQR (3–9) yrs		3562			
					Uveitis	676/3484	0.1940299	32.33830846	
					MAS: macrophage activation syndrome (systemic JIA only)	62/366	0.1693989	28.23315118	
					Varicella	1166/2312	0.5043253	84.05420992	
					Tuberculosis	54/3006	0.0179641	2.994011976	
Krause, et al, 2017, USA[37]	F: 57	Mean 8.6 (±5.1) yrs at index	Median 1.3, range (0.2–4.0) yrs	Mean 6.3 (±4.4) yrs		89			Caucasian 75, Hispanic 2, Black 4, Asian 3, Other 1, Unknown 4
					Joint surgeries	6	0.0674157	10.70090958	
					Non-joint surgeries	14	0.1573034	24.96878901	
					Non-surgical hospitalisations (due to infection, JIA management etc.)	12	0.1348315	21.40181915	
					Hospitalisations for depression	5	0.0561798	8.917424648	
					Pregnancies	9/28	0.3214286	51.02040816	
					Depression	7	0.0786517	12.48439451	
Lee, et al, 2020, USA[38]	F: 10,233	Mean 13.86 ±3.44 yrs		Mean 2.68 yrs SD 2.34		15210			
					Type 1 diabetes	18	0.0011834	44/100,000 person yrs (95% CI 28-70)	
					Type 2 diabetes	27	0.0017751	66/100,000 person yrs (95% CI 45-96)	
					Hyperlipidaemia	191.646	0.0126	4.701492537	
					Hypertension	158.184	0.0104	3.880597015	
					Obesity	305.721	0.0201	7.5	

Horneff, et al, 2022, Germany[39]	aged 2 to 15 yrs		546	
		Allergic rhinitis	76	0.1391941
		Predominantly allergic asthma	15	0.0274725
		Amyloidosis	<5	<5/546
		Anaemia	15	0.0274725
		Phobic anxiety disorders	17	0.0311355
		Essential hypertension	<5	<5/546
		Atopic dermatitis	80	0.1465201
		Persistent somatoform pain disorder	46	0.0842491
		Ulcerative colitis	<5	<5/546
		Depression	21	0.0384615
		Diabetes mellitus	5	0.0091575
		Iron deficiency	8	0.014652
		Post-viral fatigue syndrome	0	0
		Fibromyalgia	13	0.0238095
		Autoimmune thyroiditis	6	0.010989
		Thyrotoxicosis	0	0
		Hypothyroidism	8	0.014652
		Lack of expected normal physiological development	26	0.047619
		Crohn’s disease	12	0.021978
		Migraine	14	0.025641
		Kidney disease (chronic)	<5	<5/546
		Psoriasis	21	0.0384615
		Sicca syndrome	5	0.0091575
		Uveitis	59	0.1080586
			849	
		Allergic rhinitis	94	0.1107185
		Predominantly allergic asthma	25	0.0294464
		Amyloidosis	11	0.0129564
		Anaemia	24	0.0282686
		Phobic anxiety disorders	30	0.0353357

					Essential (primary) hypertension	<5	<5/849
					Atopic dermatitis	105	0.1236749
					Persistent somatoform pain disorder	71	0.0836278
					Ulcerative colitis	<5	<5/849
					Depression	19	0.0223793
					Diabetes mellitus	7	0.008245
					Iron deficiency	7	0.008245
					Post-viral fatigue syndrome	<5	<5/849
					Fibromyalgia	9	0.0106007
					Autoimmune thyroiditis	13	0.0153121
					Thyrotoxicosis	<5	<5/849
					Hypothyroidism	18	0.0212014
					Lack of expected normal physiological development	31	0.0365135
					Crohn‘s disease	<5	<5/849
					Migraine	29	0.0341578
					Kidney disease (chronic)	0	0
					Osteoporosis	5	0.0058893
					Psoriasis	39	0.0459364
					Sicca syndrome	<5	<5/849
					Uveitis	90	0.1060071
Bolt, IB. et al, 2008, Switzerland[40]	F: 174	≤18 yrs (see comments)	Mean 6.49, range (0.67–15.67) yrs	Mean 4.82, range (0.08–16.58) yrs		265	
					Synechiae	10	0.0377358	7.829014327
					Cataracts	9	0.0339623	7.046112894
					Band keratopathy	5	0.0188679	3.914507164
					Glaucoma	3	0.0113208	2.348704298
					Macular oedema	2	0.0075472	1.565802865
					Uveitis	35	0.1320755	27.40155014
					Impaired visual acuity	1	0.0037736	0.782901433
					Legal blindness in one eye	2	0.0075472	1.565802865

Simon, et al, 2020, USA[41]		< 18 yrs						
	F: 6,173	Mean 10.6±4.5 yrs						
					Multiple sclerosis	1	0.0001158	
					Asthma	777	0.0001158	
					Anxiety	295	0.0001158	
					Depression	208	0.0001158	
					Diabetes mellitus	92	0.0001158	
					Anaemia	498	0.0001158	
					Hypertension	2	0.0001158	
					Hyperlipidaemia	174	0.0001158	
					Malignancy	56	0.0001158	
	F: 6,587	Mean 10.5±4.5 yrs						
					Multiple sclerosis	1	0.0001092	
					Asthma	887	0.0001092	
					Anxiety	317	0.0001092	
					Depression	214	0.0001092	
					Diabetes mellitus	84	0.0001092	
					Anaemia	479	0.0001092	
					Hypertension	6	0.0001092	
					Hyperlipidaemia	202	0.0001092	
					Malignancy	49	0.0001092	
Schulz, C. et al, 2022, Germany[42]	F 333	Mean 6.8 yrs (range: 2-22)	Median 6.0 yrs (<1-16.4)	Median 4.4 yrs (range: 0.9–14.1)				Europe 413, Asia 31, Turkish 28 , Africa Arab 2, mixed countries of origin 30 and
					Hypothyroidism	6	0.012024	
					Autoimmune thyroiditis	8	0.0160321	
					Other thyroid disorder	2	0.004008	
					Coeliac disease	3	0.006012	
					Scleroderma	1	0.002004	

Swart, et al, 2018, Italy[43]	F: 5,584	≤ 18 yrs	Median 5.4 yrs (IQR : 2.4–10.0)	Med 6.2 (IQR 2.8–10.9) yrs	Med 5.3 (IQR 2.7–8.8) yrs	Overlap syndrome	1	0.002004	unknown 23
							8274		
						Infections	1523 (split below)		
						Serious/targeted infections (Epstein– Barr virus, cytomegalovirus, papilloma virus, herpes zoster primary and reactivation, and opportunistic infections)	674	0.08146	
						Tuberculosis	27	0.0032632	
						Other infections	822	0.0993474	
						Macrophage activation syndrome	75	0.0090645	
						Autoimmune disease	50 (split below)		
						Inflammatory bowel disease (IBD)	21	0.0025381	
						Multiple sclerosis	2	0.0002417	
						Other autoimmune diseases excluding IBD, uveitis, and demyelinisation disorder	18	0.0021755	
						Malignancies	16 (split below)		
						Leukaemia’s	3	0.0003626	
						Lymphomas	2	0.0002417	
						Hematopoietic neoplasms (excluding leukaemia and lymphoma)	1	0.0001209	
						Neoplasm (other)	10	0.0012086	
						Gastrointestinal (GI) ulcer/GI bleed/GI perforation	17	0.0020546	
						Pregnancy	9	0.0010877	
						Congestive heart failure	1	0.0001209	

	F: 2,670	Median 7.6 yrs (IQR 3.2–11.7)	Med 6.1 (IQR 3.5–9.5) yrs		3990	
				Infections	1509 (split below)	
				Serious/targeted infections (Epstein– Barr virus, cytomegalovirus, papilloma virus, herpes zoster primary and reactivation, and opportunistic infections)	171	0.0428571
				Tuberculosis	0	0
				Other infections	1338	0.3353383
				Macrophage activation syndrome	11	0.0027569
				Autoimmune disease	50 (split below)	
				Inflammatory bowel disease (IBD)	23	0.0057644
				Multiple sclerosis	0	0
				Other autoimmune diseases excluding IBD, uveitis, and demyelinisation disorder	24	0.006015
				Malignancies	13 (split below)	
				Leukaemia	2	0.0005013
				Lymphoma	5	0.0012531
				Hematopoietic neoplasms (excluding leukaemia and lymphoma)	2	0.0005013
				Neoplasm (other)	4	0.0010025
				Gastrointestinal (GI) ulcer/GI bleed/GI perforation	4	0.0010025
				Pregnancy	7	0.0017544
				Congestive heart failure	0	0
Giancane, et al, 2019, Italy[44]	F: 189	Med 3.1 yrs (IQR 1.8–5.8)	Med 7.7 yrs (IQR 6.1–10.2)		239	
				Ocular	21	0.0878661

					Muscular atrophy	19	0.0794979
					Osteoporosis	3	0.0125523
					Subcutaneous atrophy	9	0.0376569
					Growth failure	19	0.0794979
					Pubertal delay	5	0.0209205
					Avascular necrosis	0	0
					Amyloidosis	0	0
					Diabetes	0	0
	F: 219	Med 2.5 yrs (IQR 1.6–4.7)	Med 6.9 yrs (IQR 5.4–9.7)			269	
					Ocular	7	0.0260223
					Muscular atrophy	8	0.0297398
					Osteoporosis	1	0.0037175
					Avascular necrosis	1	0.0037175
					Subcutaneous atrophy	6	0.0223048
					Growth failure	1	0.0037175
					Diabetes	0	0
					Amyloidosis	0	0
Grassi, et al, 2007, Italy[45]	F: 244	mean 4.9 ± 3.6 yrs		mean 7.6 ± 5.6 yrs		309	
					Synechiae	12	0.038835	5.109862034
					Band keratopathy	8	0.02589	3.406574689
					Cataract	6	0.0194175	2.554931017
					Glaucoma	2	0.0064725	0.851643672
					Retinal detachment	1	0.0032362	0.425821836
					Iris atrophy	1	0.0032362	0.425821836
					Blindness*	4	0.012945	1.703287345
					Uveitis	62	0.2006472	26.40095384
					Severe visual loss	8	0.02589	3.406574689
Horneff, et al, 2016, Germany[46]				13,198 observation yrs		3695	

Lovell, DJ. et al, 2021, USA[47]	F: 867	Mean 6.6 yrs (SD 4.5)	Mean 8.0 (SD 4.8)

Suspected Malignancies	12 (split below)		
Hodgkin's lymphoma	2	0.0005413	4.10117E-05
non-Hodgkin's lymphoma	1	0.0002706	2.05058E-05
acute lymphatic leukaemia (ALL)	2	0.0005413	4.10117E-05
lymphoproliferative disorder	1	0.0002706	2.05058E-05
thyroid carcinoma	1	0.0002706	2.05058E-05
yolk sac carcinoma	1	0.0002706	2.05058E-05
anaplastic ependymoma	1	0.0002706	2.05058E-05
	1332		
Autoimmune diseases	1085 (split below)		
Addison disease	11	0.0082583	
Autoimmune/Hashimoto's thyroiditis	16	0.012012	
Graves' disease	1	0.0007508	
Morphoea	12	0.009009	
Multiple sclerosis	1	0.0007508	
Takayasu arteritis	1	0.0007508	
Type 1 diabetes mellitus	17	0.0127628	
Vitiligo	5	0.0037538	
Myasthenia Gravis	0	0	
Pernicious Anaemia	0	0	
Primary biliary cirrhosis	0	0	
Associated conditions		0	
Autoimmune disease not elsewhere classified	8	0.006006	
Chronic glomerulonephritis	1	0.0007508	
Chronic urticaria	49	0.0367868	
Episcleritis/scleritis	5	0.0037538	
Erythema nodosum	1	0.0007508	
Haemolytic anaemia	1	0.0007508	
Immune thrombocytopenia purpura	2	0.0015015	
Leukocytoclastic vasculitis	2	0.0015015	

						Pulmonary fibrosis/interstitial lung disease	15	0.0112613
						Raynaud's syndrome	19	0.0142643
						Sjögren's syndrome/sicca syndrome	3	0.0022523
						Thrombocytopenia purpura	1	0.0007508
						Vasculitis	8	0.006006
Barthel, et al, 2015, Germany[48]	F: 2,073	Mean 6.1 ± 3.9 yrs			8389 patient-yrs		3071	
						Inflammatory bowel disease	11 (split below)	
						Crohn Disease	8	0.002605	0.000310527
						Ulcerative colitis	3	0.0009769	0.000116448
Sahin, et al, 2021, Turkey[49]	F: 289		8.8 ± 4.7 yrs (mean) (range 9 months to 17 yrs)	4.0 ± 1.9 yrs (mean)			500	
						Uveitis	34	0.068
						Osteoporosis	8	0.016
						Delayed puberty	6	0.012
						Autoimmune Thyroiditis	4	0.008
						Inflammatory Bowel Disease (IBD)	2	0.004
						Depression	2	0.004
						Hypertension	1	0.002
						Amyloidosis	1	0.002
						familial Mediterranean fever (FMF)	63	0.126
						Psoriasis	15	0.03
						Cataract	4	0.008
						Band keratopathy	1	0.002
						Posterior synechia	1	0.002
Pohjankoski, et al, 2010, Finland[50]	F: 283		Mean 7 (1–15) yrs				417	
						Uveitis	91	0.2182254
						Diabetes mellitus type 1	9	0.0215827

Cakan, et al, 2020, Turkey[51]	F: 202				Mean 29.2±18.0 months	Coeliac disease	3	0.0071942	
						Hypothyroidism	3	0.0071942	
							405		
Cosickic, et al, 2017, Bosnia and Herzegovina[52]	F: 61	Med 7.1 rs (IQR, 2.3 - 10.6 yrs)				Uveitis	27	0.0666667	2.283105023
							97		
						Synechia	3	0.0309278	
						Band keratopathy	2	0.0206186	
						Cataracts	2	0.0206186	
						Glaucoma	1	0.0103093	
						Uveitis	14	0.1443299	
Berthold, et al, 2022, Sweden[53]	F: 433			Med 8.7 yrs ((IQR) 3.7 – 12.7 yrs)			640		
						Depressive disorder	93	0.1453125	
						Anxiety disorder	111	0.1734375	
							105		
Lien, et al, 2003, Norway[54]	F: 80	Mean 17.0 ±1.8 yrs	Mean 2.8±1.4 yrs		Mean 14.2 ±1.4 yrs	Osteopenia/Low total-body BMC	42/103	0.407767	
						Osteoporosis (BMD Z score≤-2.5)	5/103	0.0485437	
							26		
Saraux, et al, 2015, France[55]	F: 18	Mean 13.5 yrs (SD 5.84)				Infections	3	0.1153846	
						Bacterial infection	3	0.1153846	
						Viral infection	1	0.0384615	

Alberdi-Saugstrup, et al, 2017, Nordic countries[56]	F: 135		Median 6.0 (IQR 2.8-10.0)				193			
						Uveitis	30	0.1554404		
Amine, et al, 2011, Morrocco[57]	F: 32	Mean 11 ± SD 3.3 yrs range [2-16]	Mean 7 ±SD 3.4 range [1–15]	Mean 3 ± SD 2.4 yrs range [1-12]			58			
						Obesity	13	0.2241379		
Adulthood / JIA										
Smitherman, et al, 2022, USA[58]	F: 144	Mean 20±1.4 yrs	CARRA (or any other) JIA registry, Age onset will have childhood onset disease.				187			White 147; Hispanic 23; Black 9; Other* 8
						Self-reported symptoms of pain or discomfort	122	0.6524064		
						Self-reported symptoms of anxiety or depression	84	0.4491979		
Mena-Vazquez, et al, 2022, Spain[14]	F: 35	Mean 25 yrs +/- 6.1		161.4 months (103.6 - 220.4 IQR)	24 months		52			Caucasian 50
						Depression	13	0.25	125	
						Arterial hypertension	1	0.0192308	9.615384615	
						Diabetes melllitus	1	0.0192308	9.615384615	

				Dyslipidaemia	4	0.0769231	38.46153846
				Obesity (BMI ≥30)	7	0.1346154	67.30769231
				Uveitis	11	0.2115385	105.7692308
Raab, et al, 2013, Germany[59]	F: 196	Mean 19.7 yrs (SD 2.8)	Mean 10.6 yrs (since symptom onset) 5.6 SD		344		
				Cardiovascular diseases	34 (split below)		
				Hypertension	25	0.0726744	6.856077227
				Cardiac insufficiency	1	0.002907	0.274243089
				Pulmonary diseases	28 (split below)		
				Asthma	26	0.0755814	7.130320316
				Gastrointestinal diseases	14 (split below)		
				Crohn's disease	4	0.0116279	1.096972356
				Ulcerative colitis	3	0.0087209	0.822729267
				Gastrointestinal ulcer	1	0.002907	0.274243089
				Liver and gallbladder diseases	3	0.0087209	0.822729267
				Diseases of the kidneys or the Urinary tract	16 (split below)		
				Renal insufficiency	1	0.002907	0.274243089
				Kidney stone	4	0.0116279	1.096972356
				Amyloidosis	3	0.0087209	0.822729267
				Gynaecological diseases	12	0.0348837	3.290917069
				Haematological diseases	23 (split below)		
				Anaemia	11	0.0319767	3.01667398
				Iron deficiency	6	0.0174419	1.645458534
				Allergic diseases	84 (split below)		
				Allergic rhinitis	50	0.1453488	13.71215445
				House dust mite allergy	16	0.0465116	4.387889425
				Thyroid/endocrine diseases	23 (split below)		

Aggarwal, et al, 2006, India[60]	M: 30	22.6 ± 5.12 yrs	Mean 11.37 ± 2.67 yrs	Mean 11.27 ± 5.39 yrs

Hypothyroidism	7	0.0203488	1.919701624
Hashimoto thyroiditis	4	0.0116279	1.096972356
Diabetes mellitus	3	0.0087209	0.822729267
Skin diseases	67 (split below)		
Psoriasis	22	0.0639535	6.03334796
Atopic dermatitis	30	0.0872093	8.227292672
Eye diseases	80 (split below)		
Uveitis	59	0.1715116	16.18034226
Cataract	3	0.0087209	0.822729267
Glaucoma	9	0.0261628	2.468187802
Other eye diseases	8 (split below)		
Benign eye tumour	1	0.002907	0.274243089
Episcleritis	1	0.002907	0.274243089
Scleritis	2	0.005814	0.548486178
Sicca syndrome/ sicca symptoms	3	0.0087209	0.822729267
Chronic conjunctivitis	1	0.002907	0.274243089
Astigmatism	1	0.002907	0.274243089
Neurologic diseases	41 (split below)		
Migraine	30	0.0872093	8.227292672
Epilepsy	5	0.0145349	1.371215445
Psychiatric disorders	32 (split below)		
Depression	17	0.0494186	4.662132514
Anxiety disorder	7	0.0203488	1.919701624
Other comorbidities	30 (split below)		
Osteoporosis	11	0.0319767	3.01667398
	30		
Osteopenia (Lumbar Spine)	14	0.4666667	41.40786749
Osteopenia (Hip)	15	0.5	44.36557232
Osteopenia (Distal third of forearm)	15	0.5	44.36557232

						Osteoporosis (Lumbar Spine)	8	0.2666667	23.66163857	
						Osteoporosis (Hip)	7	0.2333333	20.70393375	
						Osteoporosis (Distal third of forearm)	12	0.4	35.49245785	
						Joint deformities	5	0.1666667	14.78852411	
Arvidsson, et al, 2010, Norway[61]	F: 44	Mean 35.2 yrs +/- 3.6 (27.9-44.4)	Mean 5.5 yrs +/- 3.1 (0.8-14.5 yrs)	Mean 29.7 yrs +/- 3.3 yrs (24.4-38.0)	Mean 27 yrs		60			
						Craniofacial growth disturbance	34	0.5666667	20.98765432	
						Micrognathia	16	0.2666667	9.87654321	
Packham, et al, 2002, UK[62]	F: 176	Mean 35.4 yr (range 19-78, SD = 11.1)	Mean 7.1 yr (range 0.8–15.9, S.D.=4.5)	Mean 28.3 yr (range 8–73, S.D.=10. 8)			246			Asian 1, Afro-Caribbean 1, Caucasian descent 244
						Uveitis	22%	0.22	7.77385159	
						Glaucoma	4.10%	0.041	1.448763251	
						Amyloidosis	8.90%	0.089	3.144876325	
						Cataracts	10.20%	0.102	3.604240283	
						Required eye surgery	10.60%	0.106	3.745583039	
						Sicca syndrome	15.00%	0.15	5.300353357	
						At least one major prosthetic joint replacement	51.2% (split below)			
						Hip prosthetic	43.90%	0.439	15.51236749	
						Revision hip prostheses	15.40%	0.154	5.441696113	
						Knee prosthetic	27.60%	0.276	9.752650177	
						Shoulder prosthetic	4.90%	0.049	1.731448763	
						Elbow prosthetic	3.30%	0.033	1.166077739	
						Wrist prosthetic	1.60%	0.016	0.565371025	
						Ankle prosthetic	1.60%	0.016	0.565371025	
Mars, et al, 2019, Finland[63]	F: 92	Mean 32.4 ± 13.4 yrs					119			

						Cardiovascular diseases*	17	0.1428571	
						Diabetes	5	0.0420168	
						Eye disorders	16	0.1344538	
						Gastrointestinal disorders	11	0.092437	
						Infections	27	0.2268908	
						Malignancies	2	0.0168067	
						Mental disorders	12	0.1008403	
						Neurological disorders	18	0.1512605	
						Other musculoskeletal disorders	38	0.3193277	
						Rheumatic diseases	76	0.6386555	
						Skin disorders	19	0.1596639	
Dimopoulou, et al, 2017, Greece[64]	F: 72	Mean 25 yrs SD 5.9, median 24 yrs range 8-41 yrs	Median 7 yrs Mean 7.7 yrs SD 4 (range 1-15.5)		Median 17 yrs range 5-31.5	Mean 17.2 yrs SD 6.7 yrs	102		
						Uveitis	11	0.1078431	6.26994984
						Prosthetic arthroplasty	9	0.0882353	5.12995896
						Mandibular correction	1	0.0098039	0.56999544
						Arthrodesis of the radiocarpal joint	1	0.0098039	0.56999544
						Hands or foot small joint correction	4	0.0392157	2.27998176
						Blindness in one eye	2	0.0196078	1.13999088
						Leg length discrepancy ≥0.5cm	37	0.3627451	21.08983128
						Mandibular dysfunction (such as reduced mouth opening, frontal facial asymmetry or retrognathia)	29.40%	0.0028824	0.167578659
Rebane, et al, 2022, Finland[65]	F: 62	Mean 23 yrs (range 16-23)				17 yrs	97		
						Coeliac disease	1	0.0103093	0.606428138
						Crohn’s disease	2	0.0206186	1.212856277
						Irritable bowel syndrome (IBS)	2	0.0206186	1.212856277
						Hypothyroidism	1	0.0103093	0.606428138
						Depression	2	0.0206186	1.212856277

					Panic disorder	2	0.0206186	1.212856277	
					Uveitis	4	0.0412371	2.425712553	
					Ulcerative colitis	0	0	0	
Minden, et al, 2019, Germany[66]	F: 464	Mean 23.4±3.9 yrs	Mean 14.3±6.1 yrs	Mean 9.1±3.7 yrs		701			
					Arthroplasty	36	0.0513552	5.643429324	
					Eye surgery	22	0.0313837	3.448762365	
					Uveitis	63	0.0898716	9.876001317	
					Synovectomy	112	0.1597718	17.55733567	
Smith, et al, 2019, USA[67]	F: 170	Mean 30.55±0.39 yrs (Age at estimated due date)	Mean 20.32±7.32 yrs			170			White 145; African American 3; Asian/Pacific islander 2; Native America 3; Other 1
					Comorbid pregestational conditions				
					Thyroid dysfunction	13	0.0764706		
					Hypertension	21	0.1235294		
					Diabetes Mellitus (Type 1 and 2)	0	0		
					Other autoimmune diseases				
					Inflammatory bowel disease	11	0.0647059		
					Sjögren’s syndrome or UCTD	3	0.0176471		
					Antiphospholipid syndrome	1	0.0058824		
					Psoriasis	11	0.0647059		
					Lupus	4	0.0235294		
					Obstetric complications				
					Preeclampsia	15	0.0882353		

					PIH without preeclampsia	10	0.0588235	
					Placenta previa and/or abruption	4	0.0235294	
					Preterm labour	25	0.1470588	
					GDM	13	0.0764706	
					Fever during pregnancy	47	0.2764706	
					Caesarean section	70	0.4117647	
Barth, et al, 2016, Germany[68]	F: 1,617	range (18–73) yrs	Median 8, range (0–15) yrs IQR (4-11)	Median 27; range (3–70) yrs IQR (15-40)		2592		
					Pain/discomfort	(split below)		
					Moderate problems	1395	0.5381944	
					Severe problems	94	0.0362654	
					Anxiety/depression:	(split below)		
					Moderate problems	624	0.2407407	
					Severe problems	54	0.0208333	
					Severe disability	783	0.3020833	
Simon, et al, 2020, USA[41]		≥ 18 yrs			1 year			
	F: 4,506	Mean 37.0 ±16.5 yrs				5789 (split below)		
					Multiple sclerosis	20	0.0034548	3.454828122

						Asthma	472	0.0815339	81.53394369
						Anxiety	555	0.0958715	95.87148039
						Depression	653	0.1128001	112.8001382
						Diabetes mellitus	440	0.0760062	76.00621869
						Anaemia	948	0.1637589	163.758853
						Hypertension	103	0.0177924	17.79236483
						Hyperlipidaemia	1162	0.2007255	200.7255139
						Malignancy	278	0.0480221	48.0221109
	F: 4,315	Mean 34.6 ±14.7 yrs					5638 (split below)		
						Multiple sclerosis	19	0.00337	3.369989358
						Asthma	486	0.0862008	86.20078042
						Anxiety	549	0.097375	97.37495566
						Depression	622	0.1103228	110.3228095
						Diabetes mellitus	371	0.0658035	65.80347641
						Anaemia	655	0.1161759	116.1759489
						Hypertension	103	0.0182689	18.26888968
						Hyperlipidaemia	1023	0.1814473	181.4473217
						Malignancy	162	0.0287336	28.73359347
Packham and Hall, 2003, UK[69]	F: 177	Mean 35.4±11.1 yrs (19-78)	Mean 7.1±4.2 yrs (range 0.8-15.9)	Mean 28.7±10.8 yrs (range 8-73)			177		
						Primary ovarian failure	9 (split below)		
						Primary ovarian failure (unrelated to medication use)	6	0.0338983	1.181125613
Ma, et al, 2022, Taiwan[70]	F: 1,155	0-30 yrs			249 593 months		2791		
						Obstructive sleep apnoea (OSA)	31	0.0111071	0.64 per 10000 person yrs

							95% CI 0.5-0.82)
					Allergic rhinitis	401	0.1436761	0.006907699
					Obesity	19	0.0068076	0.000327297
					Osteoporosis	7	0.0025081	0.000120583
					Disorders of mineral metabolism	3	0.0010749	5.16785E-05
					Hypocalcaemia/hypomagnesemia	0	0	0
Zhang-Jian, et al, 2020, Taiwan[71]	F: 549	Age at pregnancy is 24.4 ± 3.4 yrs				549		
					Pregnancy outcomes			
					Stillbirth	4	0.007286	
					Foetal distress	32	0.0582878	
					Foetal abnormalities, any	38	0.0692168	
					Preeclampsia	17	0.0309654	
					Gestational diabetes	24	0.0437158	
					Pregnancy-related hypertension	24	0.0437158	
					Antepartum haemorrhage	55	0.1001821	
					Severe postpartum haemorrhage	25	0.0455373	
					Chorioamnionitis	5	0.0091075	
					Caesarean delivery	155	0.2823315	
					Low birth weight <2500g	64	0.1165756	
					Prematurity <37 weeks	49	0.0892532	
Rypdal, et al, 2021, USA[72]	F: 297	≥19 yrs	Mean 5.7 IQR 2.6-9.8	17.6 yrs IQR (16.7-18.4)		434		
					Uveitis	96	0.2211982	12.56807708
					At least 1 ocular complication	31/80	0.3875	22.01704545
					Cataract	25/80	0.3125	17.75568182
					Glaucoma	22/80	0.275	15.625
					Synechiae	14/80	0.175	9.943181818
					Macular oedema	8/80	0.1	5.681818182
					Band keratopathy	7/80	0.0875	4.971590909
					Epiretinal membrane	3/80	0.0375	2.130681818

				Hypotony	3/80	0.0375	2.130681818	
				Phthisis	3/80	0.0375	2.130681818	
				Blindness <3/60 - binocular	2/80	0.025	1.420454545	
				Blindness <3/60 - monocular	3/80	0.0375	2.130681818	
Childhood / jSLE								
Davis, et al, 2023, USA[73]	F: 571	18.4±3.7 yrs (10-24 yrs)			650			White 367; Black 110; Hispanic 91; Asian 41; Unknown 41
				Depression	117	0.18		
				Anxiety	78	0.12		
				Adjustment disorder	65	0.1		
				Seizure/stroke disorder	52	0.08		
Garf, et al, 2015, Egypt[74]	F: 49	Mean 13.1±3.0 yrs	Mean 3.9±3.0 yrs (range		70			

				0.2-12 yrs)					
						Vitamin D deficiency	28	0.4	
						Vitamin D insufficiency (≤30/ng/ml)	42	0.6	
Paim-Marques, et al, 2019, Brazil[75]	F: 61	Mean 17.9±3.07 yrs	Mean 13.56 ± 2.60 yrs (corneal verticillata) 12.63 ± 2.9 (no corneal verticillata)		Mean 5.06 ± 2.04 yrs (with corneal verticillata)		76		
						Visual acuity reduction	25	0.3289474	65.00936135
						Posterior subcapsular cataracts	4	0.0526316	10.40149782
						Keratitis	40	0.5263158	104.0149782
						Crystalline abnormalities	1	0.0131579	2.600374454
						Corneal vortex keratopathy	16	0.2105263	41.60599126
Silva, et al, 2015, Brazil[76]	F: 732	Median 14.5 (6–20) yrs, IFI; 17.0 (2–25.9) non-IFI	Median 12.4 (range 5.1–17.2) yrs, IFI; 11.8 (0.2–17.8) non IFI	Median 1.0 (range 0–7.3) yrs, IFI; 4.7 (range 0–23.4) non- IFI			852			White 592/830
						Invasive Fungal Infections (Proven)	22 (split as follows)		
						Candidiasis	13	0.0152582	
						Aspergillosis	6	0.0070423	
						Cryptococcosis	2	0.0023474	
						Histoplasmosis	1	0.0011737	
						Arterial hypertension	119/843	0.1411625	
						Acute renal failure	37/841	0.0439952	
						Chronic renal failure	27/842	0.0320665	

				Renal replacement Therapy	28/842	0.0332542	
Knight, et al, 2016, USA[77]	F: 805	Mean 14.7 SD 2.0 (10-18 yrs)			970		White 141; African American 410; Latino 265; Other 154
				Depression diagnosis	181	0.1865979	
				Anxiety	72	0.0742268	
				Acute stress/adjustment	60	0.0618557	
				Other psychiatric disorder	175	0.1804124	
				Seizures/stroke	155	0.1597938	
Das Chagas Medeiros, et al, 2016, Brazil[78]	F: 52	Mean 15.5±2.0 yrs	Mean 10.4±6.4 yrs		60		
				Antiphospholipid syndrome	3/49	0.0612245	5.886970173
				Hypothyroidism	1/37	0.027027	2.598752599
				Thrombosis	7/60 (split below)		
				Deep vein thrombosis	1	0.0166667	1.602564103
				Mesenteric artery thrombosis	1	0.0166667	1.602564103
				CVA	1	0.0166667	1.602564103
				Pulmonary embolus	1	0.0166667	1.602564103
				Portal vein thrombosis	1	0.0166667	1.602564103
				Renal vein thrombosis	1	0.0166667	1.602564103
				DVT + PE	1	0.0166667	1.602564103
				Neoplasia	1/58	0.0172414	1.657824934
				hypertension	27/60	0.45	43.26923077
				Cardiovascular diseases	3/58 (split below)		
				Congestive heart failure			
				Coronary artery disease			
				Aortic and mitral valve disorder			
				Diabetes	0/60	0	0
				Seizure	7/60	0.1166667	11.21794872
				Psychosis	2/60	0.0333333	3.205128205

Frittoli, et al, 2016, Brazil[79]	F: 38	Mean 14.5±2.84 yrs				41		
					Cognitive dysfunction	17	0.4146341	
					Anxiety symptoms	8	0.195122	
					Depression symptoms	1	0.0243902	
Holland , et al, 2018, USA[80]	F: 869	Mean 12.2±3.1 yrs	Mean 3.81±2.98 yrs	Mean 3.8 yrs		1048		
					Ocular			
					Cataracts	4.20%	4.008E-05	0.010546404
					Retinal change	1.70%	1.622E-05	0.004268783
					Neuropsychiatric			
					Cognitive impairment	8.40%	8.015E-05	0.021092808
					Seizures requiring therapy	3.90%	3.721E-05	0.00979309
					Cerebrovascular accident	2.9%/0.4%	0.0069179	1.820510245
					Cranial/peripheral neuropathy	2.40%	2.29E-05	0.006026517
					Transverse myelitis	0.40%	3.817E-06	0.001004419
					Renal			
					End-stage renal disease	1.30%	1.24E-05	0.003264363
					Pulmonary			
					Pulmonary hypertension	0.40%	3.817E-06	0.001004419
					Pleural fibrosis	1.30%	1.24E-05	0.003264363
					Shrinking lung	0.80%	7.634E-06	0.002008839
					Pulmonary fibrosis	0.90%	8.588E-06	0.002259944
					Pulmonary infarction	0%	0	0
					Cardiovascular			
					Angina/artery bypass	0.10%	9.542E-07	0.000251105
					Myocardial infarction	0.1 / 0.1	0.0009542	0.251104861
					Cardiomyopathy	0.90%	8.588E-06	0.002259944
					Valvular disease	0.90%	8.588E-06	0.002259944

				Pericarditis/pericardiectomy	1.60%	1.527E-05	0.004017678
				Peripheral vascular			
				Claudication for 6 months	0.30%	2.863E-06	0.000753315
				minor tissue loss	2.40%	2.29E-05	0.006026517
				significant tissue loss	0.7/0.3%	0.0022265	0.585911343
				Venous thrombus swelling, ulcer/stasis	2.50%	2.385E-05	0.006277622
				Gastrointestinal			
				Bowel infarction or resection	1.3 / 0.3%	0.0041349	1.088121066
				Mesenteric insufficiency	0.10%	9.542E-07	0.000251105
				Peritonitis	0.20%	1.908E-06	0.00050221
				Stricture/upper GI tract surgery ever	0.40%	3.817E-06	0.001004419
				Pancreatic insufficiency	0.50%	4.771E-06	0.001255524
				Musculoskeletal			
				Muscle atrophy/weakness	7.90%	7.538E-05	0.019837284
				Deforming/erosive arthritis	4.70%	4.485E-05	0.011801928
				Osteoporosis with fracture	2.60%	2.481E-05	0.006528726
				Avascular necrosis	2.7 / 1.0	0.0025763	0.677983126
				Osteomyelitis	0.40%	3.817E-06	0.001004419
				Ruptured tendons	0.10%	9.542E-07	0.000251105
				Skin			
				Scarring chronic alopecia	9.10%	8.683E-05	0.022850542
				Extensive scar/panniculum	1.70%	1.622E-05	0.004268783
				Skin ulceration for 6 months	2.70%	2.576E-05	0.006779831
				Premature gonadal failure	2%	1.908E-05	0.005022097
				Diabetes	0.90%	8.588E-06	0.002259944
				Malignancy	0.20%	1.908E-06	0.00050221
Gad, et al, 2013, Egypt[81]	F: 17	Mean 12.90±2.67 3 yrs (8-16 yrs)	Mean 4.40±1.8 75 yrs		20		
				Sensorineural hearing loss (SNHL)	5	0.25	
				Mild Conductive hearing loss (CHL)	2	0.1	

Jongvilaikasem and Rianthavorn, 2021, Thailand[82]	F: 87	Mean 11.3 ± 2.4 yrs		26 months		97			
					Short final height	22	0.2268041	104.6788263	
					Mild CKD	88	0.9072165	418.7153053	
					Advanced CKD	9	0.0927835	42.82315623	
AlAhmed, et al, 2020, USA[83]	F: 306	Mean 11.39±3.8 yrs	Mean 12.04±3.62 year (Age first seen by paediatric rheumatologist)			386			Hispanic 102; Non-Hispanic 283, White 216, African American 105, Asia 33, Other 13
					Cerebrovascular accident	2	0.0051813		
					Cranial nerve disease	1	0.0025907		
					Renal dialysis	3	0.007772		
					Arterial thrombosis	3	0.007772		
					Venous thrombosis	10	0.0259067		
					Avascular necrosis	5	0.0129534		
					Autoimmune thyroiditis	10	0.0259067		
					Coeliac disease	1	0.0025907		
					T1DM	1	0.0025907		
					Other autoimmune disorder	22	0.0569948		
					Pulmonary hypertension	0	0		
					Coronary heart disease	0	0		
					Renal transplant	0	0		

Karve, S. et al, 2012, USA[12]	0-15 yrs	18 months		71		
			Asthma	8	0.1126761	75.11737089
			Atopic dermatitis	2	0.028169	18.77934272
			Allergic rhinitis	6	0.084507	56.33802817
			General diabetes	1	0.0140845	9.389671362
			Rheumatoid arthritis	14	0.1971831	131.4553991
			Multiple sclerosis	1	0.0140845	9.389671362
			Thyroiditis	1	0.0140845	9.389671362
			Fracture in post-index period	4	0.056338	37.55868545
			Psoriasis	0	0	0
			Type 1 diabetes	0	0	0

						Other CNS demyelinating disease	0	0	0	
						Vitiligo	0	0	0	
						Grave's disease	0	0	0	
Ye, Q., et al, 2021, China[84]	F: 82	Median age 169.02 months (range 31.00-220.53)	133.83, range (27.70–202.83) months		28.00, range (0–144.00) months		107			
						Cataracts	21	0.1962617		
						Steroid-induced osteonecrosis of femoral head	3	0.0280374		
						Osteoporosis	5	0.046729		
						Short stature	8	0.0747664		
						Obesity	7	0.0654206		
						Diabetes	6	0.0560748		
						HDL resistance	1	0.0093458		
						Upper respiratory tract infection	11	0.1028037		
						Pneumonia	13	0.1214953		
						EB virus infection	15	0.1401869		
						Progressive systemic sclerosis	1	0.0093458		
						Secondary Sjogren	6	0.0560748		
						Juvenile rheumatoid arthritis	1	0.0093458		
						Antiphospholipid syndrome	2	0.0186916		
Cann, et al, 2022, Australia[85]	F: 37			*12.5 yrs (mean) 13 yrs (median)*	Mean 4.3 yrs , IQR (2.5–6.1) yrs		42			Caucasian 16, Indigenous
						Cataract	2	0.047619	11.07419712	Australian 11, Asian
						retinal change	2	0.047619	11.07419712	

						CVA	3	0.0714286	16.61129568	10, Other 5
						Significant tissue loss	1	0.0238095	5.53709856	
						Venous thrombosis	1	0.0238095	5.53709856	
						Muscle weakness	1	0.0238095	5.53709856	
						AVN	1	0.0238095	5.53709856	
						Scarring alopecia	2	0.047619	11.07419712	
						Diabetes	1	0.0238095	5.53709856	
						Growth failure	3	0.0714286	16.61129568	
						Delayed puberty	1	0.0238095	5.53709856	
						Premature gonadal failure	0	0	0	
						Malignancy	0	0	0	
Medhat, et al, 2020, Egypt[86]	F: 86	Mean 12.7 ±2.6 yrs (range 6–16)	Mean 9.9±2.5, range (4–15) yrs	Median 24, IQR (12–48) months			106			
						Osteoporosis	6	0.0566038		
						Hypothyroidism	2	0.0188679		
						Diabetes mellitus	4	0.0377358		
						Hypertension	31	0.2924528		
						Hepatitis C virus infection	5	0.0471698		
						Chronic kidney disease	5	0.0471698		
						End-stage renal disease	1	0.009434		
						Avascular necrosis	1	0.009434		
						Sicca manifestations	2	0.0188679		
						Ischaemic heart disease	0	0		
						CVA	0	0		
						Malignancy	0	0		
Aggarwal, et al, 2018, India[87]	F: 250	≤ 18 yrs	Median 14 yrs range (1-17)		Median 3.5 yrs, Range (0.25-28 yrs) (248 patients); 25 patients were not followed up		273			

							Cushingoid habitus	35	0.1282051	36.63003663	
							Growth retardation	32	0.1172161	33.4903192	
							Avascular necrosis	9	0.032967	9.419152276	
							Cataract	8	0.029304	8.372579801	
							Osteopenia	10	0.03663	10.46572475	
							Infection	72	0.2637363	75.35321821	
							Hypertension	42	0.1538462	43.95604396	
							DVT	8	0.029304	8.372579801	
							Pulmonary thromboembolism	2	0.007326	2.09314495	
							Stroke	3	0.010989	3.139717425	
							Renal vein thrombosis	1	0.003663	1.046572475	
Al-Mayouf, et al, 2008, Saudi Arabia[88]	F: 76	F 14.4±3.9 yrs, male 15.8±4.4 yrs (mean)		Mean 9.3±2.8 yrs(F); 10±2.5 yrs (M)	F 66.5 ± 39.2 months, Male 76.2 ± 56.4 months, median 5 yrs	F 60.5 ± 38 months, Male 69.1 ± 57.5 months		89			
							Diabetes	1	0.011236	0.185718265	
							Malignancy	0	0	0	
							Dialysis + transplant (ESRD)	5	0.0561798	0.928591327	
							Premature gonadal failure	0	0	0	
							Transverse myelitis	2	0.0224719	0.371436531	
							Pyoderma gangrenosum	1	0.011236	0.185718265	
							Pancreatic pseudocyst	1	0.011236	0.185718265	
Costagliola, et al, 2018, Italy[89]	F: 24		Mean 14.6 ± 1.6 yrs			Mean 14.17 ± 8.04 yrs (1 patient excluded)		25			Caucasian 24, Hispanic 1
							Renal damage (including nephrotic syndrome or ESRD)	9	0.36	25.40578687	

					Vascular involvement (including DVT or ischaemic arterial lesions)	7	0.28	19.76005646	
					Nervous system involvement (with ischemic stroke, cranial neuropathies and cerebral vasculitis)	4	0.16	11.29146083	
					Cutaneous damage (chronic non-scarring alopecia)	3	0.12	8.468595625	
					Myositis	1	0.04	2.822865208	
					Acute myocardial infarction	1	0.04	2.822865208	
					Pulmonary (restrictive syndrome and pulmonary heart)	1	0.04	2.822865208	
					CMV reactivation	3	0.12	8.468595625	
					Herpes zoster reactivation	2	0.08	5.645730416	
					Severe EBV infection	1	0.04	2.822865208	
					Mycobacterium tuberculosis infection	1	0.04	2.822865208	
					ESRD	3	0.12	8.468595625	
					Requiring dialysis	1	0.04	2.822865208	
					Multi-infarct encephalopathy	1	0.04	2.822865208	
Torrente-Segarra, et al, 2017, Spain[90]	F: 435	Mean 31.5±10.5 Median 30 range [24-28] yrs Childhood	Mean 16.6 ± 6.3 yrs, median 16 [13-18]	Mean 10.7±7.2 yrs		484			Caucasian 441
					Articular				
					Erosive arthritis	40	0.0826446		
					Atrophy, muscle weakness	15	0.0309917		
					Tendon rupture	5	0.0103306		
					Osteoporosis	15	0.0309917		
					Fibromyalgia	11	0.0227273		
					Avascular necrosis	1.6 ± 0.6 2 [1-2]			
					Pulmonary embolism	11	0.0227273		
					Lung fibrosis	6	0.0123967		

Pulmonary hypertension	8	0.0165289
Alveolar haemorrhage	6	0.0123967
Shrinking lung	4	0.0082645
Pleural fibrosis	1	0.0020661
Pulmonary infarction	1	0.0020661
Vascular disease	25	0.0516529
Valvular dysfunction	19	0.0392562
Libman-Sachs Endocarditis	8	0.0165289
Cardiomyopathy	7	0.0144628
Angina or cardiac bypass	6	0.0123967
Myocarditis	3	0.0061983
Number of heart attacks	1.1 ± 0.0 1.0 [1-1]	
Small tissue loss	22	0.0454545
Major tissue loss	3	0.0061983
Venous thrombosis	3	0.0061983
Intermittent claudication >6 months	1	0.0020661
Renal terminal failure	27	0.0557851
Seizures (requiring therapy for 6 months)	46	0.0950413
Cognitive impairment/ Psychosis	21	0.0433884
Neuropathy	13	0.0268595
Transverse myelitis	3	0.0061983
Number of strokes	1.2±0.6 1[1-1]	
Cataract	31	0.0640496
Retinopathy	22	0.0454545
Amenorrhoea <40	22 (51.2%)	0.0454545
Dyslipidaemia	104	0.214876
High blood pressure	100	0.2066116
Autoimmune thyroiditis	31	0.0640496
Alcoholism	14	0.0289256
Congestive heart failure	8	0.0165289
Diabetes mellitus with organ involvement	8	0.0165289

						Diabetes mellitus without organ involvement	0	0	
						Chronic obstructive pulmonary disease	7	0.0144628	
						Peripheral arterial disease	3	0.0061983	
						Malignancy Solid without metastasis	15	0.0309917	
						Malignancy Solid with metastasis	0	0	
						Leukaemia	1	0.0020661	
						Lymphoma	2	0.0041322	
						Dialysis	23	0.0475207	
						Kidney transplantation	15	0.0309917	
						Splenectomy	14	0.0289256	
Ponin, et al, 2022, Thailand[91]	F: 87	Mean 20.6 ± 3.0 yrs	Mean 12.1 ± 2.3 yrs	Mean 5.5 ± 2.7 yrs			106		
						Growth impairment	25	0.2358491	
Avar Aydin, et al, 2018, USA[92]	F: 87		Mean yrs 13.2±3.1		Median 29.5 months (IQR 6.0–55.2)		110		Caucasian 54; African American 39; Asian 8; Other 9
						Hypertension	22	0.2	81.3559322
						Obesity	19	0.1727273	70.26194145
						Dyslipidaemia	14	0.1272727	51.77195686
Ahluwalia, et al, 2014, India[93]	F: 17				Mean 82 months (range 9–210 months)		21 (27 patients initially, 6 lost to follow up)		
						Thrombosis	3 (split below)		
						Cerebral infarct (death)	1	0.047619	6.968641115
						Pulmonary thromboembolism (death)	1	0.047619	6.968641115
						Spinal infarct	1	0.047619	6.968641115

Sinicato, et al, 2017, Brazil[94]	F: 37	16.7 ± 4.0 (mean)		Mean 4.8 ± 4.1 yrs			41	
					iglyceridemia		11	0.268
					Insulin resistance		2	0.049
					Arterial hypertension		4	0.097
					Albumin Exretion		5	0.122
					high LDL-c levels		19	0.463
					hypercholesterolemia		8	0.195
Cabral, et al, 2013, Portugal[95]	F: 46		Mean 12.6±4.04 yrs Median 13.5 range (1-17)	Mean 5.5±5.4 yrs Median 4 range (0-24)			56	
					Cataract		4	0.0714286	12.98701299
					Epilepsy		3	0.0535714	9.74025974
					Obesity		25	0.4464286	81.16883117
					Cardiac failure		2	0.0357143	6.493506494
					Arterial hypertension		5	0.0892857	16.23376623
					Avascular necrosis		1	0.0178571	3.246753247
					Growth delay		3	0.0535714	9.74025974
					Osteoporosis		5	0.0892857	16.23376623
					Gastrointestinal tract bleeding		1	0.0178571	3.246753247
					Macrophage activation syndrome		2	0.0357143	6.493506494
					Infection		6 (split below)		
					Pneumonia		2	0.0357143	6.493506494
					Sepsis		1	0.0178571	3.246753247
					Bacterial endocarditis		1	0.0178571	3.246753247
					Cellulitis		3	0.0535714	9.74025974
					Cognitive impairment		4	0.0714286	12.98701299

					End stage renal disease	2	0.0357143	6.493506494	
					Vascular thrombosis	5 (split below)			
					Cerebrovascular (thrombosis)	4	0.0714286	12.98701299	
					Cerebral hemorrhage	2	0.0357143	6.493506494	
					Myocardium infarct	1	0.0178571	3.246753247	
					Epilepsy	3	0.0535714	9.74025974	
Bernatsky, et al, 2017, North America[96]	F: 978	Mean 13±3.3 yrs		Mean 7.6 yrs, SD 6.4 yrs, IQ 3.4-10.0 yrs/8839 PY		1168			
					Non-Hodgkin lymphoma	3	0.0025685	0.337959625	
					Head and neck squamous cell carcinomas (tongue and nasopharynx)	2	0.0017123	0.225306417	
					Glioblastoma	1	0.0008562	0.112653208	
					Thyroid cancer (papillary adenoma)	1	0.0008562	0.112653208	
					Breast cancer (infiltrating duct carcinoma)	1	0.0008562	0.112653208	
					Adenocarcinoma not otherwise specified	1	0.0008562	0.112653208	
					Cancers whose morphology and histology codes were indeterminate	3	0.0025685	0.337959625	
Yang, et al, 2015, Canada[97]	F: 30 (AVN), F: 82 (non-AVN)		Mean 13.8 ± 2.5 (AVN), 13.8 ± 2.4 (non-AVN)	Mean 6.89 ± 5.39 yrs (AVN), 5.72 ± 5.09 yrs (non-AVN)		617			AVN: South Asian 10; *Other Asian 9; Black 9; White 8; Other 1. Non-AVN: South Asian 10, Other Asian 30, Black 12, WhIte 40,

								Hispanic 1 Other 13, Unknown 5
					Symptomatic Avascular Necrosis (AVN)	37	0.0599676	8.703568228
Lin, et al, 2018, Taiwan[98]			Mean 12.9±3.12 yrs (range 6-18 yrs)	mean 6 yrs		35		
					Vitamin D deficiency	28	0.8	133.3333333
Kahwage, et al, 2017, Brazil[99]	F: 6/7 (uveitis)	≤25 yrs, Median 10.4 range (8.8 -14.5) (Uveitis)				852		
					Uveitis	7	0.008216	
					Cataracts & bilateral irreversible blindness	1	0.0011737	
					Unilateral blindness	1	0.0011737	
Lai, et al, 2022, China[100]	F: 95	Age at enrolment - Girls mean 14.4 (SD 3.4), Boys mean 13.9 (SD 5.0)		Girls mean 6.6 (SD 4.1) Boys mean 5.2 (SD 4.0)		115		
					Hypertension	22	0.1913043	
					Chronic kidney disease	14	0.1217391	
Adulthood / jSLE								

Al Hamzi, et al, 2014, Saudi Arabia[101]	F: 45	Mean 23.6±4 yrs	Mean 10.3±6 yrs	Mean 15±4 yrs		48		
					Growth retardation/Short stature	24	0.5	
					Delayed puberty	13	0.2708333	
					Renal disease	12 (as split below)		
					Progressive renal disease and required dialysis	7	0.1458333	
					Renal transplant	5	0.1041667	
					Neuropsychiatric	9	0.1875	
					Cardiovascular	6	0.125	
					Musculoskeletal	9	0.1875	
					Skin	2	0.0416667	
					Seizure disorder	6	0.125	
					Chorea	3	0.0625	
					Cerebrovascular accident	3	0.0625	
					Miscarriage	4	0.0833333	
Tanigava, et al, 2022, Brazil[102]	F: 65	Median 21 yrs (IQR 20-26)	Median 14 IQR 12-16	Median 9 yrs IQR 6-14		69		White 35; Black 31; Asian 3

Mental disorders	16	0.2318841
Obesity	14	0.2028986
Thyroid diseases	7	0.1014493
Fibromyalgia	6	0.0869565
Antiphospholipid syndrome	56	0.8115942
Osteoporosis and osteoporotic fracture	5	0.0724638
Hypertension	5	0.0724638
Cerebrovascular disease	4	0.057971
Allergic disorders	4	0.057971

Gastrointestinal diseases	4	0.057971
Valvular disease	4	0.057971
Chronic kidney disease	3	0.0434783
Diabetes mellitus	2	0.0289855
Dyslipidaemia	2	0.0289855
Retinal change or optic atrophy	5	0.0724638
Cataract	2	0.0289855
Cognitive impairment or major psychosis	3	0.0434783
Seizures requiring therapy for ≥6 months	2	0.0289855
Cerebrovascular accident	3	0.0434783
Cranial or peripheral neuropathy	3	0.0434783
Transverse myelitis	0	0
End-stage renal disease	3	0.0434783
Pulmonary hypertension	0	0
Pulmonary fibrosis	2	0.0289855
Shrinking lung	0	0
Pleural fibrosis	0	0
Pulmonary infarction	0	0
Angina or coronary artery bypass	0	0
Myocardial infarction	0	0
Cardiomyopathy	0	0
Pericarditis or pericardiectomy	0	0
Venous thrombosis	1	0.0144928
Infarction or resection of bowel	1	0.0144928
Mesenteric insufficiency	0	0
Chronic peritonitis	0	0

					Stricture or upper gastrointestinal tract surgery	0	0	
					Pancreatic insufficiency	1	0.0144928	
					Muscle atrophy or weakness	0	0	
					Deforming or erosive arthritis	0	0	
					Osteoporosis with fracture or vertebral collapse	2	0.0289855	
					Avascular necrosis	2	0.0289855	
					Osteomyelitis	0	0	
					Ruptured tendon	0	0	
					Scarring chronic alopecia	0	0	
					Extensive scarring	1	0.0144928	
					Skin ulceration (excluding thrombosis)	4	0.057971	
					Premature gonadal failure	0	0	
					Diabetes requiring therapy	0	0	
					Malignancy	0	0	
Hersh, et al,2009, USA[103]	F: 78	Mean 30.5±12.8 yrs	14.0 +/-2.7 range (2–17)	Mean 16.5 yrs +/- 8.5 (range 3-47)		90		White 48.9%; Asian/Pacific Islander 16.7%; Hispanic/ Latino 15.6 %; African American 7.8%; Other 11%
					Pulmonary	23.60%	0.236	
					Clotting disorders	11.30%	0.113	
					Seizures	23.60%	0.236	
					Myocardial infarction	7.90%	0.079	
					Transplant	12.4	0.1377778	
					Dialysis	19.1	0.1377778	
Sousa, et al, 2016, Portugal[104]	F: 77	Mean 25.1 ± 9.1 yrs	13.4 ± 5.5			89		Caucasian 78, African 3, Others 4

						Hypertension	17	0.1910112	
						Diabetes	3	0.0337079	
						Hypercholesterolemia	2	0.0224719	
						Thyroid disease	3	0.0337079	
						Antiphospholipid syndrome	10	0.1123596	
						Sjögren's syndrome	2	0.0224719	
Koutsonikoli, et al, 2015, Greece[105]	F: 39	Mean 22±7.8 yrs at the end of follow up	Mean 12.5 ± 3.1 (SD)	Median 7.4 yrs and IQR (3.9–14.2) yrs	440 person-yrs (people*avg follow up time)		47		All Caucasians
						Cataract	4	0.0851064	0.193423598
						Retinopathy	1	0.0212766	0.048355899
						Optic atrophy	1	0.0212766	0.048355899
						Cataract and retinopathy	1	0.0212766	0.048355899
						Seizures	2	0.0425532	0.096711799
						Cognitive impairment and seizures	1	0.0212766	0.048355899
						Major psychosis and seizures	1	0.0212766	0.048355899
						Cerebral vascular accident	1	0.0212766	0.048355899
						Transverse myelitis	1	0.0212766	0.048355899
						Single venous thrombosis	5	0.106383	0.241779497
						Significant tissue loss and venous thrombosis	1	0.0212766	0.048355899
						Avascular necrosis	3	0.0638298	0.145067698
						Persistent impairment of renal function or end-stage renal disease	0	0	0
						Pulmonary fibrosis	1	0.0212766	0.048355899
						Pulmonary infarction	1	0.0212766	0.048355899
						Premature gonadal failure	3	0.0638298	0.145067698
						Delayed growth	6	0.1276596	0.290135397

Gamal, et al, 2022, Egypt[106]	F: 163	25 yrs (median); range, 20 to 30.3 yrs	14 (median) (range 12-15)		12 yrs (median); (range, 6.75 to 17.25) yrs			
						186		
					Hypertension	82	0.4408602	
					Diabetes mellitus	12	0.0645161	
					Dyslipidaemia	75	0.4032258	
					Thyroid dysfunction	10	0.0537634	
					Malignancy	1	0.0053763	
					Osteoporosis	27/150	0.18	
					Avascular necrosis	23	0.1236559	
					End stage renal disease	17	0.0913978	
					Thrombosis	22	0.1182796	
					Raynaud's phenomenon	51	0.2741935	
					Pulmonary hypertension	24	0.1290323	
Groot, et al, 2019, Netherlands[4]	F: 101	Median 33 yrs (range 18-65)		14 (range 4-17)	Median 20 yrs range (1-55 yrs)			
						111		White 80, Non-White 31
					Muscle atrophy or weakness	1	0.009009	
					Deforming or erosive arthritis	10	0.0900901	
					Osteoporosis with fracture	6	0.0540541	
					Avascular necrosis	14	0.1261261	
					Osteomyelitis	1	0.009009	
					Ruptured tendon	1	0.009009	
					Cognitive impairment or psychosis	12	0.1081081	
					Seizures	10	0.0900901	
					CVA	7	0.0630631	
					Cranial or Peripheral Neuropathy	4	0.036036	
					Transverse Myelitis	0	0	
					ESRD	8	0.0720721	
					Claudication	0	0	
					Minor tissue loss	0	0	
					Significant tissue loss	4	0.036036	

		Venous thrombosis	8	0.0720721
		Scarring alopecia	8	0.0720721
		Scarring/ panniculum	0	0
		Skin ulceration	3	0.027027
		Cataracts	8	0.0720721
		Retinal change	3	0.027027
		Angina/bypass	4	0.036036
		MI	5	0.045045
		Cardiomyopathy	1	0.009009
		Valvular disease	2	0.018018
		Pericarditis	0	0
		PAH	1	0.009009
		Pulmonary fibrosis	4	0.036036
		Shrinking lung	1	0.009009
		Pleural Fibrosis	0	0
		Pulmonary Infarction	0	0
		Infarction or resection of bowel	4	0.036036
		Mesenteric insufficiency	0	0
		Chronic peritonitis	0	0
		Stricture or upper GI surgery	0	0
		Pancreatitis	0	0
		Premature gonadal failure	2	0.018018
		Malignancy	2	0.018018
		IDDM	1	0.009009
Sinicato, et al, 2017, Brazil[94]	F:32		35	
		Low HDL-cholesterol	5	0.143
		Hypertriglyceridemia	6	0.171
		Insulin Resistance	0	0.00
		Arterial Hypertension	4	0.14
		Albumin excretion	4	0.114
Childhood / JDM				

Huber, et al, 2000, Canada[1]	F: 46	Median 13 (range 7-26)	Median 5.8 yrs Range (1-16)	Median 7.2 yrs range (3.2 - 13.9 yrs)		65		Caucasian 56; Asian 7, African Canadian 1, Indian 1
					Calcinosis	22	0.3384615	47.00854701
					Insulin-dependent diabetes mellitus	3	0.0461538	6.41025641
					Connective tissue diseases	2 (split below)		
					Scleroderma	1	0.0153846	2.136752137
					Overlap connective tissue disease	1	0.0153846	2.136752137
					Malignancies	0	0	
Cho, et al, 2019, Korea[107]	F: 68	Mean 12.7 ± 7.1 yrs		10 yrs		130		
					Congestive heart failure	1	0.0076923	0.769230769
					Cardiac arrhythmias	1	0.0076923	0.769230769
					Valvular disease	1	0.0076923	0.769230769
					Pulmonary circulation disorders	2	0.0153846	1.538461538
					Peripheral vascular disorders	3	0.0230769	2.307692308
					Hypertension, uncomplicated	5	0.0384615	3.846153846
					Hypertension, complicated	2	0.0153846	1.538461538
					Paralysis	3	0.0230769	2.307692308
					Other neurological disorders	2	0.0153846	1.538461538
					Chronic pulmonary disease	36	0.2769231	27.69230769
					Diabetes, uncomplicated	8	0.0615385	6.153846154
					Diabetes, complicated	2	0.0153846	1.538461538
					Renal failure	1	0.0076923	0.769230769
					Liver disease	15	0.1153846	11.53846154
					Peptic ulcer disease excluding bleeding	15	0.1153846	11.53846154
					AIDS/HIV	1	0.0076923	0.769230769
					Solid tumour without metastasis	1	0.0076923	0.769230769
					Coagulopathy	4	0.0307692	3.076923077
					Weight loss	1	0.0076923	0.769230769

							Fluid and electrolyte disorders	8	0.0615385	6.153846154	
							Deficiency anaemia	17	0.1307692	13.07692308	
							Depression	6	0.0461538	4.615384615	
							Pulmonary disease specified				
							Interstitial lung disease	2	0.0153846	1.538461538	
Concannon and Han, 2021, New Zealand[108]	F: 18	≤15 yrs		Mean 8.6 yrs SD 3.7 yrs		Mean 6.5, range (0.4–16.7) yrs		31			Asian 5. Maori 7, pacific island 3, European 16
							Calcinosis	9	0.2903226	44.66501241	
							Cutaneous vasculopathy	5	0.1612903	24.81389578	
							Interstitial lung disease	6	0.1935484	29.77667494	
							Pneumomediastinum	2	0.0645161	9.925558313	
							Pneumothorax	3	0.0967742	14.88833747	
							Surgical emphysema	1	0.0322581	4.962779156	
Okong'o, et al, 2016, South Africa[109]	F: 16		Median 6.75, range (2.0–9.7) yrs	Median 7.9 , range (3.4 – 9.75) year	median 4, range (0.5-84) months	Minimum of 12 months, median 55.5 months (inactive disease) 49.1 months (active disease)		25			Indigenous African 13, mixed race 11, European ancestry 1
							Skin				
							Calcinosis	8	0.32		
							Skin ulcers	1	0.04		
							Oedema (Generalized, Periorbital or limb)	0	0		
							Alopecia	3	0.12		
							Lipodystrophy	1	0.04		
							Musculoskeletal:		0		
							Arthritis	2	0.08		

				Contractures	5	0.2	
				Osteoporosis with fracture	1	0.04	
				Endocrine:		0	
				Growth failure	8	0.32	
				Diabetes mellitus	1	0.04	
				Adrenal insufficiency	0	0	
				Respiratory:		0	
				Interstitial lung disease (ILD)	2	0.08	
				Pulmonary haemorrhage	1	0.04	
				Gastrointestinal:		0	
				Abdominal pain or bleeding	0	0	
				Ocular:		0	
				Cataracts	2	0.08	
				Nervous system:		0	
				Sensorineural hearing loss	1	0.04	
				Seizures, neuropathy		0	
				Cardiovascular		0	
				Raynaud’s phenomenon	6	0.24	
				Cardiomyopathy	0	0	
				Infections:		0	
				Fungal (skin)	7	0.28	
				Bacterial (staph aureus 2, gram negative 1)	3	0.12	
				TB	2	0.08	
Ravelli, A. et al, 2010, Italy, UK[2]	F: 321	Mean 6.9, range (0.9–17.8) yrs	Mean 7.7, range (2–25.2) yrs		490		White 320
				Cutaneous scarring/atrophy	195/443	0.4401806	
				Calcinosis	106/406	0.2610837	
				Joint contractures	82/448	0.1830357	
				Lipodystrophy	43/444	0.0968468	
				Hirsutism	44/431	0.1020882	
				Growth failure	34/424	0.0801887	
				Osteoporosis with fractures	26/435	0.0597701	

					Cataract	9/430	0.0209302		
					Gastrointestinal infarction/resection	7/444	0.0157658		
					Malignancy	0/447	0		
Tabarki, et al, 1998, France[110]	F: 22	Mean 6.9 yrs (range 2-14 yrs)		Mean 4.9 yrs (range: 2.5-13 yrs)		36			
					Calcinosis	15	0.4166667	85.03401361	
					Impaired function	8	0.2222222	45.35147392	
					Cataracts	1	0.0277778	5.66893424	
					Vertebral collapse	2	0.0555556	11.33786848	
					Contractures	1	0.0277778	5.66893424	
Chevalier, et al, 2021, France[111]	F: 4	Median 6 yrs range 2-11 yrs	Median 5.5 months	Mean 14.9±8.8 yrs		7			Caucasian 100%
					Polycystic ovary syndrome	1	0.1428571	9.587727709	
					Depressive disorder	1	0.1428571	9.587727709	
					mild physical dysfunction (CHAQ/HAQ >0)	2	0.2857143	19.17545542	
					Calcinosis	2	0.2857143	19.17545542	
					Lipodystrophy	2	0.2857143	19.17545542	
					Osteopenia/osteoporosis	0	0	0	
					Cardiovascular damage	0	0	0	
					Peripheral vascular damage	0	0	0	
					Ocular damage	0	0	0	
					Malignancy	0	0	0	
					Fracture	0	0	0	
					Moderate or severe disability CHAQ/HAQ>1	0	0	0	

Al-Mayouf, et al, 2017, Saudi Arabia[112]	F: 58	Mean 6 ± 3 yrs		Mean 5 ± 4.4 yrs		92			
					Permanent skin changes	16	0.173913	34.7826087	
					Residual muscle weakness	12	0.1304348	26.08695652	
					Calcinosis	27	0.2934783	58.69565217	
					Growth failure	37	0.4021739	80.43478261	
					Osteoporosis	15	0.1630435	32.60869565	
					Lipodystrophy	3	0.0326087	6.52173913	
					Mucinous cystic ovarian tumour	1	0.0108696	2.173913043	
Sag, et al, 2021, Turkey[113]	F: 35	Mean 8.1±4.3 yrs	Mean 8.7±4.4	Median 4.85 yrs IQR 3.7, Mean 5.66±3.59 yrs		58			
					Calcinosis	19	0.3275862	67.54354781	
					Pulmonary involvement	8	0.137931	28.43938855	
					Skin ulcers	4	0.0689655	14.21969428	
					Gastrointestinal involvement	4	0.0689655	14.21969428	
					Severe muscle disease (wheel-chair dependency)	3	0.0517241	10.66477071	
					Cardiac involvement	2	0.0344828	7.109847138	
Ramanan, et al, 2005, Canada[114]									
	F: 20		Mean 8.4 yrs	Mean 34.6 months	JDM patients after June 1997	31			Black 2; East Indian 4; Native Canadian 1; Asian 1; White 23
					Calcinosis	2	0.0645161	22.37553608	
					Cataracts	3	0.0967742	33.56330412	

	F:16			6.7 yrs (mean)	4 yrs (1056 patient months)	JDM patients pre-June 1997	22			Black 1; East Indian 1; Native Canadian 1; Asian 2; White 17
						Calcinosis	5	0.2272727	56.81818182	
						Cataracts	8	0.3636364	90.90909091	
Sun, et al, 2015, Taiwan[115]	F: 26		Mean 81.97 ± 46.63 months (25-216)	Mean 87.97± 45.90 months (28-213)	For those receiving IVIG: Mean 79 months		39			
						Calcinosis	15	0.3846154		
						Pulmonary involvement	4	0.1025641	Follow up time and cases were not reported coherently. Hence, no incidence rate was calculated	
						Malignancy	0	0		
						Disability	4	0.1025641		
						GI perforation	1	0.025641		
						Avascular necrosis of the hip joint	1	0.025641		
Silverberg, et al, 2018, USA[116]	F: 915	0-17					1407			White 543; Black 203; Hispanic 260; Asian 19;
						Obesity	38	0.0270078		
						Hypertension	119	0.0845771		
						Diabetes, uncomplicated	18	0.0127932		
						Late effects of cerebrovascular disease	≤10	≤10/1407		

						Pulmonary circulatory disorder	15	0.010661	Multiracial/other 71
						Valvular heart disease	17	0.0120824	
						Peripheral and visceral atherosclerosis	≤10	≤10/1407	
						Any lipid abnormalities	≤10	≤10/1407	
						Lipodystrophy	≤10	≤10/1407	
						Personal history of TIA and cerebral infarction without residual deficits	≤10	≤10/1407	
						Arrythmia	47	0.0334044	
						Valvular heart disease	17	0.0120824	
Adulthood / JDM									
Boros, et al, 2022, UK[117]	F:68	Mean 20.6 yrs ±3.9 (SD)	Mean 9.2 yrs ± 4.3 (SD)	Mean 12.4 yrs ± 5.0			84		White 69
						Feel shorter than peers	20 /80	0.25	
						Feel weight was affected by myositis	25/80	0.3125	
						Puberty affected by myositis	16/80	0.2	
						Calcinosis	18/82	0.225	
						Lipodystrophy	6/82	0.0731707	
						Pulmonary hypertension	2/83	0.0240964	
Mathiesen, et al, 2014, Denmark[118]	F: 37	Mean 21.7 yrs (SD 11.1) (Range 4.2-45.9) (Non-restrictive lung); 14.2 (4.4) (10.0-19.7) (restrictive lung)	Mean 7.7 yrs (SD 4.7) (Range 1.4-17.8) (non-restrictive lung); 7.7 (4.2) (1.5-10.5) (restrictive lung)	Mean 4.2 yrs (SD 2.9) (Range 0.610.6) (non-restrictive lung); 5.3 (2.5) (2.6- 8.5) (restrictive lung)	Mean 14.3 yrs (SD 10.7) (Range 1.1-35.1) (non-restrictive lung); 6.5 (3.0) (2.6-9.2) (restrictive lung)		51		
						Restrictive interstitial lung disease	4	0.0784314	12.06636501
						Obstructive lung function pattern	7	0.1372549	9.598244892

Tsaltskan, et al, 2020, USA[5]	F: 39	Median 24.0 yrs [IQR 19.0-28.6] yrs	Median 11.0 [IQR 8.5-15.7]	median 11.5 yrs, [IQR 4.5-18.9] yrs		49		Caucasian 59.2%, African American 11.1% , Hispanic 17.7%, Other 12.0%
					Cutaneous Damage:			
					Cutaneous scarring or atrophy	28	0.5714286	
					Calcinosis	27	0.5510204	
					Lipodystrophy	16	0.3265306	
					Alopecia	8	0.1632653	
					Poikiloderma	6	0.122449	
					Skeletal Damage:			
					Joint Contracture	31	0.6326531	
					Deforming Arthropathy	8	0.1632653	
					Osteoporosis with fracture	4	0.0816327	
					Avascular Necrosis	2	0.0408163	
					Endocrine Damage:			
					Irregular Menses	10/40	0.25	
					Hirsutism	7/40	0.175	
					Diabetes	6	0.122449	
					Amenorrhea	4/40	0.1	
					Infertility	1	0.0204082	
					Pulmonary Damage:			
					Impaired lung function	7	0.1428571	
					Dysphonia	4	0.0816327	
					Pulmonary Fibrosis	1	0.0204082	
					Ocular Damage:			
					Cataract	4	0.0816327	
					Vision Loss	3	0.0612245	
					Gastrointestinal Damage:			
					Dysmotility	5	0.1020408	
					Dysphagia	3	0.0612245	
					Infection:			
					Chronic Infection	5	0.1020408	
					Multiple Infections	4	0.0816327	

					Cardiovascular Damage:			
					Hypertension	4	0.0816327	
					Peripheral Vascular Damage:			
					Tissue or pulp loss	4	0.0816327	
					Digit or limb loss	1	0.0204082	
					Malignancy	1	0.0204082	
Mathiesen, et al, 2012, Denmark[11] [9]	F: 38	Mean 21.5 yrs SD 11.1 Range 4.2-45.9)	Mean 4.3 yrs SD 2.9 range (0.6-10.6)	Mean 13.9 SD 10.5 (range 2.1–35.1) yrs		53		
					Deforming arthropathy	1	0.0188679	1.357404642
					Pulmonary fibrosis	3	0.0566038	4.072213927
					Growth failure	6/49	0.122449	8.809279107
					Diabetes	1	0.0188679	1.357404642
					Scarring or atrophy	21	0.3962264	28.50549749
					Calcinosis	11	0.2075472	14.93145107
					Lipodystrophy	7	0.1320755	9.501832496
					Tissue or pulp loss	1	0.0188679	1.357404642
					Alopecia, poikiloderma	0	0	0
					Joint contractures	11	0.2075472	14.93145107
					Osteoporosis/osteopenia estimated by DXA	5	0.0943396	6.787023212
					Osteoporotic fractures	2	0.0377358	2.714809285
					Delay in secondary sexual characteristics*	3/14	0.2142857	15.41623844
					Hypertriglyceridemia	9	0.1698113	12.21664178
					Hirsutism, irregular menses, amenorrhoea	0	0	0
					Paraesthesia	2	0.0377358	2.714809285
					Arthritis	3	0.0566038	4.072213927
					Impaired pulmonary function test	4	0.0754717	5.429618569
					Pulmonary hypertension	0	0	0
					Severe abdominal pain	3	0.0566038	4.072213927
					Dysphagia, gastrointestinal dysfunction	1	0.0188679	1.357404642

							Gastrointestinal surgery	1	0.0188679	1.357404642	
							Hypertension, cardiomyopathy, angina, myocardial infarction	0	0	0	
							Tissue or pulp loss, claudication	0	0	0	
							Cataract, visual loss	0	0	0	
							Chronic infection, multiple infections	0	0	0	
							Malignancy	0	0	0	
Sanner, et al, 2010, Norway[120]	F: 36	Median 21.5 (range 6.7–55.4) yrs	Mean 8.3 ±4.5 yrs	Mean 9.1 ± 4.5 yrs	minimum 24 months	Mean 16.9 ± 10.6		59			European 57
							Calcinosis	14	0.2372881	14.04071808	

Supplementary Table S4.2: Studies comparing comorbidity incidence between cases and controls of JIA/jSLE patients reported during childhood and adulthood. NR=p value not reported, NS=P value described as NS.

Study	Measure of effect	Control group	Comorbidity	Effect size	95% CI	p-value
Childhood/ JIA						
Schenck, et al, 2018, Germany[33]	Prevalence ratio	Diabetes registry data	Type 1 diabetes	1.76	(1.34-2.28)	<0.001
Krause, et al, 2017, USA[37]	Rate Ratio	Non-JIA comparator subjects. Age-sex matched	Joint surgeries	3.93	(1.18-24.94)	NR
			Non-joint surgeries	1.9	(1.05-3.67)	NR
			Hospitalizations	2.25	(1.04-5.53)	NR
			Hospitalizations for depression	1.41	(0.47-4.70)	NR
Lee, et al, 2020, USA[38]	Adjusted Hazard ratios	Healthy children with at least 1 Wellcare visit	Type 1 diabetes	1.81	(1.03–3.17)	NR
			Type 2 diabetes	1.22	(0.79–1.90)	NR
			Composite diabetes	1.4	(0.94–2.09)	NR
	Age, sex, index date-matched Hazard ratios	Healthy children with at least 1 Wellcare visit	Type 1 diabetes	1.89	(1.11-3.19)	NR

			Type 2 diabetes	1.33	(0.88–2.02)	NR
			Composite diabetes	1.52	(1.05–2.21)	NR
	Age, sex, index date-matched Hazard ratios	Children with asthma	Type 1 diabetes	1.46		NR
			Type 2 diabetes	0.95	(0.64–1.43)	NR
			Composite diabetes	1.11	(0.77–1.60)	NR
	Adjusted Hazard ratios	Children with asthma	Type 1 diabetes	1.48	(0.86–2.56)	NR
			Type 2 diabetes	1.01	(0.66–1.54)	NR
			Composite diabetes	1.18	(0.80–1.73)	NR
Beukelman, et al, 2012, USA[28]	Incidence rate ratio	Children with ADHD	Coccidioides	101	(8.1–5319)	NR
			Salmonella	3.8	(1.20–9.50)	NR
			Herpes zoster	2.1	(1.40–3.00)	NR
			All infections	2.4	(1.70-3.30)	NR
			All infections except Herpes zoster	4.2	(2.10-7.60)	NR
Berthold, et al, 2022, Sweden[53]	Hazard ratios	Non-JIA comparator subjects	**Females**			
		Sex, year of birth and residential region matched	Depression	1.1	(0.90–1.50)	NS
			Anxiety	1.2	(0.90–1.50)	NS
			Males			
			Depression	0.8	(0.50–1.40)	NS
			Anxiety	0.6	(0.40–1.10)	NS
Lovell, et al, 2021, USA[47]	Log risk ratio	General paediatric patients without JIA	Addison's disease	1.65	(1.02-2.21)	P<0.05
			Autoimmune/Hashimoto's thyroiditis	2.25	(1.72 to 2.71)	P<0.05
			Graves' disease	0.38	(–2.77-2.26)	NS
			Morphoea	3.15	(2.54-3.70)	P<0.05
			Multiple sclerosis	1.9	(–1.27-3.83)	NS
			Takayasu's arteritis	1.45	(–1.71-3.35)	NS
			Type 1 diabetes	1.25	(0.75-1.70)	P<0.05
			Vitiligo	1.76	(0.74-2.59)	P<0.05

			Autoimmune disease not elsewhere classified	2.55	(1.78-3.21)	P<0.05
			Chronic glomerulonephritis	0.73	(–2.45-2.64)	NS
			Chronic urticaria	0.58	(0.28-0.84)	P<0.05
			Episcleritis/scleritis	3.18	(2.14-4.03)	P<0.05
			Erythema nodosum	1.12	(–2.03-3.02)	NS
			Haemolytic anaemia	–0.36	(–3.49-1.54)	NS
			Immune thrombocytopenia purpura	0.4	(–1.42-1.68)	NS
			Leukocytoclastic vasculitis	4.08	(2.19-5.49)	P<0.05
			Pulmonary fibrosis/interstitial lung disease	2.46	(2.28-2.62)	P<0.05
			Raynaud's syndrome	3.37	(2.80-3.86)	P<0.05
			Sjögren's syndrome/sicca syndrome	3.27	(2.40-5.01)	P<0.05
			Thrombocytopenia purpura	0.57	(–2.61-2.46)	NS
			Vasculitis	3.71	(2.90-4.37)	P<0.05
Simon, TA et al, 2020, USA[41]	Odd Ratio (OR)	Patients with ADHD 1:5 matched for age (± 2 years), sex, number of medical encounters (in-patient, out-patient, and emergency room visits), and calendar year of diagnosis (± 1 year)	**MarketScan**			
			Addison disease	6.16	(2.83-13.42)	NR
			Alopecia areata	2.03	(0.93-4.46)	NR
			Celiac disease	3.99	(2.66-5.98)	NR
			Chronic glomerulonephritis	Infinite		NR
			Chronic urticaria	3.4	(2.74-4.22)	NR
			Crohn's disease	16.84	(10.91-25.99)	NR
			Erythema nodosum	12.05	(3.20-45.4)	NR
			Giant cell arteritis	4.51	(0.28-72.18	NR
			Grave's disease	3.23	(1.43-7.27)	NR
			Hashimoto's thyroiditis	3.57	(2.48-5.14)	NR
			Haemolytic anaemia	6.9	(3.98-11.97)	NR
			Immune thrombocytopenia	4.52	(1.88-10.86	NR
			Interstitial lung disease	24.86	(5.51-112.15)	NR
			Morphoea	21.36	(9.45-48.29)	NR
			Multiple sclerosis	2.26	(0.21-24.90)	NR
			Myasthenia gravis	4.51	(0.64-32.06)	NR

Pernicious anaemia	0		NR
Polymyalgia rheumatica	Infinite		NR
Polymyositis	Infinite		NR
Primary biliary cirrhosis	0		NR
Psoriasis	8.42	(5.54-12.78)	NR
Raynaud's syndrome	5.7	(3.44-9.46)	NR
Scleritis and episcleritis	17.19	(6.42-46.04)	NR
Sjogren's syndrome/sicca syndrome	Infinite		NR
Systemic sclerosis	24.89	(8.57-72.23)	NR
Type 1 diabetes	1.7	(1.27-2.27)	NR
Ulcerative colitis	8.34	4.77-14.59)	NR
Uveitis	165.1	(108.97-250.14)	NR
Vasculitis	22.64	(9.42-54.42)	NR
Vitiligo	2.85	(1.39-5.88)	NR
Patients with ADHD	**Pharmetrics Data**		
Addison disease	2.51	(1.16-5.43)	NR
Alopecia areata	2.43	(1.274.65)	NR
Celiac disease	3.82	(2.61-5.60)	NR
Chronic glomerulonephritis	31.57	(3.88-256.56)	NR
Chronic urticaria	2.04	(1.65-2.51)	NR
Crohn's disease	12.99	(8.78-19.2)	NR
Erythema nodosum	9.02	(2.72-29.96)	NR
Giant cell arteritis	Infinite		NR
Grave's disease	1.69	(0.79-3.64)	NR
Hashimoto's thyroiditis	4.26	(3.03-6.01)	NR
Haemolytic anaemia	4.52	(2.50-8.16)	NR
Immune thrombocytopenia	4.89	(2.23-10.72)	NR
Interstitial lung disease	12.03	(3.19-45.33)	NR
Morphoea	18.97	(7.15-50.33)	NR
Multiple sclerosis	3.38	(0.76-15.11)	NR

			Myasthenia gravis	Infinite		NR
			Pernicious anaemia	0		NR
			Polymyalgia rheumatica	9.02	(0.82-99.43)	NR
			Polymyositis	22.55	(2.63-193.01)	NR
			Primary biliary cirrhosis	Infinite		NR
			Psoriasis	5.89	(4.07-8.53)	NR
			Raynaud's syndrome	12.16	(7.45-19.84	NR
			Scleritis and episcleritis	13.55	(5.76-31.88)	NR
			Sjogren's syndrome/sicca syndrome	72.24	(9.58-544.36)	NR
			Systemic sclerosis	90.34	(12.12-673.22)	NR
			Type 1 diabetes	1.82	(1.37-2.41)	NR
			Ulcerative colitis	6.49	(3.81-11.05)	NR
			Uveitis	133.23	(92.47-191.95)	NR
			Vasculitis	10.32	(4.24-25.08)	NR
			Vitiligo	2.61	(1.24-5.49)	NR
Horneff, et al, 2016, Germany[46]	Risk ratios	German Childhood Cancer Registry report 2015				
			Methotrexate Drug Cohort			
			All malignancy	3	(0.75-12.0)	0.12
			Lymphoma & Leukaemia	6.7	(1.67-26.82)	0.007
			Etanercept Drug Cohort			
			All malignancy	6.3	(3.00-13.20)	<0.0001
			Lymphoma & Leukaemia	8	(2.99-21.38)	<0.0001
Adulthood / JIA						
Raab, et al, 2013, Germany[59]	% difference (absolute)	Community sample in Germany	Cardiovascular diseases (as split below)	1.60		0.343
		Age and gender matched by propensity scores	Hypertension	-0.60		0.759
			Cardiac insufficiency	-0.90		0.111

			Asthma	-2.50		0.098
			Gastrointestinal diseases	-6.40		<0.001
			Gastrointestinal ulcer	-1.00		0.065
			Liver and gallbladder diseases	0.10		0.125
			Renal insufficiency	-1.00		0.064
			Diabetes mellitus	-0.70		0.332
			Depression	-4.10		0.004
Aggarwal, et al, 2006, India[60]	% difference (absolute)	Healthy male hospital employees	Osteopenia (Lumbar Spine)	-5.50		NS
			Osteopenia (Hip)	28.30		< 0.05
			Osteopenia (Distal third of forearm)	10.90		< 0.001
			Osteoporosis (lumbar spine)	22.40		NS
			Osteoporosis (hip)	5.90		< 0.05
			Osteoporosis (Distal third of forearm)	35.70		< 0.001
Smith, et al, 2019, USA[67]	Unadjusted Risk ratio	Women without autoimmune disease who delivered live born infants	Pre-eclampsia	2.48	(1.30–4.74)	P<0.05
			Pregnancy induced hypertension without pre-eclampsia	1.75	(0.83–3.69)	NR
			Placenta previa and/or abruption	1.46	(0.46–4.68)	NR
			Preterm labour	2.03	(1.27–3.23)	P<0.05
			Gestational diabetes	1.66	(0.87–3.16)	NR
			Fever during pregnancy	1.19	(0.89–1.58)	NR
			Caesarean section	1.55	(1.23–1.94)	P<0.05
	Adjusted risk ratio	Adjusted risk ratio computed using Poisson regression	Pre-eclampsia	1.87	(0.71–4.96)	NR
			Pregnancy induced hypertension without preeclampsia	0.60		(0.08–4.78)
			Gestational diabetes	0.91	(0.25–3.38)	NR
			Fever during pregnancy	2.61	(1.21–5.63)	P<0.05
	Adjusted risk ratio	Multivariable adjustment for the prior factors	Pre-eclampsia	2.19	(0.82–5.88)	NR
			Gestational diabetes	0.86	(0.24–2.97)	NR

			Fever during pregnancy	3.06	(1.42–6.56)	P<0.05
Simon, et al, 2020, USA[41]	Odds ratios	MarketScan patients with ADHD 1:5 matched age (± 2 years), sex, number of medical encounters (in-patient, out-patient, and emergency room visits), and calendar year of diagnosis (± 1 year).	Addison disease	4.87	(3.09-7.66)	NR
			Alopecia areata	1.09	(0.45-2.64)	NR
			Coeliac disease	1.32	(0.73-2.39)	NR
			Chronic glomerulonephritis	2.36	(0.59-9.43)	NR
			Chronic urticaria	2.53	1.94-3.30)	NR
			Crohn's disease	3.88	(2.93-5.16)	NR
			Erythema nodosum	0.94	(0.21-4.30)	NR
			Giant cell arteritis	Infinite		NR
			Grave's disease	2.15	(1.26-3.65)	NR
			Hashimoto's thyroiditis	1.56	(1.09-2.25)	NR
			Haemolytic anaemia	6.86	(3.88-12.14)	NR
			Immune thrombocytopenia	3.97	(2.35-6.69)	NR
			Interstitial lung disease	11.54	(6.94-19.2)	NR
			Morphoea	3.63	(1.59-8.28)	NR
			Multiple sclerosis	1.50	(0.96-2.33)	NR
			Myasthenia gravis	2.36	(0.59-9.43)	NR
			Pernicious anaemia	1.30	(0.75-2.27)	NR
			Polymyalgia rheumatica	8.20	(4.34-15.5)	NR
			Polymyositis	7.55	(2.47-23.09)	NR
			Primary biliary cirrhosis	2.83	(0.68-11.84)	NR
			Psoriasis	2.31	(1.73-3.08)	NR
			Raynaud's syndrome	3.24	(2.33-4.50)	NR
			Scleritis and episcleritis	5.25	(2.78-9.94)	NR
			Sjogren's syndrome/sicca syndrome	6.82	(4.76-9.77)	NR
			Systemic sclerosis	10.80	(4.44-26.26)	NR
			Type 1 diabetes	2.50	(1.98-3.14)	NR
			Ulcerative colitis	3.69	(2.69-5.06)	NR
			Uveitis	18.46	(13.8-24.69)	NR
			Vasculitis	12.13	(7.03-20.94)	NR
		PharMetrics patients with ADHD	Addison disease	4.67	(2.83-7.73)	NR

			Alopecia areata	1.66	(0.81-3.41)	NR
			Coeliac disease	2.63	(1.68-4.11)	NR
			Chronic glomerulonephritis	24.06	(2.81-205.96)	NR
			Chronic urticaria	2.03	(1.55-2.66)	NR
			Crohn's disease	3.23	(2.36-4.42)	NR
			Erythema nodosum	7.70	(2.52-23.55)	NR
			Giant cell arteritis	3.21	(0.54-19.19)	NR
			Grave's disease	1.45	(0.83-2.54)	NR
			Hashimoto's thyroiditis	1.99	(1.42-2.80)	NR
			Haemolytic anaemia	5.46	(2.73-10.94)	NR
			Immune thrombocytopenic	4.82	(2.51-9.27)	NR
			Interstitial lung disease	12.89	(6.63-25.04)	NR
			Morphoea	4.81	(1.69-13.73)	NR
			Multiple sclerosis	1.52	(0.89-2.58)	NR
			Myasthenia gravis	3.85	(1.03-14.34)	NR
			Pernicious anaemia	1.24	(0.70-2.20)	NR
			Polymyalgia rheumatica	3.78	(1.72-8.34)	NR
			Polymyositis	4.01	(1.22-13.14)	NR
			Primary biliary cirrhosis	2.40	(0.22-26.52)	NR
			Psoriasis	2.93	(2.22-3.86)	NR
			Raynaud's syndrome	2.53	(1.76-3.65)	NR
			Scleritis and episcleritis	3.66	(2.02-6.66)	NR
			Sjogren's syndrome/sicca syndrome	6.86	(4.70-10.02)	NR
			Systemic sclerosis	8.83	(3.26-23.88)	NR
			Type 1 diabetes	2.15	(1.66-2.78)	NR
			Ulcerative colitis	2.41	(1.63-3.57)	NR
			Uveitis	29.54	(21.51-40.58)	NR
			Vasculitis	7.63	(4.29-13.56)	NR
			Vitiligo	2.16	(1.12-4.15)	NR
Ma, et al, 2022, Taiwan[70]	Relative risk	Individuals without JIA	Obstructive sleep apnoea	1.936	(1.26-2.97)	NR
	Adjusted hazard ratios	Adjusted for demographic variables including age, sex, length of hospital stays, and comorbidities at baseline.	Obstructive sleep apnoea	1.922	(1.24–2.97)	NR

	Adjusted hazard ratios	Adjusted for demographic variables including age, sex, length of hospital stays, and comorbidities at baseline and allergic rhinitis	Obstructive sleep apnoea	1.363	(0.70-2.65)	NS
Zhang-Jian, et al, 2020, Taiwan[71]	Adjusted odds ratio (for infant sex, Charlson comorbidity index, urbanization, income, occupation, maternal nationality)	Non-JIA patients 1:5 matched maternal age and birth year To avoid possible confounding factors and for case control selection, only the first child from either group was included for further analysis.	Stillbirth	0.92	(0.32–2.69)	NR
			Foetal distress	0.96	(0.65–1.41)	NR
			Foetal abnormalities, any	0.91	(0.64–1.29)	NR
			Pre-eclampsia	1.42	(0.82–2.46)	NR
			Gestational diabetes	0.75	(0.48–1.16)	NR
			Pregnancy-related hypertension	1.24	(0.79–1.96)	NR
			Antepartum haemorrhage	1.09	(0.81–1.47)	NR
			Severe postpartum haemorrhage	1.20	(0.77–1.87)	NR
			Chorioamnionitis	1.34	(0.49–3.66)	NR
			Caesarean delivery	1.07	(0.89–1.27)	NR
			Low birth weight <2500g	1.35	(1.02–1.79)	<0.05
			Prematurity <37 weeks	1.19	(0.87–1.64)	NR
Krause, et al, 2017, USA[37]	Rate ratio	Non-JIA comparator Age and sex matched	Joint surgeries	8.5	(2.27-120.10)	NR
			Non-joint surgeries	1.92	(0.89-4.57)	NR
			Hospitalizations	1.79	(0.69-5.25)	NR
			Hospitalizations for depression	0.26	(0.02-1.11)	NR
			Pregnancies	0.68	(0.38-1.20)	NS
Childhood / jSLE						
Frittoli, et al, 2016, Brazil[79]	% difference (absolute)	Patients with cognitive dysfunction. Gender, age and socioeconomic status matched	Cognitive dysfunction	4.86		0.080
Gad, et al, 2013, Egypt[81]	% difference (absolute)	Healthy Children. Age and sex matched	Sensorineural hearing loss	25.00		NR
			Conductive hearing loss	0		NR
Karve, et al, 2012, USA[12]	% difference (absolute)	Comparator without IBD or SLE. Age (calculated as of year 2005) and sex, and with a minimum health plan enrolment of 6 months before and 12 months after the assigned index date of the corresponding patient were matched.	Asthma	8.50	NA	NR
			Atopic dermatitis	0.70	NA	NR
			Allergic rhinitis	2.90	NA	NR
			Thyroiditis	0.70	NA	NR

			Fracture in post-index period	2.10	NA	NR
			Psoriasis	0.00	NA	NR
			Type 1 diabetes	-0.70	NA	NR
			General diabetes	0.00	NA	NR
			Multiple sclerosis	1.40	NA	NR
			Other CNS demyelinating disease	0.00	NA	NR
			Vitiligo	0.00	NA	NR
			Rheumatoid arthritis	19.7	NA	NR
Sinicato, et al, 2017, Brazil[94]	% difference (absolute)	Volunteers with no history of chronic, cardiovascular, and autoimmune disease. Age, gender, and demographic background matched	High LDL-c levels	33.00		0.300
			Hypercholesterolemia	16.17		0.210
			Insulin resistance	4.90		0.177
			Arterial hypertension	9.80		0.074
			Albumin excretion	12.20		0.080
Adulthood/ jSLE						
Sinicato, et al, 2017, Brazil[94]	% difference (absolute)	Volunteers with no history of chronic, cardiovascular, and autoimmune disease. Age, gender, and demographic background matched	Insulin resistance	0		0.141
			Hypertension	11.4		0.086
			Albumin excretion	11.4		0.086

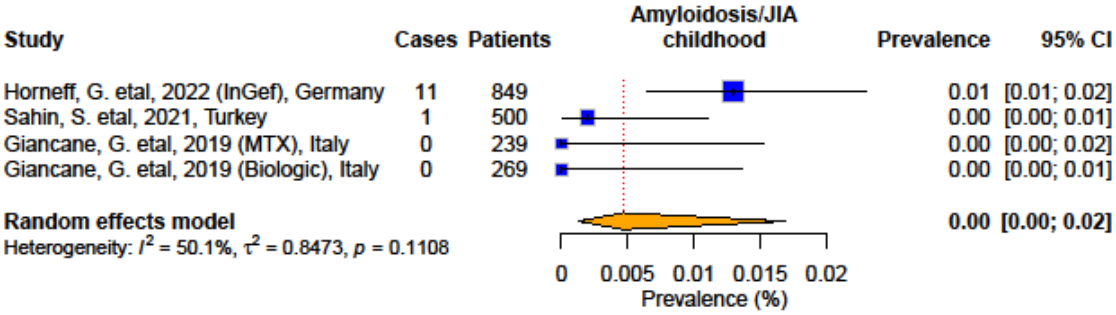
Supplementary Figures S4.1: Meta-analysis: Pooled estimates, Forest plots and Funnel plots

Childhood/JIA

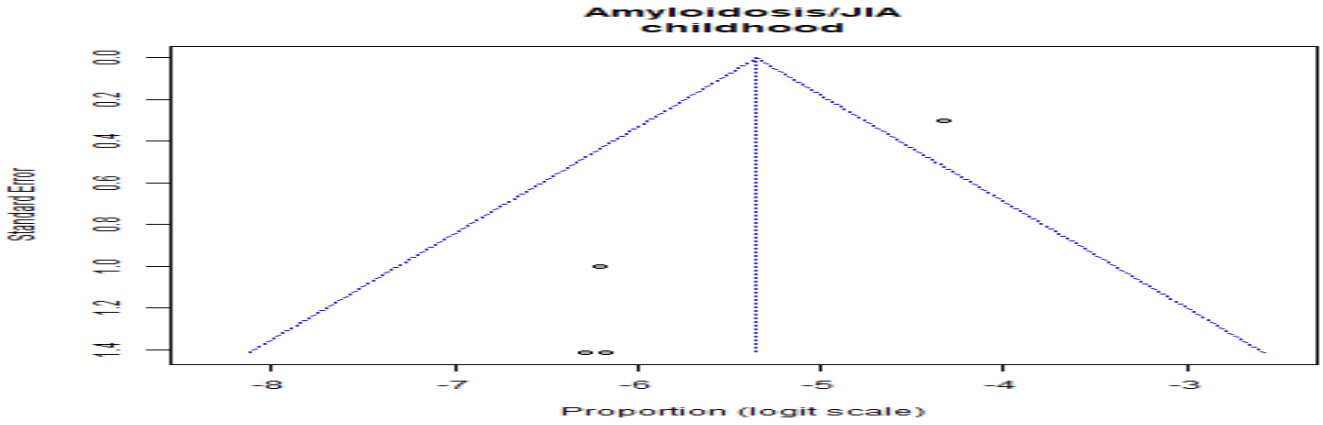
Supplementary Figure S4.1: Forest and Funnel Plots of studies reporting the prevalence of amyloidosis in children with Juvenile Idiopathic Arthritis

Amyloidosis

Forest



Funnel



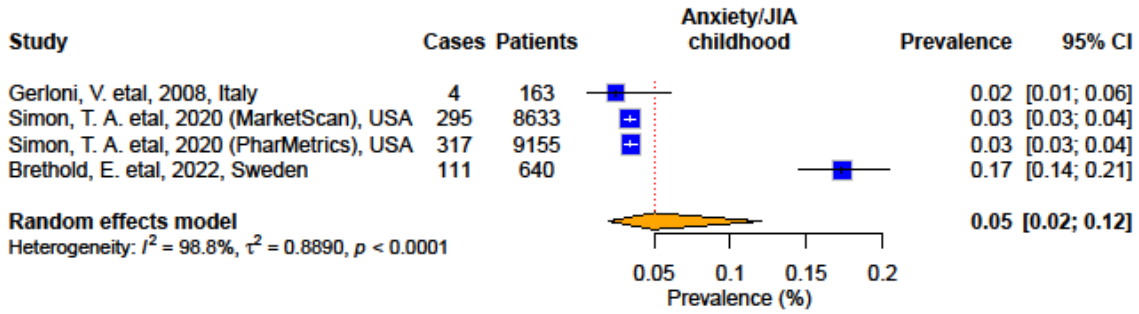
Summary statistics

```
                                        proportion           95%-CI     %W(random)
Horneff G. etal, 2022 (InGef), Germany      0.0130 [0.0065; 0.0231]     46.1
Sahin, S. etal, 2021, Turkey                0.0020 [0.0001; 0.0111]     23.4
Giancane G etal, 2019 (MTX), Italy          0.0000 [0.0000; 0.0153]     15.2
Giancane G etal, 2019 (Biologic), Italy     0.0000 [0.0000; 0.0136]     15.2
Number of studies: k = 4; Number of observations: o = 1857; Number of events: e = 12
                    proportion           95%-CI
Random effects model     0.0047 [0.0013; 0.0170]
Quantifying heterogeneity (with 95%-CIs): tau^2 = 0.8473 [0.0000; 11.9786]; tau = 0.9205 [0.0000; 3.4610]
 I^2 = 50.1% [0.0%; 83.5%]; H = 1.42 [1.00; 2.46]
Test of heterogeneity:
    Q d.f. p-value
 6.02    3  0.1108
```

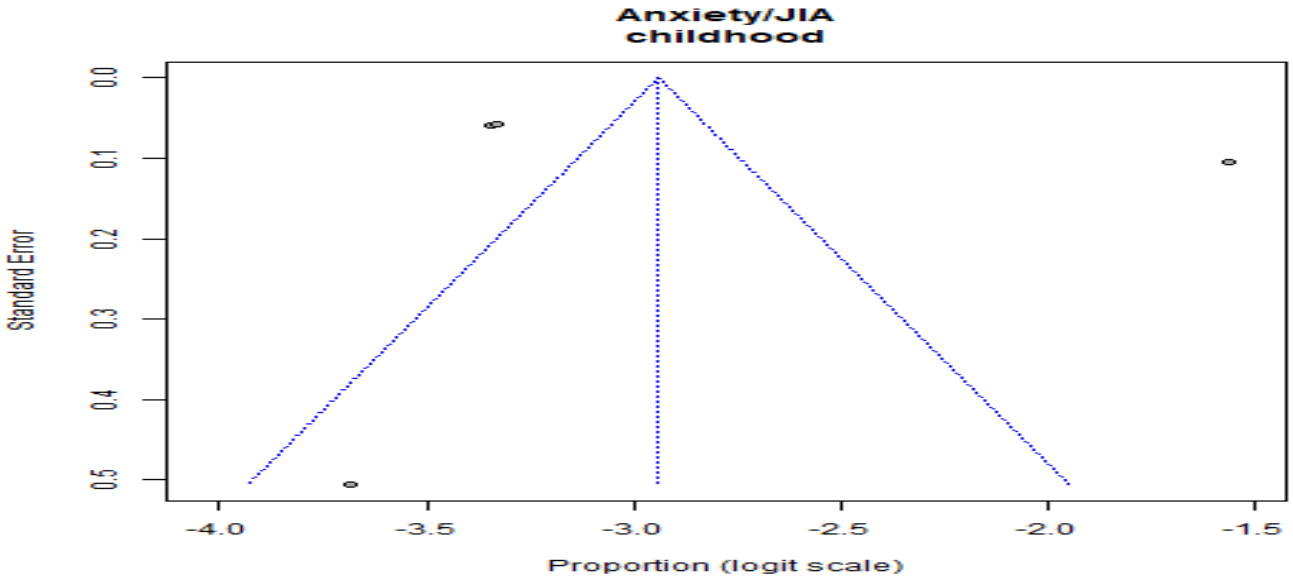
Supplementary Figure S4.2: Forest and Funnel Plots of studies reporting the prevalence of anxiety in children with Juvenile Idiopathic Arthritis

Anxiety

Forest



Funnel



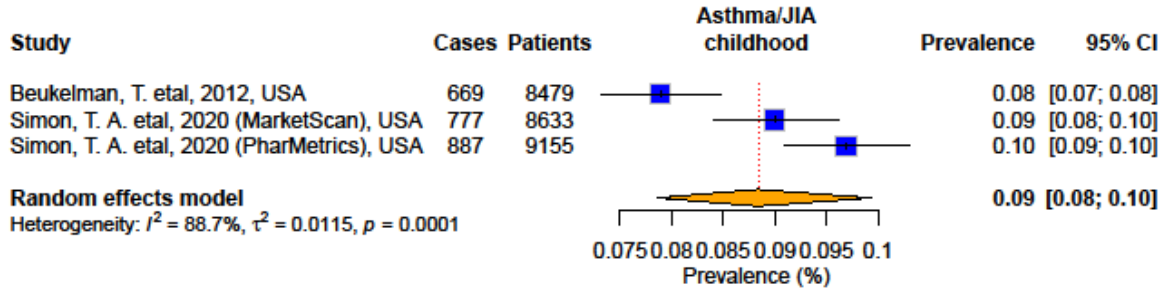
Summary statistics

```
                                          proportion           95%-CI  %W(random)
Gerloni, V. etal, 2008, Italy                 0.0245 [0.0067; 0.0616]  20.7
Simon, T. A. etal, 2020 (MarketScan), USA     0.0342 [0.0304; 0.0382]  26.5
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0346 [0.0310; 0.0386]  26.5
Brethold E. etal, 2022, Sweden                0.1734 [0.1449; 0.2050]  26.3
Number of studies: k = 4; Number of observations: o = 18591; Number of events: e = 727
                    proportion           95%-CI
Random effects model     0.0502 [0.0200; 0.1206]
Quantifying heterogeneity (with 95%-CIs): tau^2 = 0.8890 [0.2565; 12.7255]; tau = 0.9429 [0.5065; 3.5673]
 I^2 = 98.8% [98.2%; 99.2%]; H = 9.15 [7.38; 11.34]
Test of heterogeneity:
      Q d.f.  p-value
 251.03    3 < 0.0001
```

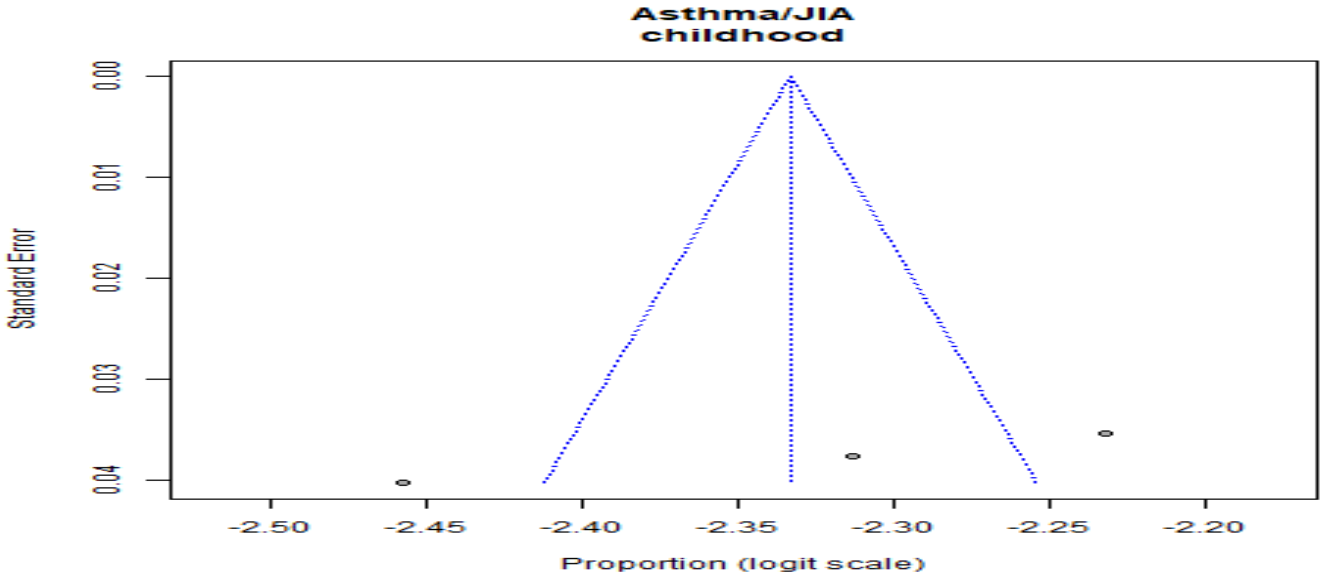
Supplementary Figure S4.3: Forest and Funnel Plots of studies reporting the prevalence of asthma in children with Juvenile Idiopathic Arthritis

Asthma

Forest



Funnel



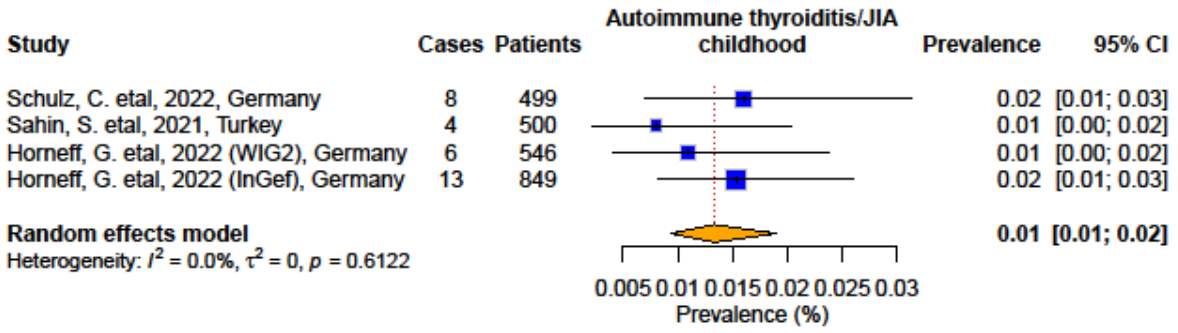
Summary statistics

```
                                        proportion           95%-CI  %W(random)
Beukelman, T. etal, 2012, USA               0.0789 [0.0732; 0.0848]  32.8
Simon, T. A. etal, 2020 (MarketScan), USA   0.0900 [0.0840; 0.0962]  33.4
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0969 [0.0909; 0.1031]  33.8
Number of studies: k = 3; Number of observations: o = 26267; Number of events: e = 2333
                     proportion           95%-CI
Random effects model     0.0884 [0.0786; 0.0993]
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.0115 [0.0021; 0.5116]; tau = 0.1072 [0.0454; 0.7153]
 I^2 = 88.7% [69.0%; 95.9%]; H = 2.98 [1.80; 4.94]
Test of heterogeneity:     Q d.f. p-value
                       17.77    2  0.0001
```

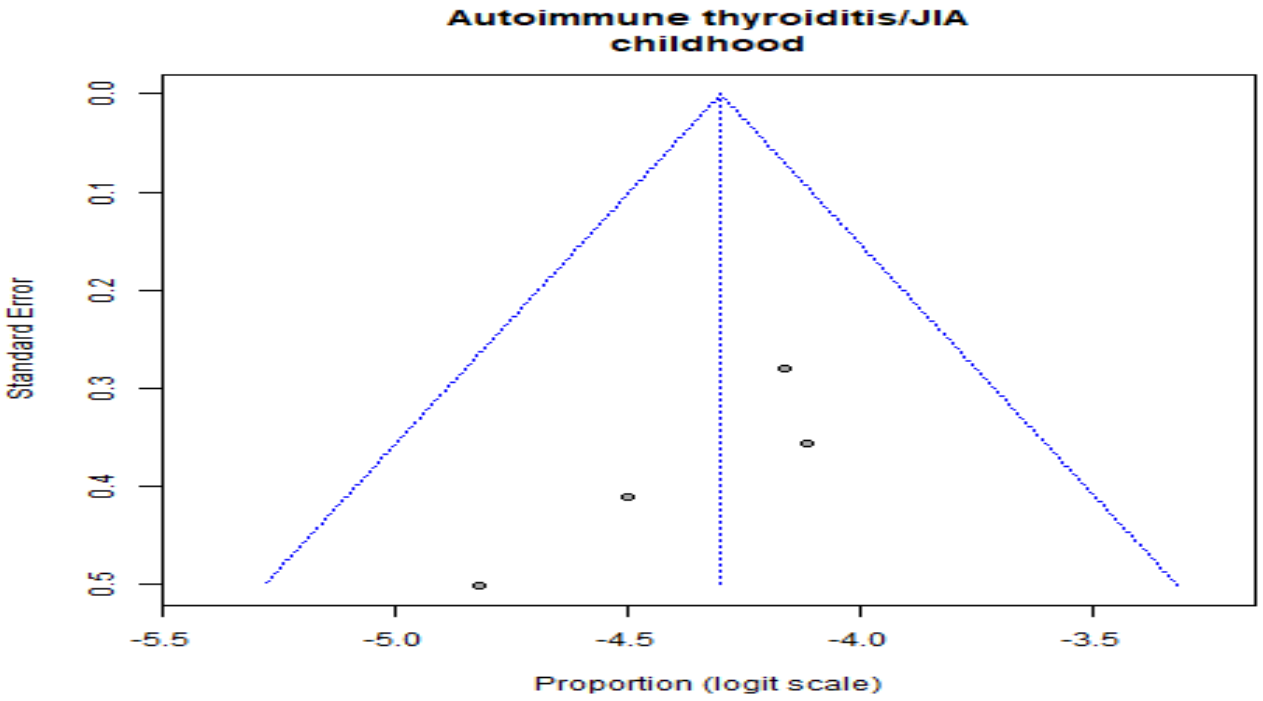
Supplementary Figure S4.4: Forest and Funnel Plots of studies reporting the prevalence of autoimmune thyroiditis in children with Juvenile Idiopathic Arthritis

Autoimmune thyroiditis

Forest



Funnel



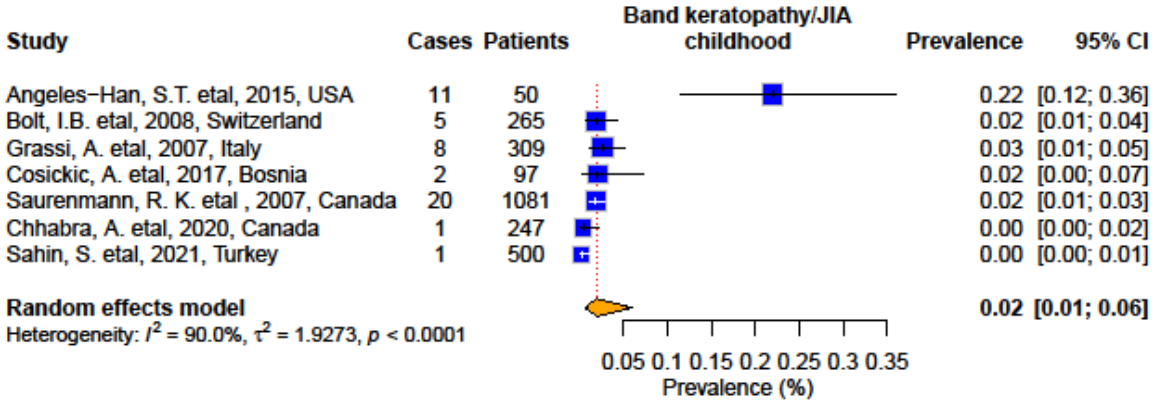
Summary statistics

```
                                    proportion           95%-CI      %W(random)
Schulz, C. etal, 2022, Germany          0.0160 [0.0069; 0.0313]      25.7
Sahin, S. etal, 2021, Turkey            0.0080 [0.0022; 0.0204]      13.0
Horneff G. etal, 2022 (WIG2), Germany   0.0110 [0.0040; 0.0238]      19.4
Horneff G. etal, 2022 (InGef), Germany  0.0153 [0.0082; 0.0260]      41.9
Number of studies: k = 4; Number of observations: o = 2394; Number of events: e = 31
                    proportion          95%-CI
Random effects model     0.0134 [0.0094; 0.0189]
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0 [0.0000; 1.3075]; tau = 0 [0.0000; 1.1435]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.81    3  0.6122
```

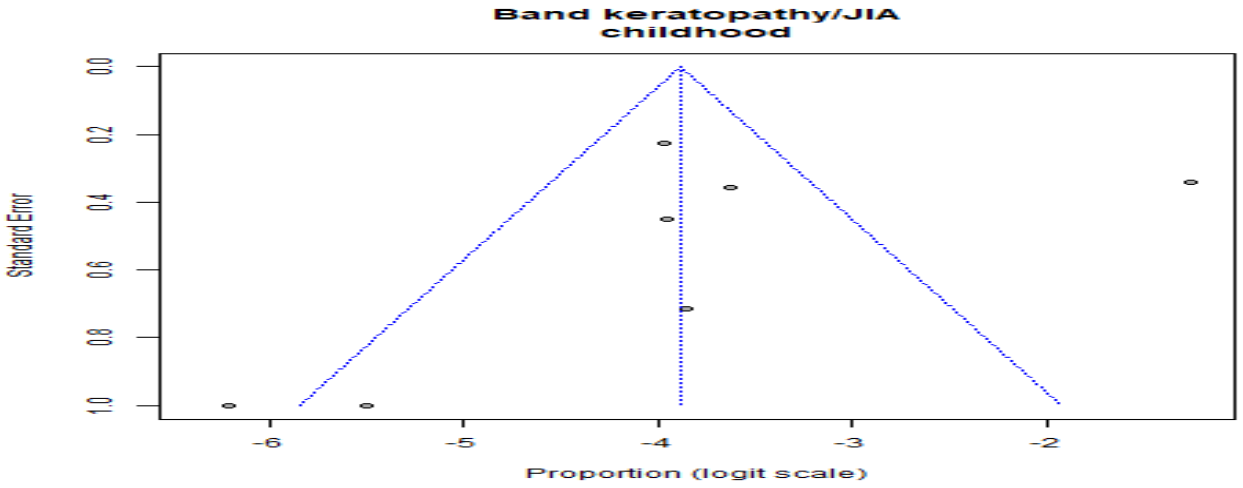
Supplementary Figure S4.5: Forest and Funnel Plots of studies reporting the prevalence of band keratopathy in children with Juvenile Idiopathic Arthritis

Band keratopathy

Forest



Funnel



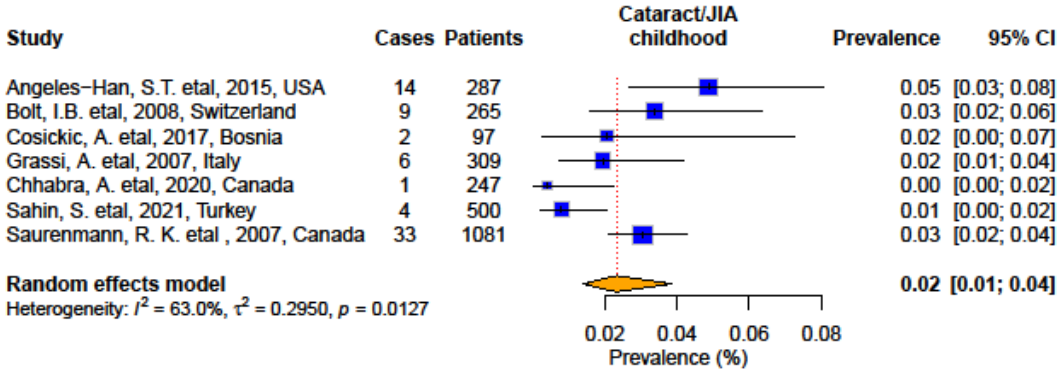
Summary statistics

```
                                    proportion           95%-CI       %W(random)
Angeles-Han S.T. etal, 2015, USA        0.2200 [0.1153; 0.3596]       16.1
Bolt I.B. etal, 2008, Switzerland       0.0189 [0.0062; 0.0435]       15.4
Grassi, A. etal, 2007, Italy            0.0259 [0.0112; 0.0504]       16.0
Cosickic, A. etal, 2017, Bosnia         0.0206 [0.0025; 0.0725]       13.5
Saurenmann, R. K. etal , 2007, Canada   0.0185 [0.0113; 0.0284]       16.6
Chhabra, A. etal, 2020, Canada          0.0040 [0.0001; 0.0223]       11.2
Sahin, S. etal, 2021, Turkey            0.0020 [0.0001; 0.0111]       11.2
Number of studies: k = 7; Number of observations: o = 2549; Number of events: e = 48
                     proportion           95%-CI
Random effects model     0.0201 [0.0066; 0.0593]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.9273 [0.6009; 11.4288]; tau = 1.3883 [0.7752; 3.3806]
 I^2 = 90.0% [81.9%; 94.5%]; H = 3.16 [2.35; 4.25]
Test of heterogeneity:      Q d.f.  p-value
                        | 59.88    6 < 0.0001
```

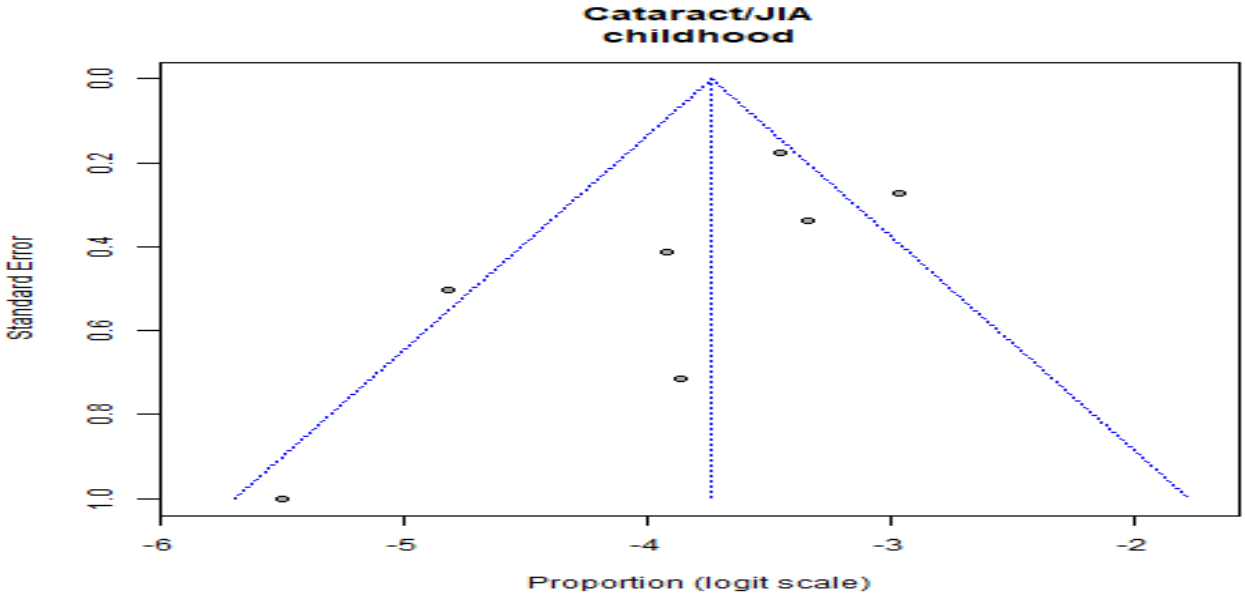
Supplementary Figure S4.6: Forest and Funnel Plots of studies reporting the prevalence of cataracts in children with Juvenile Idiopathic Arthritis

Cataract

Forest



Funnel



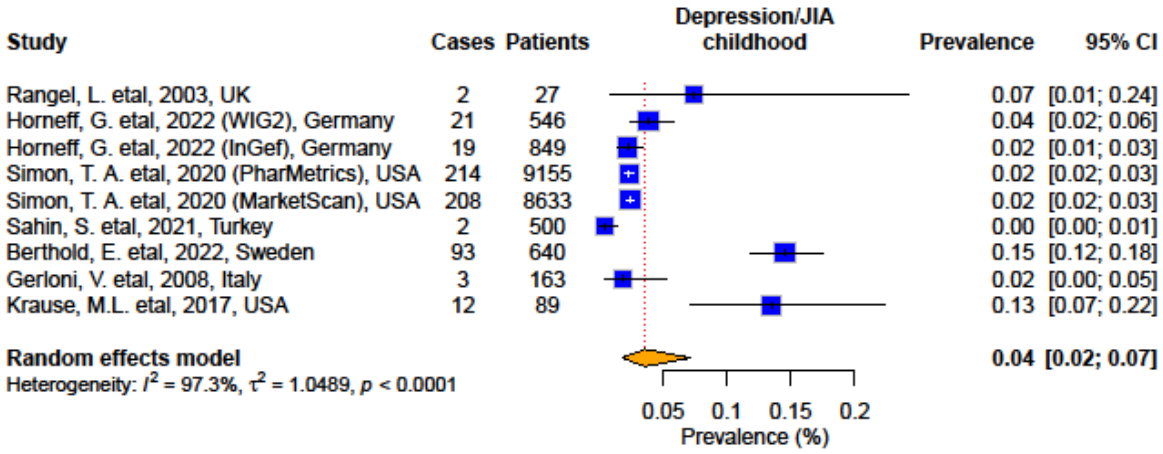
Summary statistics

```
                                    proportion           95%-CI       %W(random)
Angeles-Han S.T. etal, 2015, USA        0.0488 [0.0269; 0.0805]       19.0
Bolt I.B. etal, 2008, Switzerland       0.0340 [0.0156; 0.0635]       17.2
Cosickic, A. etal, 2017, Bosnia         0.0206 [0.0025; 0.0725]       8.7
Grassi, A. etal, 2007, Italy            0.0194 [0.0072; 0.0418]       15.2
Chhabra, A. etal, 2020, Canada          0.0040 [0.0001; 0.0223]       5.4
Sahin, S. etal, 2021, Turkey            0.0080 [0.0022; 0.0204]       12.9
Saurenmann, R. K. etal , 2007, Canada   0.0305 [0.0211; 0.0426]       21.6
Number of studies: k = 7; Number of observations: o = 2786; Number of events: e = 69
                     proportion           95%-CI
Random effects model     0.0232 [0.0139; 0.0385]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2950 [0.0217; 3.3077]; tau = 0.5432 [0.1474; 1.8187]
 I^2 = 63.0% [16.0%; 83.7%]; H = 1.64 [1.09; 2.48]
Test of heterogeneity:      Q d.f. p-value
                        16.21    6  0.0127
```

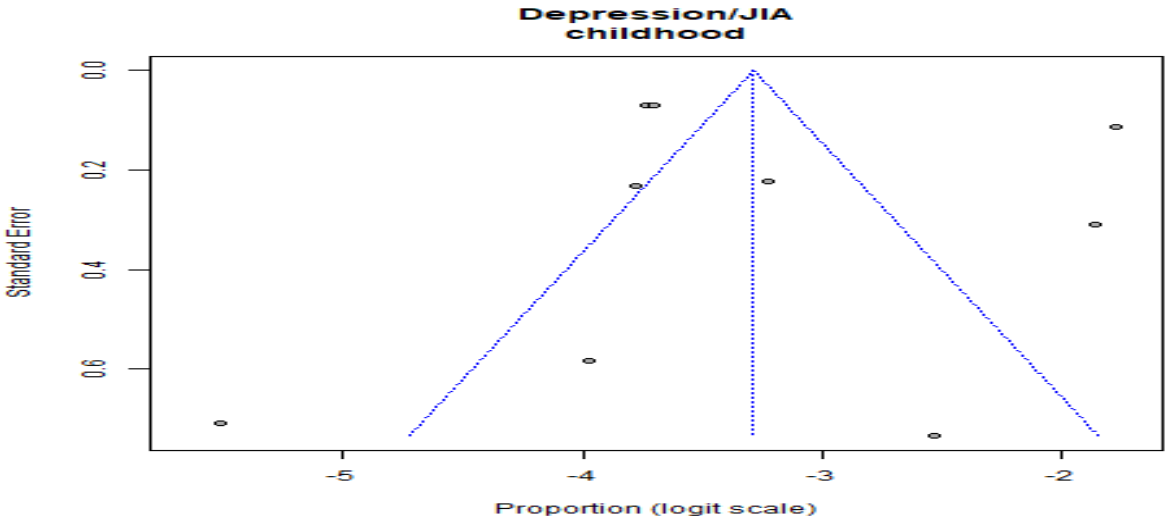
Supplementary Figure S4.7: Forest and Funnel Plots of studies reporting the prevalence of depression in children with Juvenile Idiopathic Arthritis

Depression

Forest



Funnel



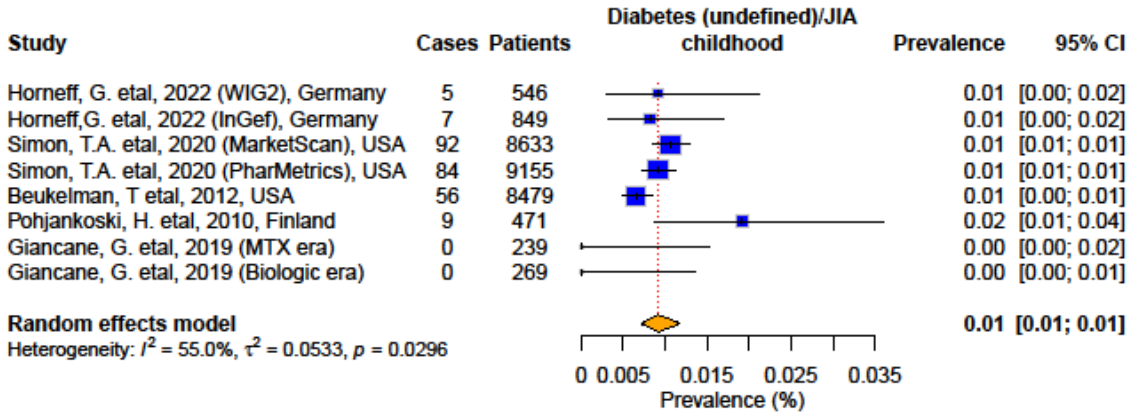
Summary statistics

```
                                          proportion           95%-CI   %W(random)
Rangel, L. etal, 2003, UK                     0.0741 [0.0091; 0.2429]  8.4
Horneff G. etal, 2022 (WIG2), Germany         0.0385 [0.0240; 0.0582]  12.1
Horneff G. etal, 2022 (InGef), Germany        0.0224 [0.0135; 0.0347]  12.1
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0234 [0.0204; 0.0267]  12.6
Simon, T. A. etal, 2020 (MarketScan), USA     0.0241 [0.0210; 0.0276]  12.6
Sahin, S. etal, 2021                          0.0040 [0.0005; 0.0144]  8.6
Berthold, E. etal, 2022                       0.1453 [0.1189; 0.1750]  12.5
Gerloni, V. etal, 2008, Italy                 0.0184 [0.0038; 0.0528]  9.6
Krause ML et al, 2017                         0.1348 [0.0717; 0.2237]  11.6
Number of studies: k = 9; Number of observations: o = 20602; Number of events: e = 574
                     proportion           95%-CI
Random effects model     0.0361 [0.0180; 0.0711]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0489 [0.4100; 4.7484]; tau = 1.0242 [0.6403; 2.1791]
 I^2 = 97.3% [96.1%; 98.1%]; H = 6.04 [5.07; 7.18]
Test of heterogeneity:       Q d.f.  p-value
                        291.44    8 < 0.0001
```

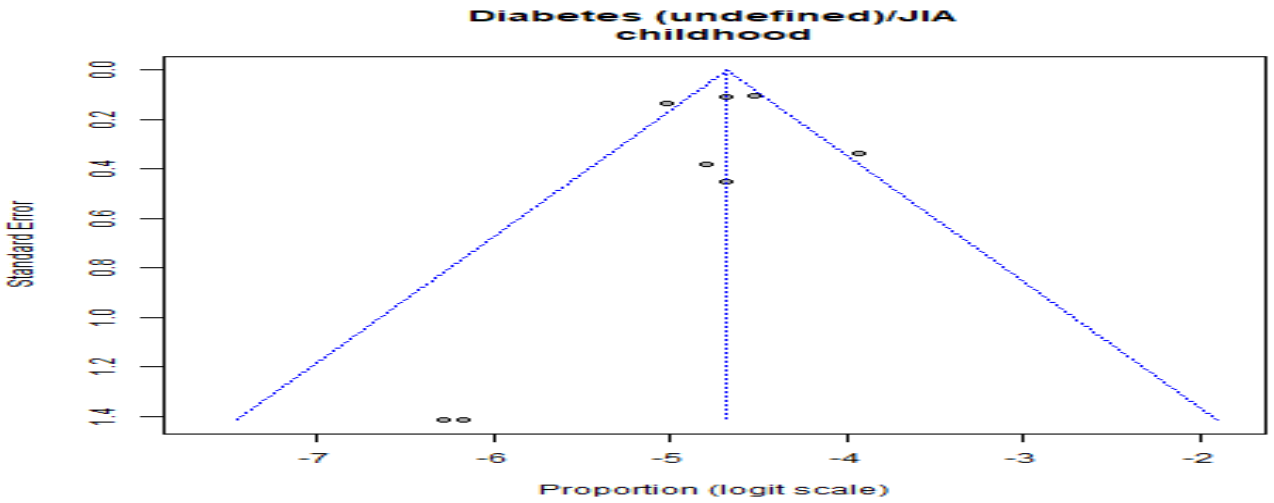
Supplementary Figure S4.8: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with Juvenile Idiopathic Arthritis

Diabetes (undefined)

Forest



Funnel



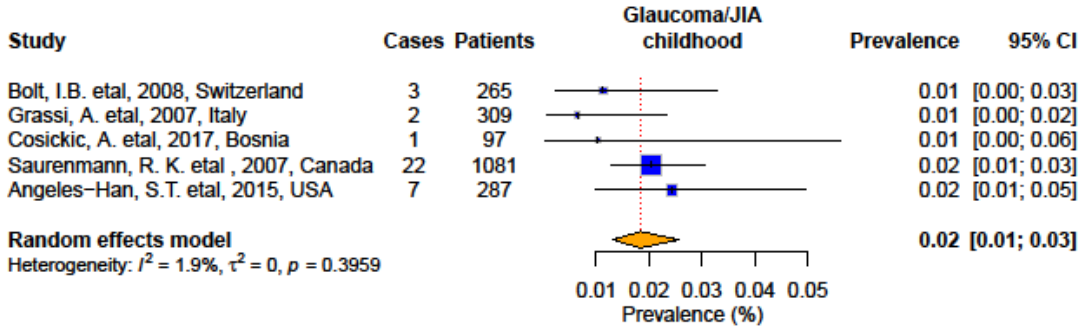
Summary statistics

```
                                        proportion           95%-CI    %W(random)
Horneff, G. etal, 2022 (WIG2), Germany      0.0092 [0.0030; 0.0212]    6.4
Horneff,G. etal, 2022 (InGef), Germany      0.0082 [0.0033; 0.0169]    8.3
Simon, TA. etal, 2020 (MarketScan), USA     0.0107 [0.0086; 0.0131]   25.6
Simon, TA. etal, 2020 (PharMetrics), USA    0.0092 [0.0073; 0.0113]   25.2
Beukelman, T etal, 2012, USA                0.0066 [0.0050; 0.0086]   23.1
Pohjankoski, H. etal, 2010, Finland         0.0191 [0.0088; 0.0360]    9.9
Giancane G etal, 2019 (MTX era)             0.0000 [0.0000; 0.0153]    0.8
Giancane G etal, 2019 (Biologic era)        0.0000 [0.0000; 0.0136]    0.8
Number of studies: k = 8; Number of observations: o = 28641; Number of events: e = 253
                    proportion           95%-CI
Random effects model     0.0092 [0.0072; 0.0118]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0533 [0.0000; 1.5500]; tau = 0.2309 [0.0000; 1.2450]
 I^2 = 55.0% [0.4%; 79.6%]; H = 1.49 [1.00; 2.22]
Test of heterogeneity:      Q d.f. p-value
                        15.55    7  0.0296
```

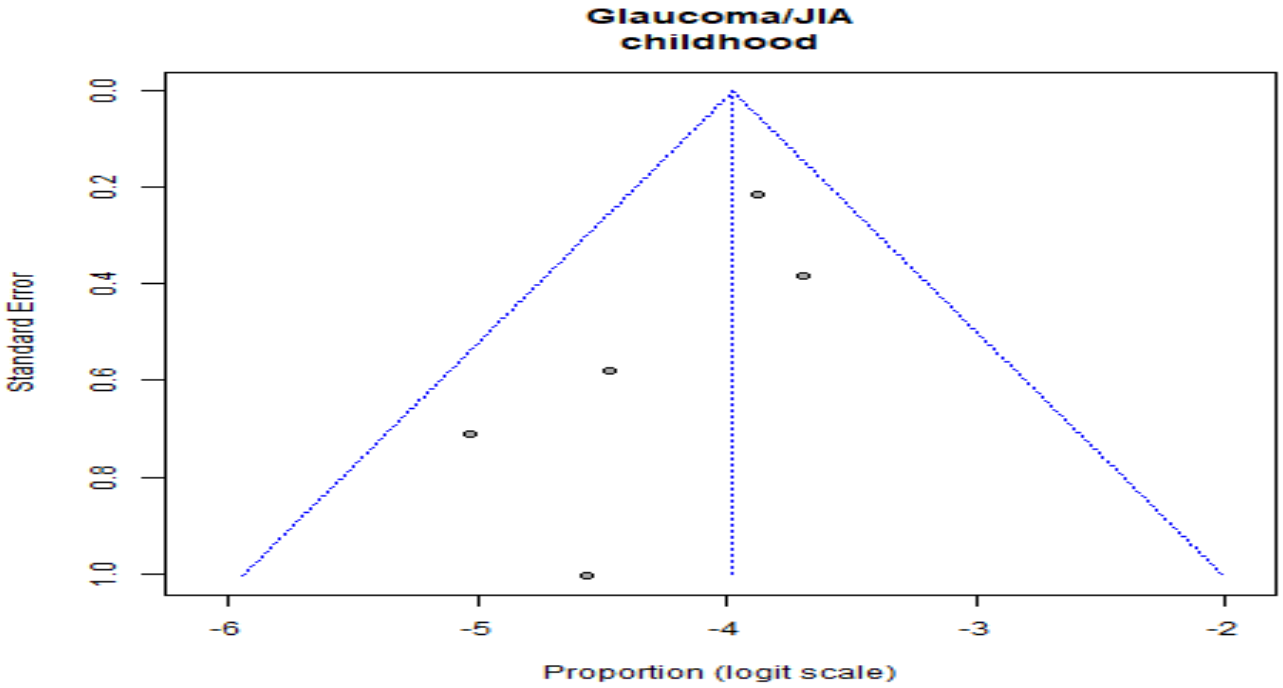
Supplementary Figure S4.9: Forest and Funnel Plots of studies reporting the prevalence of glaucoma in children with Juvenile Idiopathic Arthritis

Glaucoma

Forest



Funnel



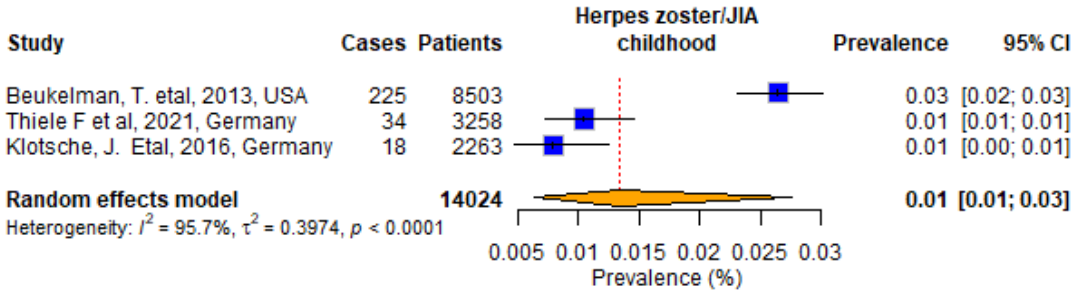
Summary statistics

```
                                        proportion           95%-CI       %W(random)
Bolt I.B. etal, 2008, Switzerland           0.0113 [0.0023; 0.0327]       8.6
Grassi, A. etal, 2007, Italy                0.0065 [0.0008; 0.0232]       5.8
Cosickic, A. etal, 2017, Bosnia             0.0103 [0.0003; 0.0561]       2.9
Saurenmann, R. K. etal , 2007, Canada       0.0204 [0.0128; 0.0307]       62.8
Angeles-Han ST et al, 2015, USA             0.0244 [0.0099; 0.0496]       19.9
Number of studies: k = 5; Number of observations: o = 2039; Number of events: e = 35
                     proportion           95%-CI
Random effects model     0.0184 [0.0133; 0.0256]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 2.1342]; tau = 0 [0.0000; 1.4609]
 I^2 = 1.9% [0.0%; 79.6%]; H = 1.01 [1.00; 2.21]
Test of heterogeneity:     Q d.f. p-value
                        4.08    4  0.3959
```

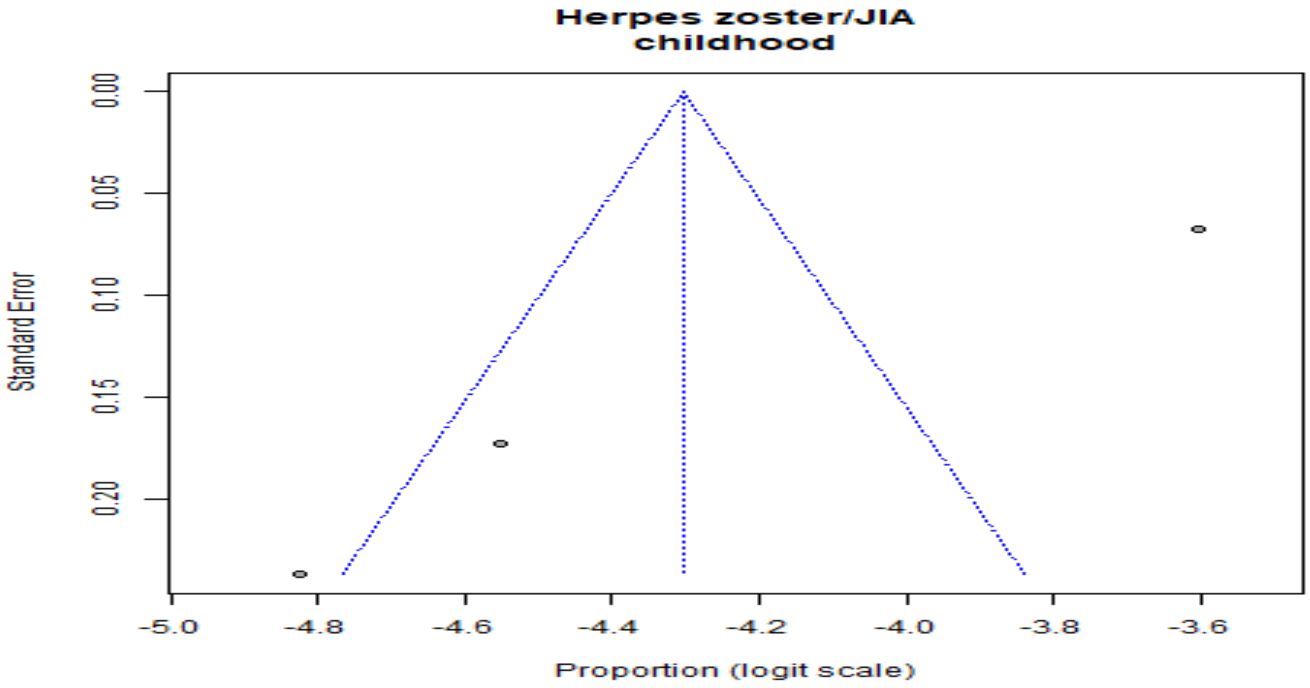
Supplementary Figure S4.10: Forest and Funnel Plots of studies reporting the prevalence of herpes zoster in children with Juvenile Idiopathic Arthritis

Herpes zoster

Forest



Funnel



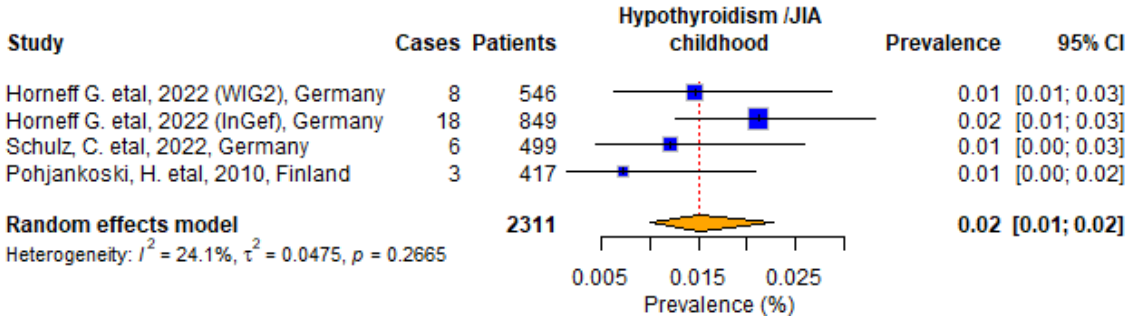
Summary statistics

```
                                proportion           95%-CI   %W(random)
Beukelman, T. etal, 2013, USA       0.0265 [0.0232; 0.0301]       35.4
Thiele F et al, 2021, Germany       0.0104 [0.0072; 0.0146]       33.3
Klotsche, J.  Etal, 2016, Germany   0.0080 [0.0047; 0.0125]       31.4
Number of studies: k = 3; Number of observations: o = 14024; Number of events: e = 277
                     proportion           95%-CI
Random effects model     0.0133 [0.0064; 0.0275]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3974 [0.0907; 16.1850]; tau = 0.6304 [0.3012; 4.0231]
 I^2 = 95.7% [90.5%; 98.0%]; H = 4.80 [3.25; 7.10]
Test of heterogeneity:      Q d.f.  p-value
                        46.13    2 < 0.0001
```

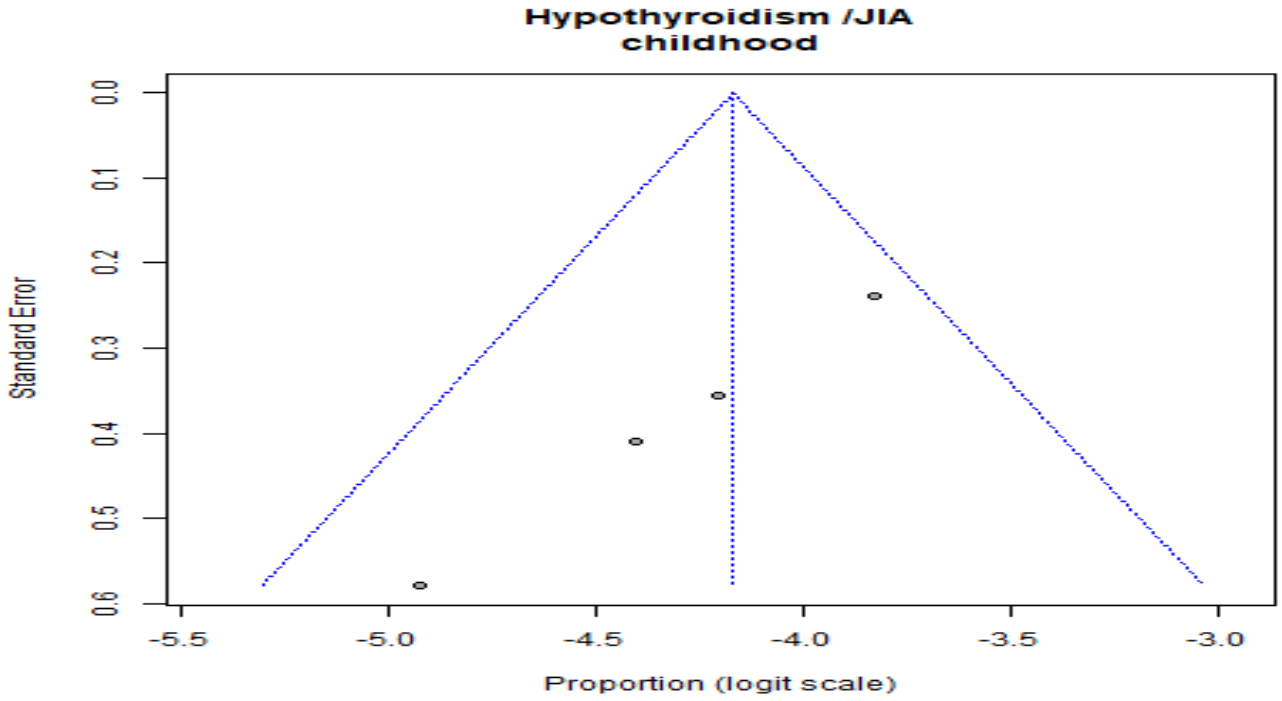
Supplementary Figure S4.11: Forest and Funnel Plots of studies reporting the prevalence of hypothyroidism in children with Juvenile Idiopathic Arthritis

Hypothyroidism

Forest



Funnel



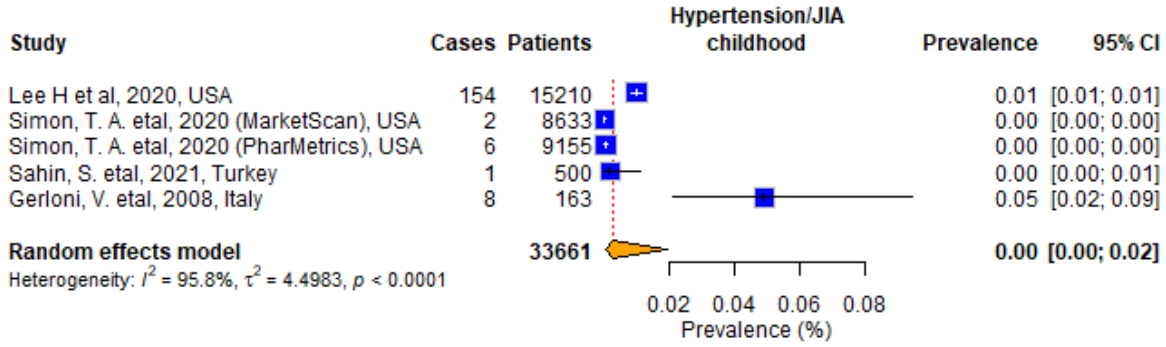
Summary statistics

```
                                        proportion           95%-CI      %W(random)
Horneff G. etal, 2022 (WIG2), Germany       0.0147 [0.0063; 0.0287]      25.4
Horneff G. etal, 2022 (InGef), Germany      0.0212 [0.0126; 0.0333]      42.5
Schulz, C. etal, 2022, Germany              0.0120 [0.0044; 0.0260]      20.5
Pohjankoski, H. etal, 2010, Finland         0.0072 [0.0015; 0.0209]      11.6
Number of studies: k = 4; Number of observations: o = 2311; Number of events: e = 35
                     proportion           95%-CI
Random effects model     0.0152 [0.0101; 0.0228]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0475 [0.0000; 2.6861]; tau = 0.2179 [0.0000; 1.6389]
 I^2 = 24.1% [0.0%; 88.4%]; H = 1.15 [1.00; 2.93]
Test of heterogeneity:     Q d.f. p-value
                         | 3.95    3  0.2665
```

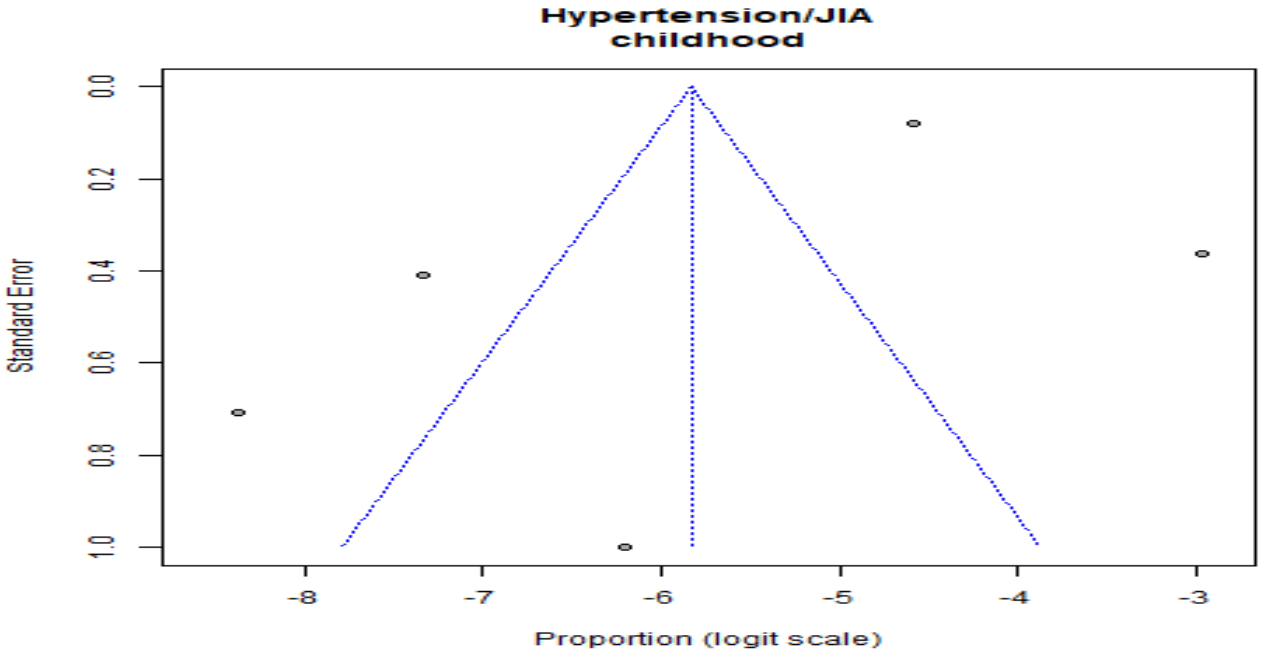
Supplementary Figure S4.12: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with Juvenile Idiopathic Arthritis

Hypertension

Forest



Funnel



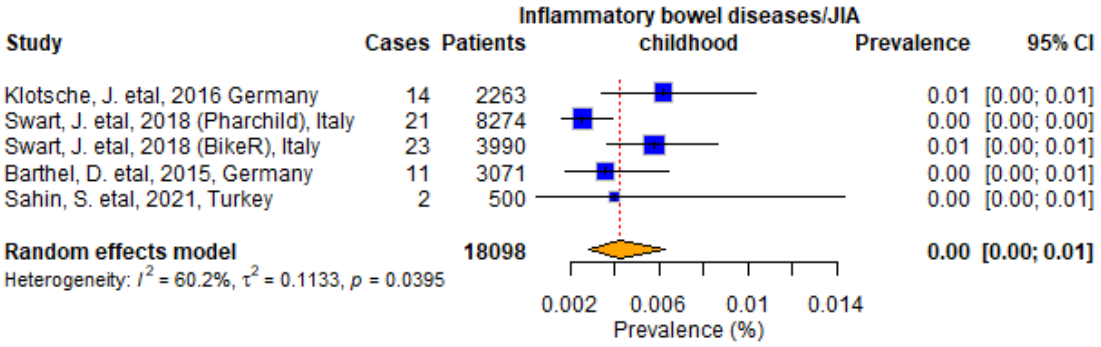
Summary statistics

```
                                          proportion           95%-CI  %W(random)
Lee H et al, 2020, USA                        0.0101 [0.0086; 0.0118]  21.5
Simon, T. A. etal, 2020 (MarketScan), USA     0.0002 [0.0000; 0.0008]  19.3
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0007 [0.0002; 0.0014]  20.7
Sahin, S. etal, 2021, Turkey                  0.0020 [0.0001; 0.0111]  17.6
Gerloni, V. etal, 2008, Italy                 0.0491 [0.0214; 0.0944]  20.9
Number of studies: k = 5; Number of observations: o = 33661; Number of events: e = 171
                     proportion           95%-CI
Random effects model     0.0029 [0.0004; 0.0197]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 4.4983 [1.4273; 38.1419]; tau = 2.1209 [1.1947; 6.1759]
 I^2 = 95.8% [92.7%; 97.6%]; H = 4.89 [3.70; 6.45]
Test of heterogeneity:      Q d.f.  p-value
                        95.46    4 < 0.0001
```

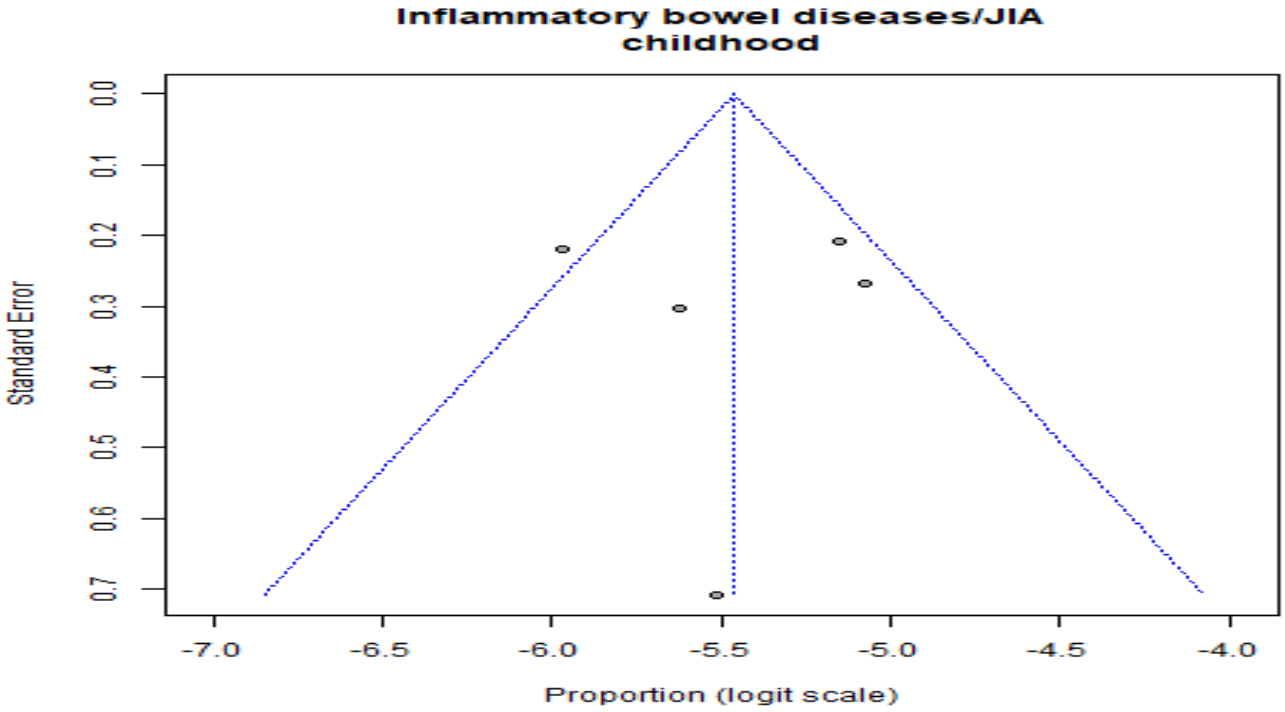
Supplementary Figure S4.13: Forest and Funnel Plots of studies reporting the prevalence of Inflammatory Bowel Diseases in children with Juvenile Idiopathic Arthritis

Inflammatory bowel diseases

Forest



Funnel



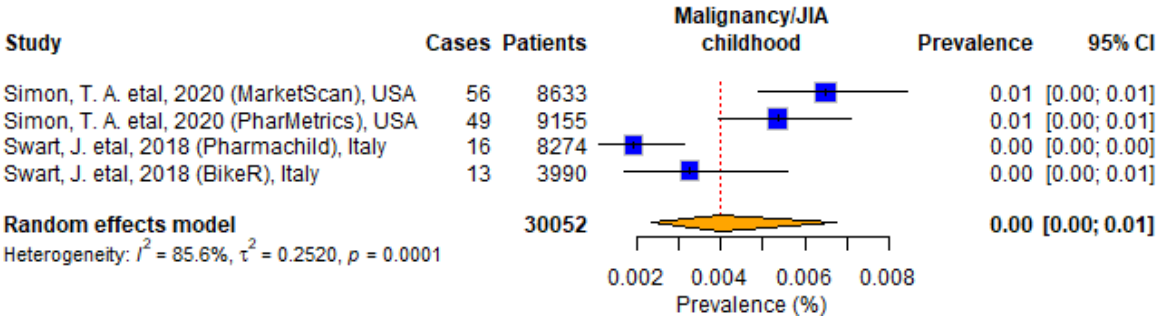
Summary statistics

```
                                        proportion           95%-CI     %W(random)
Klotsche, J. etal, 2016 Germany             0.0062 [0.0034; 0.0104]     22.0
Swart, J. etal, 2018 (Pharchild), Italy     0.0025 [0.0016; 0.0039]     25.4
Swart, J. etal, 2018 (BikeR), Italy         0.0058 [0.0037; 0.0086]     26.0
Barthel, D. etal, 2015, Germany             0.0036 [0.0018; 0.0064]     20.0
Sahin, S. etal, 2021, Turkey                0.0040 [0.0005; 0.0144]      6.6
Number of studies: k = 5; Number of observations: o = 18098; Number of events: e = 71
                    proportion           95%-CI
Random effects model     0.0042 [0.0028; 0.0063]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1133 [0.0000; 1.0495]; tau = 0.3366 [0.0000; 1.0245]
 I^2 = 60.2% [0.0%; 85.1%]; H = 1.59 [1.00; 2.59]
Test of heterogeneity:      Q d.f. p-value
                        10.06    4  0.0395
```

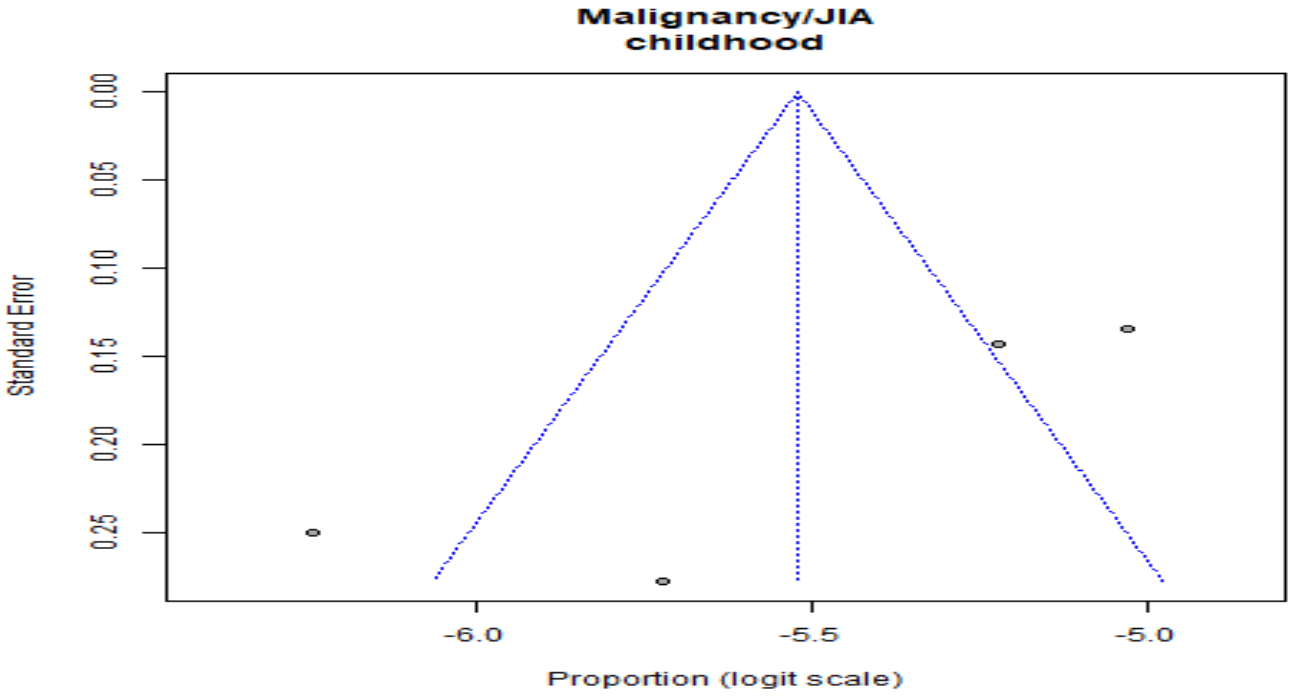
Supplementary Figure S4.14: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with Juvenile Idiopathic Arthritis

Malignancy

Forest



Funnel



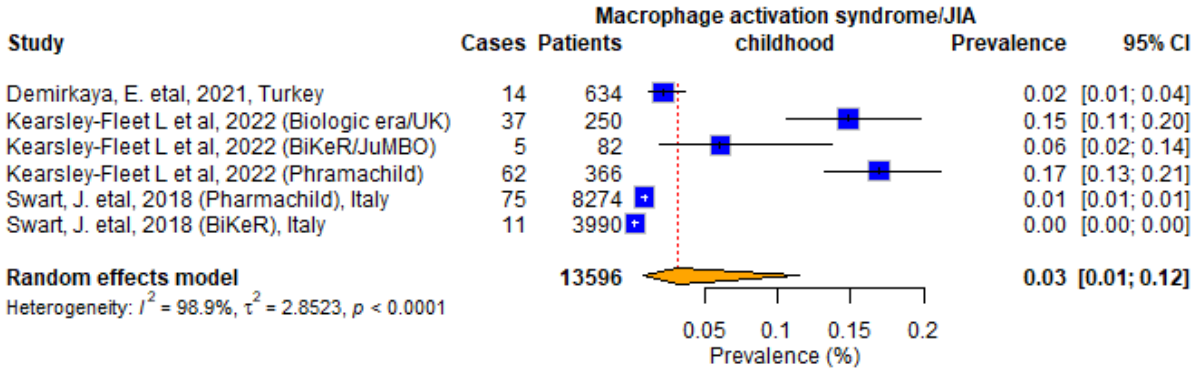
Summary statistics

```
                                        proportion           95%-CI  %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0065 [0.0049; 0.0084]  27.3
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0054 [0.0040; 0.0071]  27.0
Swart, J. etal, 2018 (Pharmachild), Italy     0.0019 [0.0011; 0.0031]  23.4
Swart, J. etal, 2018 (BikeR), Italy           0.0033 [0.0017; 0.0056]  22.4
Number of studies: k = 4; Number of observations: o = 30052; Number of events: e = 134
                    proportion           95%-CI
Random effects model     0.0040 [0.0023; 0.0068]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2520 [0.0517; 4.0772]; tau = 0.5020 [0.2273; 2.0192]
 I^2 = 85.6% [64.7%; 94.2%]; H = 2.64 [1.68; 4.14]
Test of heterogeneity:      Q d.f. p-value
                        20.88    3  0.0001
```

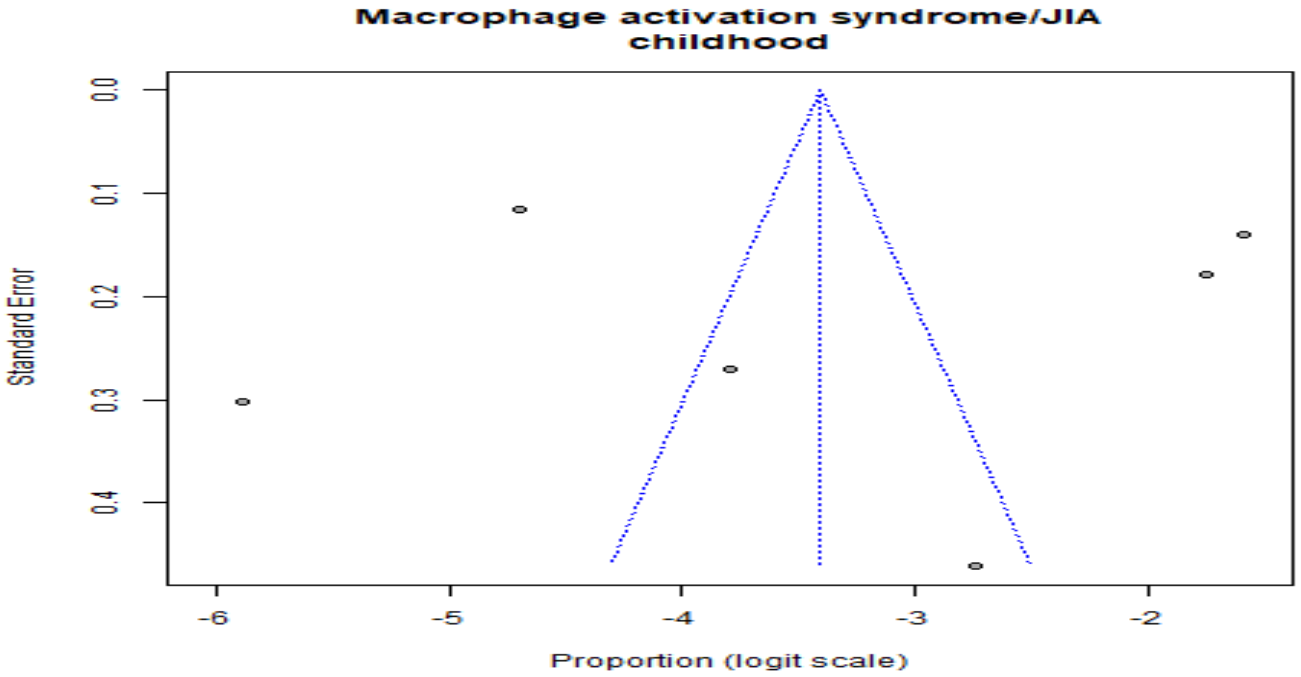
Supplementary Figure S4.15: Forest and Funnel Plots of studies reporting the prevalence of macrophage activation syndrome in children with Juvenile Idiopathic Arthritis

Macrophage activation syndrome

Forest



Funnel



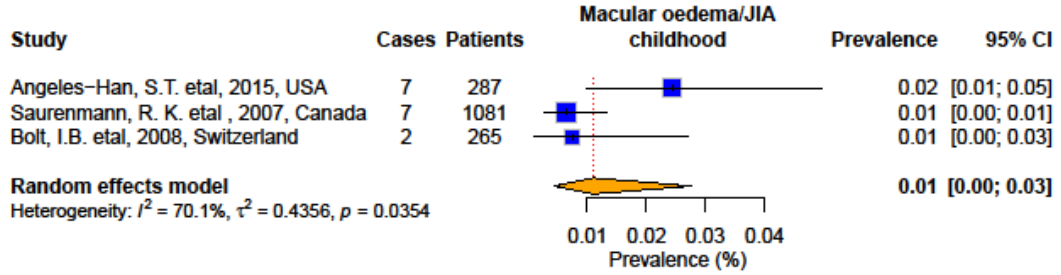
Summary statistics

```
                                          proportion           95%-CI %W(random)
Demirkaya, E. etal, 2021, Turkey              0.0221 [0.0121; 0.0368]       16.7
Kearsley-Fleet L et al, 2022 (Biologic era/UK)  0.1480 [0.1064; 0.1982]       16.9
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)    0.0610 [0.0201; 0.1366]       15.9
Kearsley-Fleet L et al, 2022 (Phramachild)    0.1694 [0.1324; 0.2118]       17.0
Swart, J. etal, 2018 (Pharmachild), Italy     0.0091 [0.0071; 0.0113]       17.0
Swart, J. etal, 2018 (BiKeR), Italy           0.0028 [0.0014; 0.0049]       16.6
Number of studies: k = 6; Number of observations: o = 13596; Number of events: e = 204
                    proportion           95%-CI
Random effects model     0.0321 [0.0084; 0.1153]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 2.8523 [1.0712; 17.3534]; tau = 1.6889 [1.0350; 4.1657]
 I^2 = 98.9% [98.5%; 99.2%]; H = 9.48 [8.05; 11.16]
Test of heterogeneity:       Q d.f.  p-value
                      449.21    5 < 0.0001
```

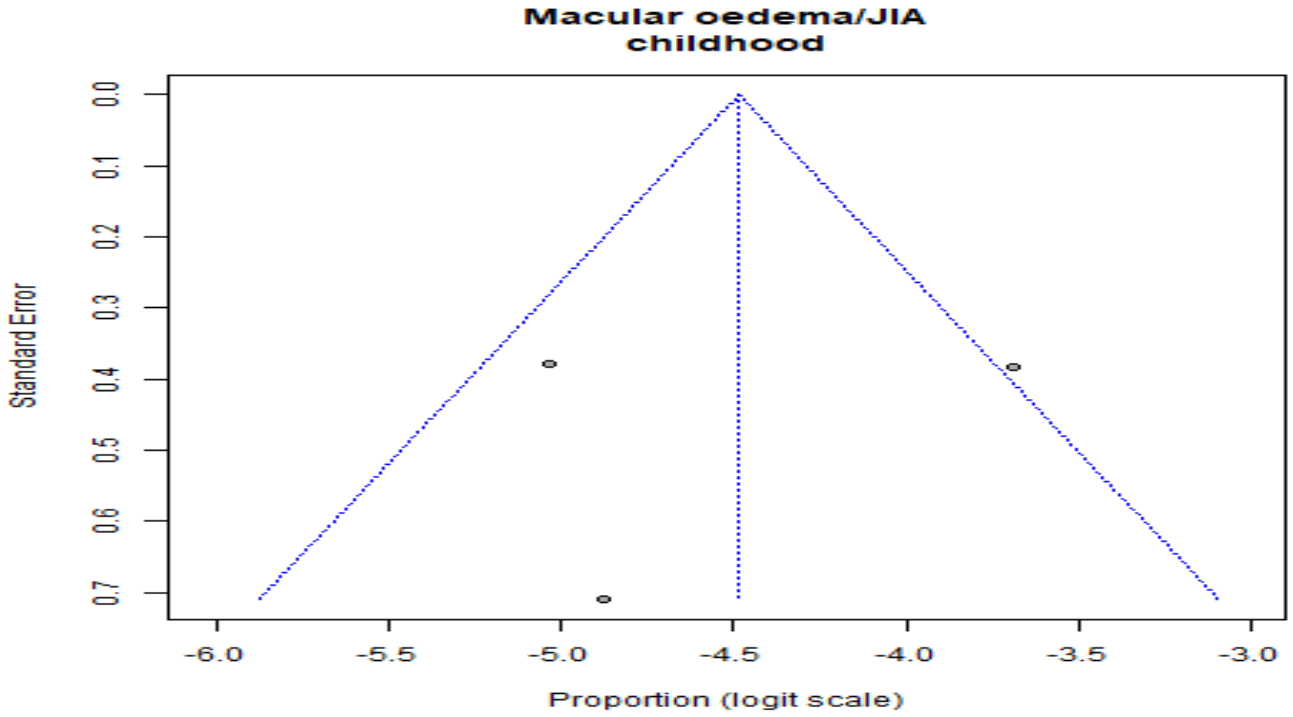
Supplementary Figure S4.16: Forest and Funnel Plots of studies reporting the prevalence of macular oedema in children with Juvenile Idiopathic Arthritis

Macular Oedema

Forest



Funnel



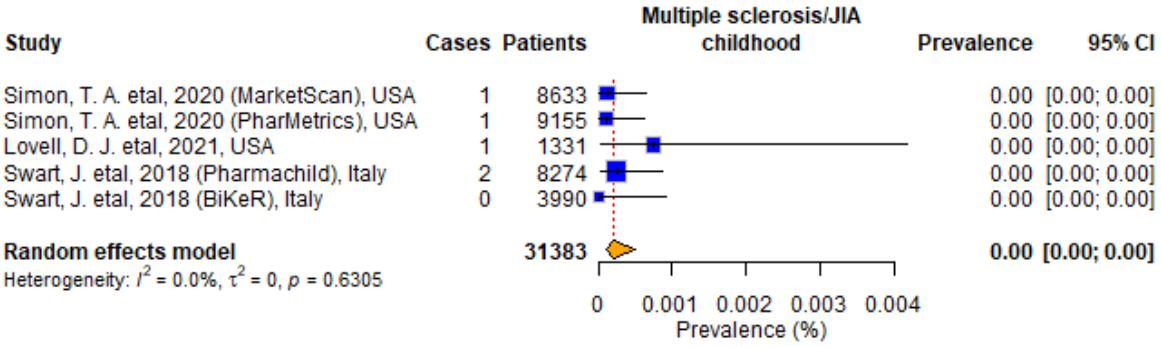
Summary statistics

```
                                    proportion           95%-CI %W(random)
Angeles-Han S.T. etal, 2015, USA        0.0244 [0.0099; 0.0496]       38.1
Saurenmann, R. K. etal , 2007, Canada   0.0065 [0.0026; 0.0133]       38.3
Bolt I.B. etal, 2008, Switzerland       0.0075 [0.0009; 0.0270]       23.6
Number of studies: k = 3; Number of observations: o = 1633; Number of events: e = 16
                     proportion           95%-CI
Random effects model     0.0112 [0.0045; 0.0276]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4356 [0.0000; 21.1933]; tau = 0.6600 [0.0000; 4.6036]
 I^2 = 70.1% [0.0%; 91.2%]; H = 1.83 [1.00; 3.38]
Test of heterogeneity:     Q d.f. p-value
                        6.68    2  0.0354
```

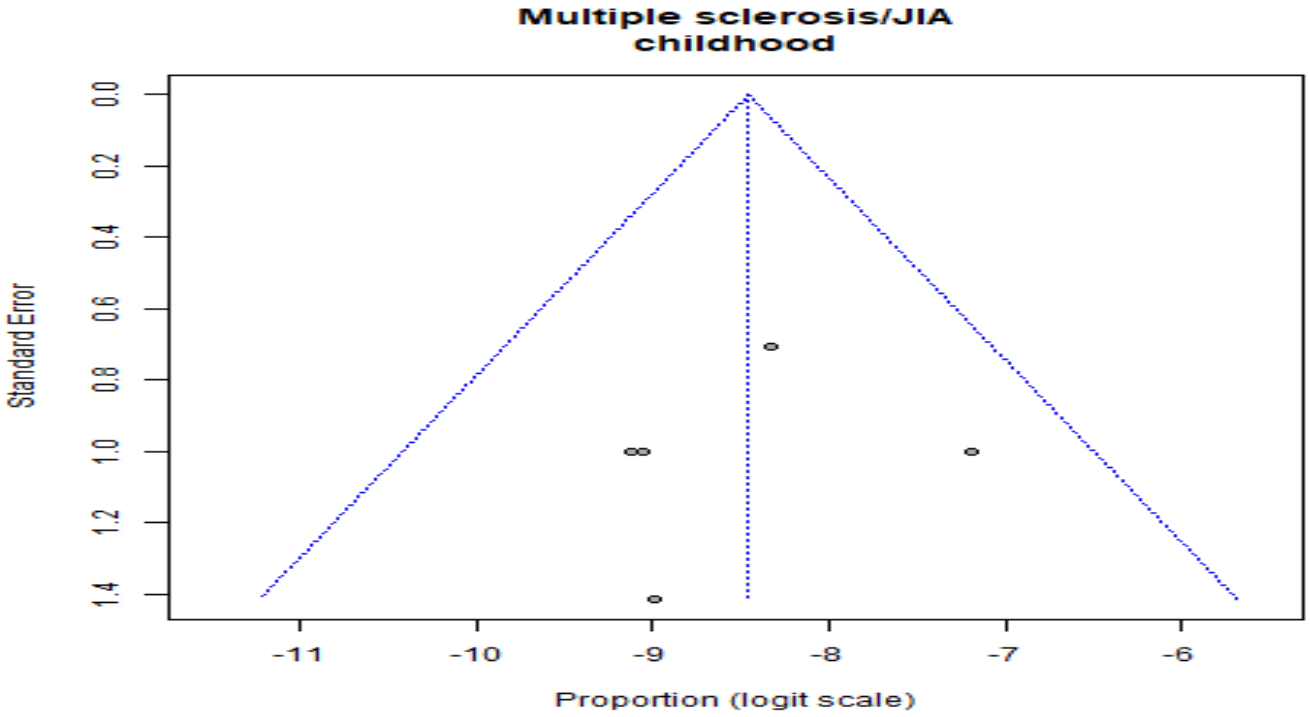
Supplementary Figure S4.17: Forest and Funnel Plots of studies reporting the prevalence of Multiple Sclerosis in children with Juvenile Idiopathic Arthritis

Multiple sclerosis

Forest



Funnel



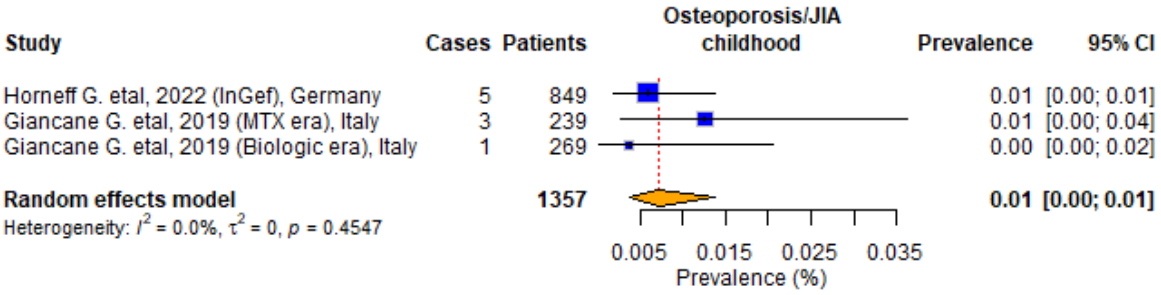
Summary statistics

```
                                        proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0001 [0.0000; 0.0006]       18.2
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0001 [0.0000; 0.0006]       18.2
Lovell, D. J. etal, 2021, USA                 0.0008 [0.0000; 0.0042]       18.2
Swart, J. etal, 2018 (Pharmachild), Italy     0.0002 [0.0000; 0.0009]       36.4
Swart, J. etal, 2018 (BiKeR), Italy           0.0000 [0.0000; 0.0009]        9.1
Number of studies: k = 5; Number of observations: o = 31383; Number of events: e = 5
                     proportion           95%-CI
Random effects model     0.0002 [0.0001; 0.0005]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 4.4533]; tau = 0 [0.0000; 2.1103]
 I^2 = 0.0% [0.0%; 79.2%]; H = 1.00 [1.00; 2.19]
Test of heterogeneity:     Q d.f. p-value
                        2.58    4  0.6305
```

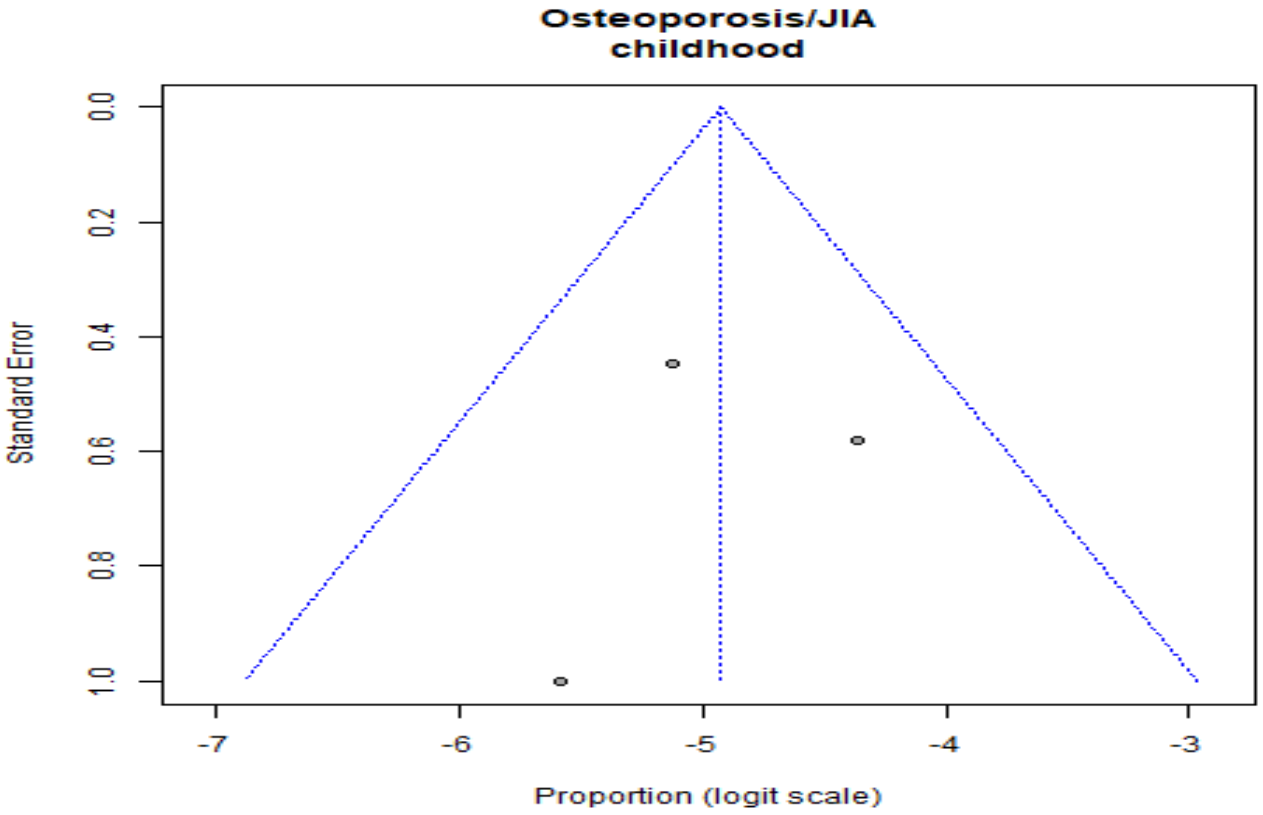
Supplementary Figure S4.18: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in children with Juvenile Idiopathic Arthritis

Osteoporosis

Forest



Funnel



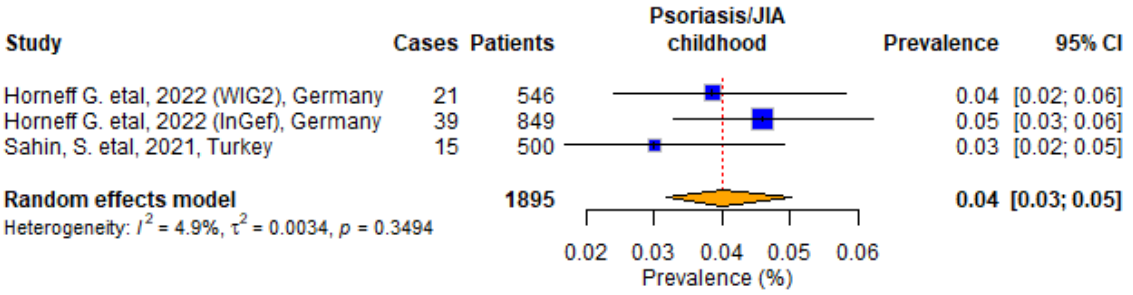
Summary statistics

```
                                          proportion           95%-CI %W(random)
Horneff G. etal, 2022 (InGef), Germany        0.0059 [0.0019; 0.0137]       55.7
Giancane G. etal, 2019 (MTX era), Italy       0.0126 [0.0026; 0.0362]       33.2
Giancane G. etal, 2019 (Biologic era), Italy  0.0037 [0.0001; 0.0205]       11.2
Number of studies: k = 3; Number of observations: o = 1357; Number of events: e = 9
                     proportion           95%-CI
Random effects model     0.0072 [0.0037; 0.0138]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 14.5276]; tau = 0 [0.0000; 3.8115]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        | 1.58    2  0.4547
```

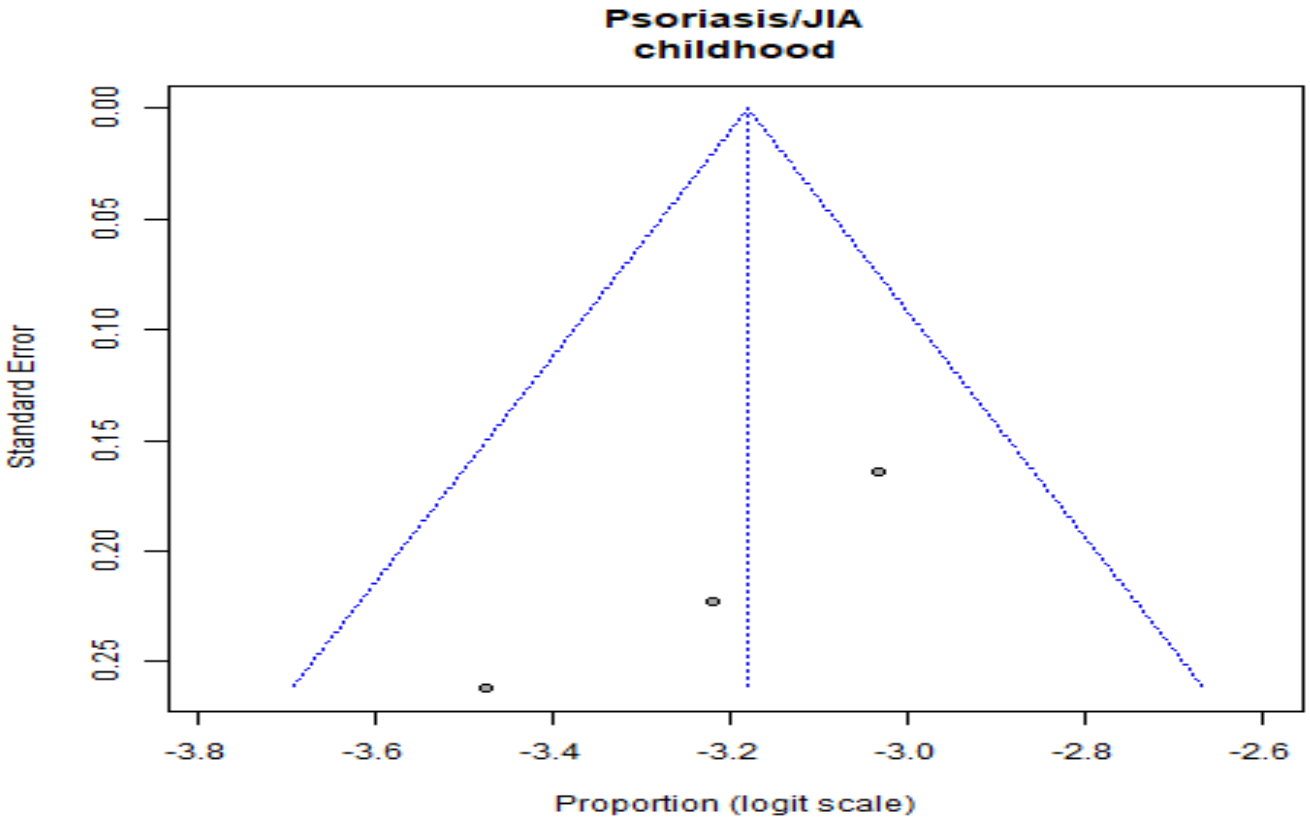
Supplementary Figure S4.19: Forest and Funnel Plots of studies reporting the prevalence of psoriasis in children with Juvenile Idiopathic Arthritis

Psoriasis

Forest



Funnel



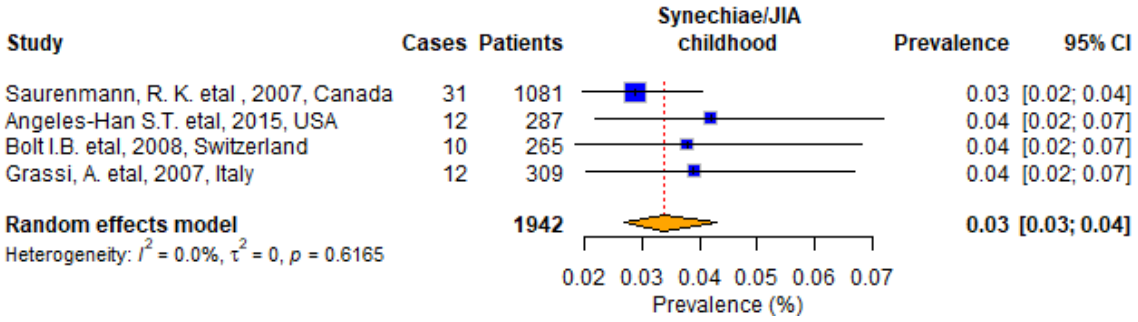
Summary statistics

```
                                    proportion           95%-CI %W(random)
Horneff G. etal, 2022 (WIG2), Germany     0.0385 [0.0240; 0.0582]       28.7
Horneff G. etal, 2022 (InGef), Germany    0.0459 [0.0329; 0.0623]       50.2
Sahin, S. etal, 2021, Turkey              0.0300 [0.0169; 0.0490]       21.1
Number of studies: k = 3; Number of observations: o = 1895; Number of events: e = 75
                     proportion           95%-CI
Random effects model     0.0399 [0.0316; 0.0503]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0034 [0.0000; 1.9016]; tau = 0.0579 [0.0000; 1.3790]
 I^2 = 4.9% [0.0%; 90.1%]; H = 1.03 [1.00; 3.18]
Test of heterogeneity:     Q d.f. p-value
                        2.10    2  0.3494
```

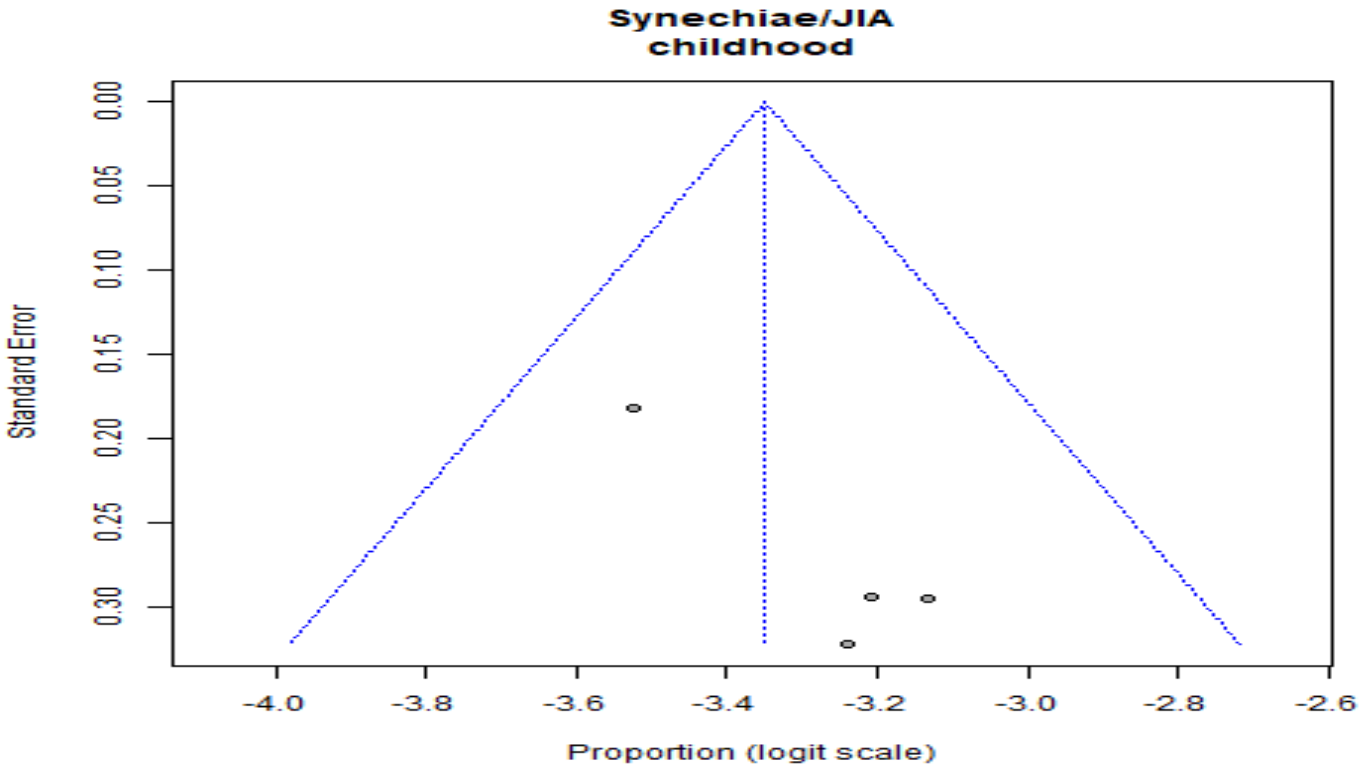
Supplementary Figure S4.20: Forest and Funnel Plots of studies reporting the prevalence of synechiae in children with Juvenile Idiopathic Arthritis

Synechiae

Forest



Funnel



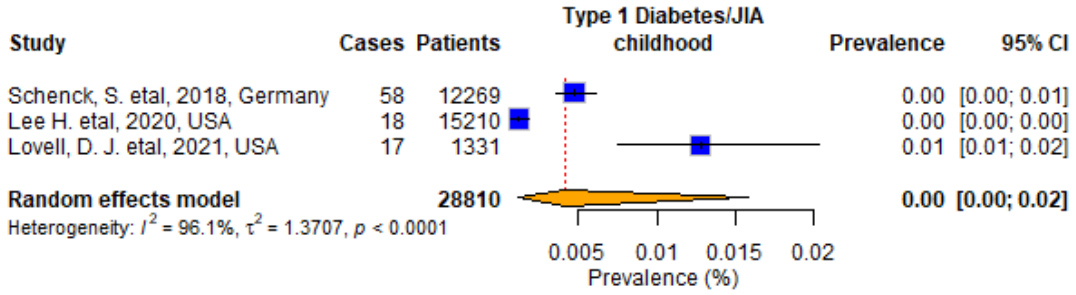
Summary statistics

```
                                    proportion           95%-CI %W(random)
Saurenmann, R. K. etal , 2007, Canada     0.0287 [0.0196; 0.0405]       48.0
Angeles-Han S.T. etal, 2015, USA          0.0418 [0.0218; 0.0719]       18.3
Bolt I.B. etal, 2008, Switzerland         0.0377 [0.0182; 0.0683]       15.3
Grassi, A. etal, 2007, Italy              0.0388 [0.0202; 0.0669]       18.4
Number of studies: k = 4; Number of observations: o = 1942; Number of events: e = 65
                     proportion           95%-CI
Random effects model     0.0339 [0.0267; 0.0430]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 0.3558]; tau = 0 [0.0000; 0.5965]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                       1.79    3  0.6165
```

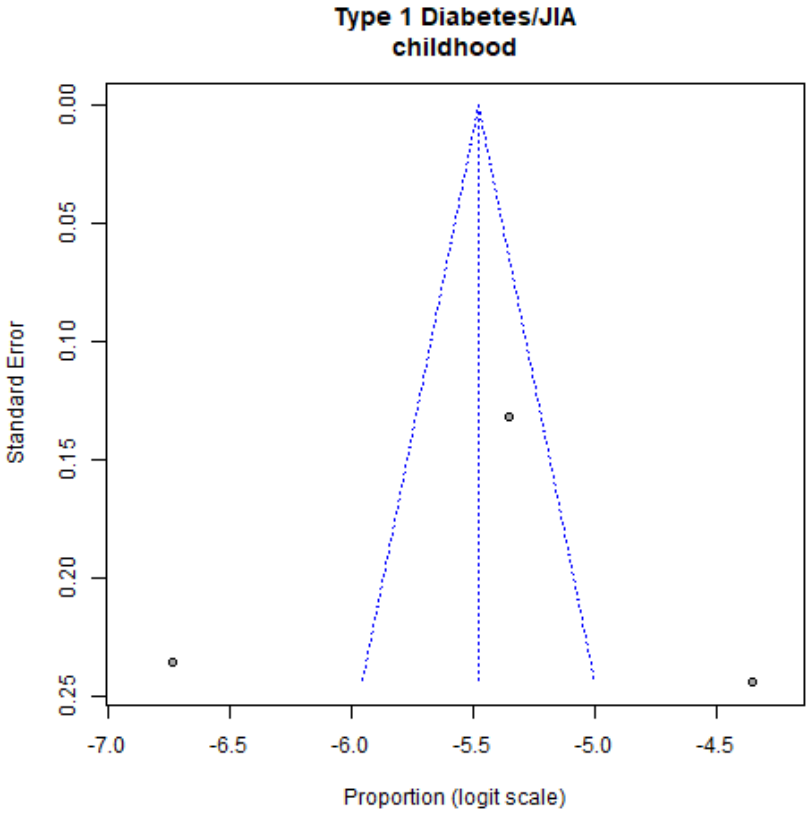
Supplementary Figure S4.21: Forest and Funnel Plots of studies reporting the prevalence of Type 1 diabetes in children with Juvenile Idiopathic Arthritis

Type 1 Diabetes

Forest



Funnel



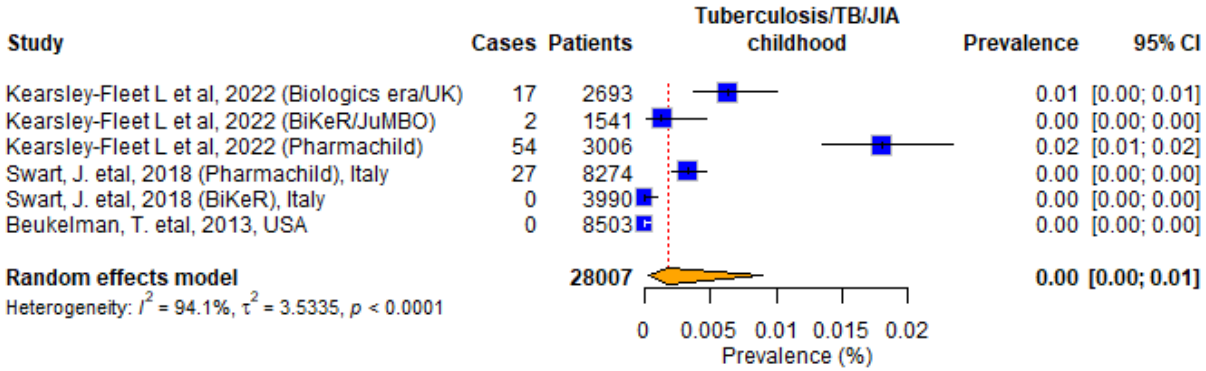
Summary statistics

```
                                proportion           95%-CI %W(random)
Schenck, S. etal, 2018, Germany     0.0047 [0.0036; 0.0061]       34.0
Lee H. etal, 2020, USA              0.0012 [0.0007; 0.0019]       33.1
Lovell, D. J. etal, 2021, USA       0.0128 [0.0075; 0.0204]       33.0
Number of studies: k = 3; Number of observations: o = 28810; Number of events: e = 93
                        proportion           95%-CI
Random effects model     0.0042 [0.0011; 0.0158]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.3707 [0.3335; 56.8637]; tau = 1.1708 [0.5775; 7.5408]
 I^2 = 96.1% [91.6%; 98.2%]; H = 5.04 [3.45; 7.37]
Test of heterogeneity:      Q d.f.  p-value
                         50.80    2 < 0.0001
```

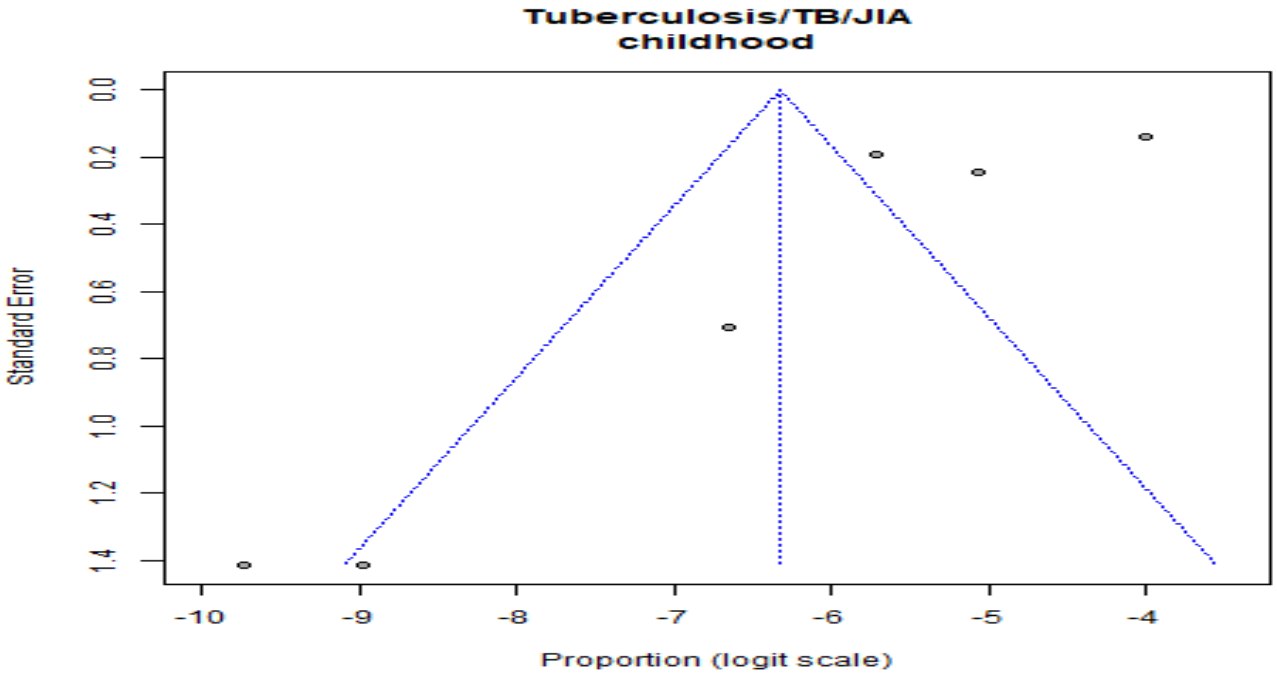
Supplementary Figure S4.22: Forest and Funnel Plots of studies reporting the prevalence of tuberculosis in children with Juvenile Idiopathic Arthritis

Tuberculosis/TB

Forest



Funnel



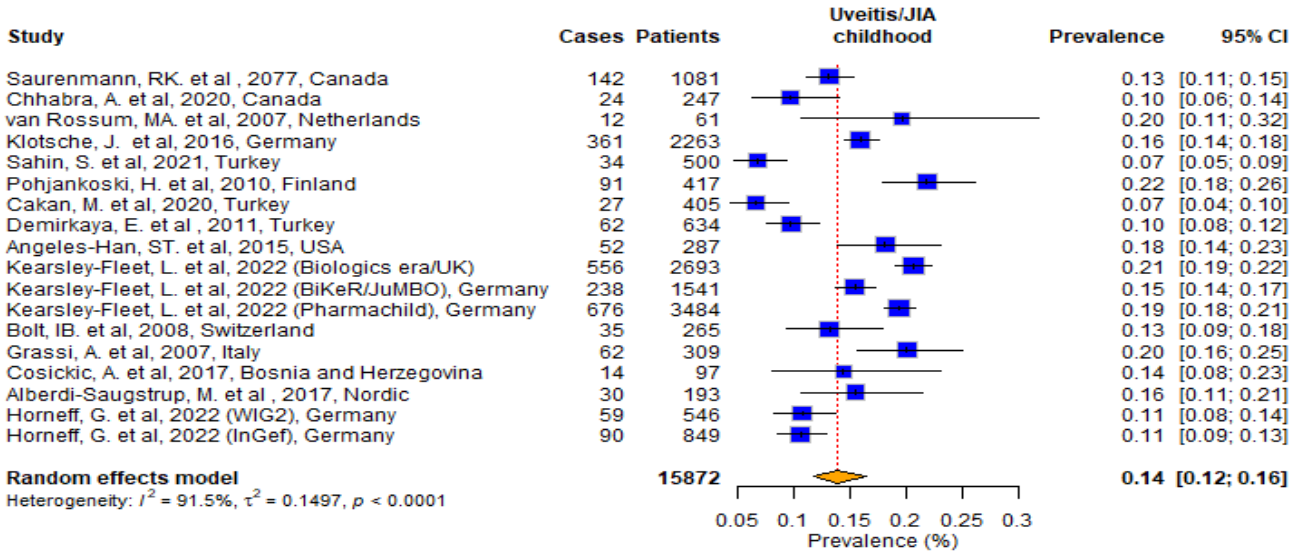
Summary statistics

```
                                          proportion           95%-CI %W(random)
Kearsley-Fleet L et al, 2022 (Biologics era/UK)     0.0063 [0.0037; 0.0101]       19.2
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)          0.0013 [0.0002; 0.0047]       17.1
Kearsley-Fleet L et al, 2022 (Pharmachild)          0.0180 [0.0135; 0.0234]       19.4
Swart, J. etal, 2018 (Pharmachild), Italy           0.0033 [0.0022; 0.0047]       19.3
Swart, J. etal, 2018 (BiKeR), Italy                 0.0000 [0.0000; 0.0009]       12.5
Beukelman, T. etal, 2013, USA                       0.0000 [0.0000; 0.0004]       12.5
Number of studies: k = 6; Number of observations: o = 28007; Number of events: e = 100
                     proportion           95%-CI
Random effects model     0.0018 [0.0004; 0.0090]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 3.5335 [0.9812; 29.4146]; tau = 1.8797 [0.9905; 5.4235]
 I^2 = 94.1% [89.8%; 96.6%]; H = 4.13 [3.13; 5.45]
Test of heterogeneity:      Q d.f.  p-value
                        85.36    5 < 0.0001
```

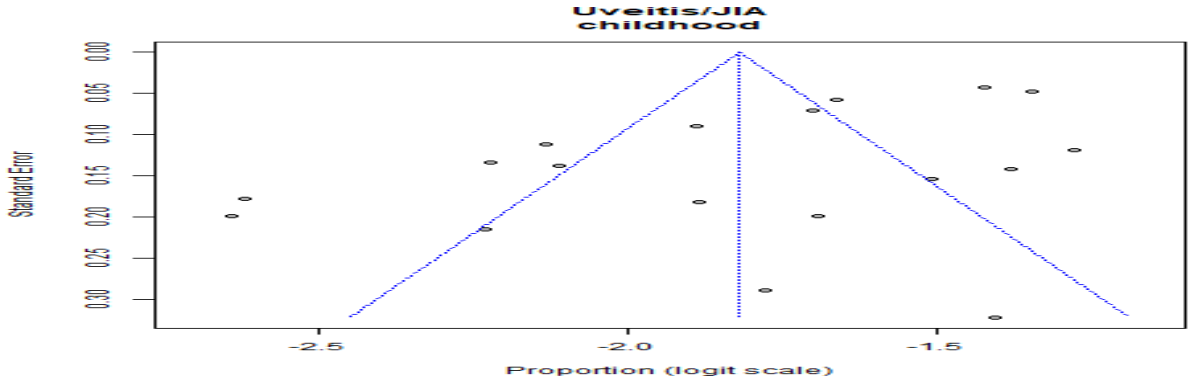
Supplementary Figure S4.23: Forest and Funnel Plots of studies reporting the prevalence of uveitis in children with Juvenile Idiopathic Arthritis

Uveitis

Forest



Funnel



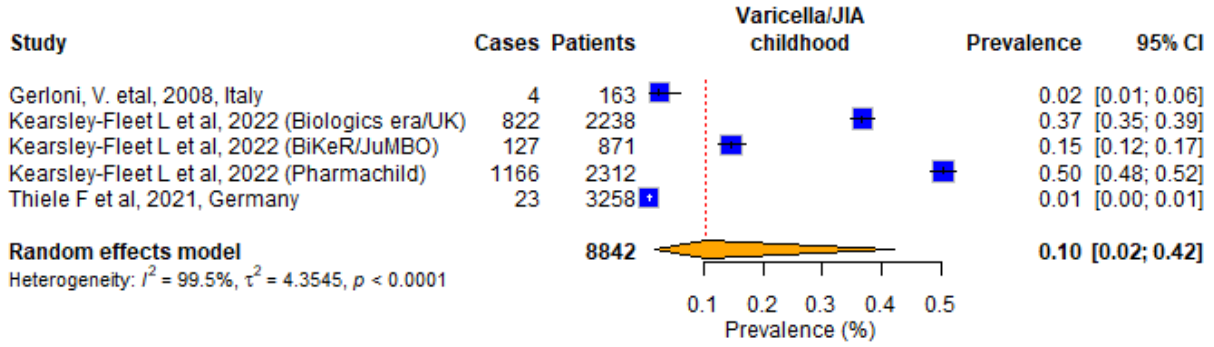
Summary statistics

```
                                                    proportion           95%-CI %W(random)
Saurenmann, RK. et al , 2077, Canada                    0.1314 [0.1118; 0.1530]        6.1
Chhabra, A. et al, 2020, Canada                         0.0972 [0.0633; 0.1411]        4.9
van Rossum, MA. et al, 2007, Netherlands                0.1967 [0.1060; 0.3184]        3.8
Klotsche, J. et al, 2016, Germany                       0.1595 [0.1447; 0.1753]        6.3
Sahin, S. et al, 2021, Turkey                           0.0680 [0.0475; 0.0937]        5.3
Pohjankoski, H. et al, 2010, Finland                    0.2182 [0.1795; 0.2610]        5.9
Cakan, M. et al, 2020, Turkey                           0.0667 [0.0444; 0.0955]        5.1
Demirkaya, E. et al , 2011, Turkey                      0.0978 [0.0758; 0.1236]        5.8
Angeles-Han, ST. et al, 2015, USA                       0.1812 [0.1384; 0.2307]        5.6
Kearsley-Fleet, L. et al, 2022 (Biologics era/UK)       0.2065 [0.1913; 0.2222]        6.4
Kearsley-Fleet, L. et al, 2022 (BiKeR/JuMBO), Germany   0.1544 [0.1367; 0.1735]        6.3
Kearsley-Fleet, L. et al, 2022 (Pharmachild), Germany   0.1940 [0.1810; 0.2076]        6.4
Bolt, IB. et al, 2008, Switzerland                      0.1321 [0.0937; 0.1789]        5.3
Grassi, A. et al, 2007, Italy                           0.2006 [0.1574; 0.2497]        5.7
Cosickic, A. et al, 2017, Bosnia and Herzegovina        0.1443 [0.0812; 0.2303]        4.2
Alberdi-Saugstrup, M. et al , 2017, Nordic              0.1554 [0.1074; 0.2144]        5.1
Horneff, G. et al, 2022 (WIG2), Germany                 0.1081 [0.0833; 0.1372]        5.7
Horneff, G. et al, 2022 (InGef), Germany                0.1060 [0.0861; 0.1287]        6.0
Number of studies: k = 18; Number of observations: o = 15872; Number of events: e = 2565
                    proportion           95%-CI
Random effects model     0.1395 [0.1179; 0.1643]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1497 [0.0741; 0.3706]; tau = 0.3870 [0.2721; 0.6088]
 I^2 = 91.5% [88.0%; 93.9%]; H = 3.42 [2.89; 4.05]
Test of heterogeneity:       Q d.f.  p-value
                        | 198.93    17 < 0.0001
```

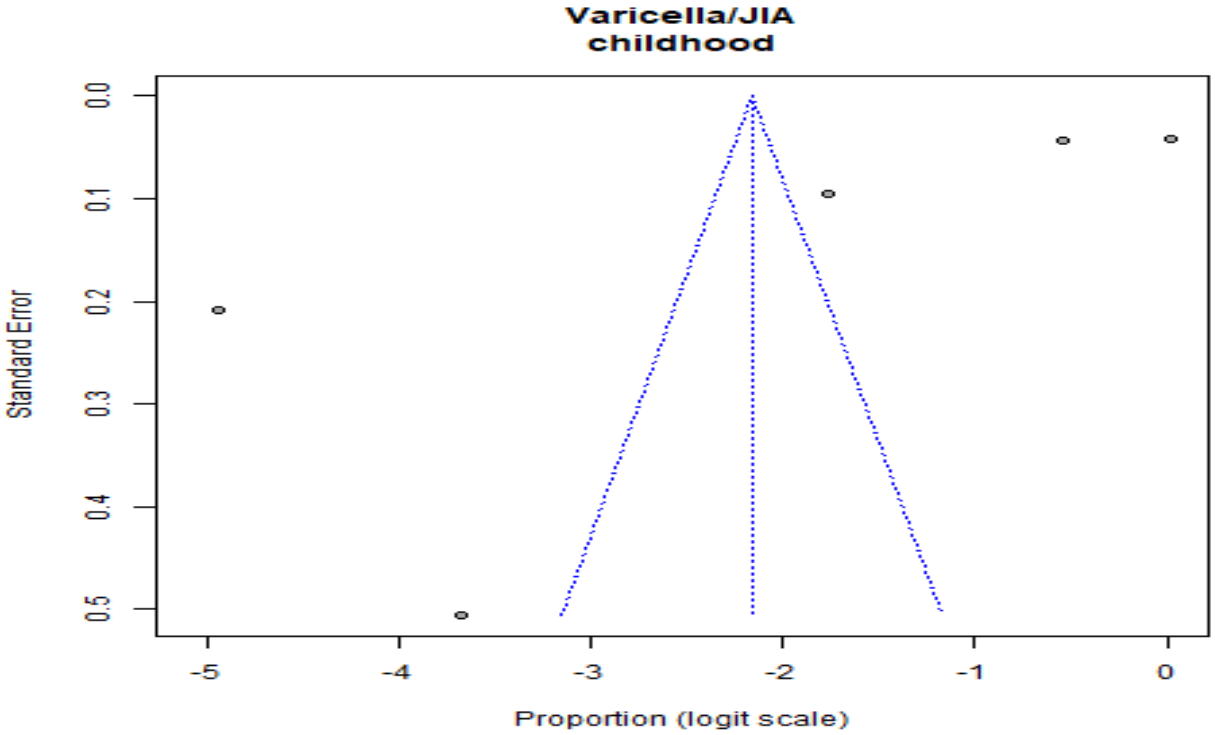
Supplementary Figure S4.24: Forest and Funnel Plots of studies reporting the prevalence of varicella in children with Juvenile Idiopathic Arthritis

Varicella

Forest



Funnel



Summary statistics

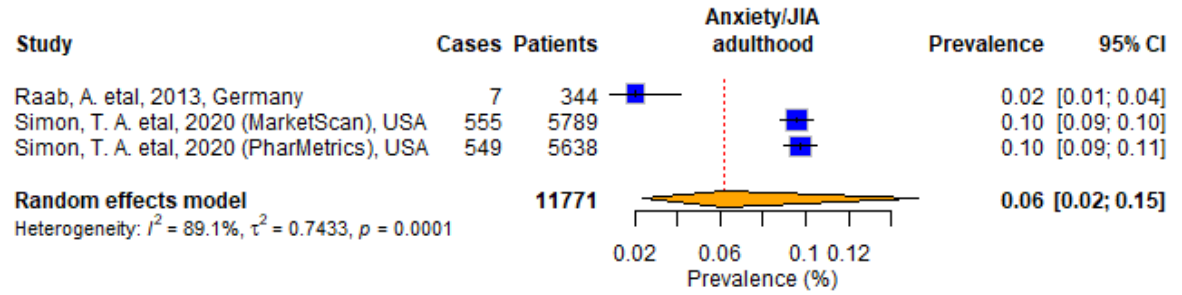
```
                                          proportion           95%-CI %W(random)
Gerloni, V. etal, 2008, Italy                 0.0245 [0.0067; 0.0616]       19.2
Kearsley-Fleet L et al, 2022 (Biologics era/UK)     0.3673 [0.3473; 0.3877]       20.3
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)          0.1458 [0.1230; 0.1710]       20.2
Kearsley-Fleet L et al, 2022 (Pharmachild)          0.5043 [0.4837; 0.5249]       20.3
Thiele F et al, 2021, Germany                 0.0071 [0.0045; 0.0106]       20.1
Number of studies: k = 5; Number of observations: o = 8842; Number of events: e = 2142
                     proportion           95%-CI
Random effects model     0.1031 [0.0179; 0.4204]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 4.3545 [1.5277; 36.2489]; tau = 2.0868 [1.2360; 6.0207]
 I^2 = 99.5% [99.4%; 99.6%]; H = 14.32 [12.48; 16.44]
Test of heterogeneity:       Q d.f.  p-value
                        820.56    4 < 0.0001
```

Adulthood/JIA

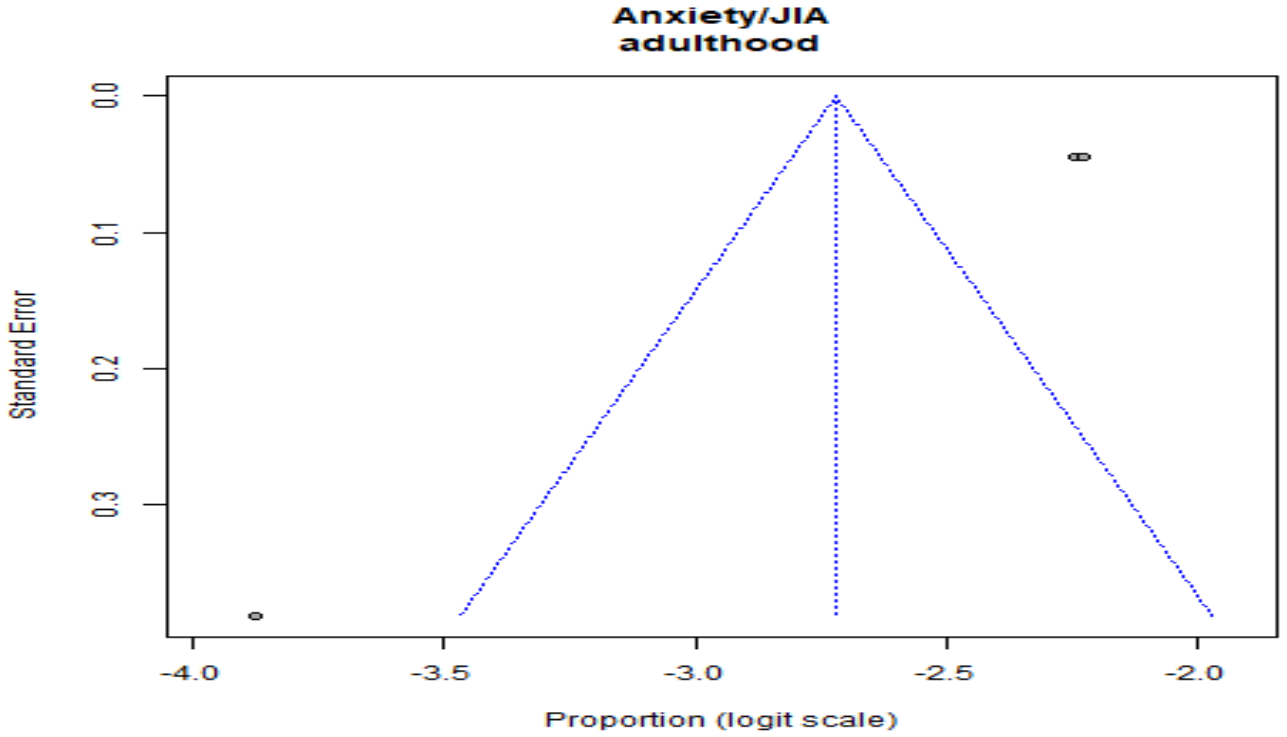
Supplementary Figure S4.25: Forest and Funnel Plots of studies reporting the prevalence of anxiety in adults with Juvenile Idiopathic Arthritis

Anxiety

Forest



Funnel



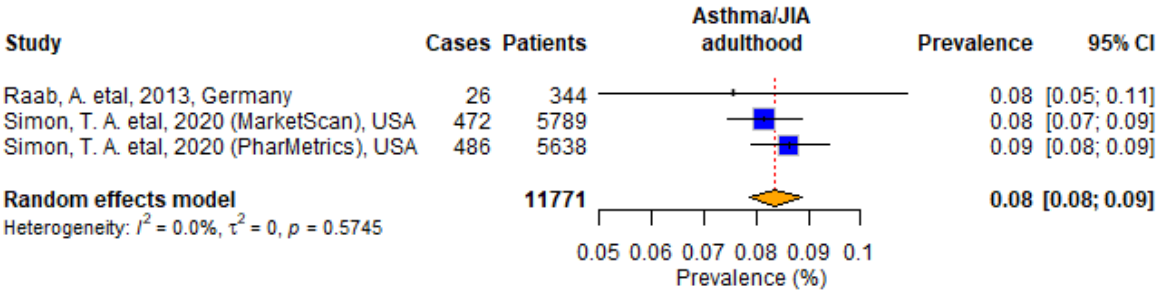
Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0203 [0.0082; 0.0415]       29.5
Simon, T. A. etal, 2020 (MarketScan), USA   0.0959 [0.0884; 0.1037]       35.2
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0974 [0.0898; 0.1054]       35.2
Number of studies: k = 3; Number of observations: o = 11771; Number of events: e = 1111
                    proportion           95%-CI
Random effects model     0.0618 [0.0236; 0.1525]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7433 [0.1448; 35.2654]; tau = 0.8622 [0.3806; 5.9385]
 I^2 = 89.1% [70.3%; 96.0%]; H = 3.03 [1.83; 5.01]
Test of heterogeneity:      Q d.f. p-value
                        18.36    2  0.0001
```

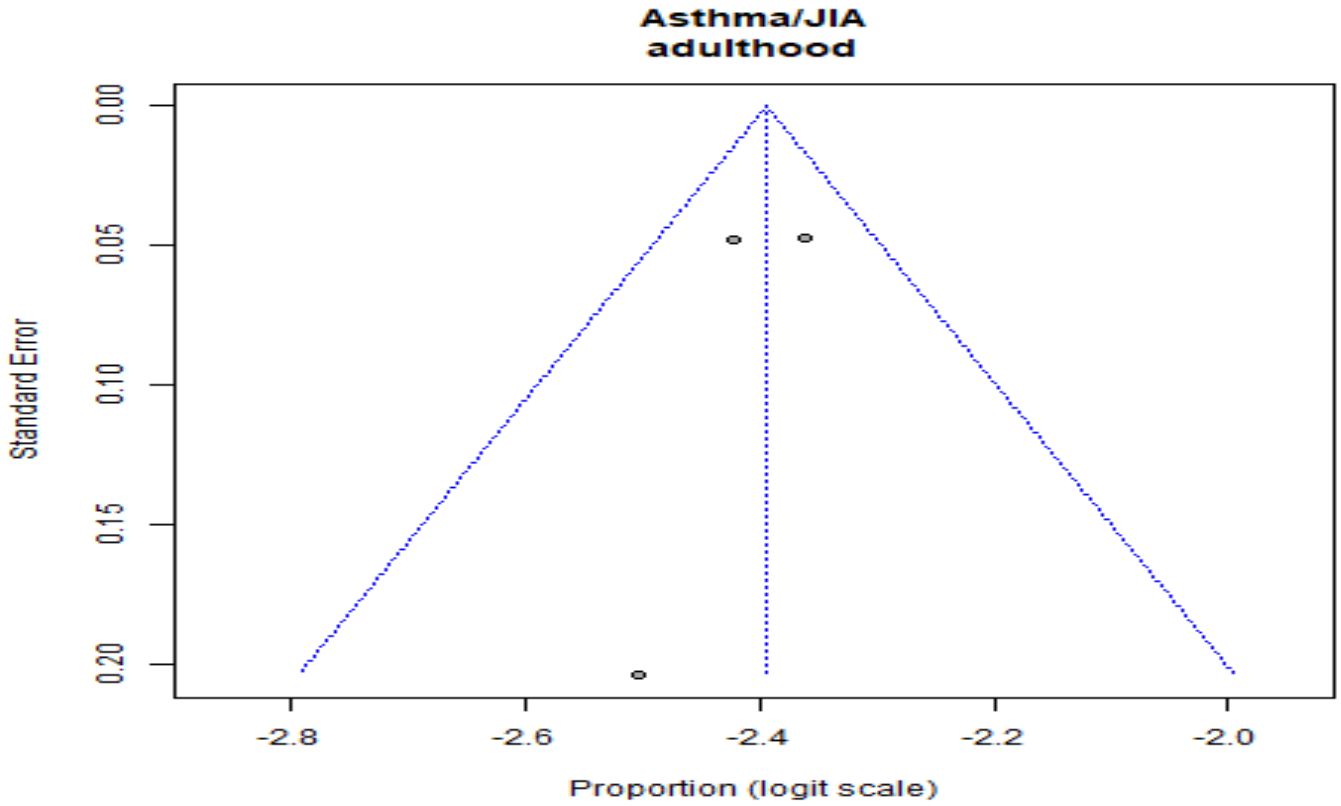
Supplementary Figure S4.26: Forest and Funnel Plots of studies reporting the prevalence of asthma in adults with Juvenile Idiopathic Arthritis

Asthma

Forest



Funnel



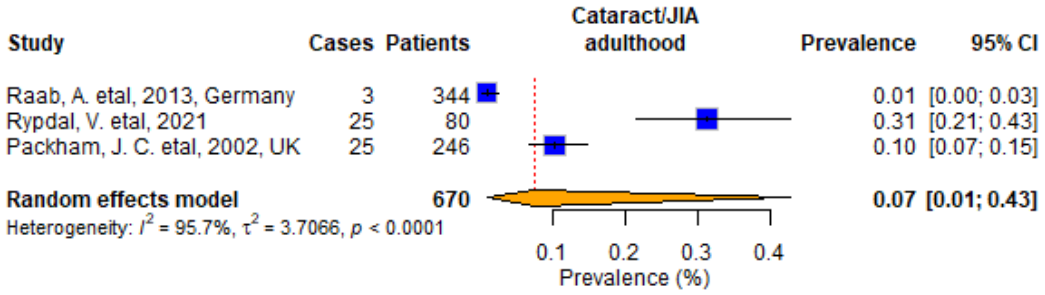
Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0756 [0.0500; 0.1088]        2.7
Simon, T. A. etal, 2020 (MarketScan), USA   0.0815 [0.0746; 0.0889]       48.1
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0862 [0.0790; 0.0938]       49.3
Number of studies: k = 3; Number of observations: o = 11771; Number of events: e = 984
                    proportion           95%-CI
Random effects model     0.0836 [0.0788; 0.0888]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 0.1803]; tau = 0 [0.0000; 0.4246]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        | 1.11    2  0.5745
```

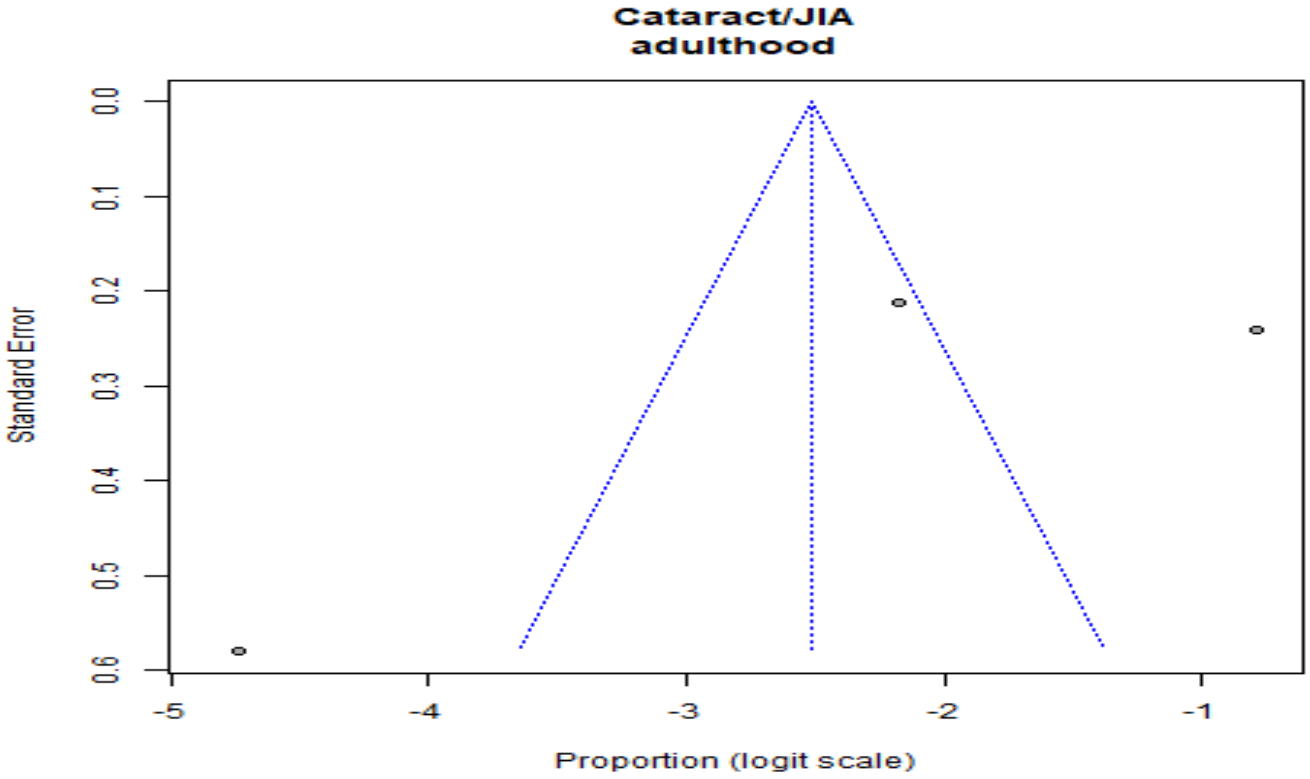
Supplementary Figure S4.27: Forest and Funnel Plots of studies reporting the prevalence of cataract in adults with Juvenile Idiopathic Arthritis

Cataract

Forest



Funnel



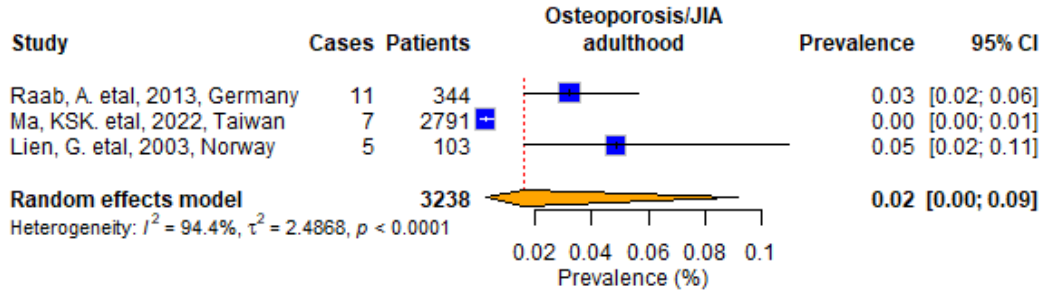
Summary statistics

```
                              proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany      0.0087 [0.0018; 0.0253]       31.7
Rypdal, V. etal, 2021             0.3125 [0.2135; 0.4259]       34.1
Packham, J. C. etal, 2002, UK     0.1016 [0.0669; 0.1464]       34.2
Number of studies: k = 3
Number of observations: o = 670
Number of events: e = 53
                     proportion           95%-CI
Random effects model     0.0748 [0.0087; 0.4266]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 3.7066 [0.8680; >100.0000]; tau = 1.9252 [0.9316; >10.0000]
 I^2 = 95.7% [90.6%; 98.0%]; H = 4.82 [3.26; 7.11]
Test of heterogeneity:      Q d.f.  p-value
                        46.40    2 < 0.0001
```

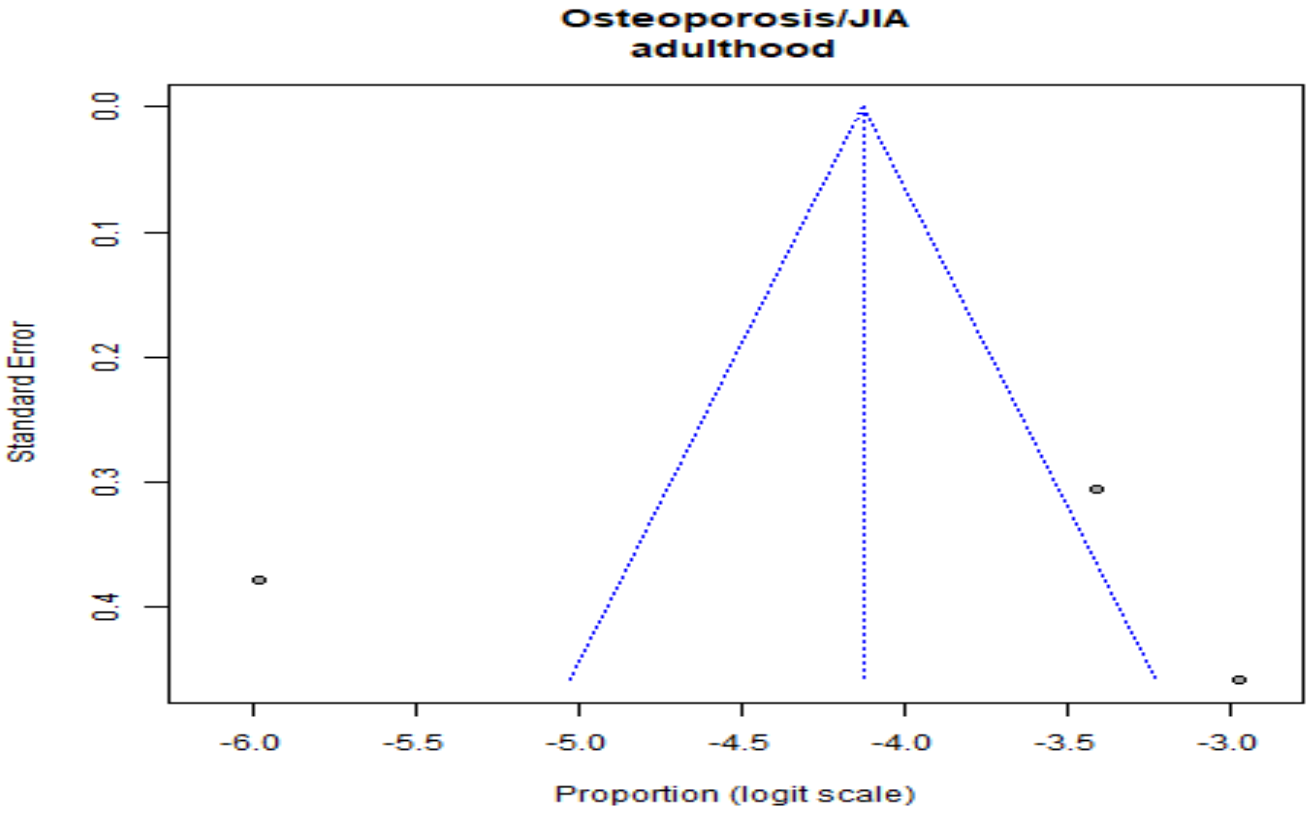
Supplementary Figure S4.28: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in adults with Juvenile Idiopathic Arthritis

Osteoporosis

Forest



Funnel



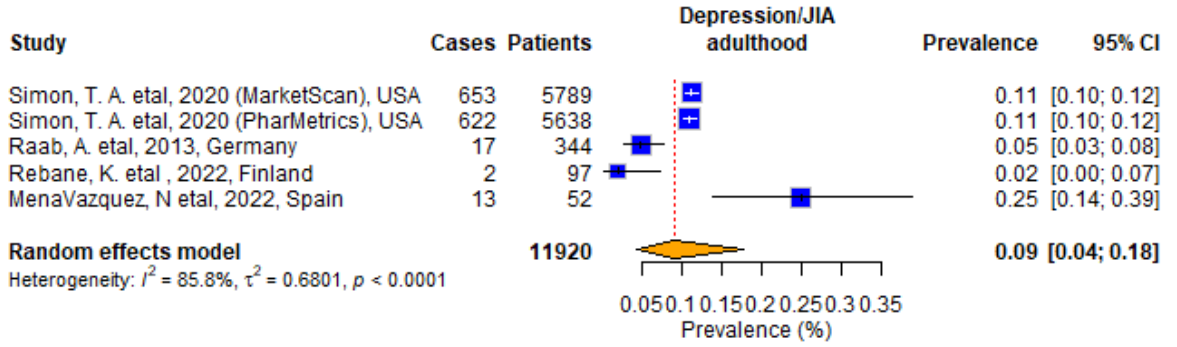
Summary statistics

```
                           proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany     0.0320 [0.0161; 0.0565]       34.0
Ma, KSK. etal, 2022, Taiwan      0.0025 [0.0010; 0.0052]       33.4
Lien, G. etal, 2003, Norway      0.0485 [0.0159; 0.1097]       32.6
Number of studies: k = 3; Number of observations: o = 3238; Number of events: e = 23
                     proportion           95%-CI
Random effects model     0.0158 [0.0026; 0.0918]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 2.4868 [0.5640; >100.0000]; tau = 1.5770 [0.7510; >10.0000]
 I^2 = 94.4% [87.1%; 97.6%]; H = 4.24 [2.78; 6.45]
Test of heterogeneity:      Q d.f.  p-value
                        35.90    2 < 0.0001
```

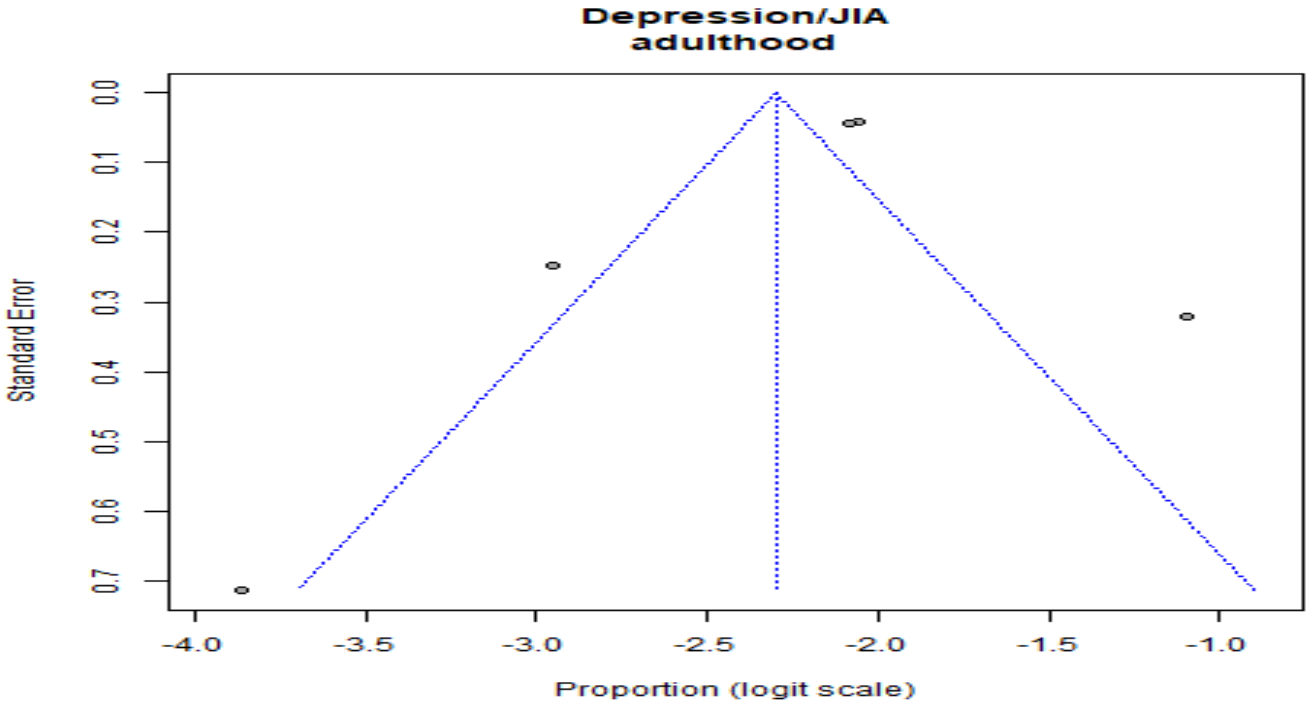
Supplementary Figure S4.29: Forest and Funnel Plots of studies reporting the prevalence of depression in adults with Juvenile Idiopathic Arthritis

Depression

Forest



Funnel



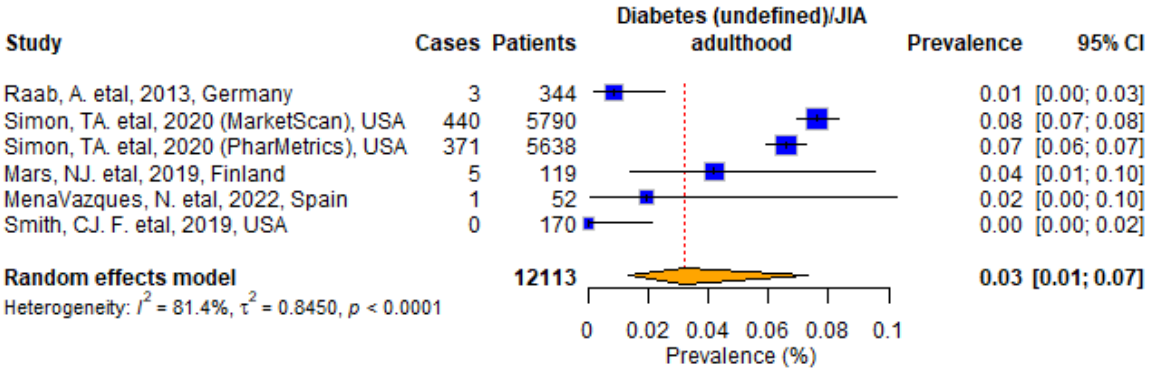
Summary statistics

```
                                          proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.1128 [0.1048; 0.1212]       22.9
Simon, T. A. etal, 2020 (PharMetrics), USA    0.1103 [0.1023; 0.1188]       22.9
Raab, A. etal, 2013, Germany                  0.0494 [0.0290; 0.0779]       21.1
Rebane, K. etal , 2022, Finland               0.0206 [0.0025; 0.0725]       13.1
MenaVazquez, N etal, 2022, Spain              0.2500 [0.1403; 0.3895]       20.0
Number of studies: k = 5; Number of observations: o = 11920; Number of events: e = 1307
                    proportion           95%-CI
Random effects model     0.0911 [0.0442; 0.1787]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.6801 [0.1703; 8.6878]; tau = 0.8247 [0.4127; 2.9475]
 I^2 = 85.8% [68.9%; 93.5%]; H = 2.66 [1.79; 3.94]
Test of heterogeneity:      Q d.f.  p-value
                        28.23    4 < 0.0001
```

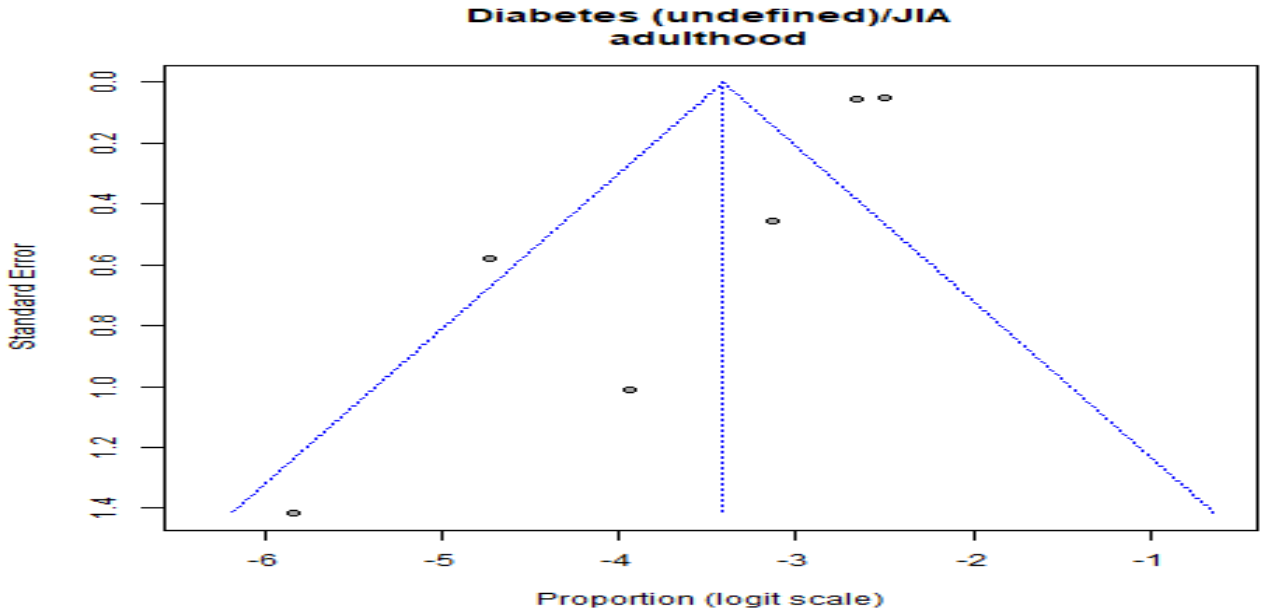
Supplementary Figure S4.30: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in adults with Juvenile Idiopathic Arthritis

Diabetes (undefined)

Forest



Funnel



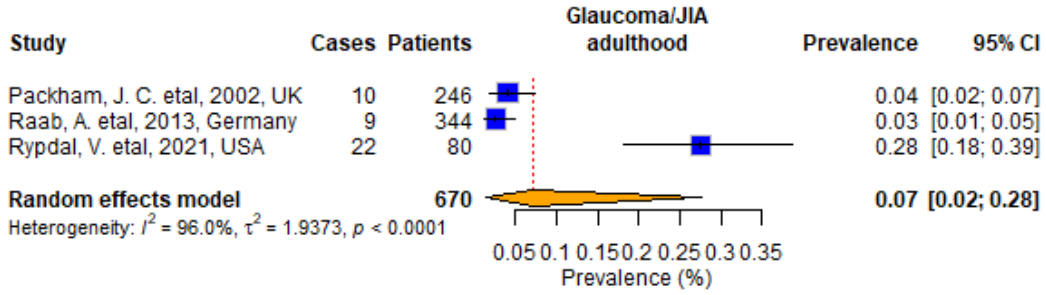
Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0087 [0.0018; 0.0253]       16.8
Simon, TA. etal, 2020 (MarketScan), USA     0.0760 [0.0693; 0.0831]       23.4
Simon, TA. etal, 2020 (PharMetrics), USA    0.0658 [0.0595; 0.0726]       23.4
Mars, NJ. etal, 2019, Finland               0.0420 [0.0138; 0.0953]       18.8
MenaVazques, N. etal, 2022, Spain           0.0192 [0.0005; 0.1026]       10.6
Smith, CJ. F. etal, 2019, USA               0.0000 [0.0000; 0.0215]        7.0
Number of studies: k = 6; Number of observations: o = 12113; Number of events: e = 820
                     proportion           95%-CI
Random effects model     0.0319 [0.0136; 0.0731]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.8450 [0.1589; 9.2267]; tau = 0.9192 [0.3986; 3.0375]
 I^2 = 81.4% [60.1%; 91.3%]; H = 2.32 [1.58; 3.39]
Test of heterogeneity:      Q d.f.  p-value
                        26.84    5 < 0.0001
```

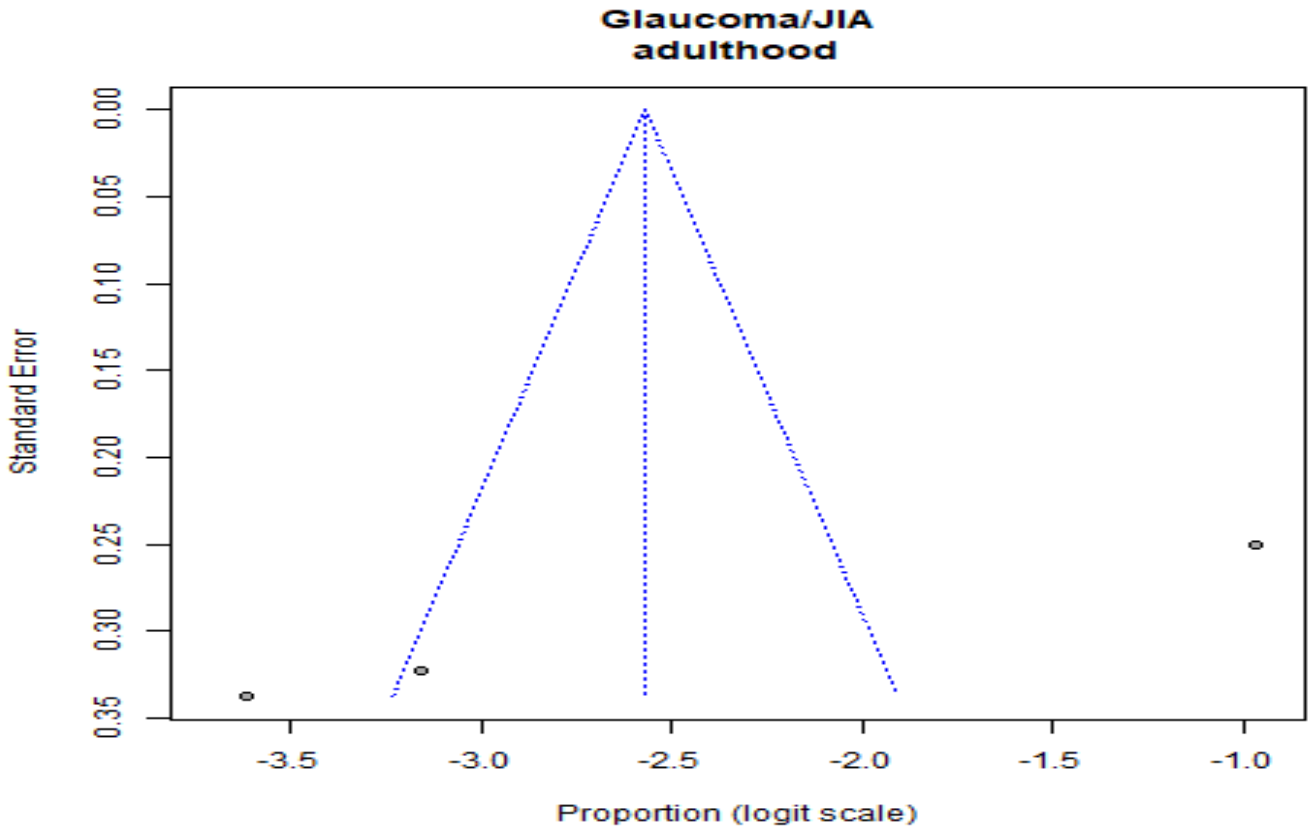
Supplementary Figure S4.31: Forest and Funnel Plots of studies reporting the prevalence of glaucoma in adults with Juvenile Idiopathic Arthritis

Glaucoma

Forest



Funnel



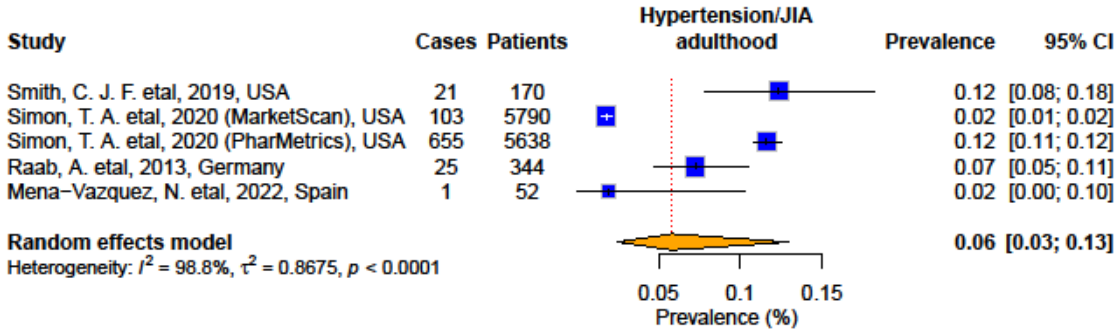
Summary statistics

```
                              proportion           95%-CI %W(random)
Packham, J. C. etal, 2002, UK     0.0407 [0.0197; 0.0735]       33.2
Raab, A. etal, 2013, Germany      0.0262 [0.0120; 0.0491]       33.0
Rypdal, V. etal, 2021, USA        0.2750 [0.1810; 0.3862]       33.8
Number of studies: k = 3; Number of observations: o = 670; Number of events: e = 41
                     proportion           95%-CI
Random effects model     0.0711 [0.0150; 0.2774]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.9373 [0.4633; 79.0548]; tau = 1.3919 [0.6807; 8.8913]
 I^2 = 96.0% [91.5%; 98.1%]; H = 5.02 [3.43; 7.35]
Test of heterogeneity:      Q d.f.  p-value
                        50.50    2 < 0.0001
```

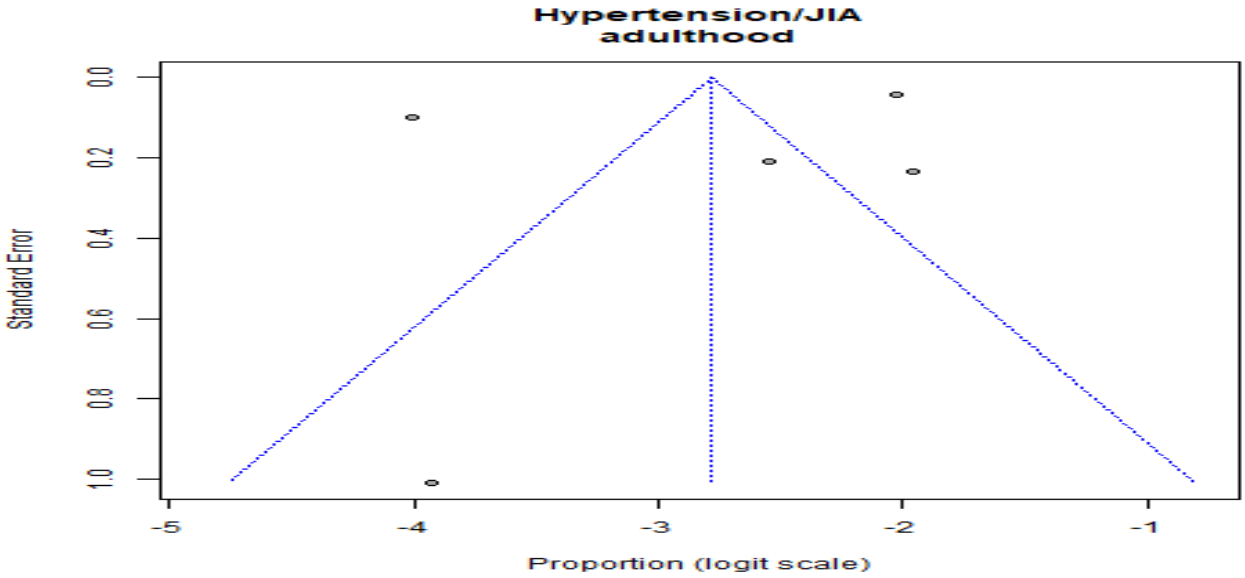
Supplementary Figure S4.32: Forest and Funnel Plots of studies reporting the prevalence of hypertension in adults with Juvenile Idiopathic Arthritis

Hypertension

Forest



Funnel



Summary statistics

```
                                          proportion   95% CI                 %W(random)
Smith, C. J. F. etal, 2019, USA               0.1235   [0.0781; 0.1826]       21.7
Simon, T. A. etal, 2020 (MarketScan), USA     0.0178   [0.0145; 0.0215]       22.8
Simon, T. A. etal, 2020 (PharMetrics), USA    0.1162   [0.1079; 0.1248]       23.0
Raab, A. etal, 2013, Germany                  0.0727   [0.0476; 0.1054]       21.9
Mena-Vazquez, N. etal, 2022, Spain            0.0192   [0.0005; 0.1026]       10.6

Number of studies: k = 5
Number of observations: o = 11994
Number of events: e = 805

                     proportion         95% CI
Random effects model     0.0584         [0.0252; 0.1296]

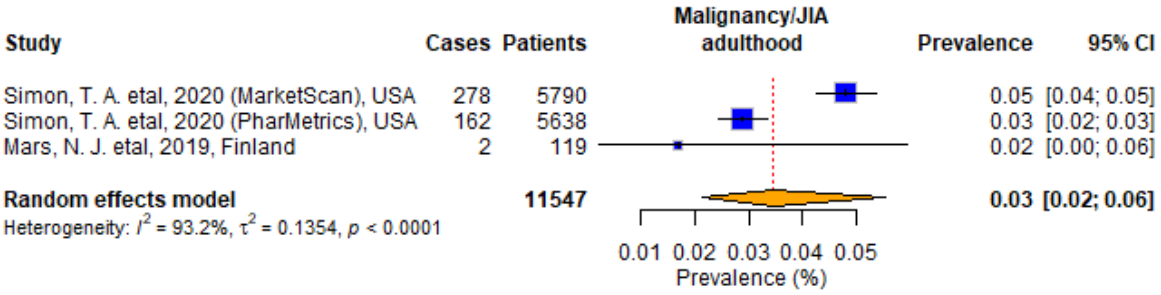
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.8675 [0.2585; 8.1278]; tau = 0.9314 [0.5084; 2.8509]
 I^2 = 98.8% [98.3%; 99.2%]; H = 9.28 [7.71; 11.17]

Test of heterogeneity:
      Q d.f.  p-value
 344.45    4 < 0.0001
```

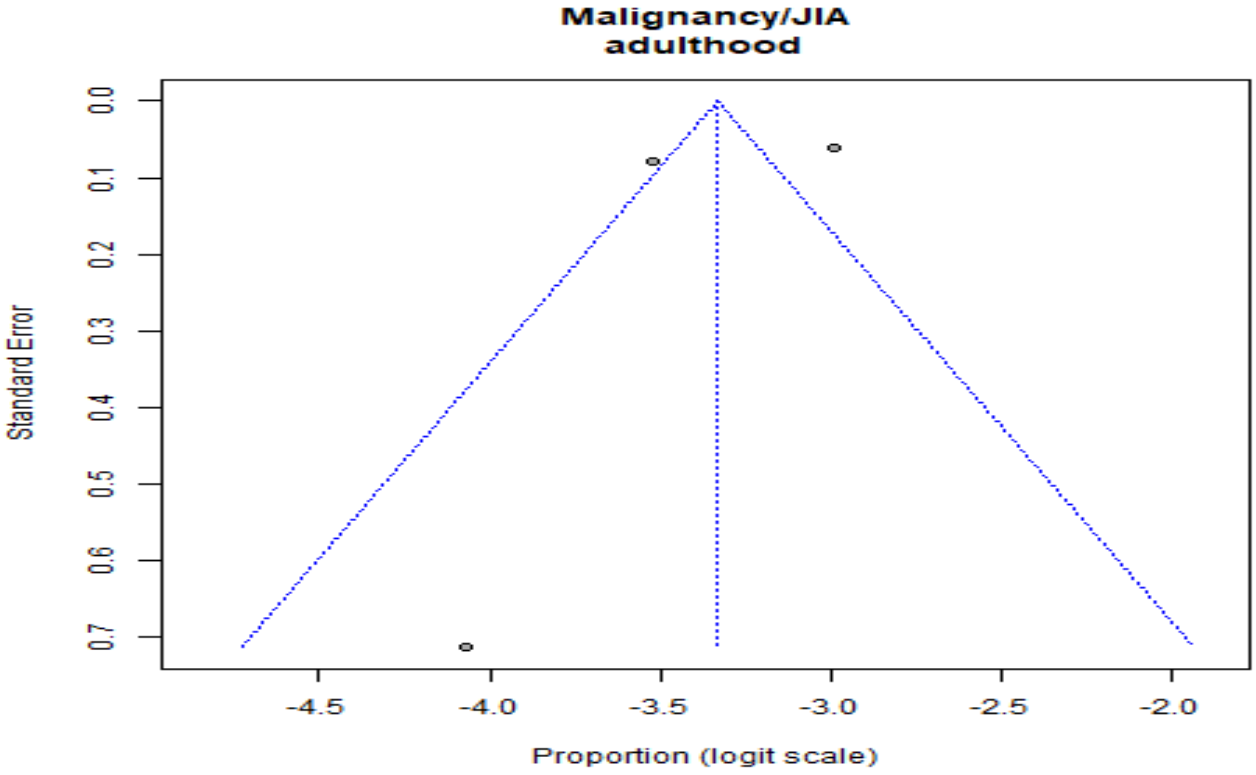
Supplementary Figure S4.33: Forest and Funnel Plots of studies reporting the prevalence of malignancy in adults with Juvenile Idiopathic Arthritis

Malignancy

Forest



Funnel



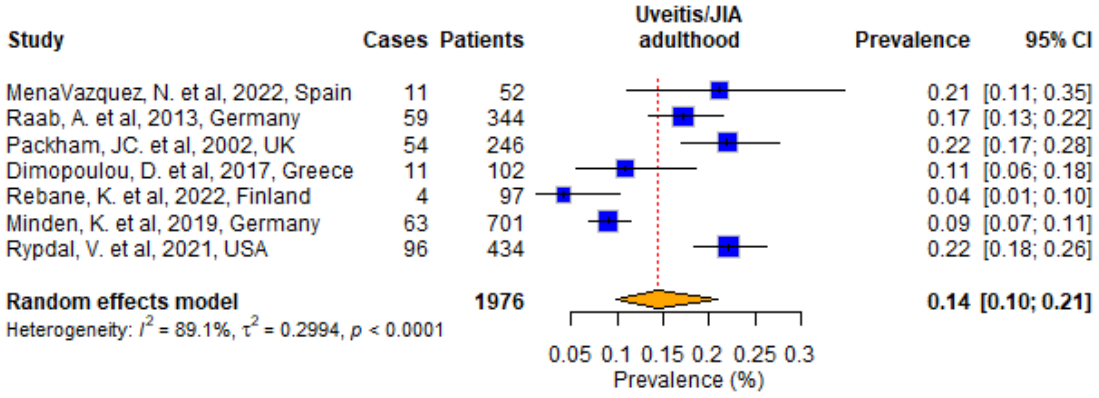
Summary statistics

```
                                        proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0480 [0.0426; 0.0538]       45.5
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0287 [0.0245; 0.0334]       44.7
Mars, N. J. etal, 2019, Finland               0.0168 [0.0020; 0.0594]        9.8
Number of studies: k = 3; Number of observations: o = 11547; Number of events: e = 442
                     proportion           95%-CI
Random effects model     0.0345 [0.0214; 0.0553]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1354 [0.0182; 11.3036]; tau = 0.3679 [0.1350; 3.3621]
 I^2 = 93.2% [83.6%; 97.2%]; H = 3.85 [2.47; 5.99]
Test of heterogeneity:      Q d.f.  p-value
                        29.61    2 < 0.0001
```

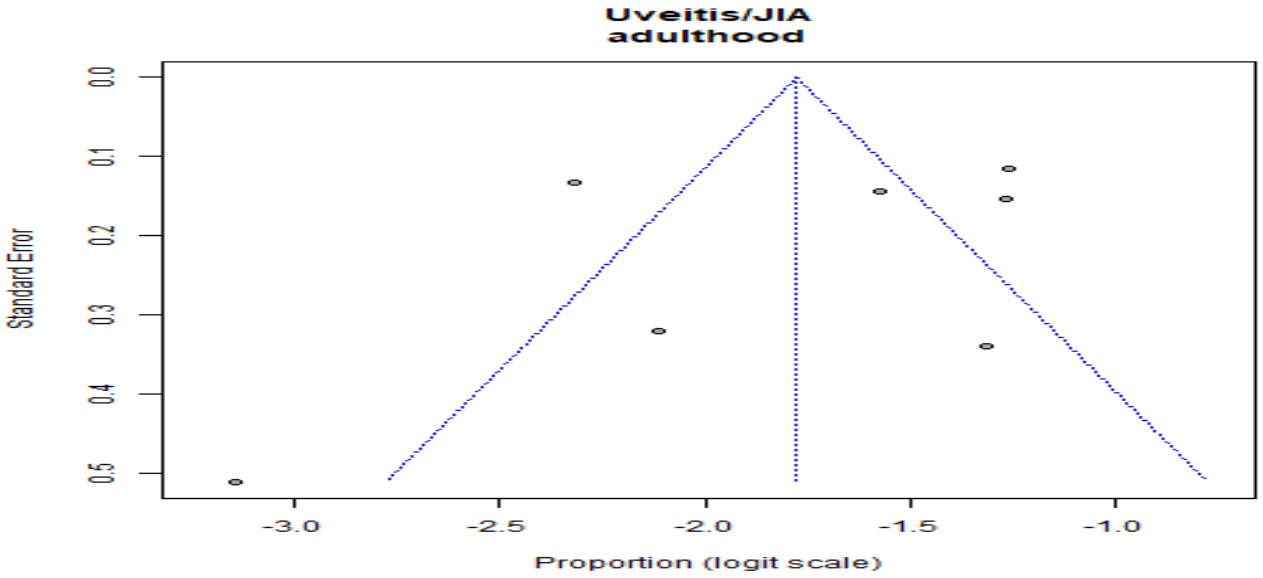
Supplementary Figure S4.34: Forest and Funnel Plots of studies reporting the prevalence of uveitis in adults with Juvenile Idiopathic Arthritis

Uveitis

Forest



Funnel



Summary statistics

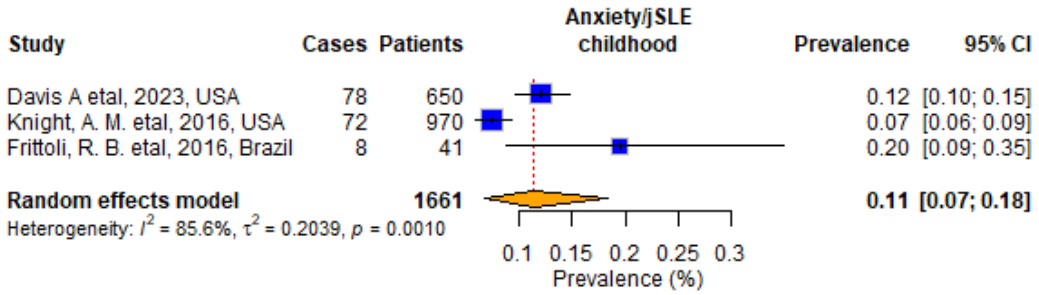
```
                                    proportion           95%-CI %W(random)
MenaVazquez, N. et al, 2022, Spain     0.2115 [0.1106; 0.3470]       12.5
Raab, A. et al, 2013, Germany          0.1715 [0.1332; 0.2156]       16.2
Packham, JC. et al, 2002, UK           0.2195 [0.1694; 0.2765]       16.1
Dimopoulou, D. et al, 2017, Greece     0.1078 [0.0551; 0.1848]       12.9
Rebane, K. et al, 2022, Finland        0.0412 [0.0113; 0.1022]        9.3
Minden, K. et al, 2019, Germany        0.0899 [0.0698; 0.1135]       16.4
Rypdal, V. et al, 2021, USA            0.2212 [0.1830; 0.2632]       16.6
Number of studies: k = 7; Number of observations: o = 1976; Number of events: e = 298
                    proportion           95%-CI
Random effects model     0.1446 [0.0976; 0.2090]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2994 [0.0932; 2.2766]; tau = 0.5472 [0.3053; 1.5089]
 I^2 = 89.1% [79.9%; 94.0%]; H = 3.02 [2.23; 4.09]
Test of heterogeneity:      Q d.f.  p-value
                        54.84    6 < 0.0001
```

Childhood/jSLE

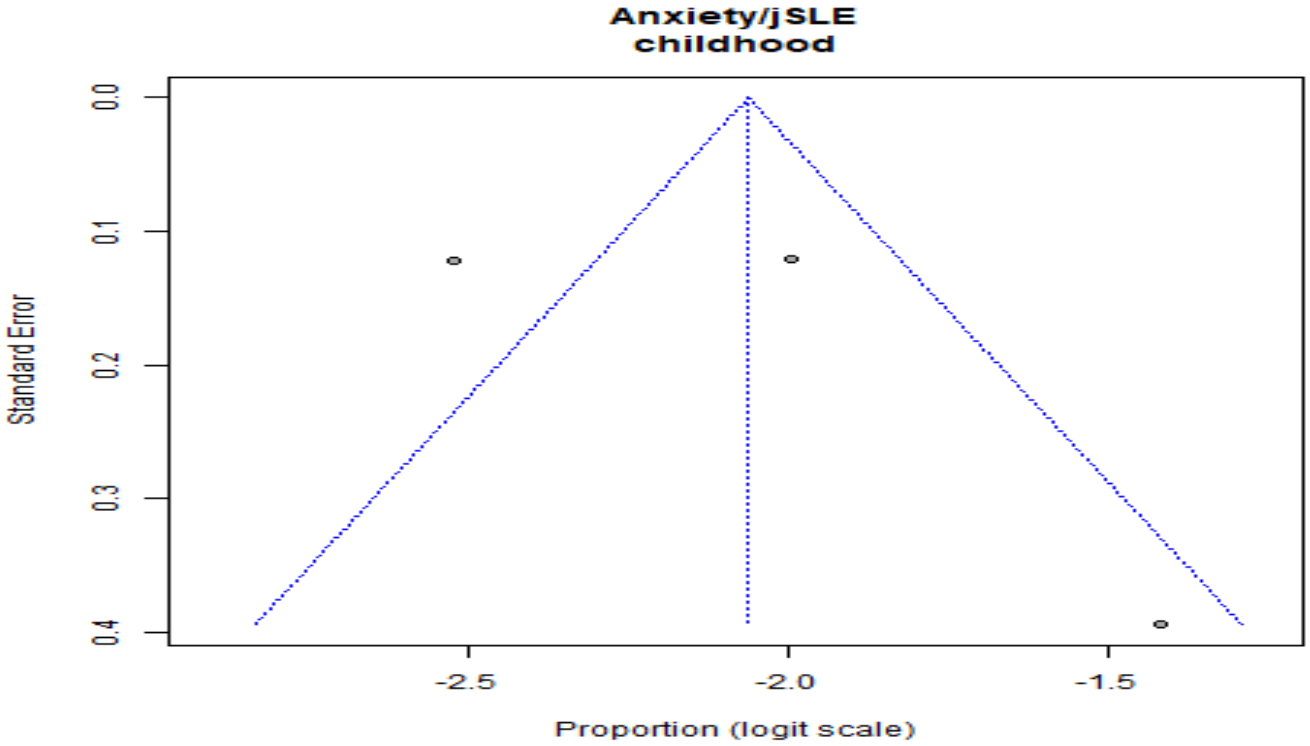
Supplementary Figure S4.35: Forest and Funnel Plots of studies reporting the prevalence of anxiety in children with juvenile-onset Systemic Lupus Erythematosus

Anxiety

Forest



Funnel



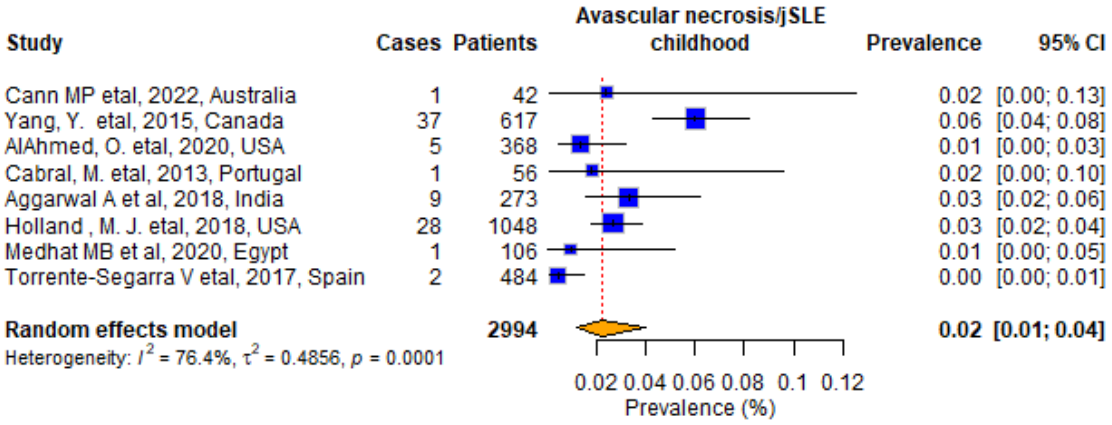
Summary statistics

```
                                proportion           95%-CI %W(random)
Davis A etal, 2023, USA             0.1200 [0.0960; 0.1475]       38.4
Knight, A. M. etal, 2016, USA       0.0742 [0.0585; 0.0926]       38.3
Frittoli, R. B. etal, 2016, Brazil  0.1951 [0.0882; 0.3487]       23.3
Number of studies: k = 3; Number of observations: o = 1661; Number of events: e = 158
                     proportion           95%-CI
Random effects model     0.1129 [0.0673; 0.1833]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2039 [0.0224; 12.0079]; tau = 0.4515 [0.1498; 3.4652]
 I^2 = 85.6% [57.7%; 95.1%]; H = 2.63 [1.54; 4.50]
Test of heterogeneity:      Q d.f. p-value
                        13.84    2  0.0010
```

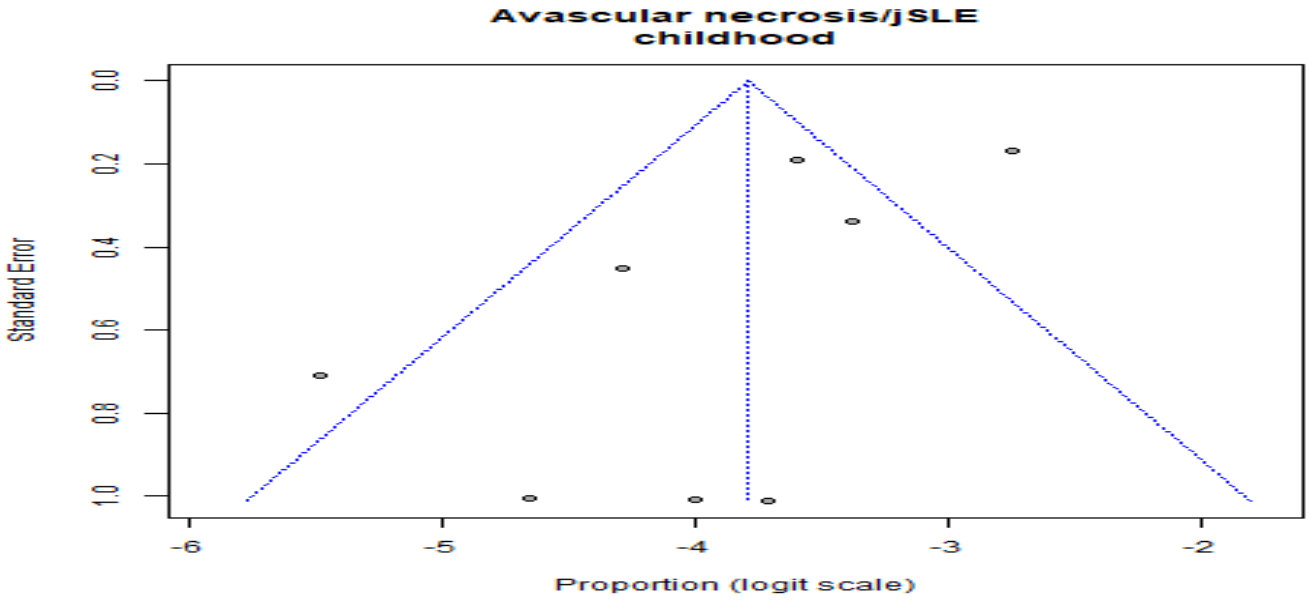
Supplementary Figure S4.36: Forest and Funnel Plots of studies reporting the prevalence of avascular necrosis in children with juvenile-onset Systemic Lupus Erythematosus

Avascular necrosis

Forest



Funnel



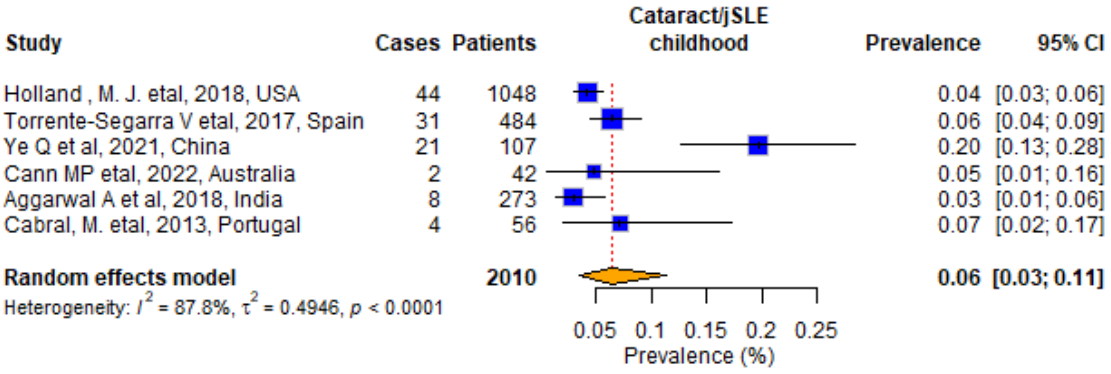
Summary statistics

```
                                  proportion           95%-CI %W(random)
Cann MP etal, 2022, Australia         0.0238 [0.0006; 0.1257]        6.6
Yang, Y.  etal, 2015, Canada          0.0600 [0.0426; 0.0817]       19.5
AlAhmed, O. etal, 2020, USA           0.0136 [0.0044; 0.0314]       14.5
Cabral, M. etal, 2013, Portugal       0.0179 [0.0005; 0.0955]        6.7
Aggarwal A et al, 2018, India         0.0330 [0.0152; 0.0617]       16.7
Holland , M. J. etal, 2018, USA       0.0267 [0.0178; 0.0384]       19.2
Medhat MB et al, 2020, Egypt          0.0094 [0.0002; 0.0514]        6.7
Torrente-Segarra V etal, 2017, Spain  0.0041 [0.0005; 0.0148]       10.1
Number of studies: k = 8; Number of observations: o = 2994; Number of events: e = 84
                    proportion           95%-CI
Random effects model     0.0220 [0.0120; 0.0402]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4856 [0.0757; 2.5590]; tau = 0.6968 [0.2752; 1.5997]
 I^2 = 76.4% [52.8%; 88.2%]; H = 2.06 [1.46; 2.91]
Test of heterogeneity:       Q d.f. p-value
                         29.62    7  0.0001
```

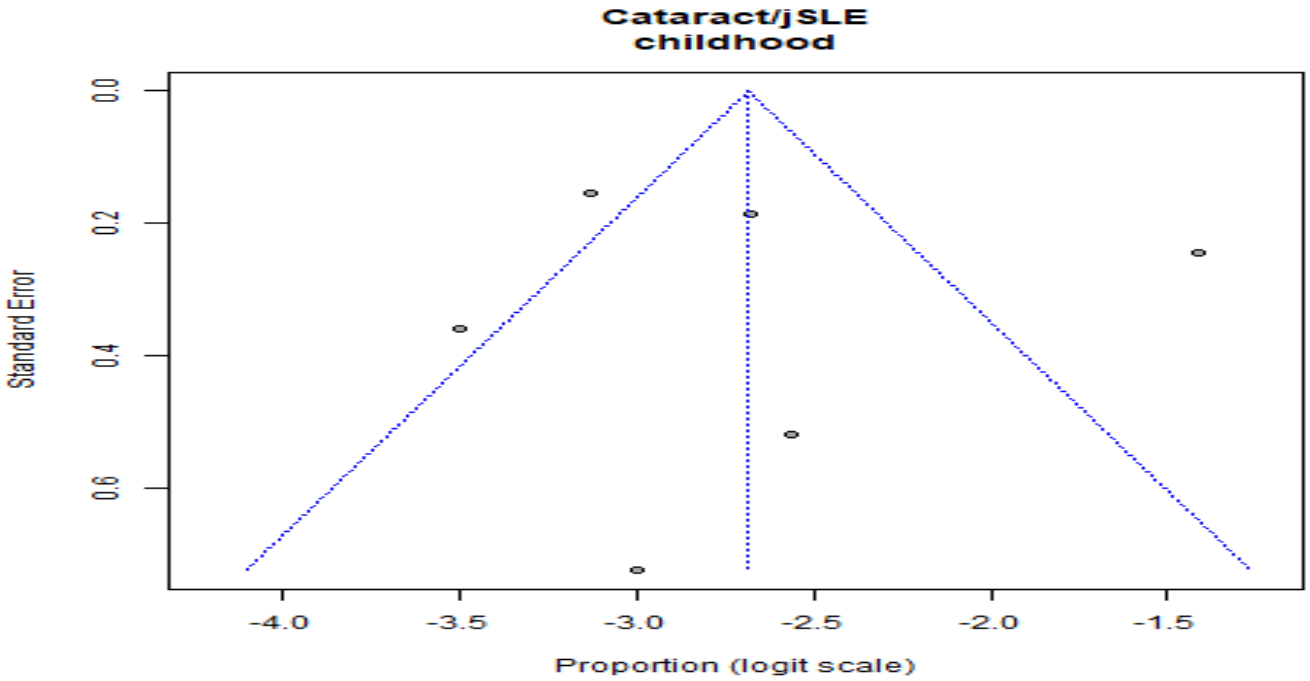
Supplementary Figure S4.37: Forest and Funnel Plots of studies reporting the prevalence of cataract in children with juvenile-onset Systemic Lupus Erythematosus

Cataract

Forest



Funnel



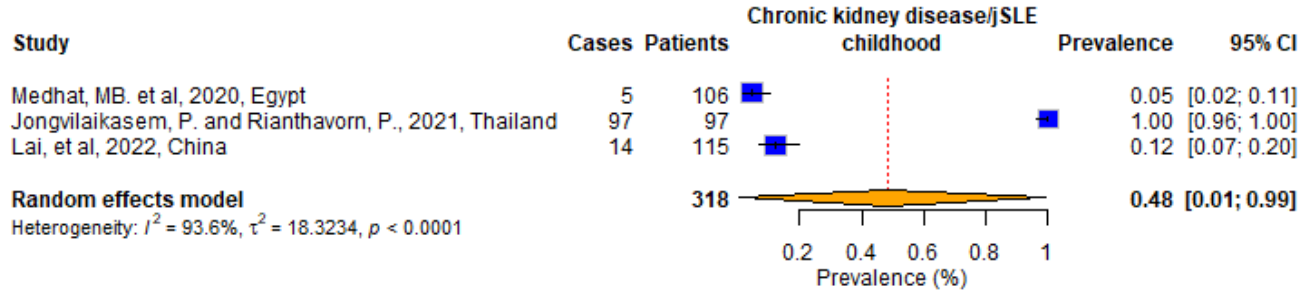
Summary statistics

```
                                    proportion           95%-CI %W(random)
Holland , M. J. etal, 2018, USA         0.0420 [0.0307; 0.0560]       20.3
Torrente-Segarra V etal, 2017, Spain    0.0640 [0.0439; 0.0897]       19.9
Ye Q et al, 2021, China                 0.1963 [0.1258; 0.2842]       19.0
Cann MP etal, 2022, Australia           0.0476 [0.0058; 0.1616]       10.3
Aggarwal A et al, 2018, India           0.0293 [0.0127; 0.0569]       16.9
Cabral, M. etal, 2013, Portugal         0.0714 [0.0198; 0.1729]       13.8
Number of studies: k = 6; Number of observations: o = 2010; Number of events: e = 110
                     proportion           95%-CI
Random effects model     0.0639 [0.0349; 0.1141]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4946 [0.1265; 3.0315]; tau = 0.7032 [0.3557; 1.7411]
 I^2 = 87.8% [75.9%; 93.8%]; H = 2.86 [2.04; 4.02]
Test of heterogeneity:      Q d.f.  p-value
                        40.98    5 < 0.0001
```

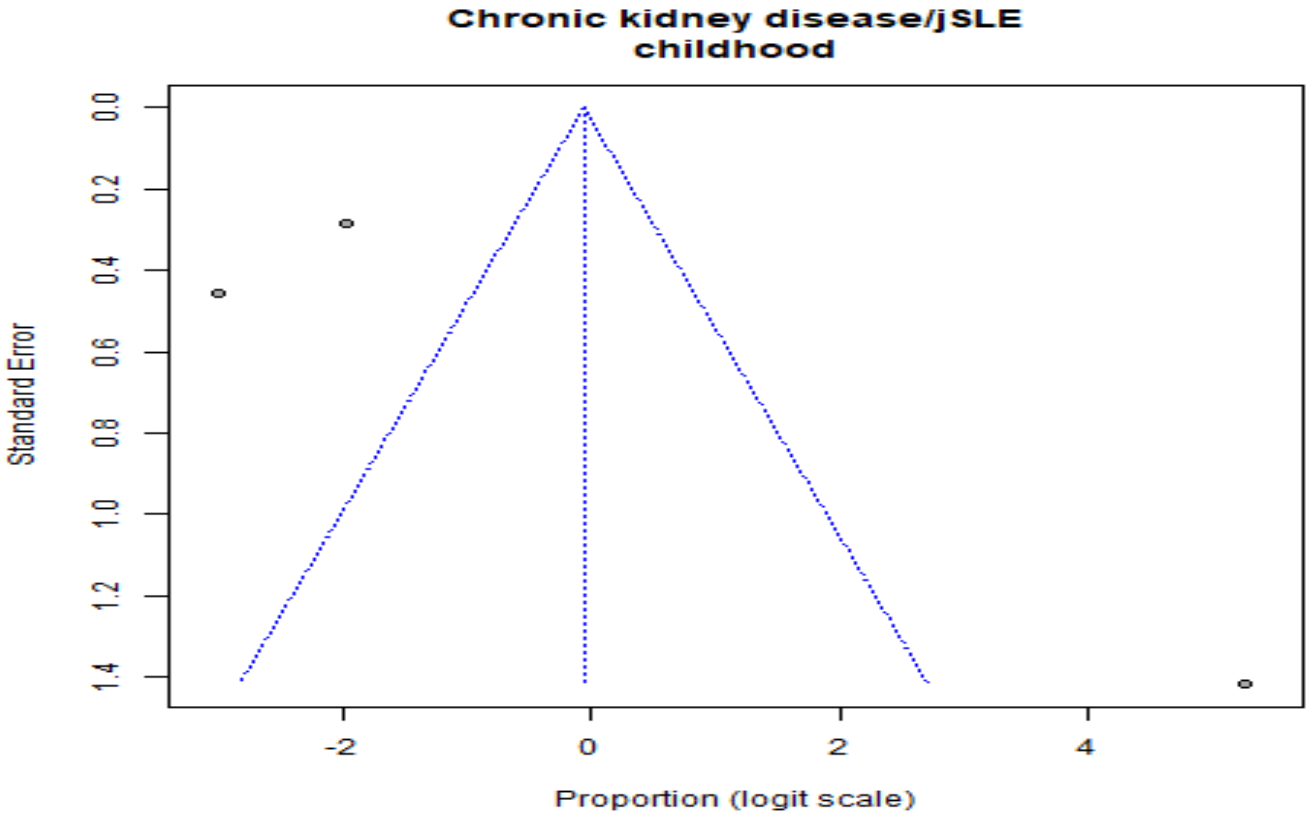
Supplementary Figure S4.38: Forest and Funnel Plots of studies reporting the prevalence of chronic kidney disease in children with juvenile-onset Systemic Lupus Erythematosus

Chronic kidney disease

Forest



Funnel



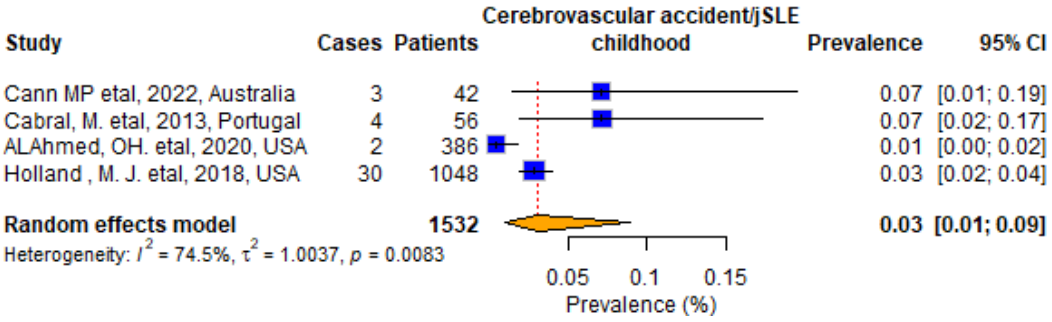
Summary statistics

```
                                                 proportion       95%-CI          %W(random)
Medhat, MB. et al, 2020, Egypt                       0.0472       [0.0155; 0.1067]        34.3
Jongvilaikasem, P. and Rianthavorn, P., 2021, Thailand     1.0000       [0.9627; 1.0000]        31.2
Lai, et al, 2022, China                              0.1217       [0.0682; 0.1958]        34.5
Number of studies: k = 3; Number of observations: o = 318; Number of events: e = 116
                     proportion           95%-CI
Random effects model     0.4838 [0.0067; 0.9924]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 18.3234 [4.1403; >183.2337]; tau = 4.2806 [2.0348; >13.5364]
 I^2 = 93.6% [84.6%; 97.3%]; H = 3.94 [2.55; 6.10]
Test of heterogeneity:      Q d.f.  p-value
                        31.09    2 < 0.0001
```

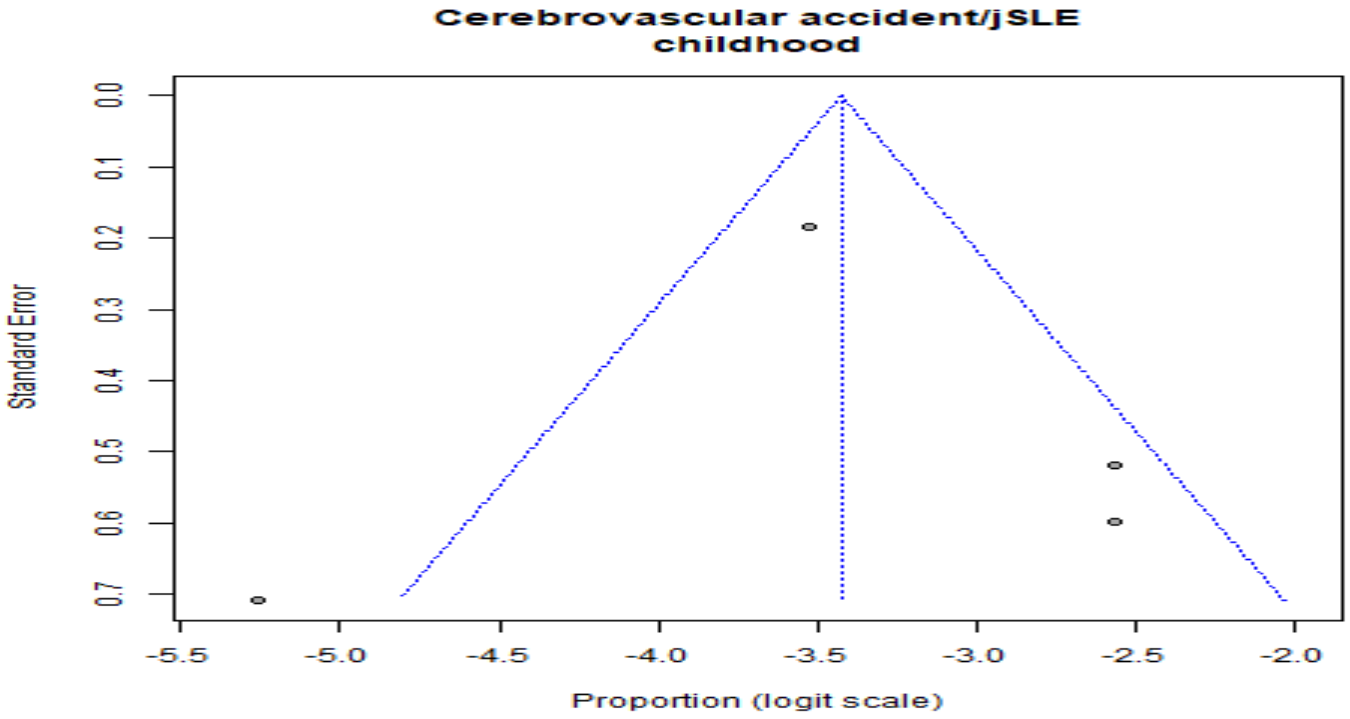
Supplementary Figure S4.39: Forest and Funnel Plots of studies reporting the prevalence of cerebrovascular accident in children with juvenile-onset Systemic Lupus Erythematosus

Cerebrovascular accident

Forest



Funnel



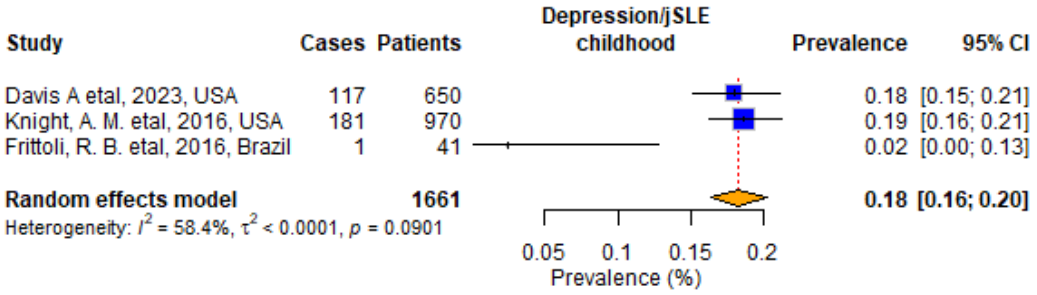
Summary statistics

```
                                proportion      95%-CI          %W(random)
Cann MP etal, 2022, Australia       0.0714      [0.0150; 0.1948]        23.3
Cabral, M. etal, 2013, Portugal     0.0714      [0.0198; 0.1729]        25.0
ALAhmed, OH. etal, 2020, USA        0.0052      [0.0006; 0.0186]        21.1
Holland , M. J. etal, 2018, USA     0.0286      [0.0194; 0.0406]        30.6
Number of studies: k = 4; Number of observations: o = 1532; Number of events: e = 39
                     proportion           95%-CI
Random effects model     0.0315 [0.0106; 0.0893]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0037 [0.0947; 21.9732]; tau = 1.0018 [0.3078; 4.6876]
 I^2 = 74.5% [28.9%; 90.8%]; H = 1.98 [1.19; 3.30]
Test of heterogeneity:      Q d.f. p-value
                        11.75    3  0.0083
```

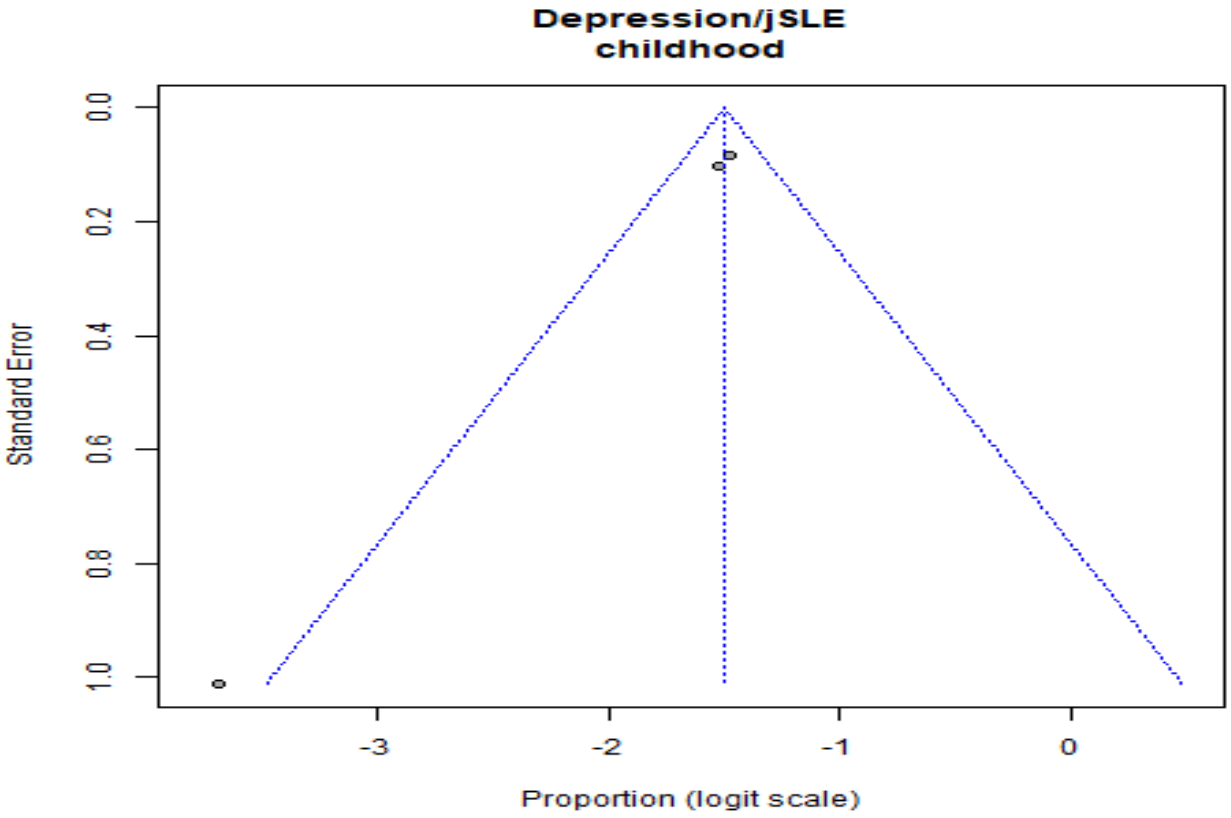
Supplementary Figure S4.40: Forest and Funnel Plots of studies reporting the prevalence of depression in children with juvenile-onset Systemic Lupus Erythematosus

Depression

Forest



Funnel



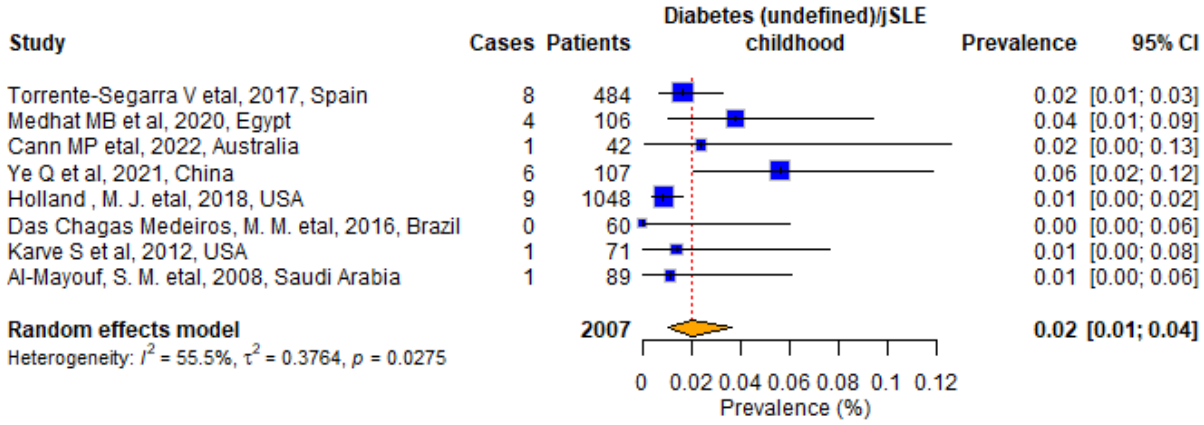
Summary statistics

```
                                proportion   95%-CI          %W(random)
Davis A etal, 2023, USA             0.1800   [0.1512; 0.2117]       39.3
Knight, A. M. etal, 2016, USA       0.1866   [0.1625; 0.2126]       60.3
Frittoli, R. B. etal, 2016, Brazil  0.0244   [0.0006; 0.1286]        0.4
Number of studies: k = 3; Number of observations: o = 1661; Number of events: e = 299
                     proportion           95%-CI
Random effects model     0.1827 [0.1647; 0.2021]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 62.7424]; tau = 0.0018 [0.0000; 7.9210]
 I^2 = 58.4% [0.0%; 88.2%]; H = 1.55 [1.00; 2.91]
Test of heterogeneity:     Q d.f. p-value
                        4.81    2  0.0901
```

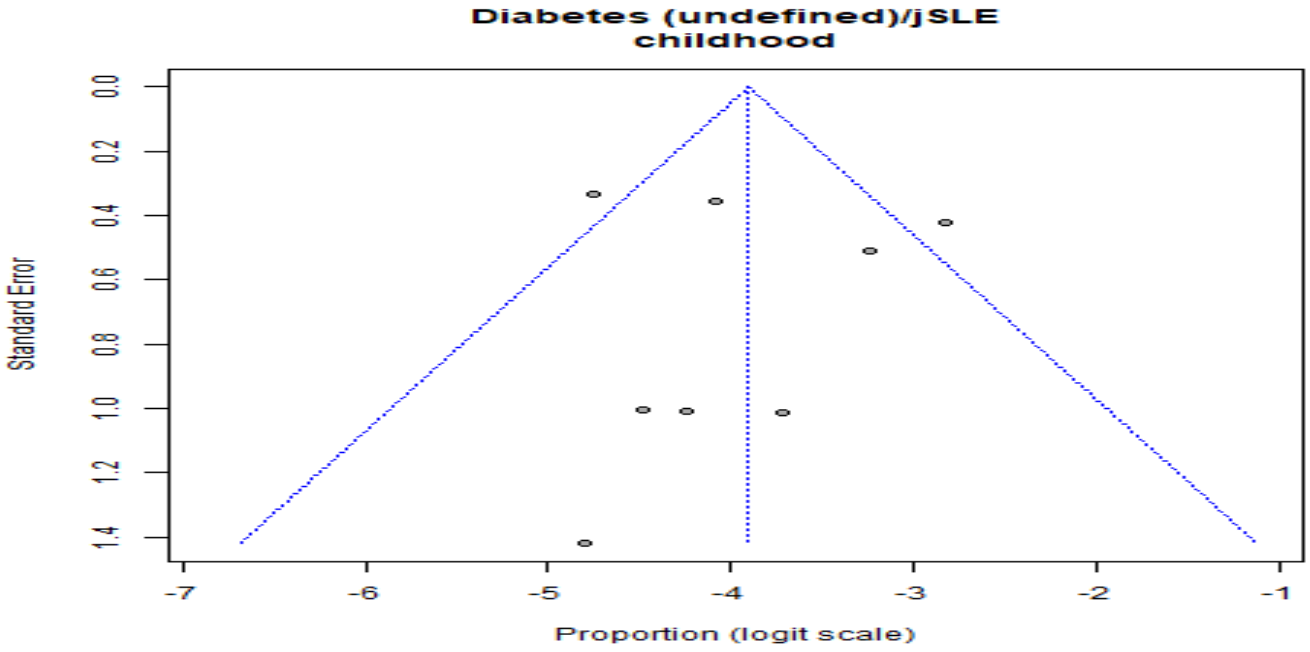
Supplementary Figure S4.41: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with juvenile-onset Systemic Lupus Erythematosus

Diabetes (undefined)

Forest



Funnel



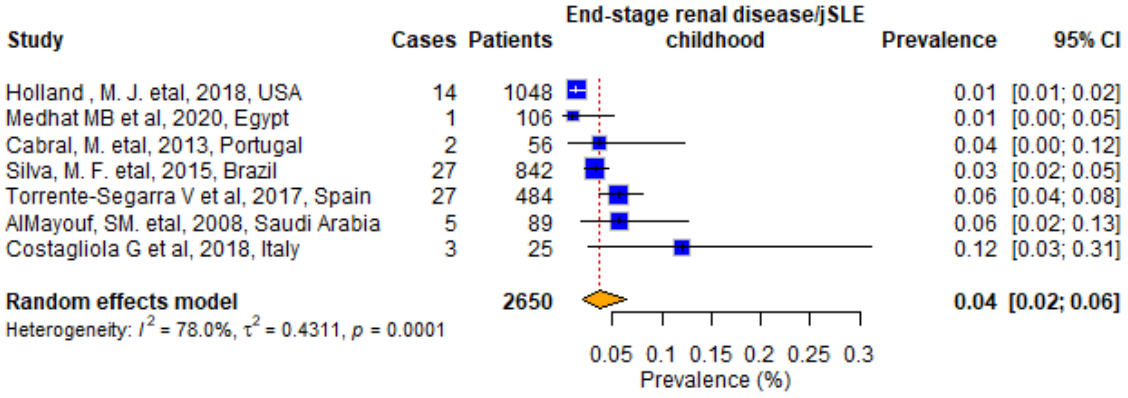
Summary statistics

```
                                                  proportion         95%-CI          %W(random)
Torrente-Segarra V etal, 2017, Spain                  0.0165        [0.0072; 0.0323]        19.9
Medhat MB et al, 2020, Egypt                          0.0377        [0.0104; 0.0938]        15.7
Cann MP etal, 2022, Australia                         0.0238        [0.0006; 0.1257]         7.1
Ye Q et al, 2021, China                               0.0561        [0.0209; 0.1181]        18.1
Holland , M. J. etal, 2018, USA                       0.0086        [0.0039; 0.0162]        20.5
Das Chagas Medeiros, M. M. etal, 2016, Brazil         0.0000        [0.0000; 0.0596]         4.2
Karve S et al, 2012, USA                              0.0141        [0.0004; 0.0760]         7.2
Al-Mayouf, S. M. etal, 2008, Saudi Arabia             0.0112        [0.0003; 0.0610]         7.2
Number of studies: k = 8; Number of observations: o = 2007; Number of events: e = 30
                     proportion           95%-CI
Random effects model     0.0198 [0.0107; 0.0362]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3764 [0.0000; 1.6458]; tau = 0.6135 [0.0000; 1.2829]
 I^2 = 55.5% [1.9%; 79.9%]; H = 1.50 [1.01; 2.23]
Test of heterogeneity:       Q d.f. p-value
                         15.75    7  0.0275
```

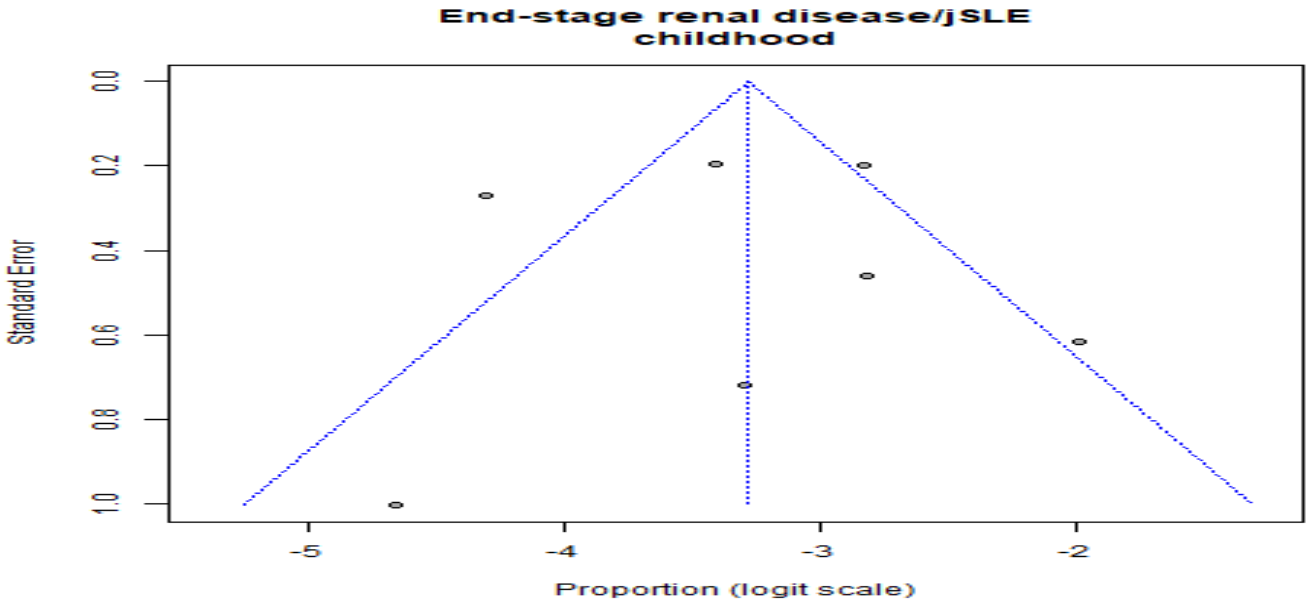
Supplementary Figure S4.42: Forest and Funnel Plots of studies reporting the prevalence of end-stage renal disease in children with juvenile-onset Systemic Lupus Erythematosus

End-stage renal disease

Forest



Funnel



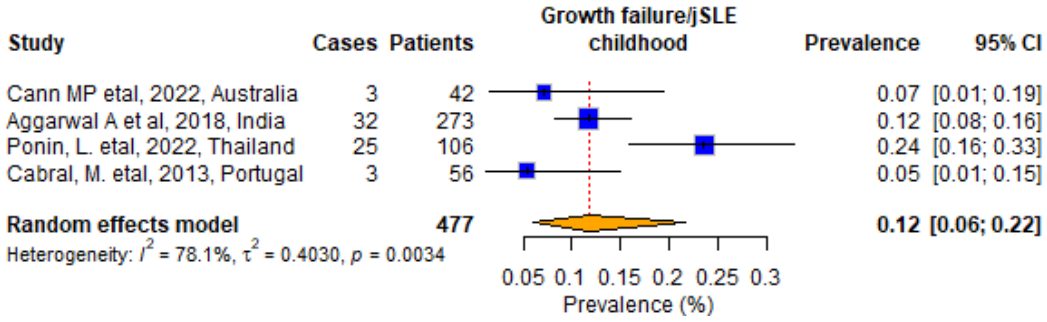
Summary statistics

```
                                    proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA         0.0134       [0.0073; 0.0223]        18.4
Medhat MB et al, 2020, Egypt            0.0094       [0.0002; 0.0514]         6.4
Cabral, M. etal, 2013, Portugal         0.0357       [0.0044; 0.1231]         9.8
Silva, M. F. etal, 2015, Brazil         0.0321       [0.0212; 0.0463]        19.8
Torrente-Segarra V et al, 2017, Spain   0.0558       [0.0371; 0.0801]        19.7
AlMayouf, SM. etal, 2008, Saudi Arabia  0.0562       [0.0185; 0.1263]        14.4
Costagliola G et al, 2018, Italy        0.1200       [0.0255; 0.3122]        11.5
Number of studies: k = 7; Number of observations: o = 2650; Number of events: e = 79
                    proportion           95%-CI
Random effects model     0.0362 [0.0203; 0.0639]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4311 [0.0842; 3.5598]; tau = 0.6566 [0.2901; 1.8867]
 I^2 = 78.0% [54.3%; 89.4%]; H = 2.13 [1.48; 3.07]
Test of heterogeneity:      Q d.f. p-value
                        27.24    6  0.0001
```

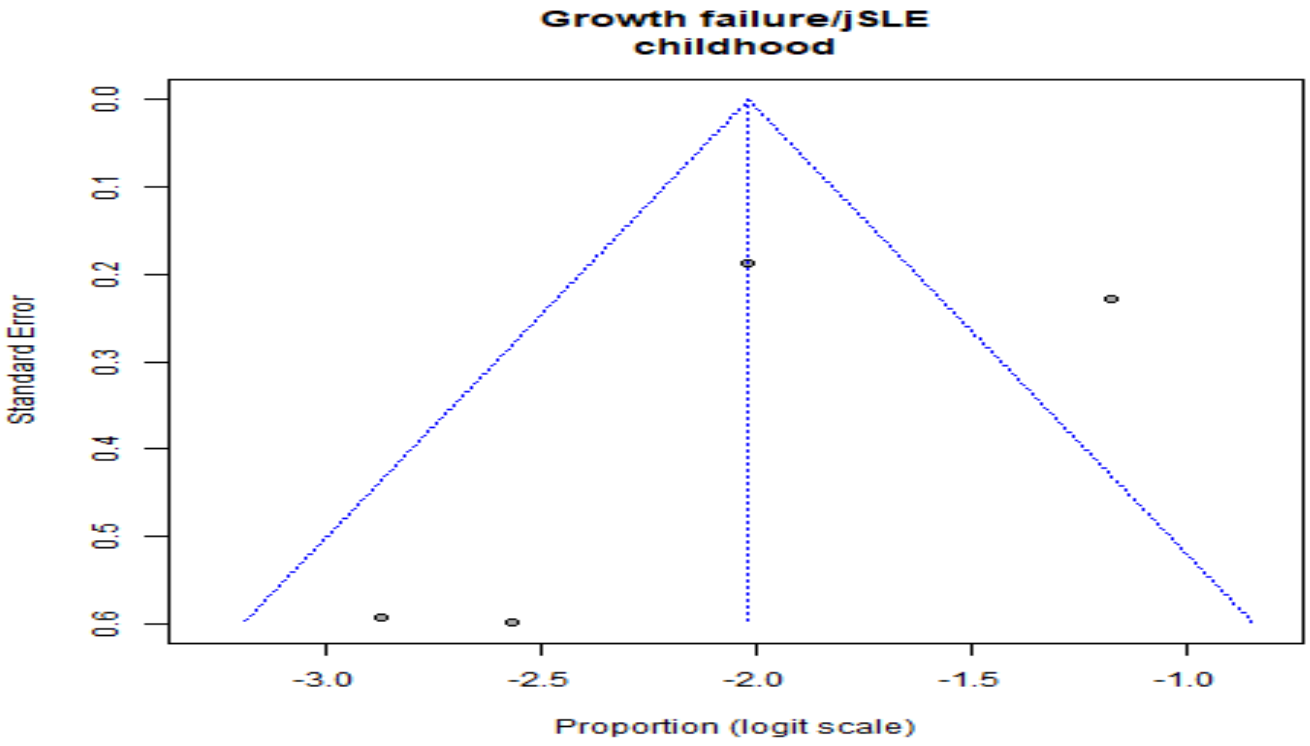
Supplementary Figure S4.43: Forest and Funnel Plots of studies reporting the prevalence of growth failure in children with juvenile-onset Systemic Lupus Erythematosus

Growth failure

Forest



Funnel



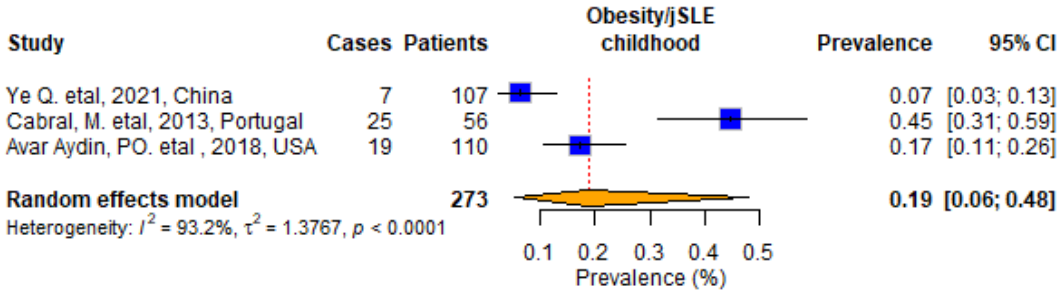
Summary statistics

```
                              proportion      95%-CI          %W(random)
Cann MP etal, 2022, Australia      0.0714      [0.0150; 0.1948]       18.4
Aggarwal A et al, 2018, India      0.1172      [0.0816; 0.1614]       32.1
Ponin, L. etal, 2022, Thailand     0.2358      [0.1588; 0.3282]       30.9
Cabral, M. etal, 2013, Portugal    0.0536      [0.0112; 0.1487]       18.6
Number of studies: k = 4; Number of observations: o = 477; Number of events: e = 63
                     proportion           95%-CI
Random effects model     0.1173 [0.0599; 0.2170]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4030 [0.0383; 7.5155]; tau = 0.6349 [0.1957; 2.7414]
 I^2 = 78.1% [40.8%; 91.9%]; H = 2.14 [1.30; 3.51]
Test of heterogeneity:      Q d.f. p-value
                        13.69    3  0.0034
```

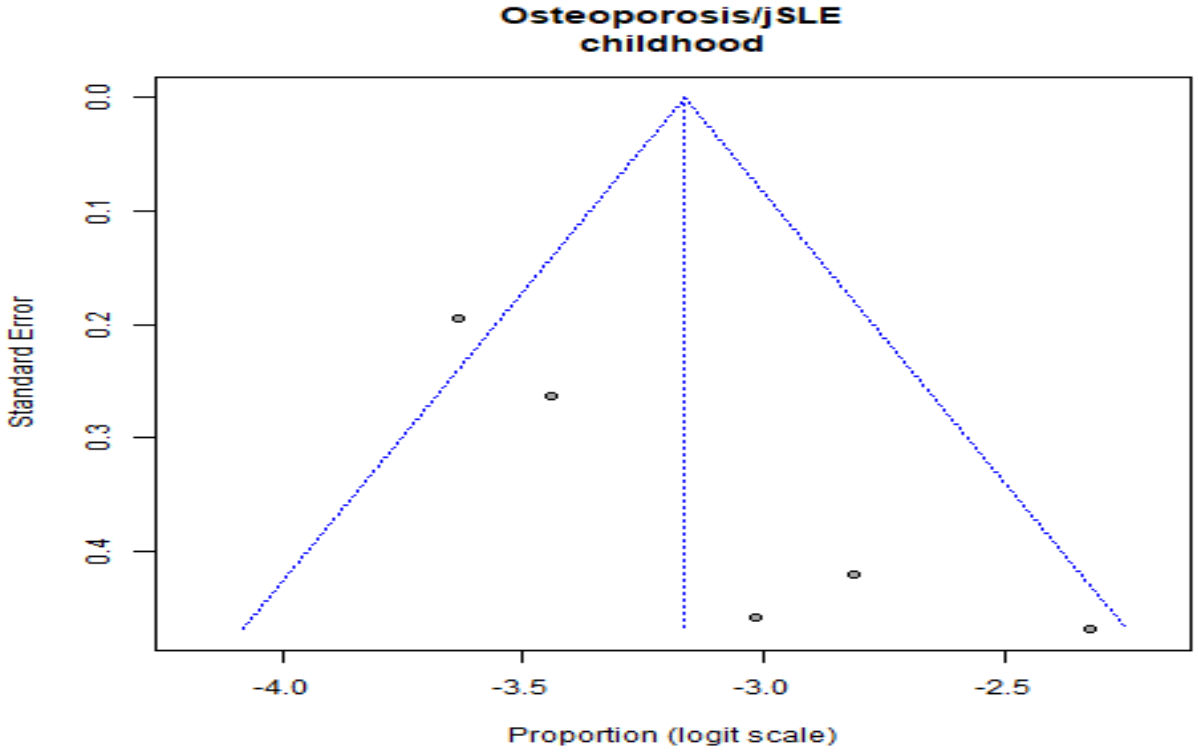
Supplementary Figure S4.44: Forest and Funnel Plots of studies reporting the prevalence of obesity in children with juvenile-onset Systemic Lupus Erythematosus

Obesity

Forest



Funnel



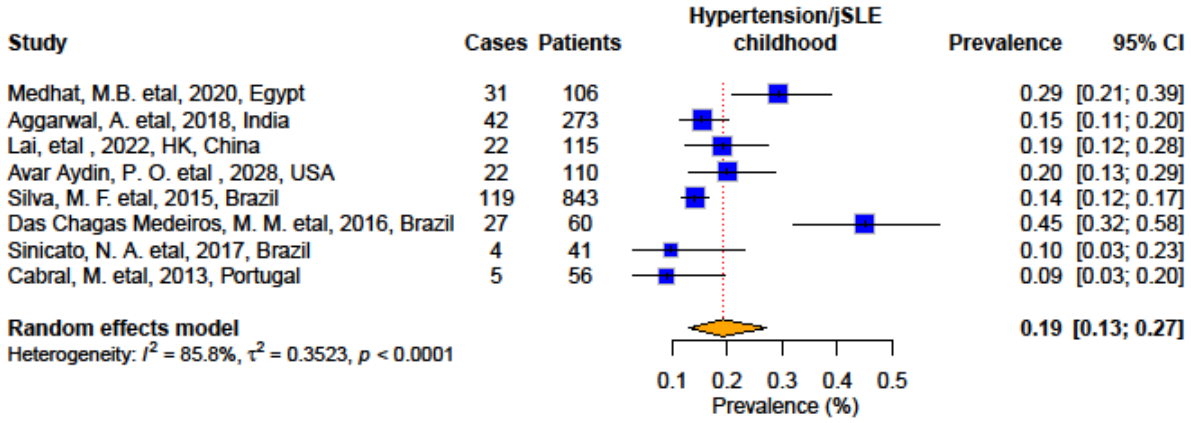
Summary statistics

```
                                proportion     95%-CI          %W(random)
Ye Q. etal, 2021, China             0.0654     [0.0267; 0.1302]        32.1
Cabral, M. etal, 2013, Portugal     0.4464     [0.3134; 0.5853]        33.9
Avar Aydin, PO. etal , 2018, USA    0.1727     [0.1073; 0.2565]        34.1
Number of studies: k = 3; Number of observations: o = 273; Number of events: e = 51
                     proportion           95%-CI
Random effects model     0.1886 [0.0556; 0.4784]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.3767 [0.2984; 59.0990]; tau = 1.1733 [0.5463; 7.6876]
 I^2 = 93.2% [83.5%; 97.2%]; H = 3.83 [2.46; 5.97]
Test of heterogeneity:      Q d.f.  p-value
                        29.39    2 < 0.0001
```

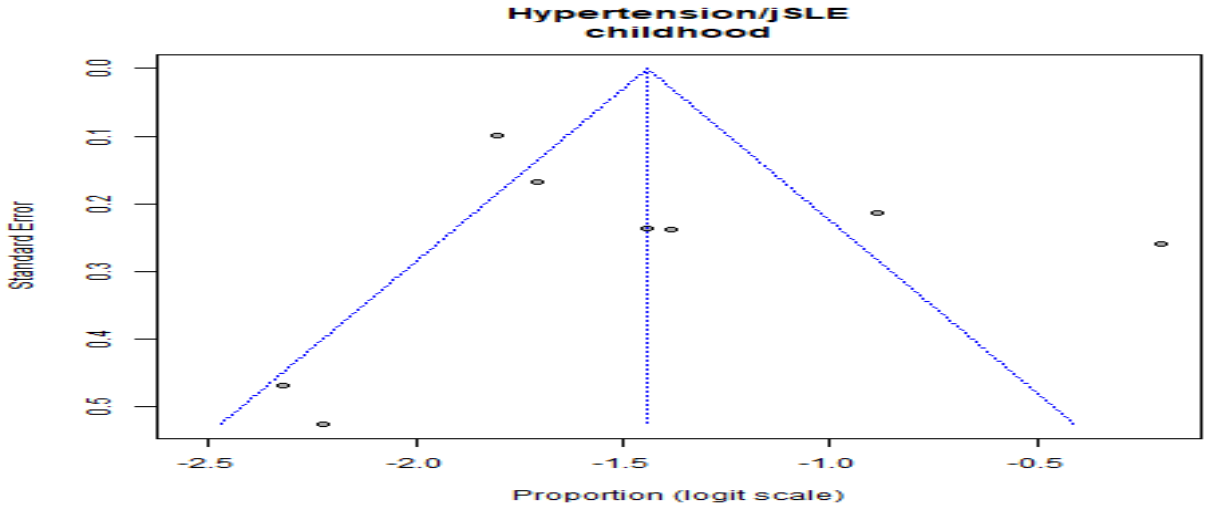
Supplementary Figure S4.45: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with juvenile-onset Systemic Lupus Erythematosus

Hypertension

Forest



Funnel



Summary statistics

```
                                          proportion        95% CI                %W(random)
Medhat, M.B. etal, 2020, Egypt                0.2925        [0.2081; 0.3887]        13.6
Aggarwal, A. etal, 2018, India                0.1538        [0.1132; 0.2022]        14.2
Lai, etal , 2022, HK, China                   0.1913        [0.1239; 0.2752]        13.2
Avar Aydin, P. O. etal , 2028, USA            0.2000        [0.1298; 0.2870]        13.2
Silva, M. F. etal, 2015, Brazil               0.1412        [0.1184; 0.1665]        14.9
Das Chagas Medeiros, M. M. etal, 2016, Brazil 0.4500        [0.3212; 0.5839]        12.9
Sinicato, N. A. etal, 2017, Brazil            0.0976        [0.0272; 0.2313]         8.6
Cabral, M. etal, 2013, Portugal               0.0893        [0.0296; 0.1962]         9.4

Number of studies: k = 8
Number of observations: o = 1604
Number of events: e = 272

                    proportion 95% CI
Random effects model     0.1914 [0.1305; 0.2718]

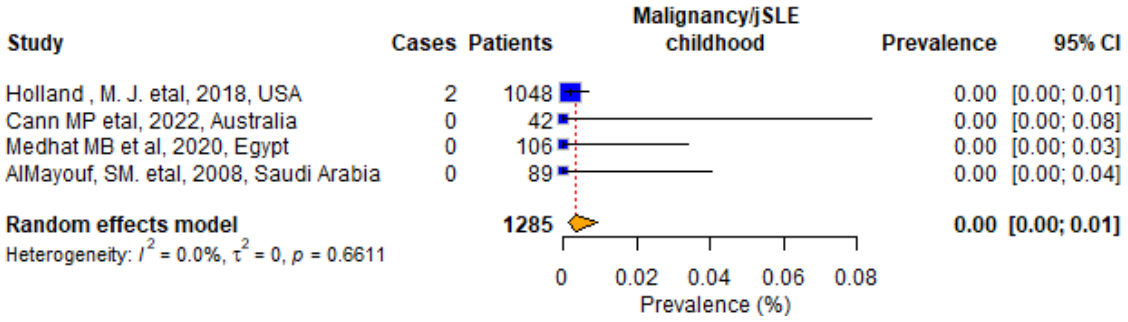
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3523 [0.1124; 1.9017]; tau = 0.5936 [0.3352; 1.3790]
 I^2 = 85.8% [74.0%; 92.3%]; H = 2.65 [1.96; 3.59]

Test of heterogeneity:
     Q d.f.  p-value
 49.34    7 < 0.0001
```

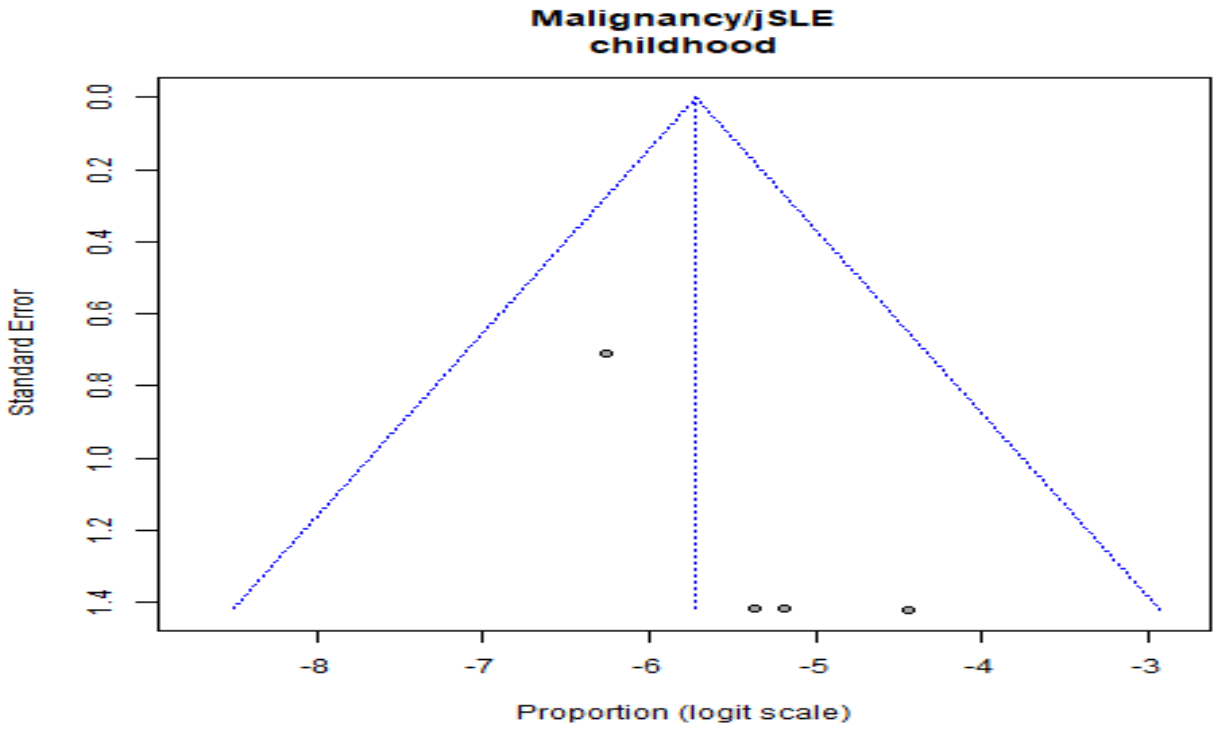
Supplementary Figure S4.46: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with juvenile-onset Systemic Lupus Erythematosus

Malignancy

Forest



Funnel



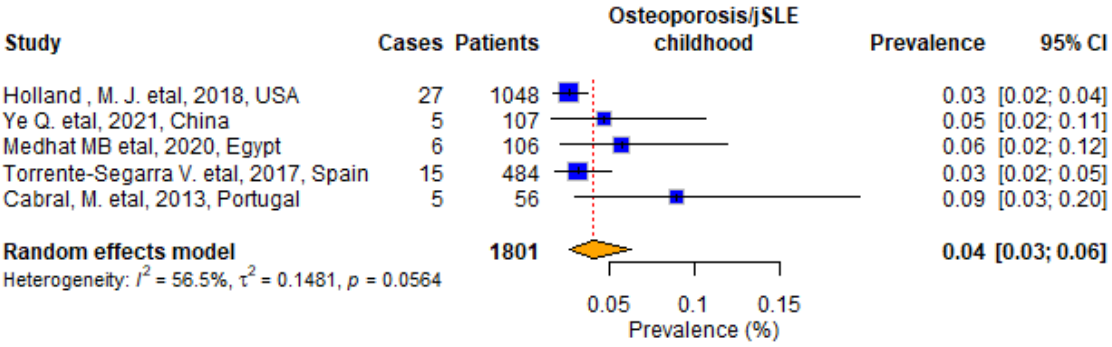
Summary statistics

```
                                        proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0019       [0.0002; 0.0069]        57.3
Cann MP etal, 2022, Australia               0.0000       [0.0000; 0.0841]        14.2
Medhat MB et al, 2020, Egypt                0.0000       [0.0000; 0.0342]        14.3
AlMayouf, SM. etal, 2008, Saudi Arabia      0.0000       [0.0000; 0.0406]        14.3
Number of studies: k = 4; Number of observations: o = 1285; Number of events: e = 2
                     proportion           95%-CI
Random effects model     0.0033 [0.0011; 0.0093]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 6.5718]; tau = 0 [0.0000; 2.5635]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.59    3  0.6611
```

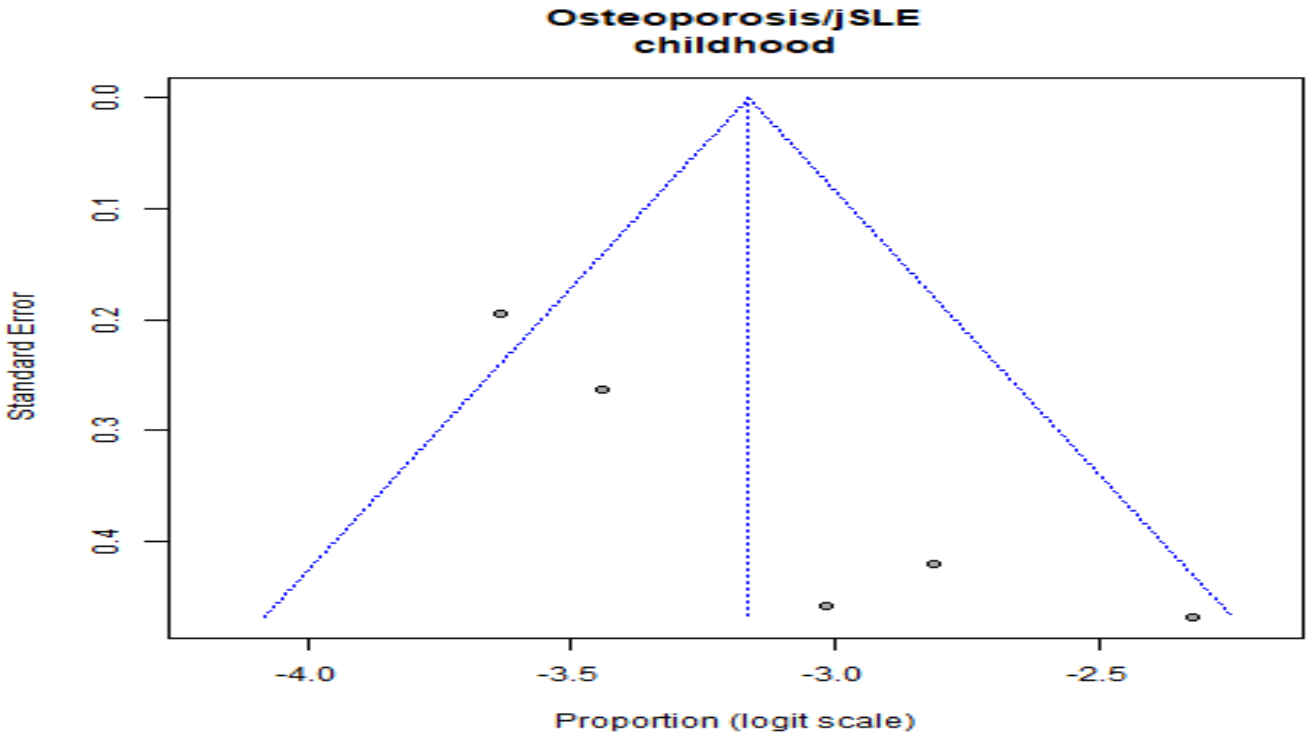
Supplementary Figure S4.47: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in children with juvenile-onset Systemic Lupus Erythematosus

Osteoporosis

Forest



Funnel



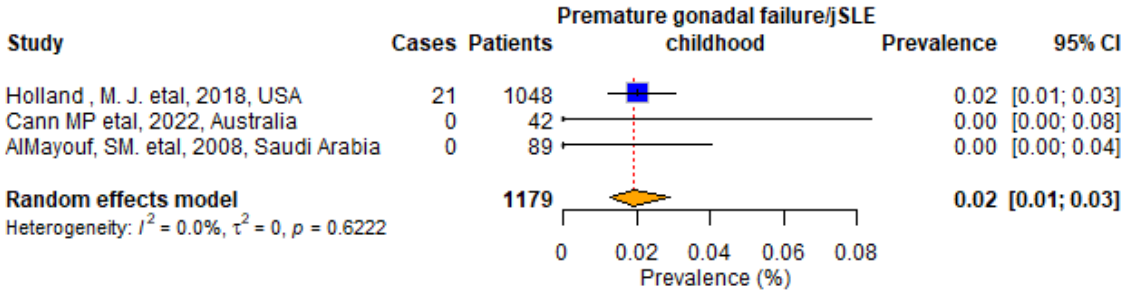
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Holland , M. J. etal, 2018, USA         0.0258       [0.0170; 0.0373]        28.9
Ye Q. etal, 2021, China                 0.0467       [0.0153; 0.1057]        15.0
Medhat MB etal, 2020, Egypt             0.0566       [0.0211; 0.1191]        16.6
Torrente-Segarra V. etal, 2017, Spain   0.0310       [0.0174; 0.0506]        24.8
Cabral, M. etal, 2013, Portugal         0.0893       [0.0296; 0.1962]        14.6
Number of studies: k = 5; Number of observations: o = 1801; Number of events: e = 58
                     proportion           95%-CI
Random effects model     0.0405 [0.0261; 0.0624]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1481 [0.0000; 2.0934]; tau = 0.3848 [0.0000; 1.4469]
 I^2 = 56.5% [0.0%; 83.9%]; H = 1.52 [1.00; 2.49]
Test of heterogeneity:     Q d.f. p-value
                        9.19    4  0.0564
```

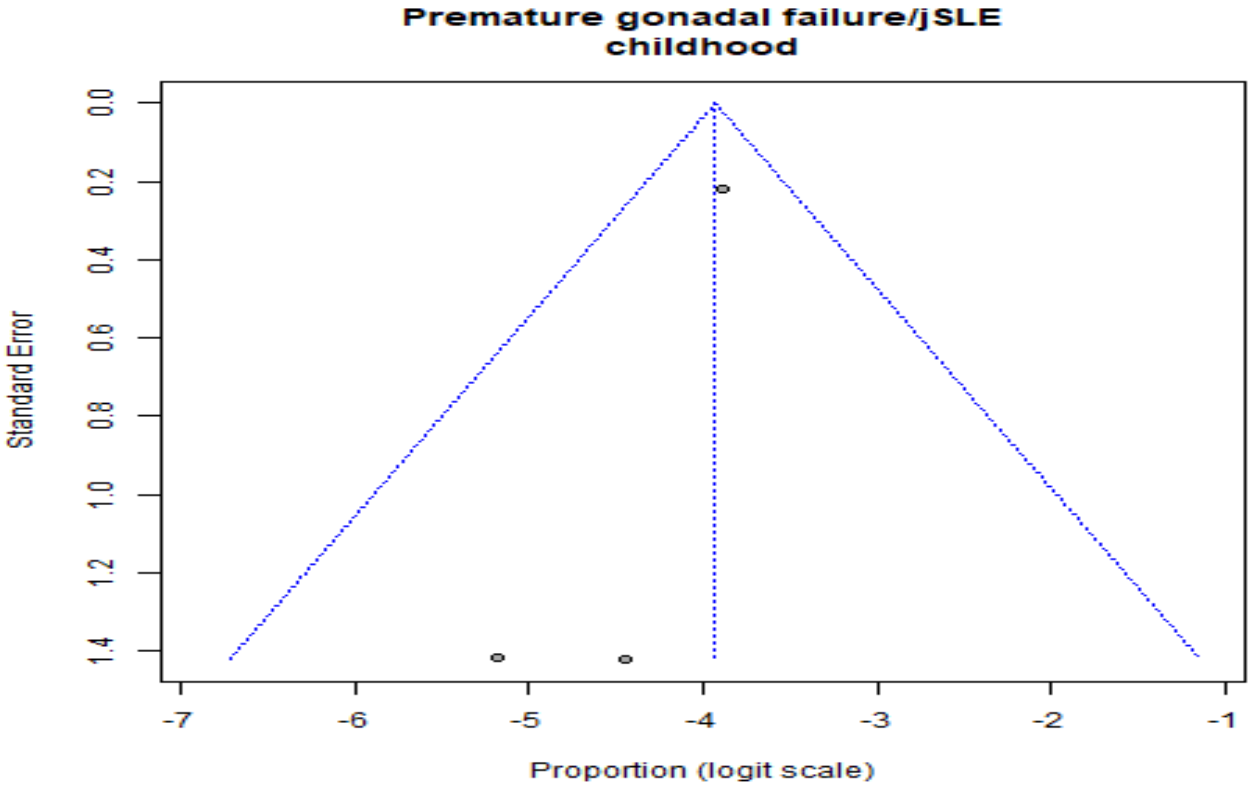
Supplementary Figure S4.48: Forest and Funnel Plots of studies reporting the prevalence of premature gonadal failure in children with juvenile-onset Systemic Lupus Erythematosus

Premature gonadal failure

Forest



Funnel



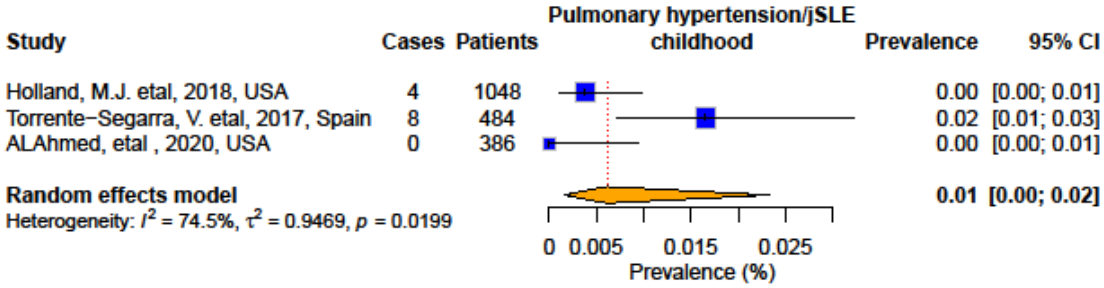
Summary statistics

```
                                      proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA           0.0200       [0.0124; 0.0305]        95.4
Cann MP etal, 2022, Australia             0.0000       [0.0000; 0.0841]         2.3
AlMayouf, SM. etal, 2008, Saudi Arabia    0.0000       [0.0000; 0.0406]         2.3
Number of studies: k = 3; Number of observations: o = 1179; Number of events: e = 21
                     proportion           95%-CI
Random effects model     0.0192 [0.0127; 0.0290]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 15.6365]; tau = 0 [0.0000; 3.9543]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.95    2  0.6222
```

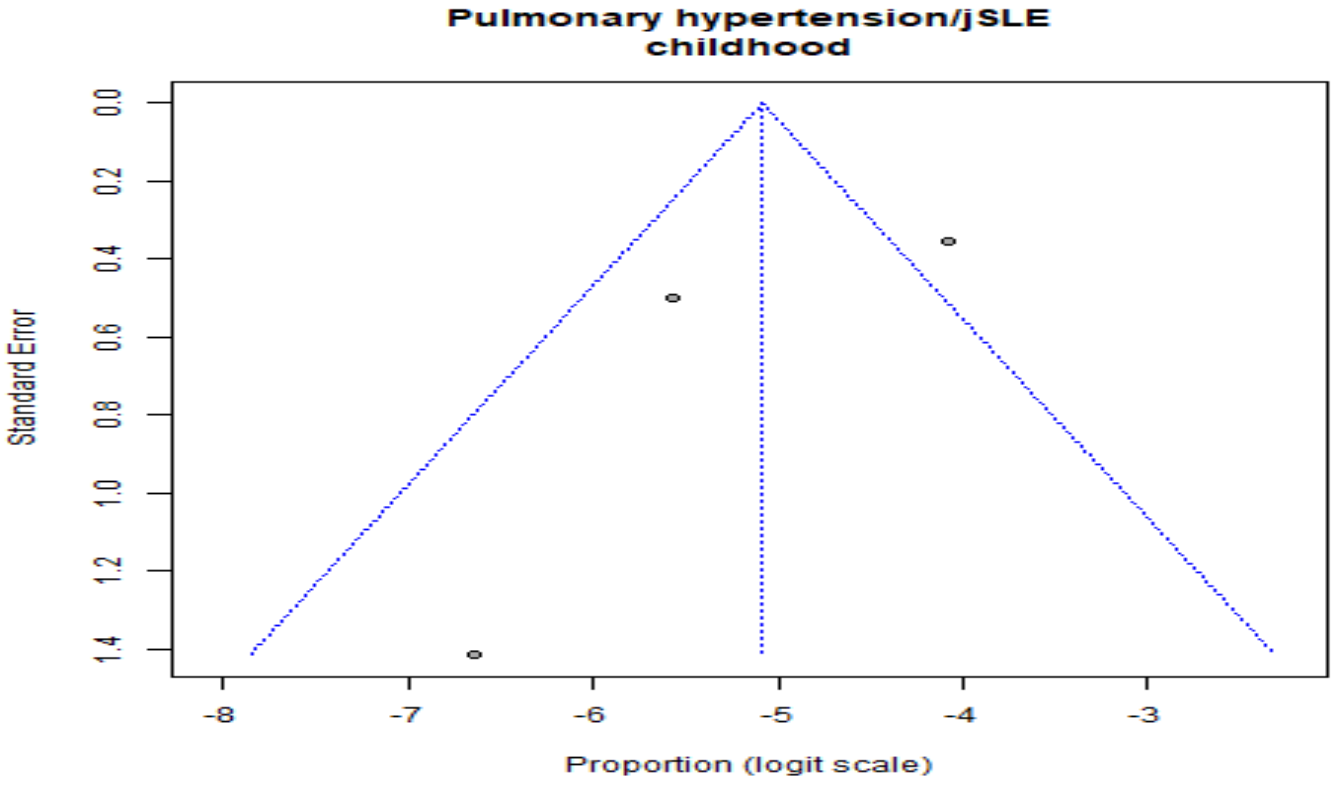
Supplementary Figure S4.49: Forest and Funnel Plots of studies reporting the prevalence of pulmonary hypertension in children with juvenile-onset Systemic Lupus Erythematosus

Pulmonary hypertension

Forest



Funnel



Summary statistics

```
                                        proportion           95%-CI       %W(random)
Holland, MJ. etal, 2018, USA                0.0038 [0.0010; 0.0097]       39.7
Torrente-Segarra, V. etal, 2017, Spain      0.0165 [0.0072; 0.0323]       44.2
ALAhmed, etal , 2020, USA                   0.0000 [0.0000; 0.0095]       16.1

Number of studies: k = 3
Number of observations: o = 1918
Number of events: e = 12

                      proportion           95%-CI
Random effects model      0.0061 [0.0016; 0.0233]

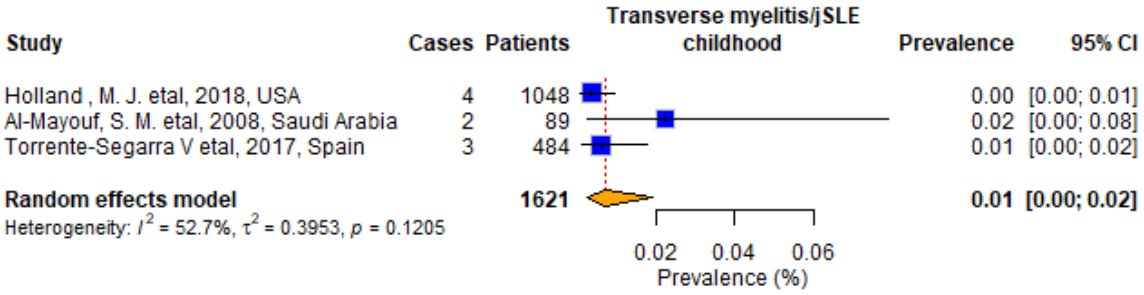
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.9469 [0.0140; 64.4784]; tau = 0.9731 [0.1183; 8.0298]
 I^2 = 74.5% [15.0%; 92.3%]; H = 1.98 [1.08; 3.61]

Test of heterogeneity:
    Q d.f. p-value
 7.83    2  0.0199
```

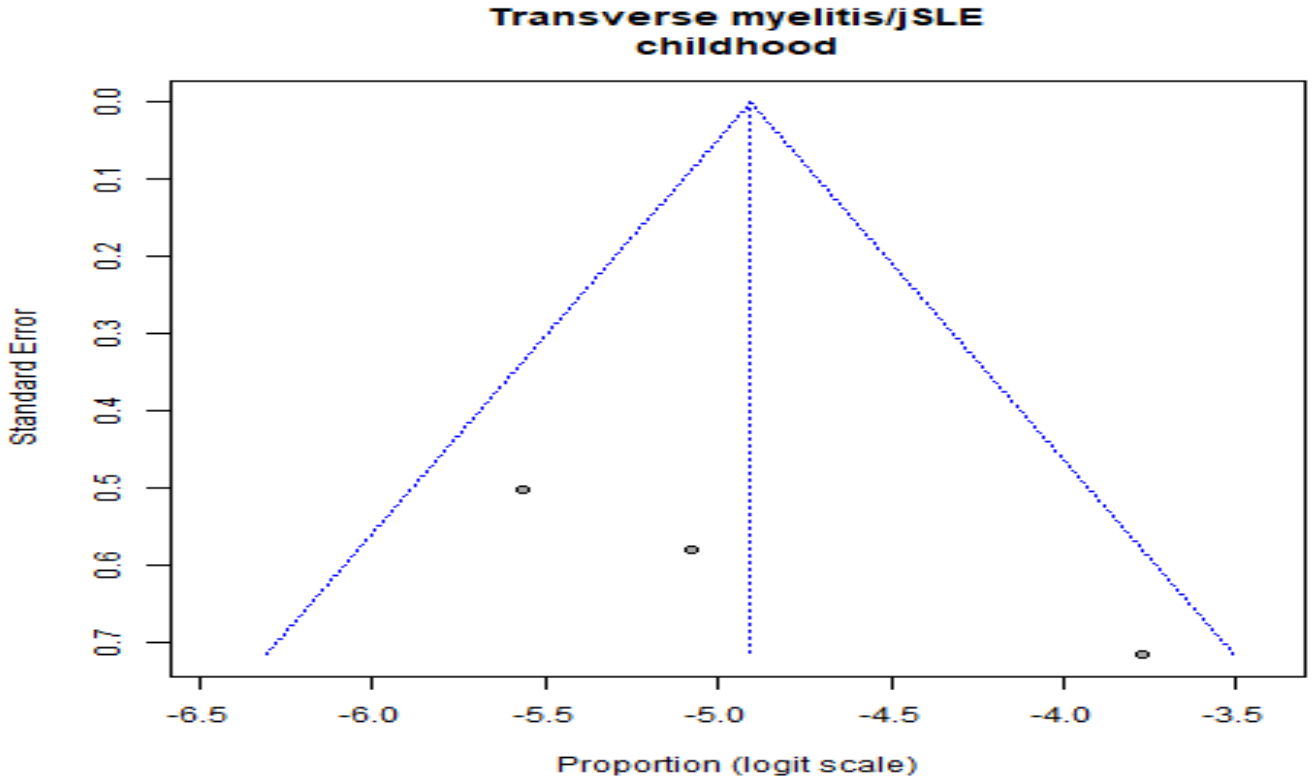
Supplementary Figure S4.50: Forest and Funnel Plots of studies reporting the prevalence of transverse myelitis in children with juvenile-onset Systemic Lupus Erythematosus

Transverse myelitis

Forest



Funnel



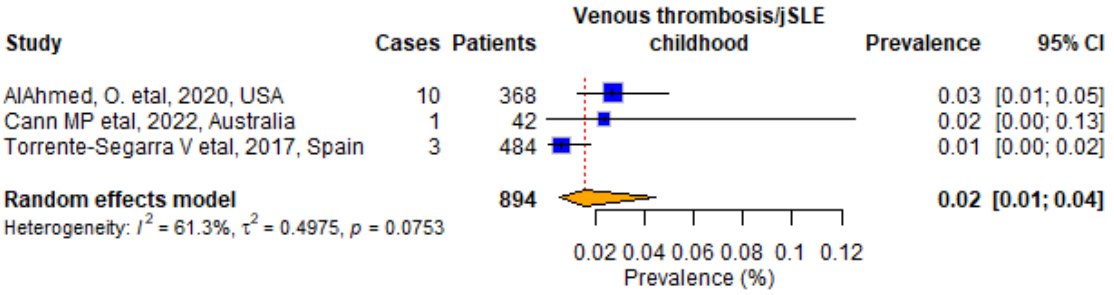
Summary statistics

```
                                        proportion    95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0038    [0.0010; 0.0097]        38.5
Al-Mayouf, S. M. etal, 2008, Saudi Arabia   0.0225    [0.0027; 0.0788]        27.4
Torrente-Segarra V etal, 2017, Spain        0.0062    [0.0013; 0.0180]        34.1
Number of studies: k = 3; Number of observations: o = 1621; Number of events: e = 9
                     proportion           95%-CI
Random effects model     0.0073 [0.0028; 0.0193]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3953 [0.0000; 33.4831]; tau = 0.6287 [0.0000; 5.7865]
 I^2 = 52.7% [0.0%; 86.4%]; H = 1.45 [1.00; 2.71]
Test of heterogeneity:     Q d.f. p-value
                        4.23    2  0.1205
```

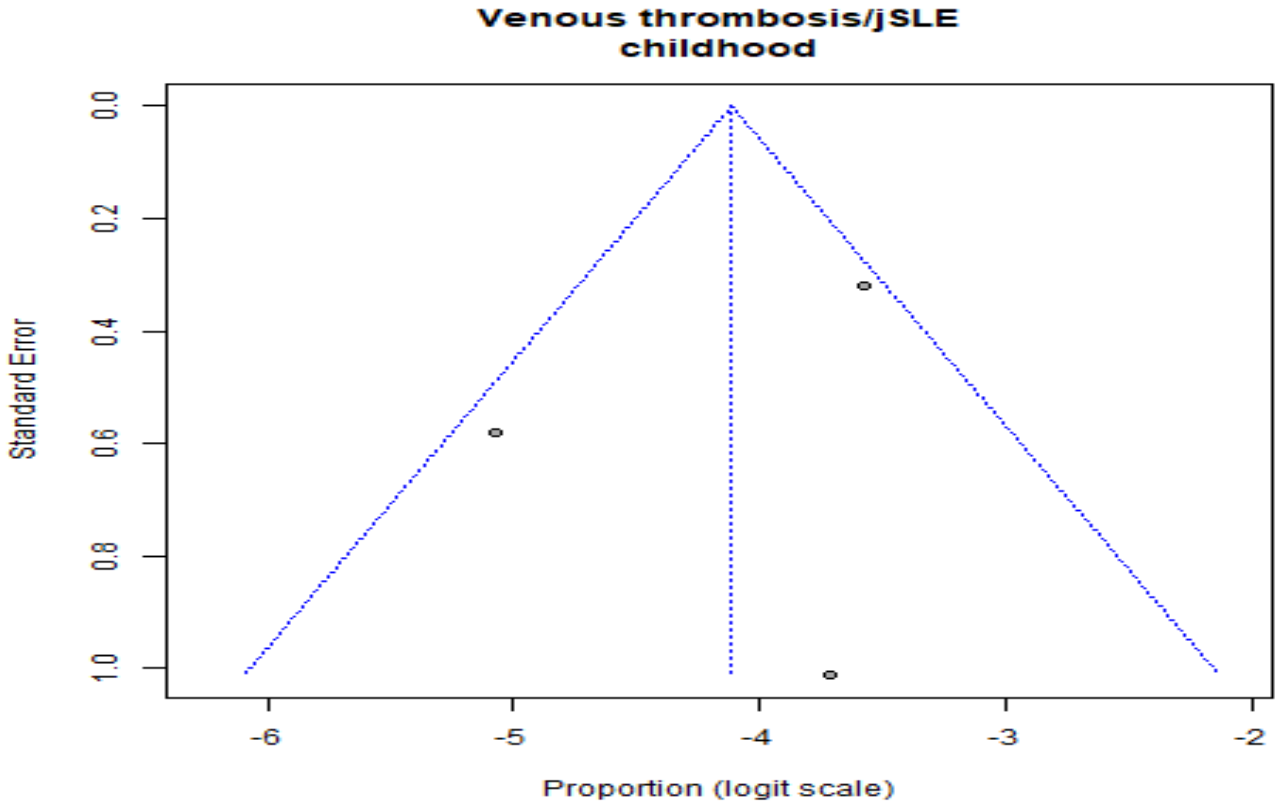
Supplementary Figure S4.51: Forest and Funnel Plots of studies reporting the prevalence of venous thrombosis in children with juvenile-onset Systemic Lupus Erythematosus

Venous thrombosis

Forest



Funnel



Summary statistics

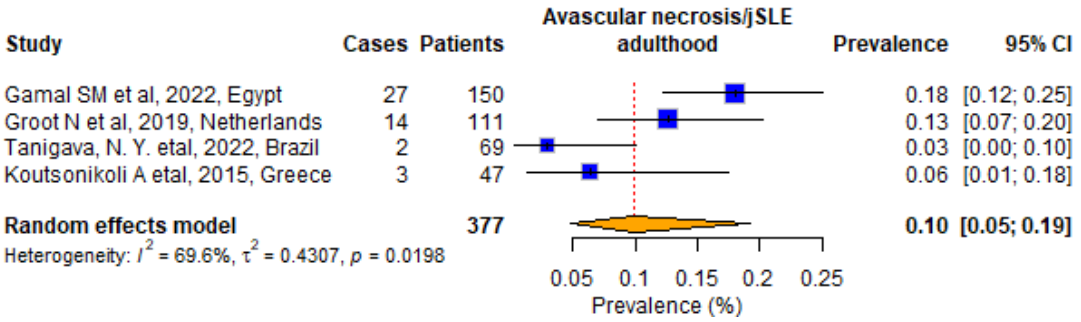
```
                                    proportion         95%-CI  %W(random)
AlAhmed, O. etal, 2020, USA             0.0272 [0.0131; 0.0494]       47.3
Cann MP etal, 2022, Australia           0.0238 [0.0006; 0.1257]       18.6
Torrente-Segarra V etal, 2017, Spain    0.0062 [0.0013; 0.0180]       34.1
Number of studies: k = 3; Number of observations: o = 894; Number of events: e = 14
                      proportion           95%-CI
Random effects model      0.0161 [0.0057; 0.0444]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4975 [0.0000; 26.7942]; tau = 0.7053 [0.0000; 5.1763]
 I^2 = 61.3% [0.0%; 89.0%]; H = 1.61 [1.00; 3.01]
Test of heterogeneity:     Q d.f. p-value
                        5.17    2  0.0753
```

Adulthood/jSLE

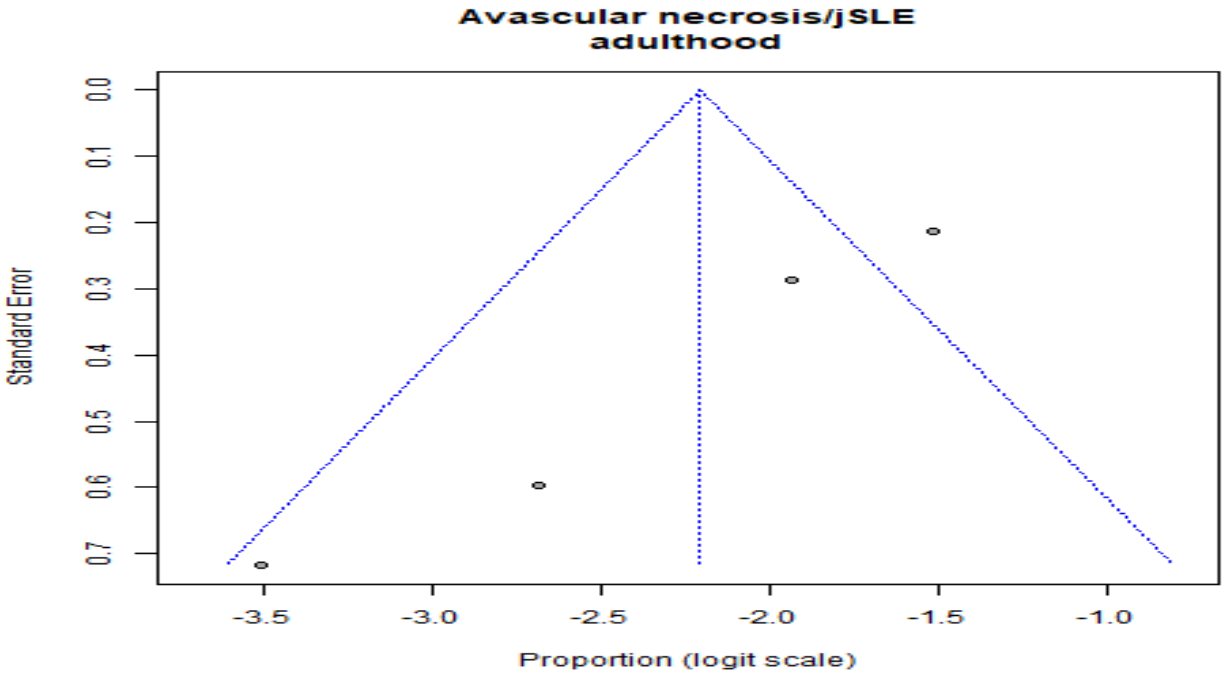
Supplementary Figure S4.52: Forest and Funnel Plots of studies reporting the prevalence of avascular necrosis in adults with juvenile-onset Systemic Lupus Erythematosus

Avascular necrosis

Forest



Funnel



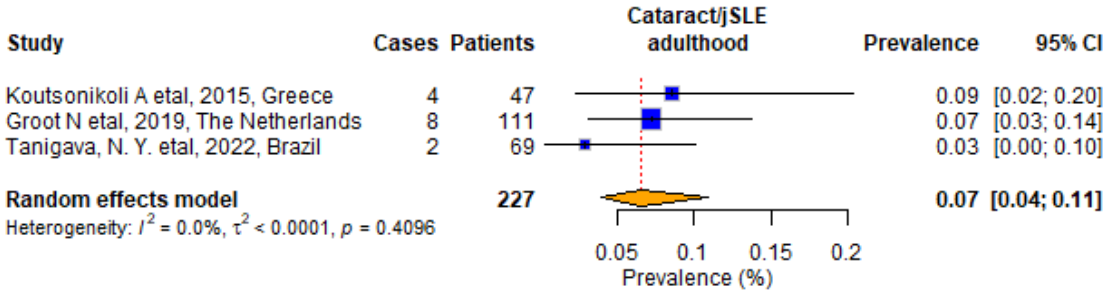
Summary statistics

```
                                  proportion   95%-CI          %W(random)
Gamal SM et al, 2022, Egypt           0.1800   [0.1221; 0.2510]       32.9
Groot N et al, 2019, Netherlands      0.1261   [0.0707; 0.2026]       30.6
Tanigava, N. Y. etal, 2022, Brazil    0.0290   [0.0035; 0.1008]       16.6
Koutsonikoli A etal, 2015, Greece     0.0638   [0.0134; 0.1754]       19.9
Number of studies: k = 4; Number of observations: o = 377; Number of events: e = 46
                     proportion           95%-CI
Random effects model     0.0990 [0.0482; 0.1928]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4307 [0.0084; 10.4154]; tau = 0.6563 [0.0918; 3.2273]
 I^2 = 69.6% [12.4%; 89.4%]; H = 1.81 [1.07; 3.08]
Test of heterogeneity:     Q d.f. p-value
                        9.86    3  0.0198
```

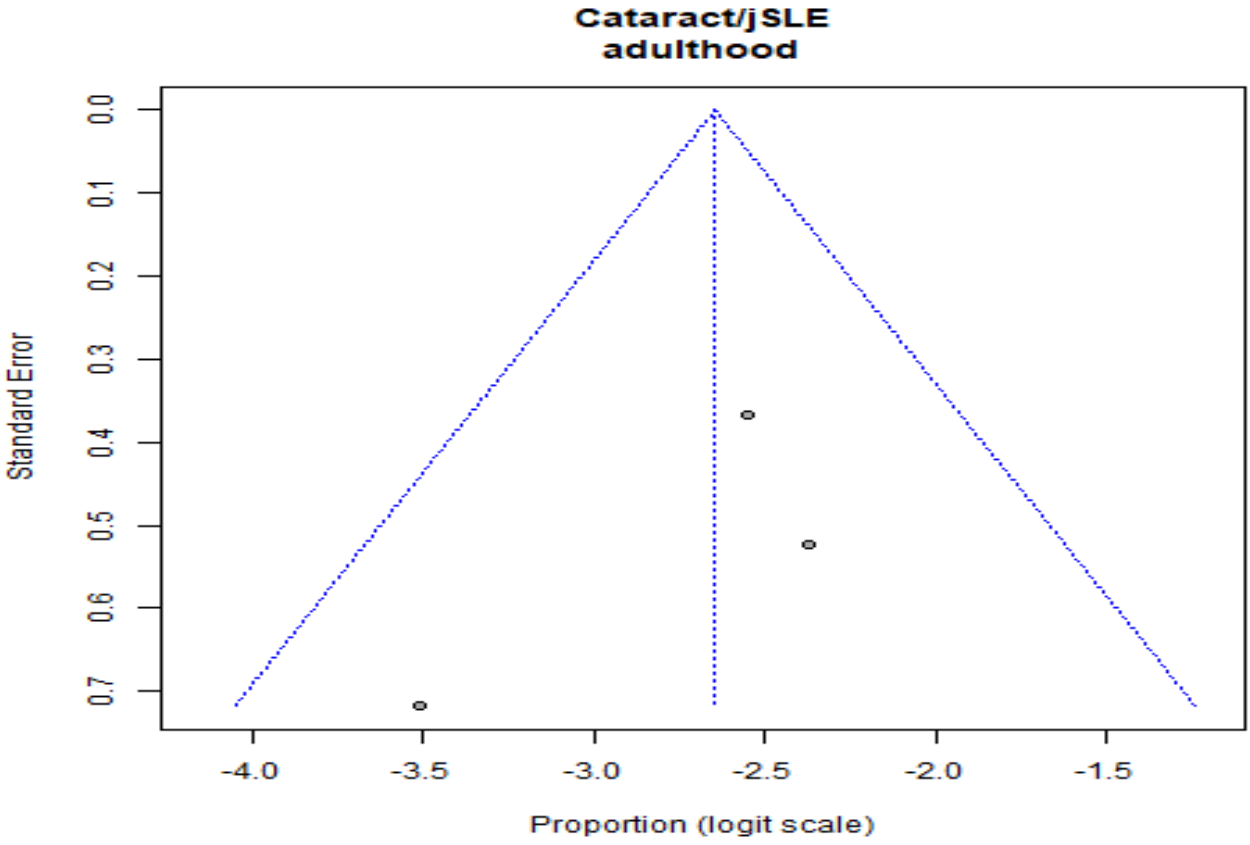
Supplementary Figure S4.53: Forest and Funnel Plots of studies reporting the prevalence of cataract in adults with juvenile-onset Systemic Lupus Erythematosus

Cataract

Forest



Funnel



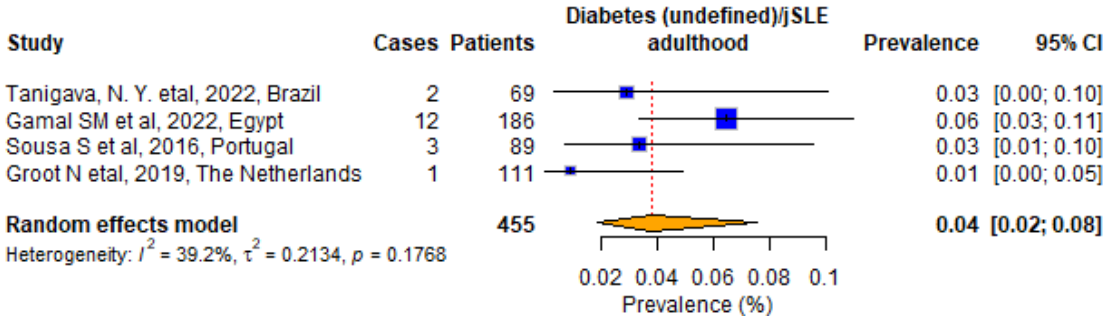
Summary statistics

```
                                proportion  95%-CI          %W(random)
Koutsonikoli A etal, 2015, Greece     0.0851  [0.0237; 0.2038]       28.1
Groot N etal, 2019, The Netherlands   0.0721  [0.0316; 0.1371]       57.0
Tanigava, N. Y. etal, 2022, Brazil    0.0290  [0.0035; 0.1008]       14.9
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 14
                    proportion           95%-CI
Random effects model     0.0662 [0.0395; 0.1087]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 14.3203]; tau = 0.0004 [0.0000; 3.7842]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        1.78    2  0.4096
```

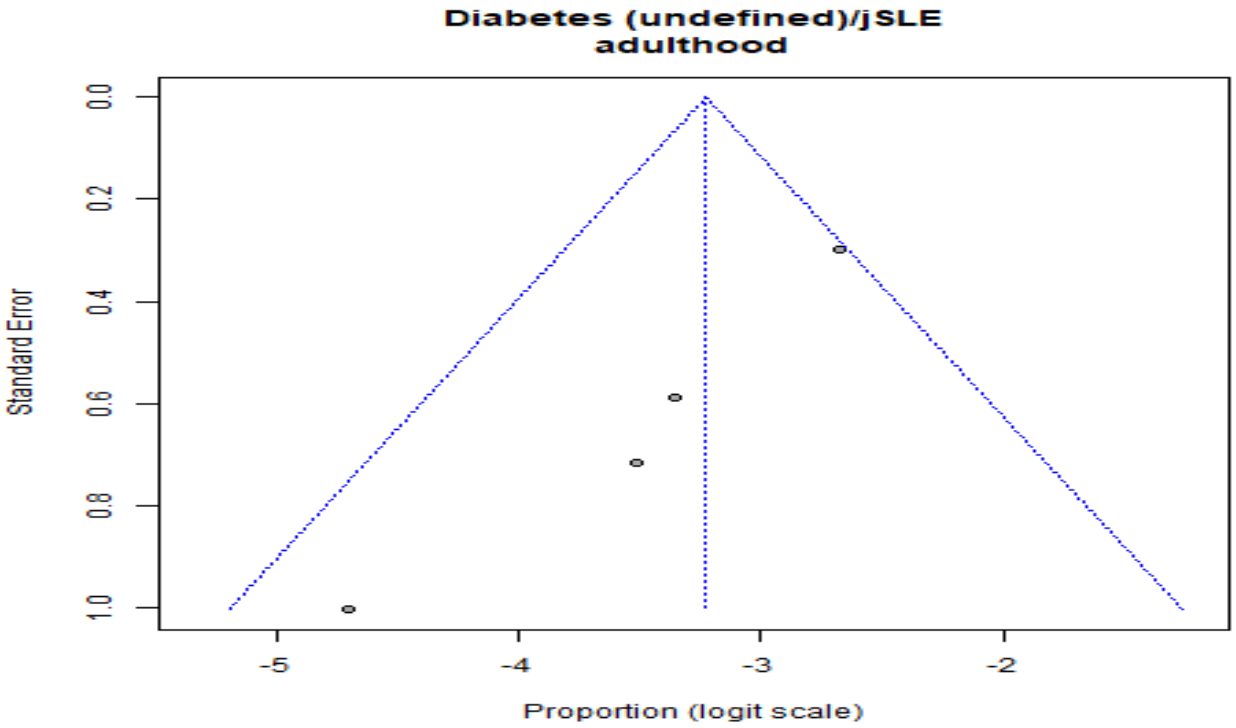
Supplementary Figure S4.54: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in adults with juvenile-onset Systemic Lupus Erythematosus

Diabetes (undefined)

Forest



Funnel



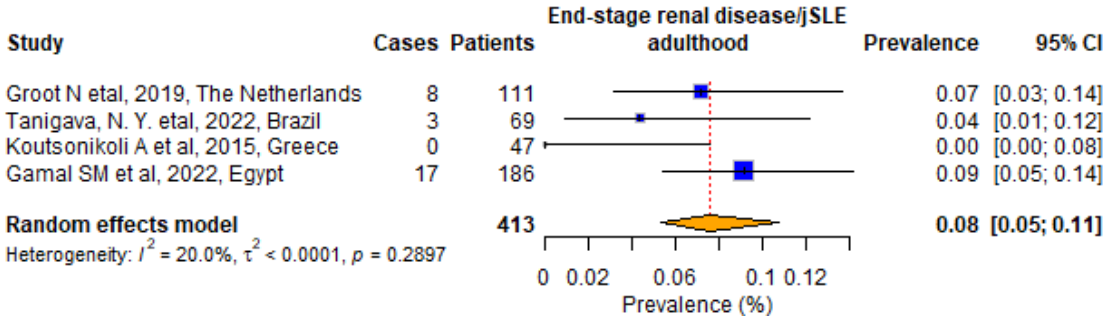
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0290  [0.0035; 0.1008]       18.8
Gamal SM et al, 2022, Egypt            0.0645  [0.0338; 0.1100]       45.4
Sousa S et al, 2016, Portugal          0.0337  [0.0070; 0.0954]       24.6
Groot N etal, 2019, The Netherlands    0.0090  [0.0002; 0.0492]       11.2
Number of studies: k = 4; Number of observations: o = 455; Number of events: e = 18
                     proportion           95%-CI
Random effects model     0.0382 [0.0188; 0.0758]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2134 [0.0000; 9.2209]; tau = 0.4620 [0.0000; 3.0366]
 I^2 = 39.2% [0.0%; 79.3%]; H = 1.28 [1.00; 2.20]
Test of heterogeneity:     Q d.f. p-value
                        4.93    3  0.1768
```

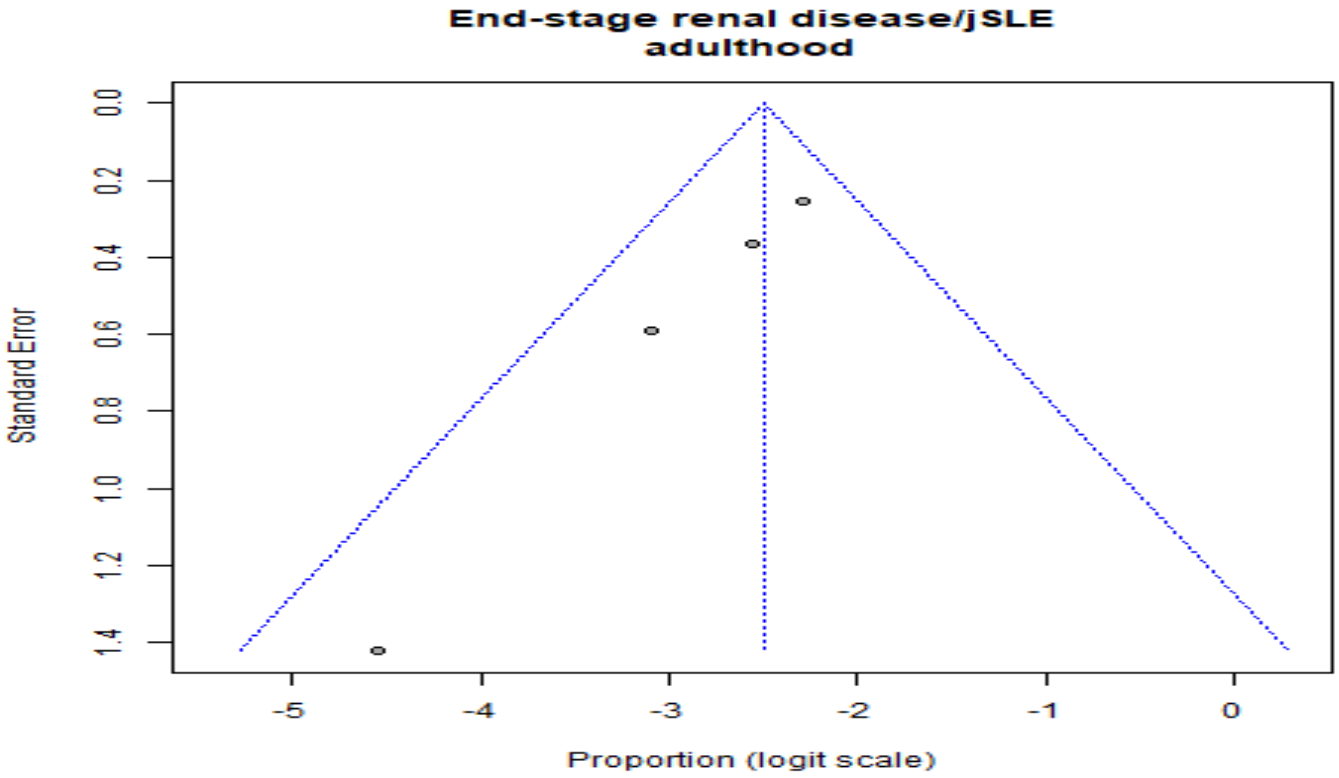
Supplementary Figure S4.55: Forest and Funnel Plots of studies reporting the prevalence of end-stage renal disease in adults with juvenile-onset Systemic Lupus Erythematosus

End-stage renal disease

Forest



Funnel



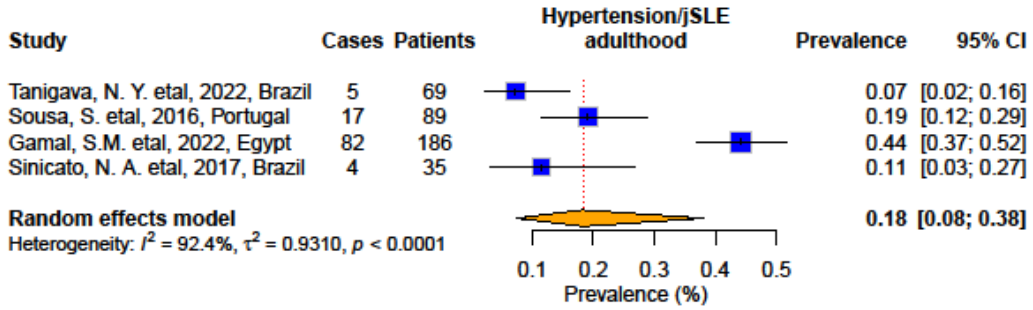
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]       28.3
Tanigava, N. Y. etal, 2022, Brazil      0.0435  [0.0091; 0.1218]       10.9
Koutsonikoli A et al, 2015, Greece      0.0000  [0.0000; 0.0755]        1.9
Gamal SM et al, 2022, Egypt             0.0914  [0.0541; 0.1423]       58.9
Number of studies: k = 4; Number of observations: o = 413; Number of events: e = 28
                     proportion           95%-CI
Random effects model     0.0759 [0.0531; 0.1075]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 12.7842]; tau = 0.0006 [0.0000; 3.5755]
 I^2 = 20.0% [0.0%; 87.8%]; H = 1.12 [1.00; 2.86]
Test of heterogeneity:     Q d.f. p-value
                        3.75    3  0.2897
```

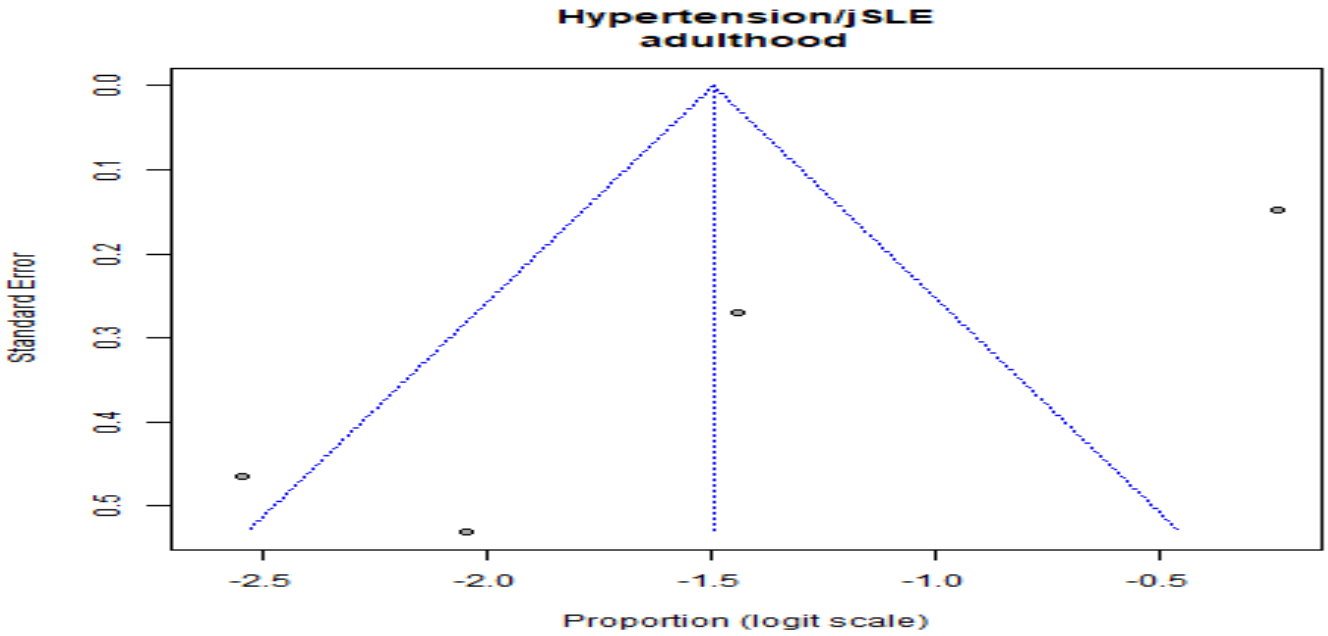
Supplementary Figure S4.56: Forest and Funnel Plots of studies reporting the prevalence of hypertension in adults with juvenile-onset Systemic Lupus Erythematosus

Hypertension

Forest



Funnel



Summary statistics

```
                                  proportion   95% CI                 %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0725   [0.0239; 0.1611]        23.3
Sousa, S. etal, 2016, Portugal         0.1910   [0.1154; 0.2881]        26.6
Gamal, S.M. etal, 2022, Egypt          0.4409   [0.3683; 0.5154]        28.0
Sinicato, N. A. etal, 2017, Brazil     0.1143   [0.0320; 0.2674]        22.0

Number of studies: k = 4
Number of observations: o = 379
Number of events: e = 108

                     proportion 95% CI
Random effects model     0.1830 [0.0752; 0.3816]

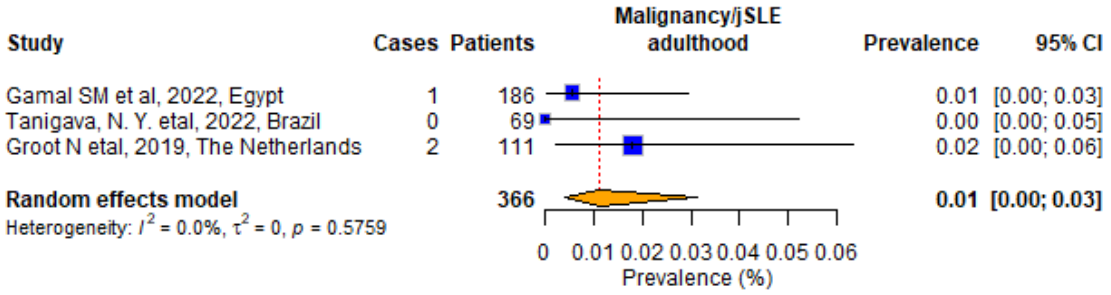
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.9310 [0.2222; 13.6986]; tau = 0.9649 [0.4714; 3.7012]
 I^2 = 92.4% [83.8%; 96.5%]; H = 3.63 [2.49; 5.31]

Test of heterogeneity:
     Q d.f.  p-value
 39.62    3 < 0.0001
```

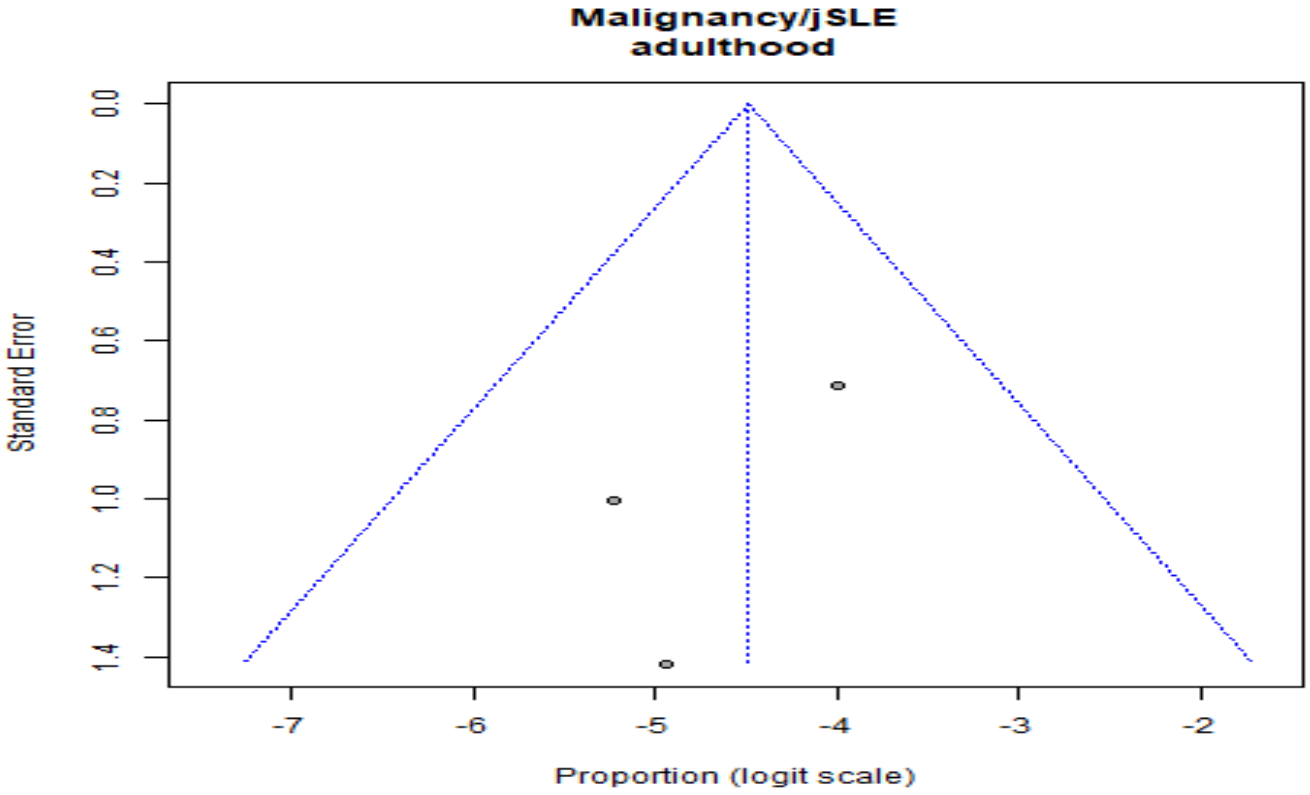
Supplementary Figure S4.57: Forest and Funnel Plots of studies reporting the prevalence of malignancy in adults with juvenile-onset Systemic Lupus Erythematosus

Malignancy

Forest



Funnel



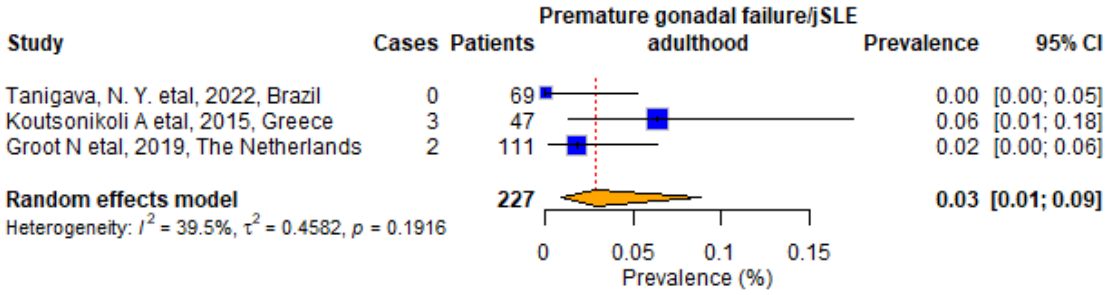
Summary statistics

```
                                    proportion  95%-CI          %W(random)
Gamal SM et al, 2022, Egypt             0.0054  [0.0001; 0.0296]        28.8
Tanigava, N. Y. etal, 2022, Brazil      0.0000  [0.0000; 0.0521]        14.4
Groot N etal, 2019, The Netherlands     0.0180  [0.0022; 0.0636]        56.8
Number of studies: k = 3; Number of observations: o = 366; Number of events: e = 3
                     proportion           95%-CI
Random effects model     0.0112 [0.0039; 0.0314]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 15.3931]; tau = 0 [0.0000; 3.9234]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        1.10    2  0.5759
```

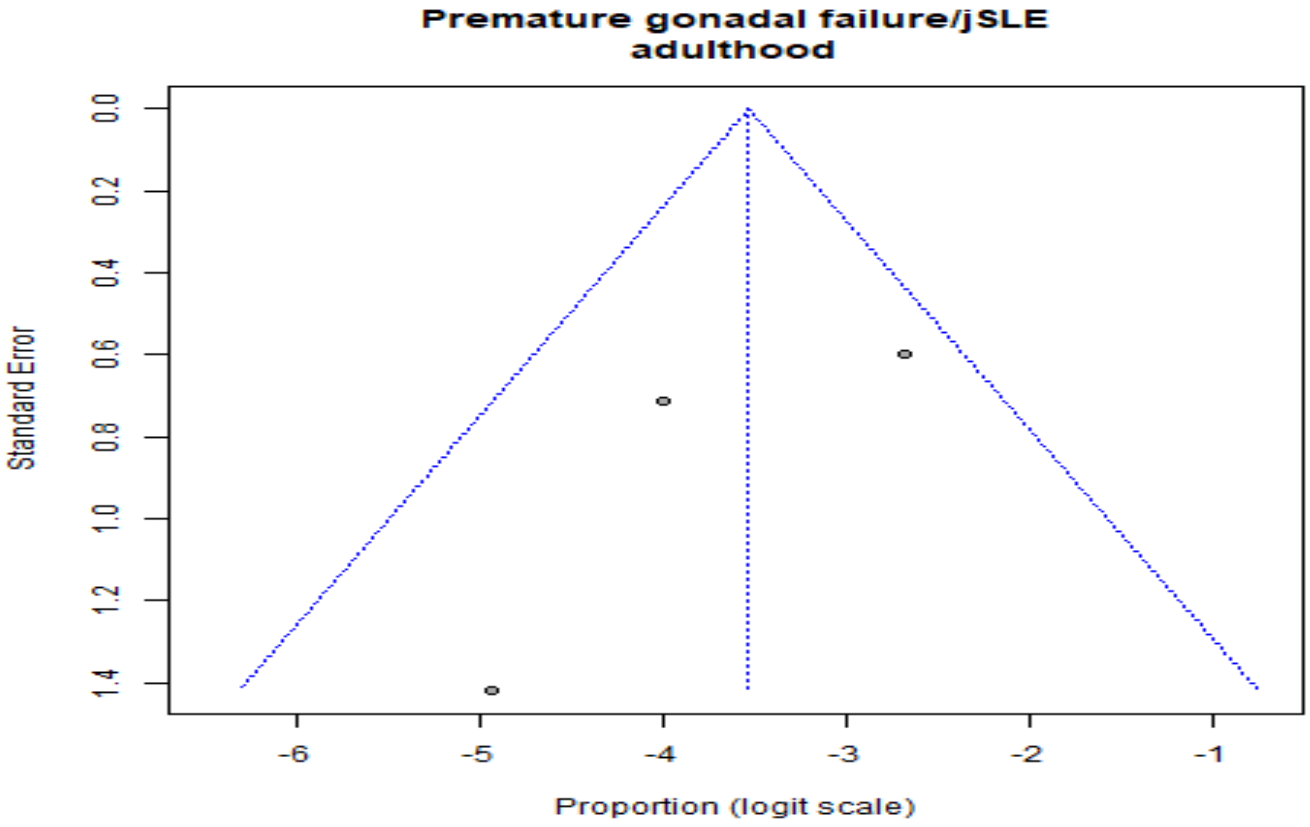
Supplementary Figure S4.58: Forest and Funnel Plots of studies reporting the prevalence of premature gonadal failure in adults with juvenile-onset Systemic Lupus Erythematosus

Premature gonadal failure

Forest



Funnel



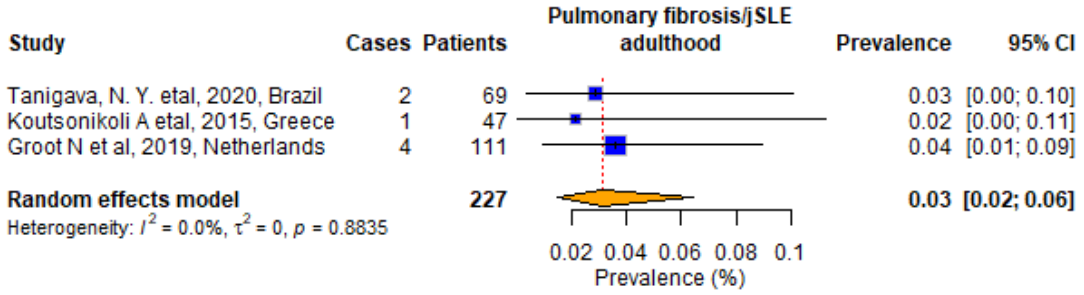
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0000  [0.0000; 0.0521]       15.2
Koutsonikoli A etal, 2015, Greece      0.0638  [0.0134; 0.1754]       46.1
Groot N etal, 2019, The Netherlands    0.0180  [0.0022; 0.0636]       38.8
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 5
                     proportion           95%-CI
Random effects model     0.0283 [0.0087; 0.0882]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4582 [0.0000; 49.3226]; tau = 0.6769 [0.0000; 7.0230]
 I^2 = 39.5% [0.0%; 81.3%]; H = 1.29 [1.00; 2.31]
Test of heterogeneity:     Q d.f. p-value
                        3.30    2  0.1916
```

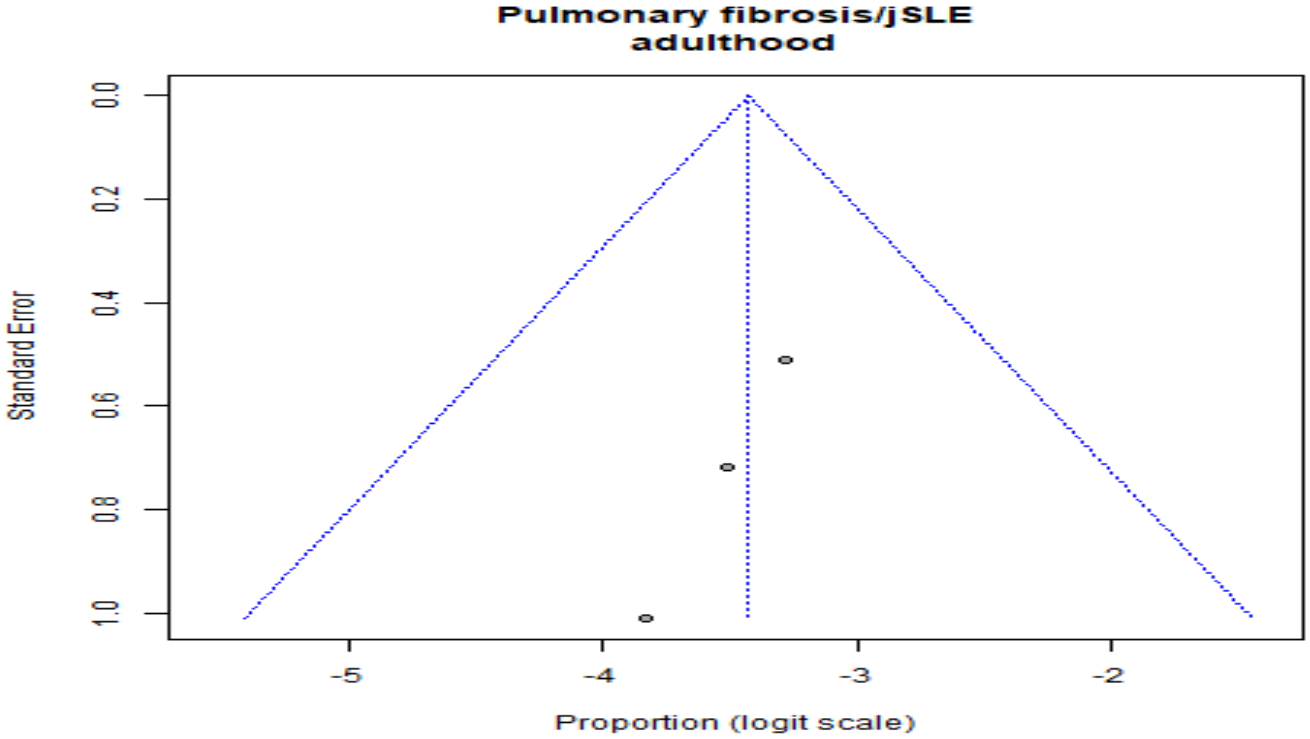
Supplementary Figure S4.59: Forest and Funnel Plots of studies reporting the prevalence of pulmonary fibrosis in adults with juvenile-onset Systemic Lupus Erythematosus

Pulmonary fibrosis

Forest



Funnel



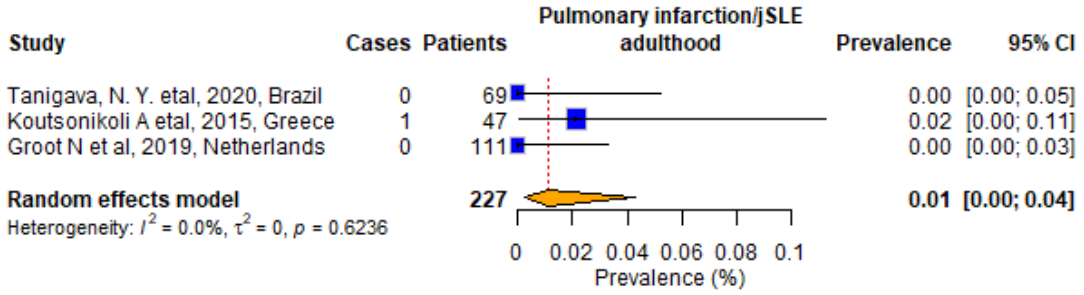
Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tanigava, N. Y. etal, 2020, Brazil     0.0290   [0.0035; 0.1008]       28.7
Koutsonikoli A etal, 2015, Greece      0.0213   [0.0005; 0.1129]       14.4
Groot N et al, 2019, Netherlands       0.0360   [0.0099; 0.0897]       56.9
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 7
                     proportion           95%-CI
Random effects model     0.0314 [0.0150; 0.0644]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 2.2629]; tau = 0 [0.0000; 1.5043]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:    Q d.f. p-value
                       0.25    2  0.8835
```

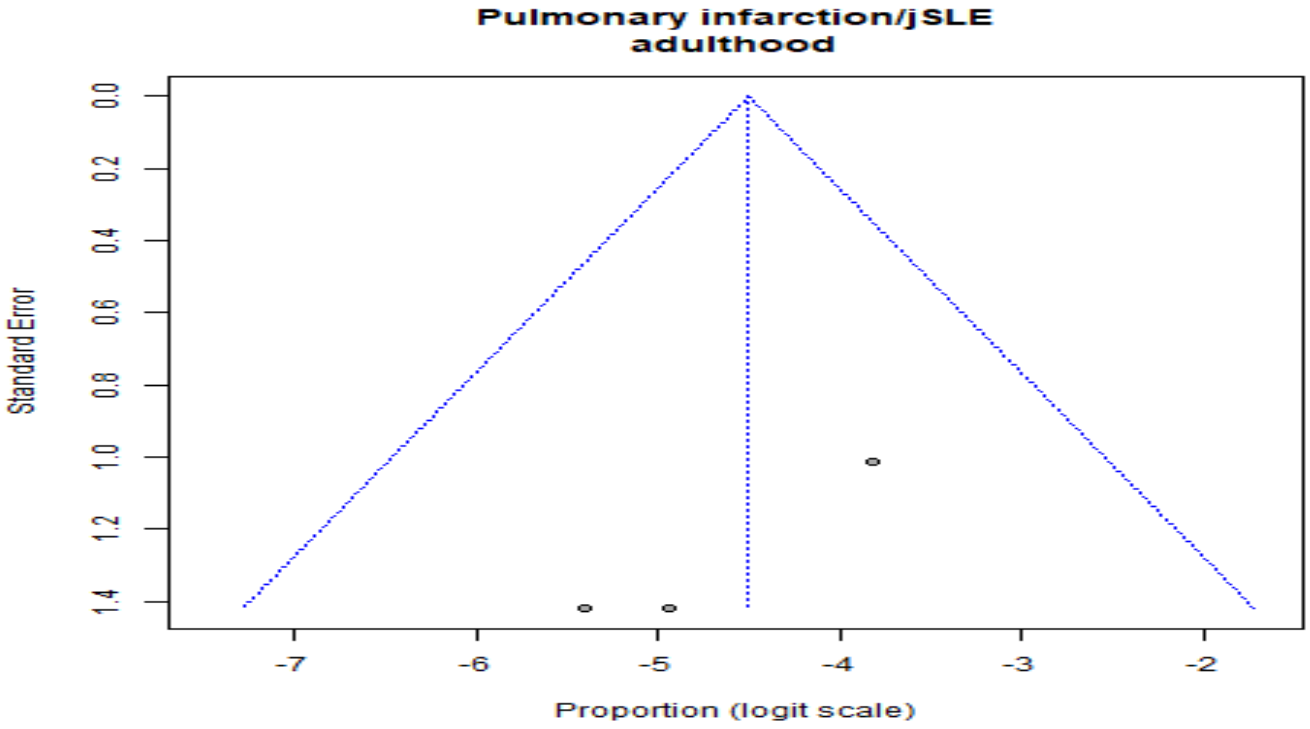
Supplementary Figure S4.60: Forest and Funnel Plots of studies reporting the prevalence of pulmonary infarction in adults with juvenile-onset Systemic Lupus Erythematosus

Pulmonary infarction

Forest



Funnel



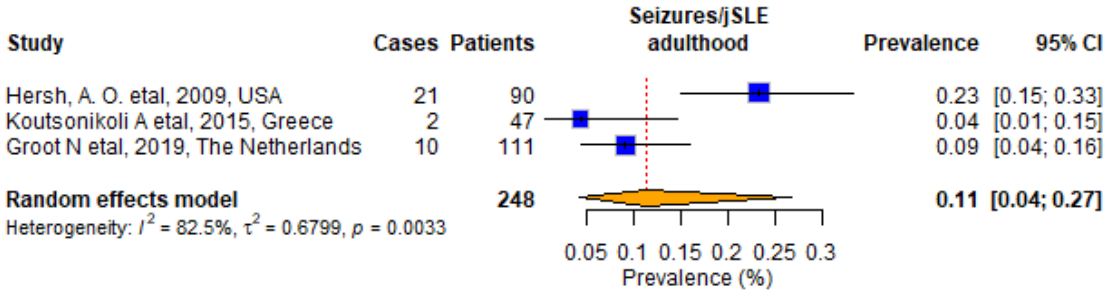
Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tanigava, N. Y. etal, 2020, Brazil     0.0000   [0.0000; 0.0521]       25.2
Koutsonikoli A etal, 2015, Greece      0.0213   [0.0005; 0.1129]       49.6
Groot N et al, 2019, Netherlands       0.0000   [0.0000; 0.0327]       25.2
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 1
                     proportion           95%-CI
Random effects model     0.0109 [0.0027; 0.0427]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 24.5185]; tau = 0 [0.0000; 4.9516]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.94    2  0.6236
```

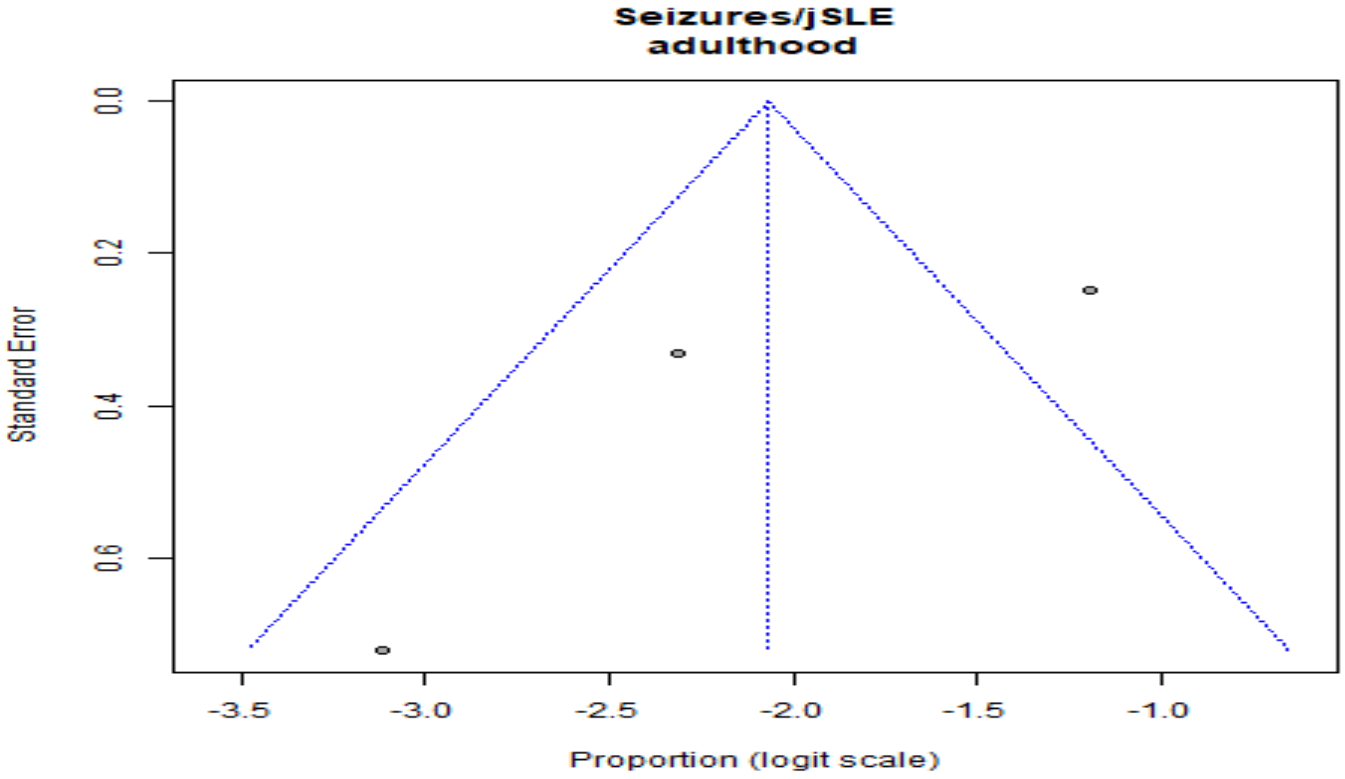
Supplementary Figure S4.61: Forest and Funnel Plots of studies reporting the prevalence of seizures in adults with juvenile-onset Systemic Lupus Erythematosus

Seizures

Forest



Funnel



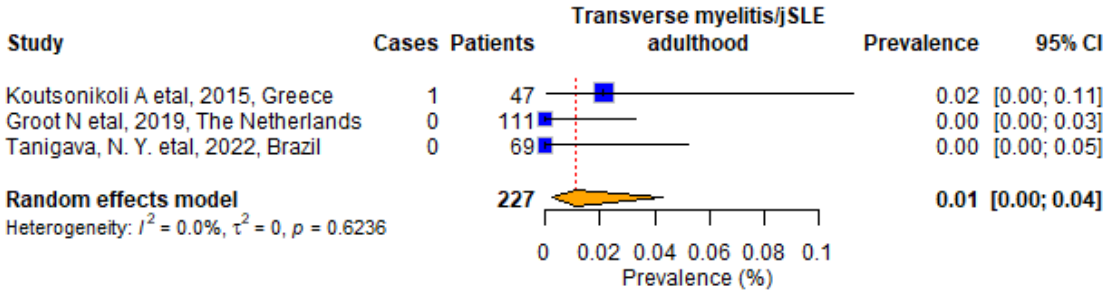
Summary statistics

```
                                  proportion  95%-CI          %W(random)
Hersh, A. O. etal, 2009, USA          0.2333  [0.1506; 0.3343]       39.1
Koutsonikoli A etal, 2015, Greece     0.0426  [0.0052; 0.1454]       24.1
Groot N etal, 2019, The Netherlands   0.0901  [0.0441; 0.1594]       36.7
Number of studies: k = 3; Number of observations: o = 248; Number of events: e = 33
                     proportion           95%-CI
Random effects model     0.1124 [0.0422; 0.2668]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.6799 [0.0646; 36.6266]; tau = 0.8246 [0.2542; 6.0520]
 I^2 = 82.5% [46.4%; 94.3%]; H = 2.39 [1.37; 4.19]
Test of heterogeneity:      Q d.f. p-value
                        11.43    2  0.0033
```

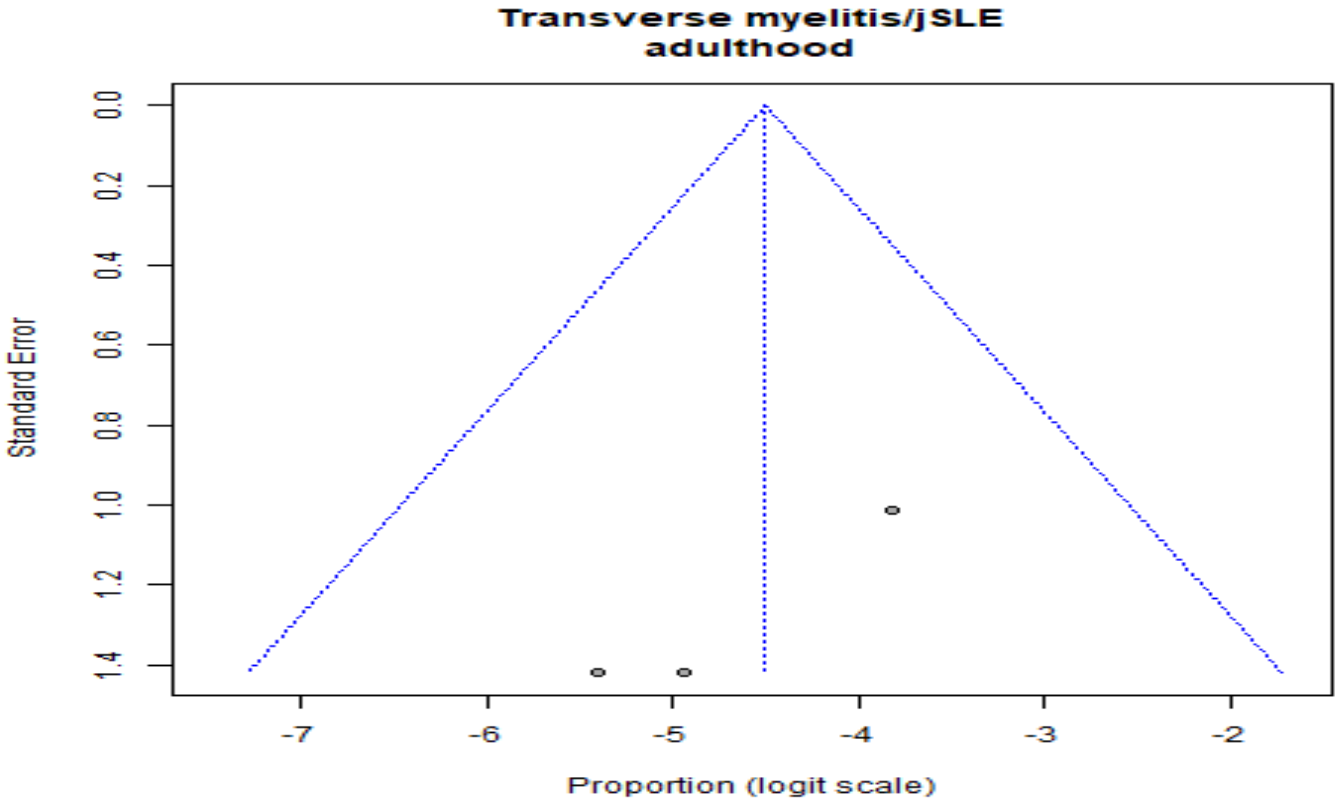
Supplementary Figure S4.62: Forest and Funnel Plots of studies reporting the prevalence of transverse myelitis in adults with juvenile-onset Systemic Lupus Erythematosus

Transverse myelitis

Forest



Funnel



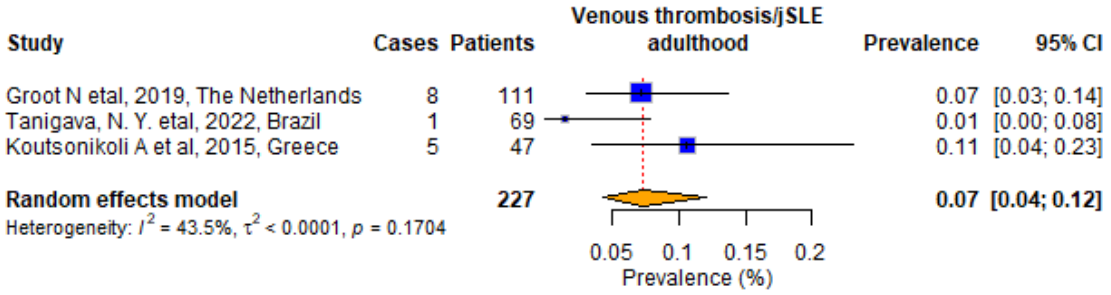
Summary statistics

```
                                    proportion  95%-CI          %W(random)
Koutsonikoli A etal, 2015, Greece       0.0213  [0.0005; 0.1129]       49.6
Groot N etal, 2019, The Netherlands     0.0000  [0.0000; 0.0327]       25.2
Tanigava, N. Y. etal, 2022, Brazil      0.0000  [0.0000; 0.0521]       25.2
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 1
                     proportion           95%-CI
Random effects model     0.0109 [0.0027; 0.0427]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 24.5185]; tau = 0 [0.0000; 4.9516]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.94    2  0.6236
```

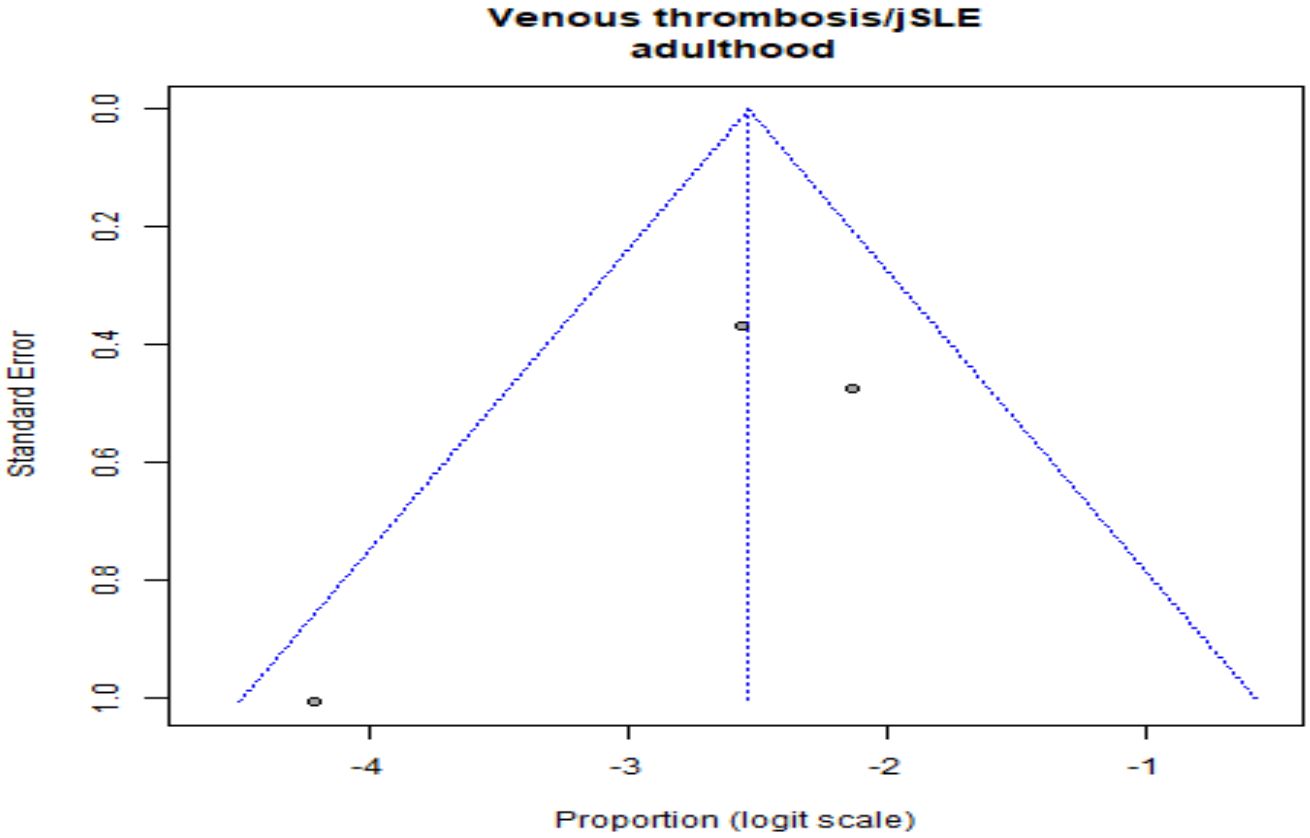
Supplementary Figure S4.63: Forest and Funnel Plots of studies reporting the prevalence of venous thrombosis in adults with juvenile-onset Systemic Lupus Erythematosus

Venous thrombosis

Forest



Funnel



Summary statistics

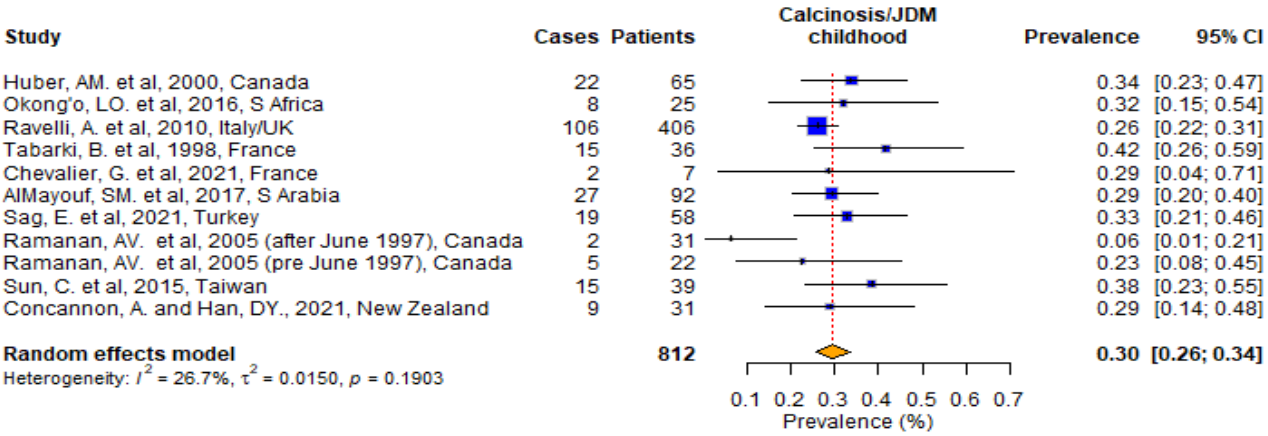
```
                                    proportion  95%-CI          %W(random)
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]        57.6
Tanigava, N. Y. etal, 2022, Brazil      0.0145  [0.0004; 0.0781]         7.7
Koutsonikoli A et al, 2015, Greece      0.1064  [0.0355; 0.2310]        34.7
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 14
                    proportion           95%-CI
Random effects model     0.0735 [0.0439; 0.1204]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 47.4982]; tau = 0.0006 [0.0000; 6.8919]
 I^2 = 43.5% [0.0%; 83.1%]; H = 1.33 [1.00; 2.43]
Test of heterogeneity:     Q d.f. p-value
                        3.54    2  0.1704
```

Childhood/JDM

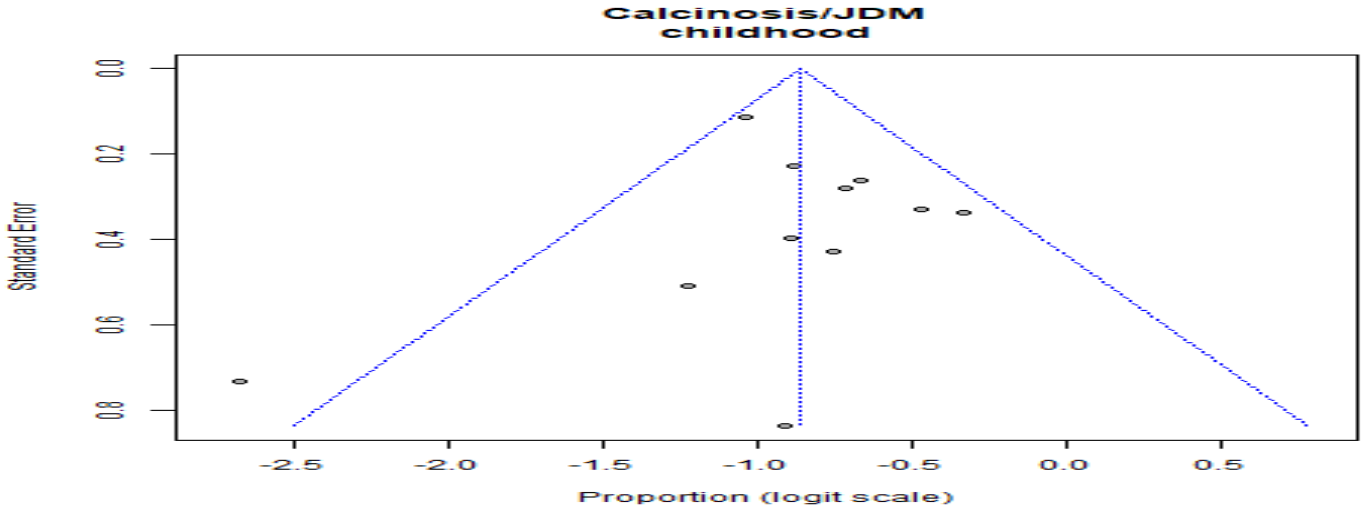
Supplementary Figure S4.64: Forest and Funnel Plots of studies reporting the prevalence of calcinosis in children with Juvenile Dermatomyositis

Calcinosis

Forest



Funnel



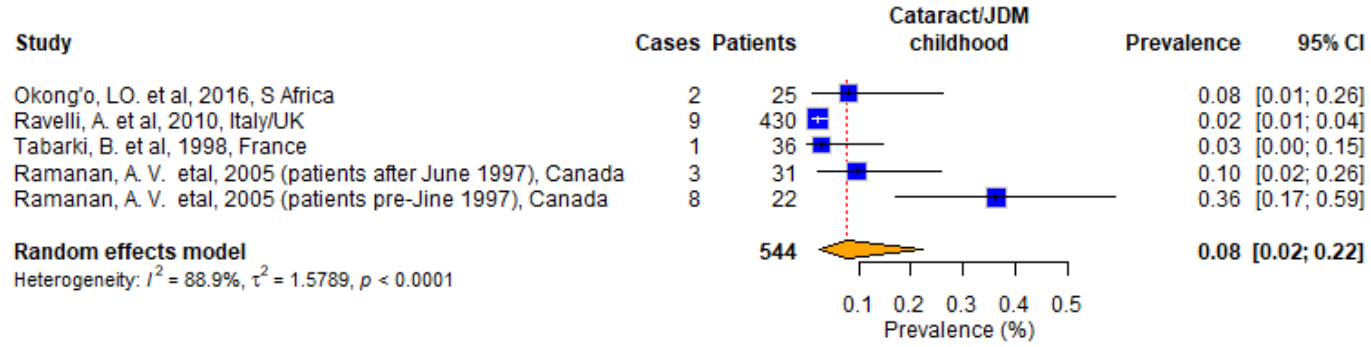
Summary statistics

```
                                                  proportion  95%-CI          %W(random)
Huber, AM. et al, 2000, Canada                        0.3385  [0.2257; 0.4665]       11.2
Okong'o, LO. et al, 2016, S Africa                    0.3200  [0.1495; 0.5350]        4.7
Ravelli, A. et al, 2010, Italy/UK                     0.2611  [0.2190; 0.3067]       33.6
Tabarki, B. et al, 1998, France                       0.4167  [0.2551; 0.5924]        7.2
Chevalier, G. et al, 2021, France                     0.2857  [0.0367; 0.7096]        1.3
AlMayouf, SM. et al, 2017, S Arabia                   0.2935  [0.2031; 0.3976]       13.9
Sag, E. et al, 2021, Turkey                           0.3276  [0.2101; 0.4634]       10.0
Ramanan, AV.  et al, 2005 (after June 1997), Canada   0.0645  [0.0079; 0.2142]        1.7
Ramanan, AV.  et al, 2005 (pre June 1997), Canada     0.2273  [0.0782; 0.4537]        3.4
Sun, C. et al, 2015, Taiwan                           0.3846  [0.2336; 0.5538]        7.6
Concannon, A. and Han, DY., 2021, New Zealand         0.2903  [0.1422; 0.4804]        5.4
Number of studies: k = 11; Number of observations: o = 812; Number of events: e = 230
                     proportion           95%-CI
Random effects model     0.2970 [0.2591; 0.3381]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0150 [0.0000; 0.7653]; tau = 0.1226 [0.0000; 0.8748]
 I^2 = 26.7% [0.0%; 63.7%]; H = 1.17 [1.00; 1.66]
Test of heterogeneity:       Q d.f. p-value
                         13.63   10  0.1903
```

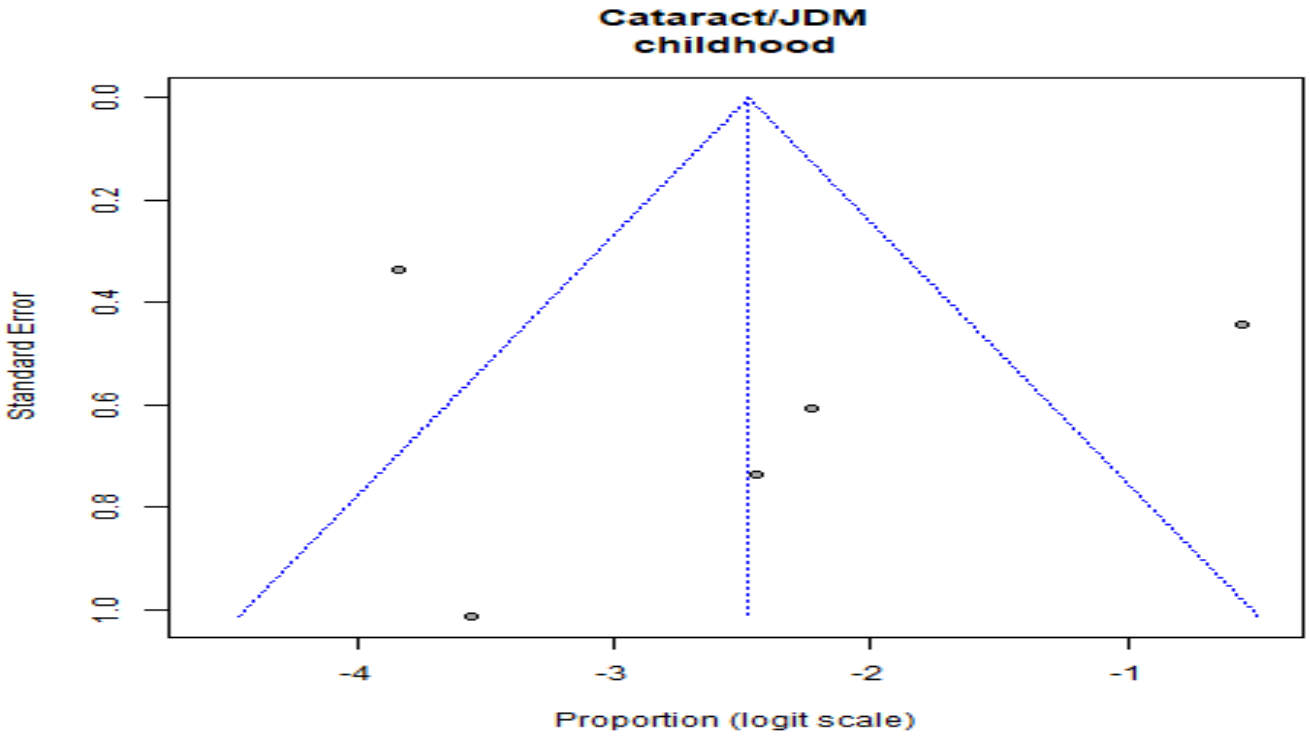
Supplementary Figure S4.65: Forest and Funnel Plots of studies reporting the prevalence of cataract in children with Juvenile Dermatomyositis

Cataract

Forest



Funnel



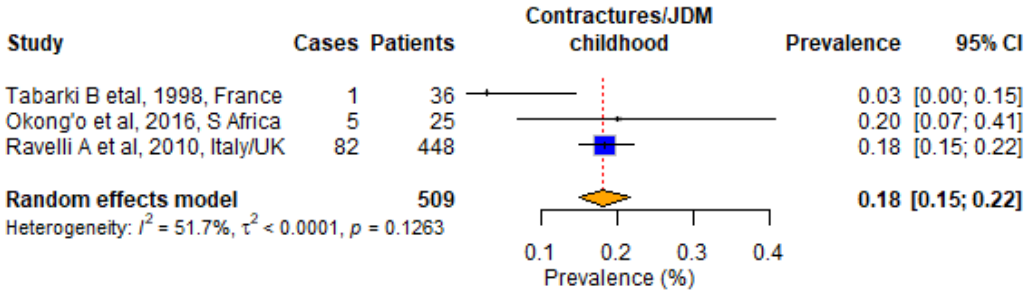
Summary statistics

```
                                                          proportion        95%-CI          %W(random)
Okong'o, LO. et al, 2016, S Africa                            0.0800        [0.0098; 0.2603]        18.7
Ravelli, A. et al, 2010, Italy/UK                             0.0209        [0.0096; 0.0394]        23.4
Tabarki, B. et al, 1998, France                               0.0278        [0.0007; 0.1453]        15.2
Ramanan, A. V.  etal, 2005 (patients after June 1997), Canada 0.0968        [0.0204; 0.2575]        20.4
Ramanan, A. V.  etal, 2005 (patients pre-Jine 1997), Canada   0.3636        [0.1720; 0.5934]        22.3
Number of studies: k = 5; Number of observations: o = 544; Number of events: e = 23
                     proportion          95%-CI
Random effects model     0.0775 [0.0239; 0.2239]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.5789 [0.3765; 13.6521]; tau = 1.2565 [0.6136; 3.6949]
 I^2 = 88.9% [76.9%; 94.7%]; H = 3.01 [2.08; 4.35]
Test of heterogeneity:      Q d.f.  p-value
                        36.16    4 < 0.0001
```

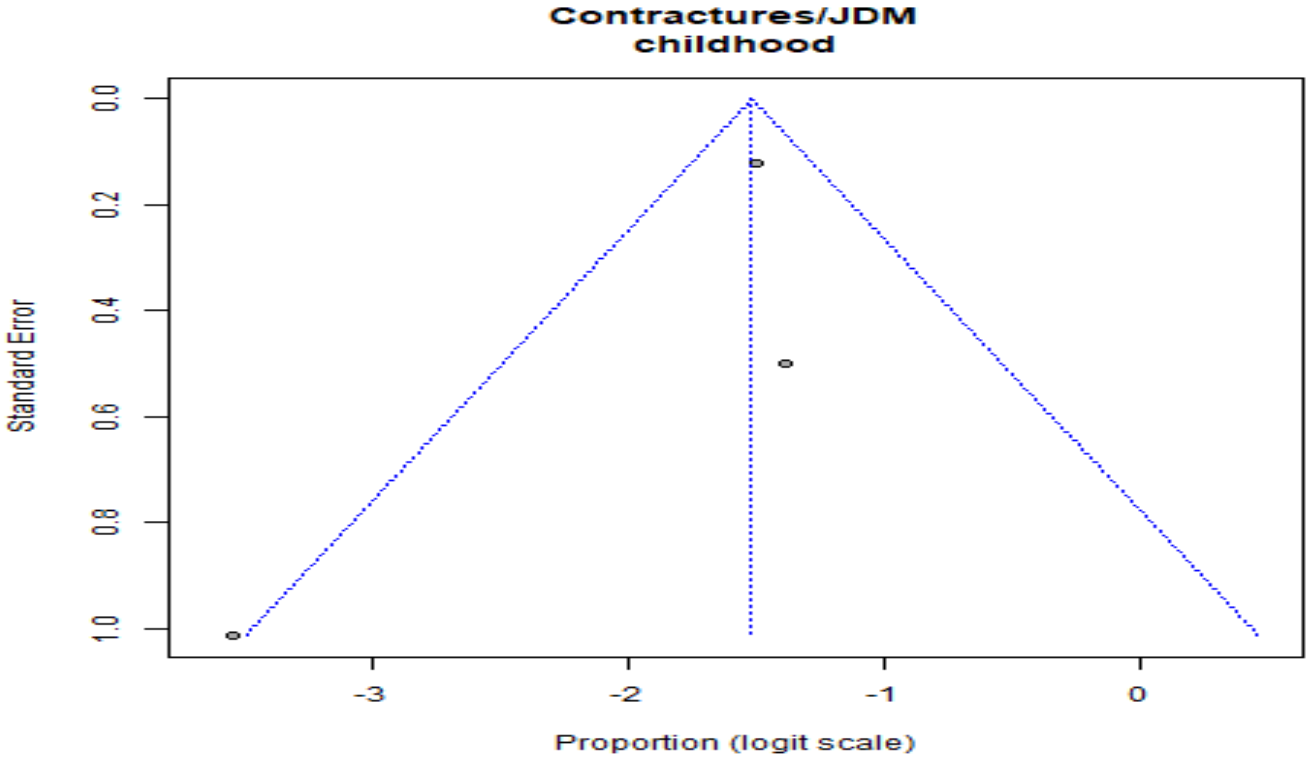
Supplementary Figure S4.66: Forest and Funnel Plots of studies reporting the prevalence of contractures in children with Juvenile Dermatomyositis

Contractures

Forest



Funnel



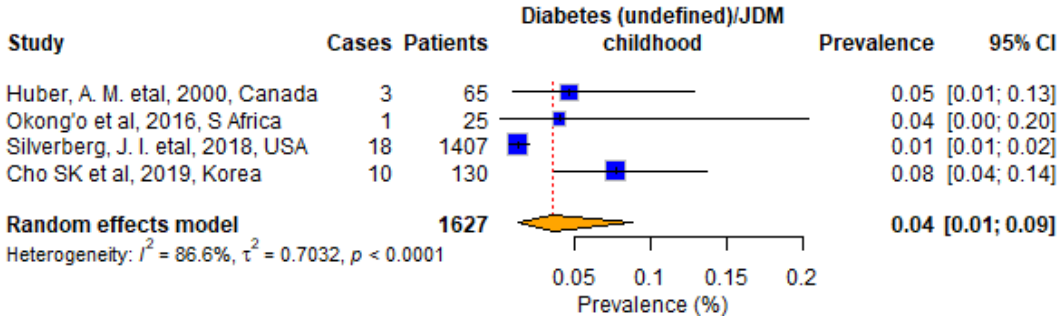
Summary statistics

```
                                proportion      95%-CI          %W(random)
Tabarki B etal, 1998, France        0.0278      [0.0007; 0.1453]        1.4
Okong'o et al, 2016, S Africa       0.2000      [0.0683; 0.4070]        5.6
Ravelli A et al, 2010, Italy/UK     0.1830      [0.1483; 0.2220]       93.1
Number of studies: k = 3; Number of observations: o = 509; Number of events: e = 88
                     proportion           95%-CI
Random effects model     0.1798 [0.1482; 0.2164]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 58.2361]; tau = 0.0020 [0.0000; 7.6313]
 I^2 = 51.7% [0.0%; 86.1%]; H = 1.44 [1.00; 2.68]
Test of heterogeneity:     Q d.f. p-value
                        4.14    2  0.1263
```

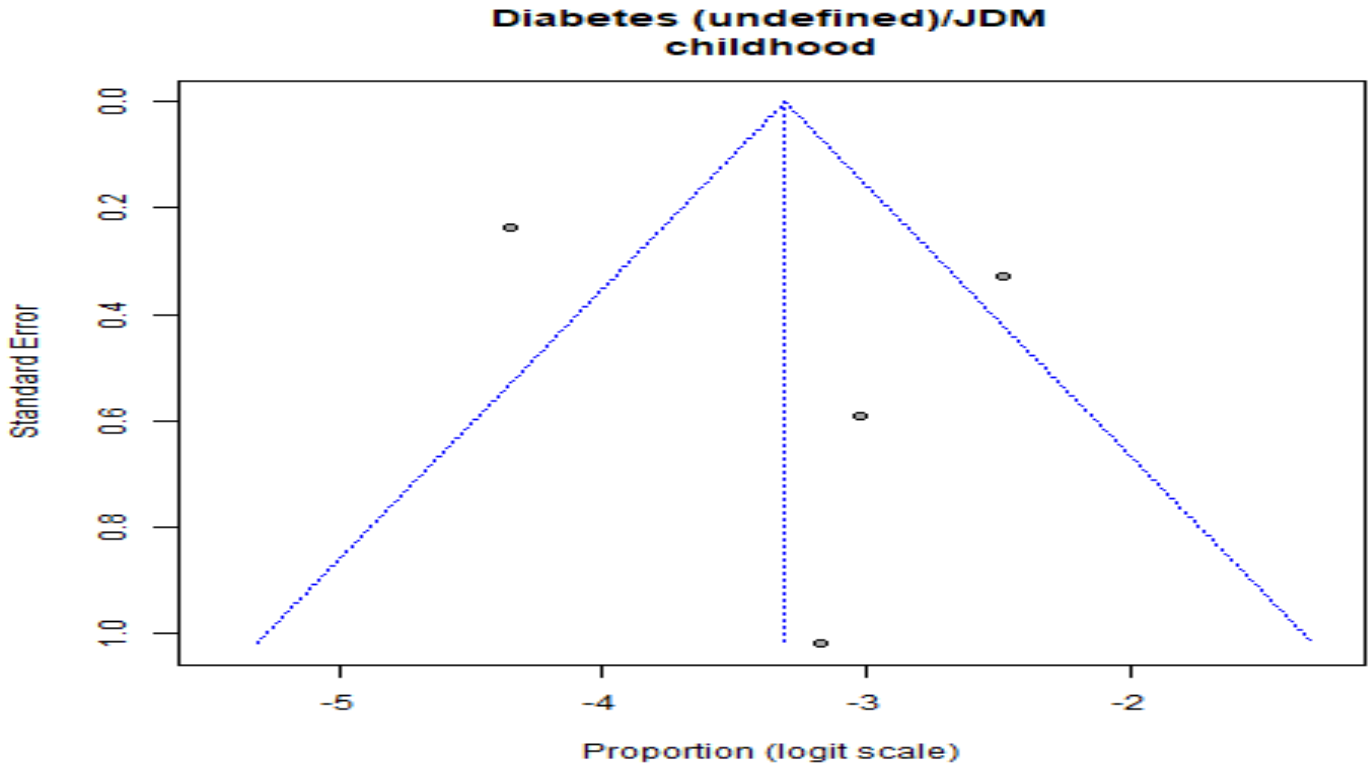
Supplementary Figure S4.67: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with Juvenile Dermatomyositis

Diabetes (undefined)

Forest



Funnel



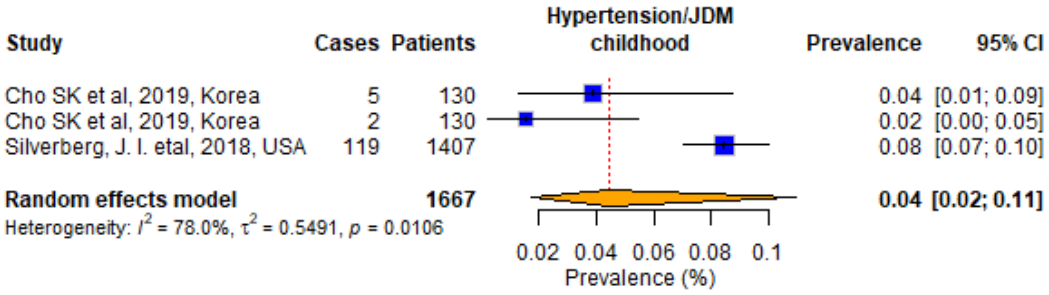
Summary statistics

```
                                proportion    95%-CI          %W(random)
Huber, A. M. etal, 2000, Canada      0.0462    [0.0096; 0.1290]        23.3
Okong'o et al, 2016, S Africa        0.0400    [0.0010; 0.2035]        14.1
Silverberg, J. I. etal, 2018, USA    0.0128    [0.0076; 0.0201]        32.3
Cho SK et al, 2019, Korea            0.0769    [0.0375; 0.1369]        30.3
Number of studies: k = 4; Number of observations: o = 1627; Number of events: e = 32
                     proportion           95%-CI
Random effects model     0.0352 [0.0136; 0.0879]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7032 [0.1139; 8.4433]; tau = 0.8386 [0.3376; 2.9057]
 I^2 = 86.6% [67.7%; 94.5%]; H = 2.74 [1.76; 4.25]
Test of heterogeneity:     Q d.f.  p-value
                       22.44    3 < 0.0001
```

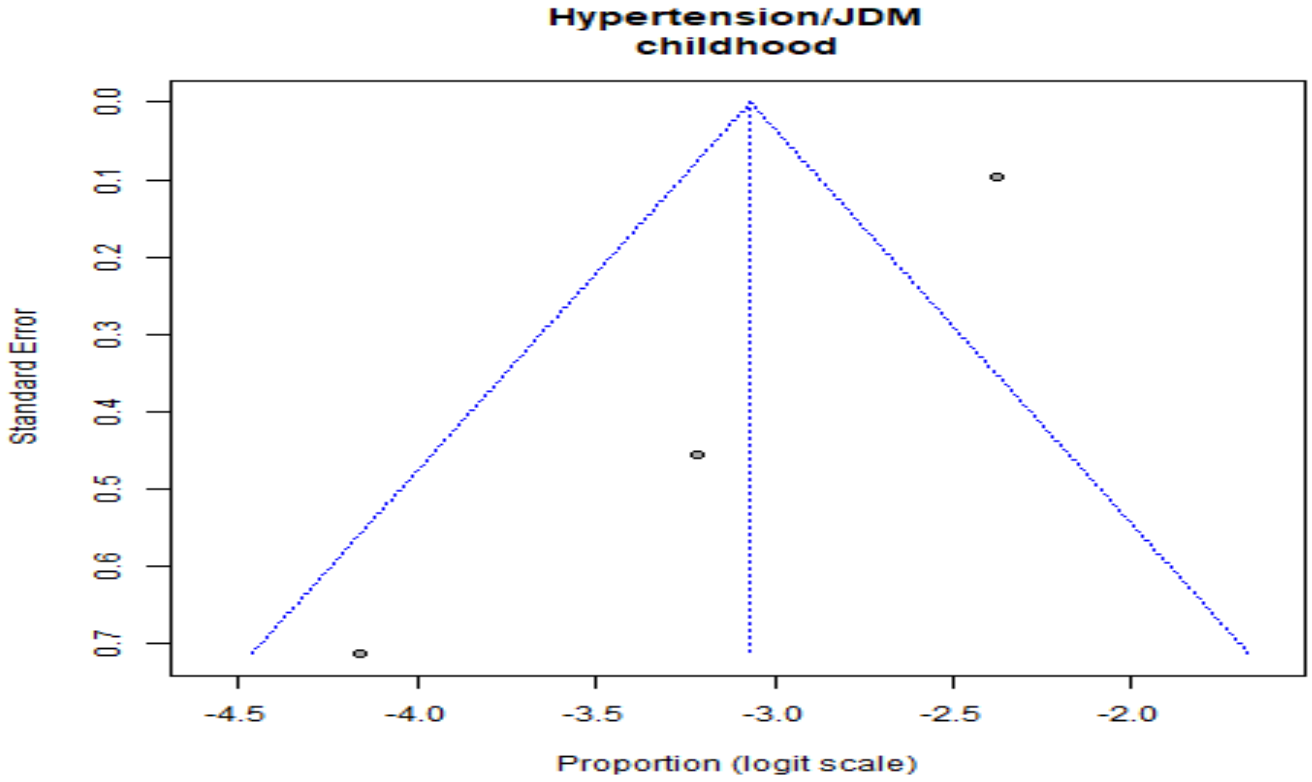
Supplementary Figure S4.68: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with Juvenile Dermatomyositis

Hypertension

Forest



Funnel



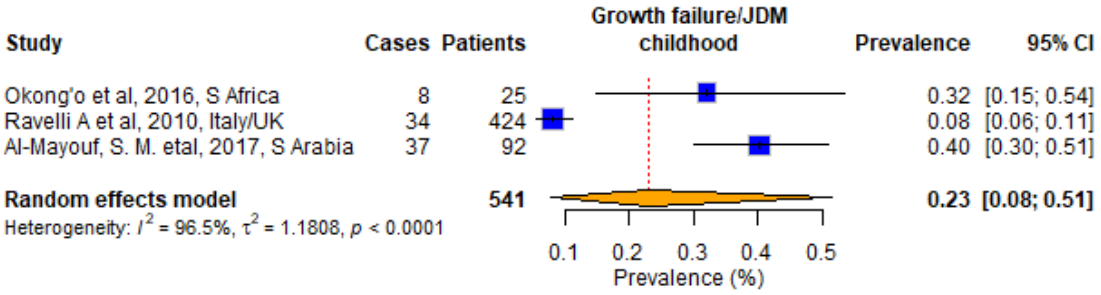
Summary statistics

```
                                  proportion    95%-CI          %W(random)
Cho SK et al, 2019, Korea             0.0385    [0.0126; 0.0875]       32.5
Cho SK et al, 2019, Korea             0.0154    [0.0019; 0.0545]       23.3
Silverberg, J. I. etal, 2018, USA     0.0846    [0.0706; 0.1004]       44.1
Number of studies: k = 3; Number of observations: o = 1667; Number of events: e = 126
                     proportion           95%-CI
Random effects model     0.0444 [0.0173; 0.1095]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.5491 [0.0293; 30.9540]; tau = 0.7410 [0.1711; 5.5636]
 I^2 = 78.0% [29.1%; 93.2%]; H = 2.13 [1.19; 3.83]
Test of heterogeneity:     Q d.f. p-value
                        9.10    2  0.0106
```

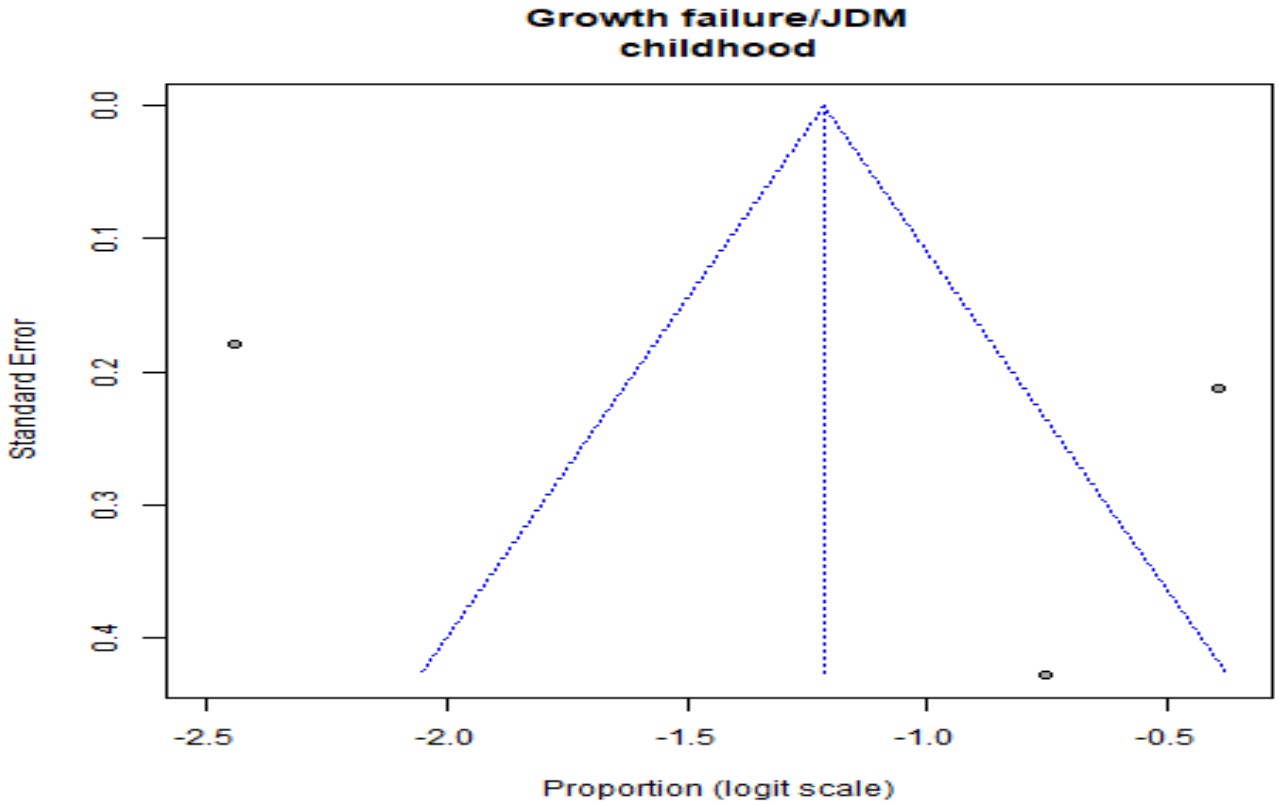
Supplementary Figure S4.69: Forest and Funnel Plots of studies reporting the prevalence of growth failure in children with Juvenile Dermatomyositis

Growth failure

Forest



Funnel



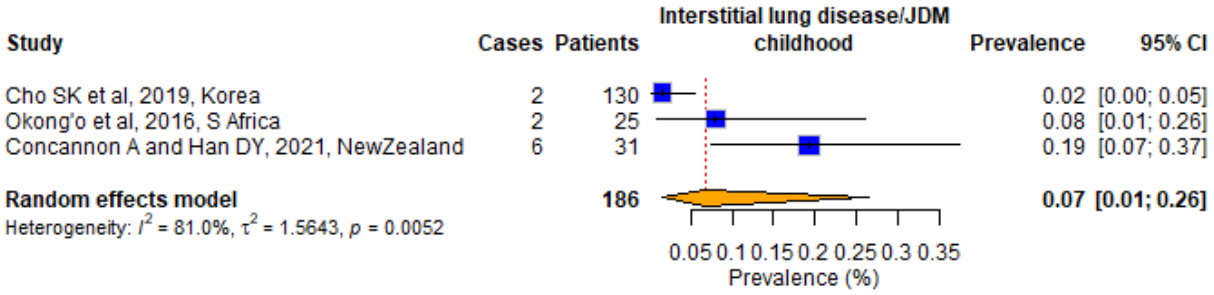
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.3200        [0.1495; 0.5350]        30.9
Ravelli A et al, 2010, Italy/UK         0.0802        [0.0562; 0.1103]        34.7
Al-Mayouf, S. M. etal, 2017, S Arabia   0.4022        [0.3012; 0.5096]        34.4
Number of studies: k = 3; Number of observations: o = 541; Number of events: e = 79
                     proportion           95%-CI
Random effects model     0.2285 [0.0766; 0.5139]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.1808 [0.2769; 46.9921]; tau = 1.0866 [0.5262; 6.8551]
 I^2 = 96.5% [92.8%; 98.3%]; H = 5.37 [3.72; 7.74]
Test of heterogeneity:      Q d.f.  p-value
                        57.58    2 < 0.0001
```

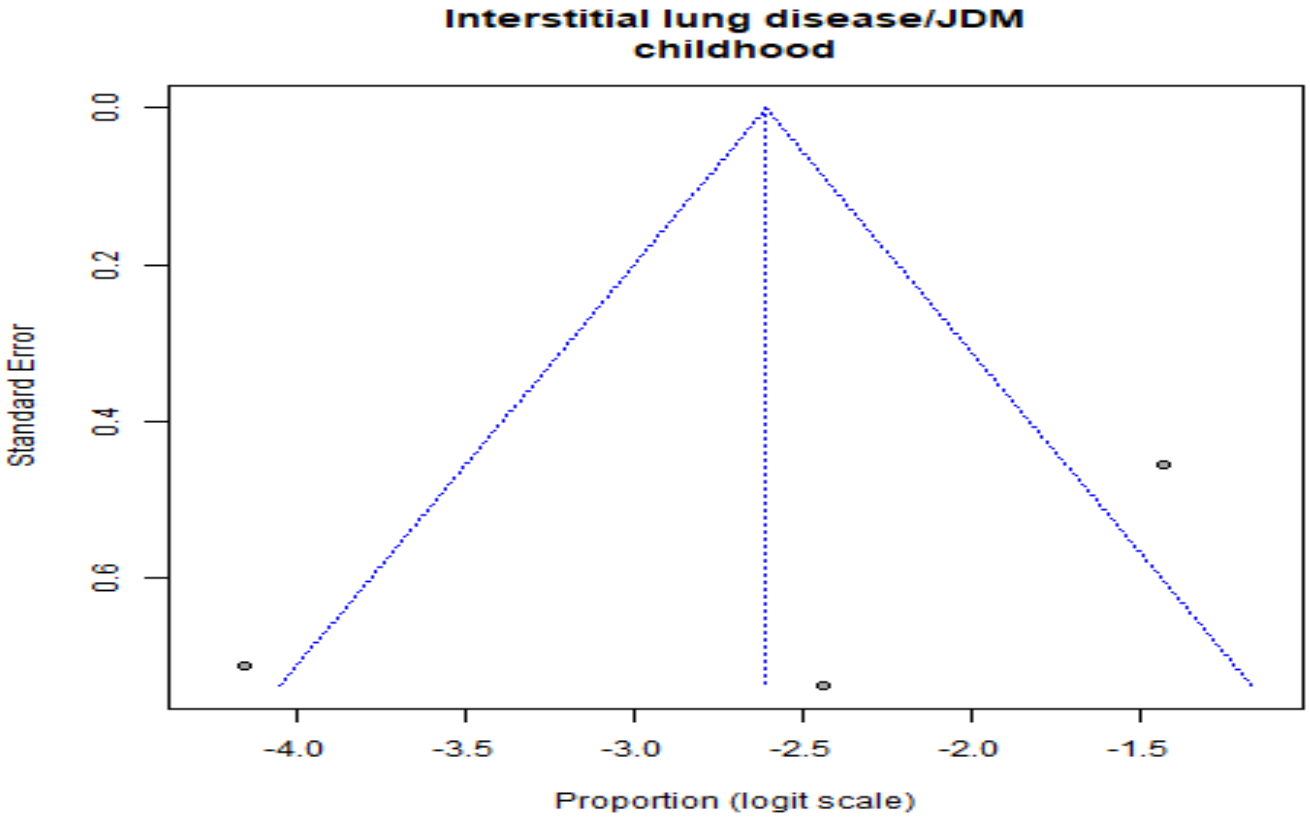
Supplementary Figure S4.70: Forest and Funnel Plots of studies reporting the prevalence of Interstitial lung disease in children with Juvenile Dermatomyositis

Interstitial lung disease

Forest



Funnel



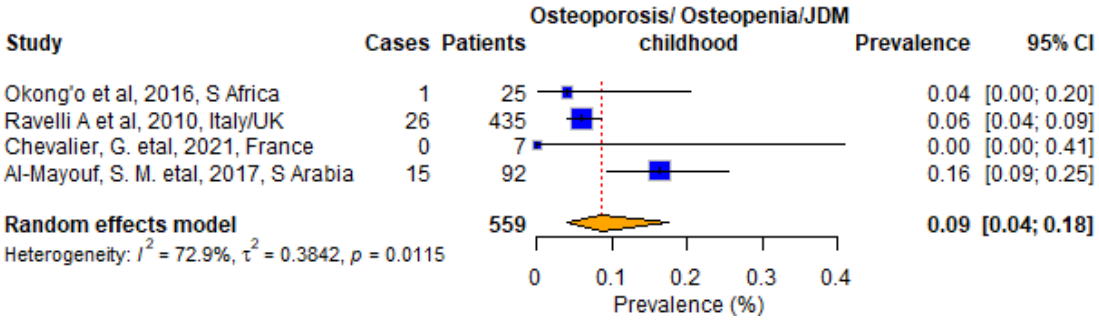
Summary statistics

```
                                        proportion    95%-CI          %W(random)
Cho SK et al, 2019, Korea                   0.0154    [0.0019; 0.0545]        31.7
Okong'o et al, 2016, S Africa               0.0800    [0.0098; 0.2603]        31.2
Concannon A and Han DY, 2021, NewZealand    0.1935    [0.0745; 0.3747]        37.1
Number of studies: k = 3; Number of observations: o = 186; Number of events: e = 10
                     proportion          95%-CI
Random effects model     0.0685 [0.0148; 0.2648]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.5643 [0.1487; 74.9226]; tau = 1.2507 [0.3856; 8.6558]
 I^2 = 81.0% [40.6%; 93.9%]; H = 2.29 [1.30; 4.05]
Test of heterogeneity:      Q d.f. p-value
                        10.52    2  0.0052
```

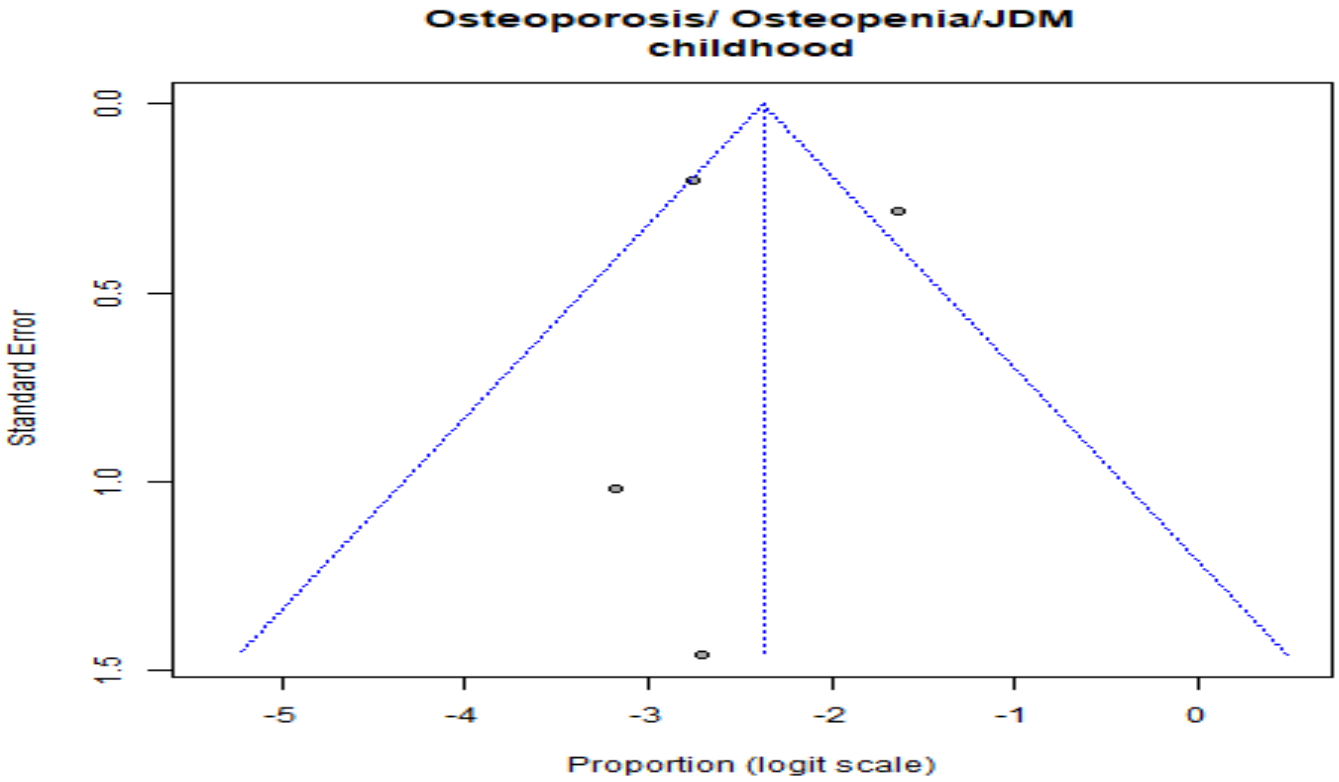
Supplementary Figure S4.71: Forest and Funnel Plots of studies reporting the prevalence of Osteoporosis/Osteopenia in children with Juvenile Dermatomyositis

Osteoporosis/ Osteopenia

Forest



Funnel



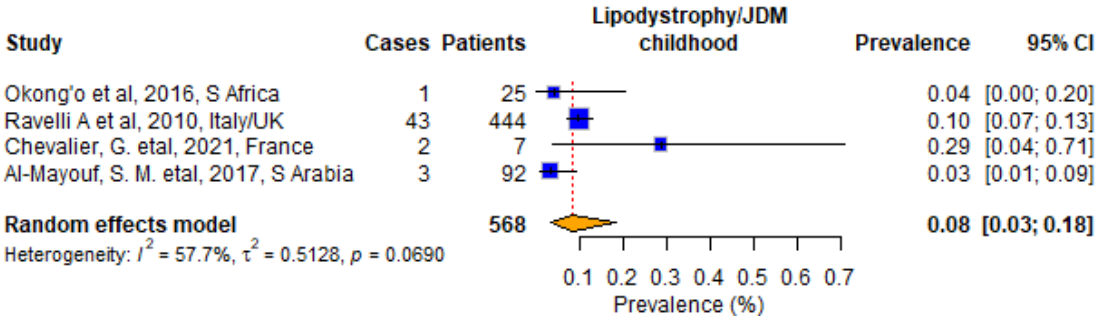
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.0400       [0.0010; 0.2035]        12.5
Ravelli A et al, 2010, Italy/UK         0.0598       [0.0394; 0.0864]        42.0
Chevalier, G. etal, 2021, France        0.0000       [0.0000; 0.4096]         7.1
Al-Mayouf, S. M. etal, 2017, S Arabia   0.1630       [0.0942; 0.2546]        38.5
Number of studies: k = 4; Number of observations: o = 559; Number of events: e = 42
                    proportion           95%-CI
Random effects model     0.0851 [0.0391; 0.1756]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3842 [0.0122; 5.6370]; tau = 0.6199 [0.1103; 2.3742]
 I^2 = 72.9% [23.4%; 90.4%]; H = 1.92 [1.14; 3.22]
Test of heterogeneity:      Q d.f. p-value
                        11.05    3  0.0115
```

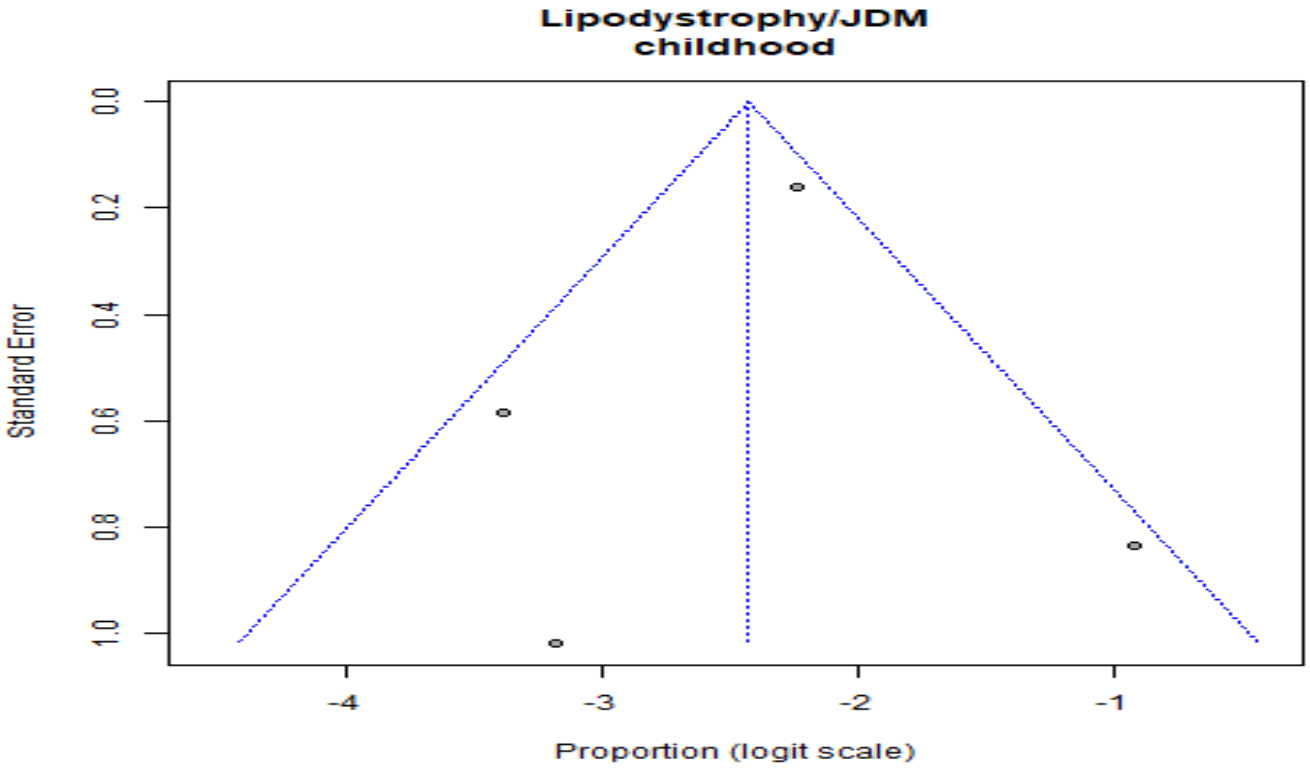
Supplementary Figure S4.72: Forest and Funnel Plots of studies reporting the prevalence of lipodystrophy in children with Juvenile Dermatomyositis

Lipodystrophy

Forest



Funnel



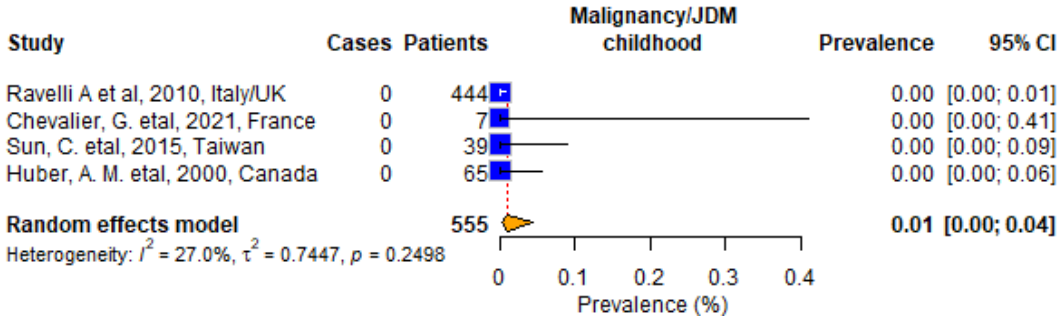
Summary statistics

```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.0400       [0.0010; 0.2035]       14.3
Ravelli A et al, 2010, Italy/UK         0.0968       [0.0710; 0.1282]       41.3
Chevalier, G. etal, 2021, France        0.2857       [0.0367; 0.7096]       18.4
Al-Mayouf, S. M. etal, 2017, S Arabia   0.0326       [0.0068; 0.0923]       26.0
Number of studies: k = 4; Number of observations: o = 568; Number of events: e = 49
                     proportion           95%-CI
Random effects model     0.0811 [0.0338; 0.1821]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.5128 [0.0000; 17.0078]; tau = 0.7161 [0.0000; 4.1241]
 I^2 = 57.7% [0.0%; 85.9%]; H = 1.54 [1.00; 2.67]
Test of heterogeneity:     Q d.f. p-value
                        7.09    3  0.0690
```

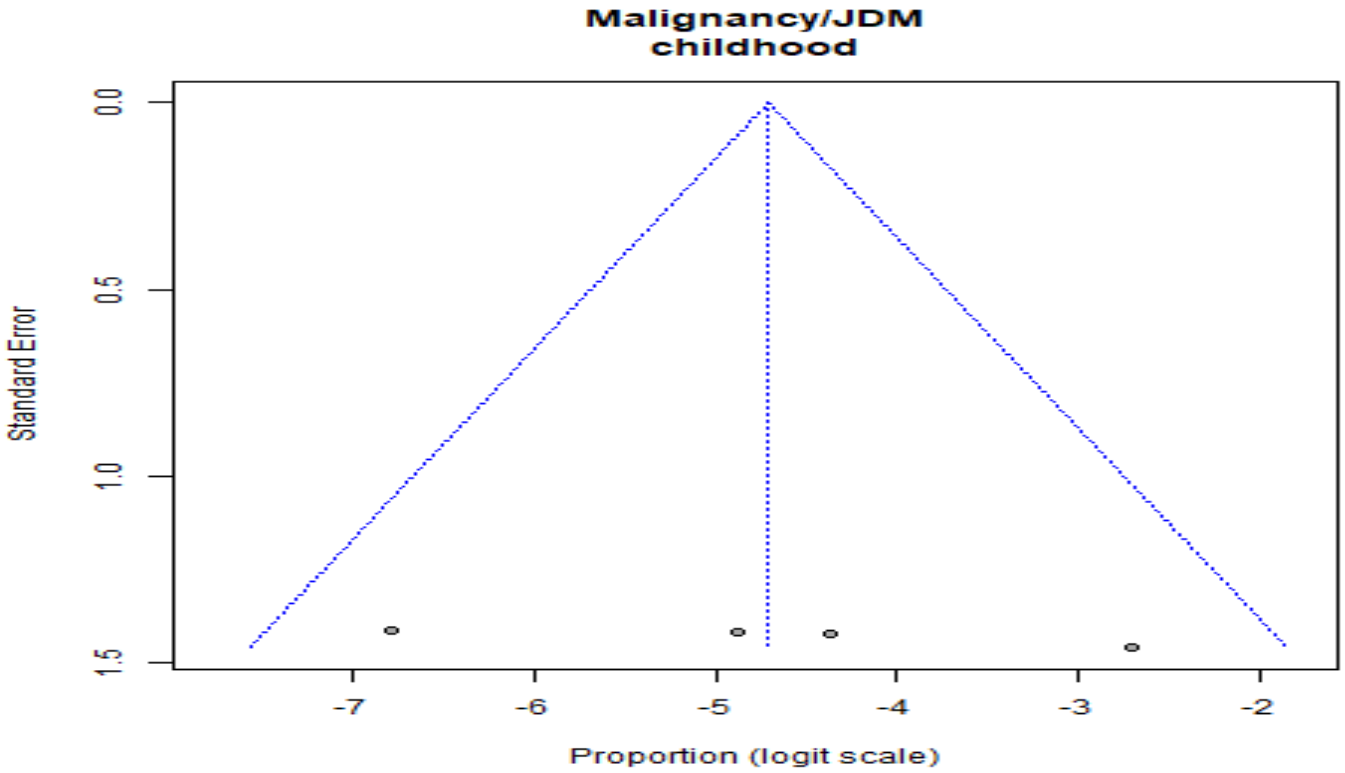
Supplementary Figure S4.73: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with Juvenile Dermatomyositis

Malignancy

Forest



Funnel



Summary statistics

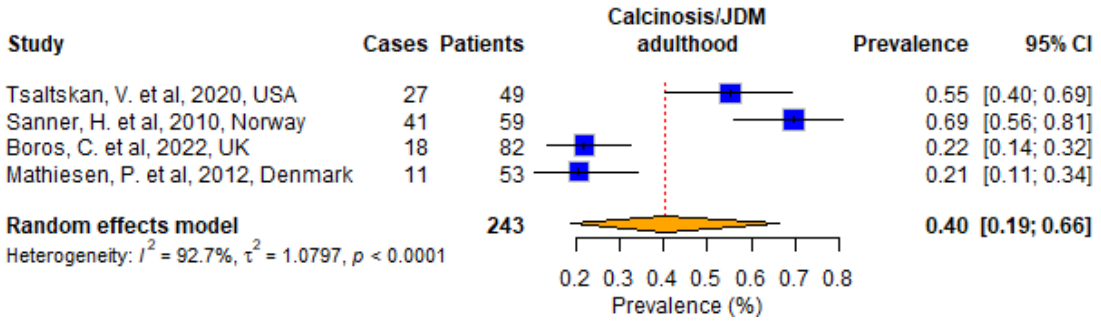
```
                                proportion     95%-CI          %W(random)
Ravelli A et al, 2010, Italy/UK     0.0000     [0.0000; 0.0083]
Chevalier, G. etal, 2021, France    0.0000     [0.0000; 0.4096]
Sun, C. etal, 2015, Taiwan          0.0000     [0.0000; 0.0903]
Huber, A. M. etal, 2000, Canada     0.0000     [0.0000; 0.0552]
Number of studies: k = 4; Number of observations: o = 555; Number of events: e = 0
                     proportion           95%-CI
Random effects model     0.0089 [0.0018; 0.0442]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7447 [0.0000; 37.2127]; tau = 0.8629 [0.0000; 6.1002]
 I^2 = 27.0% [0.0%; 72.6%]; H = 1.17 [1.00; 1.91]
Test of heterogeneity:     Q d.f. p-value
                        4.11    3  0.2498
```

Adulthood/JDM

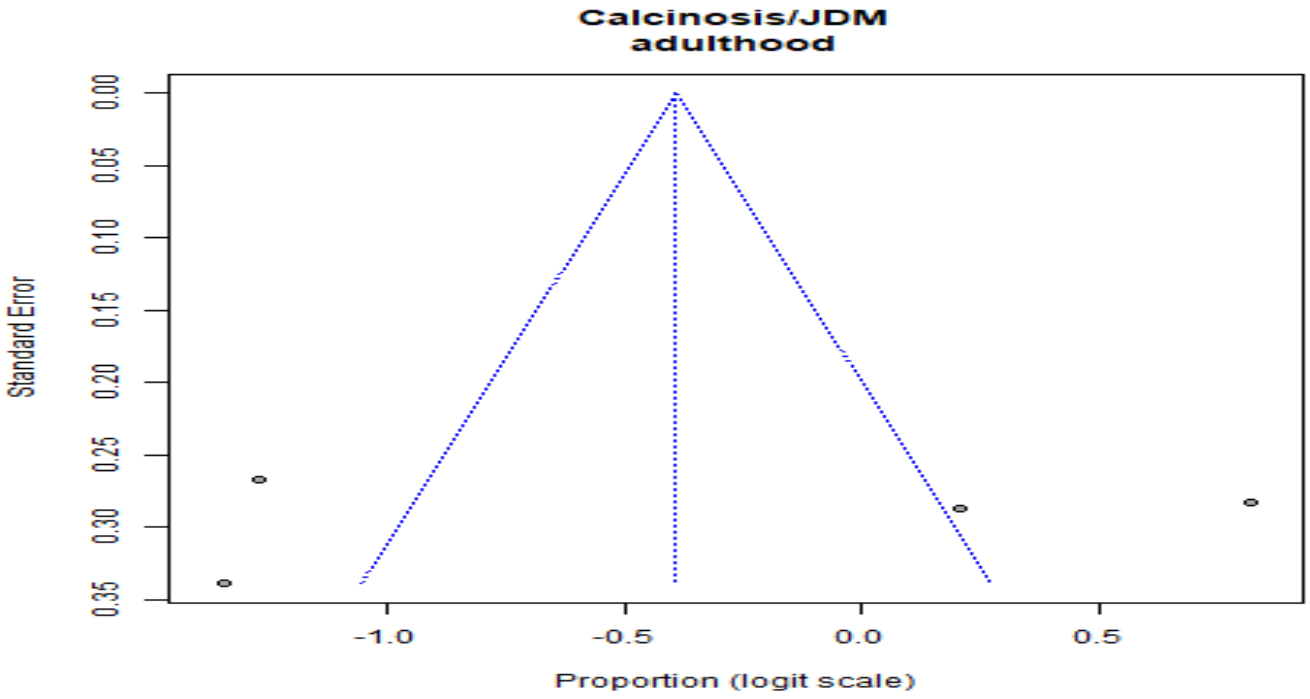
Supplementary Figure S4.74: Forest and Funnel Plots of studies reporting the prevalence of calcinosis in adults with Juvenile Dermatomyositis

Calcinosis

Forest



Funnel



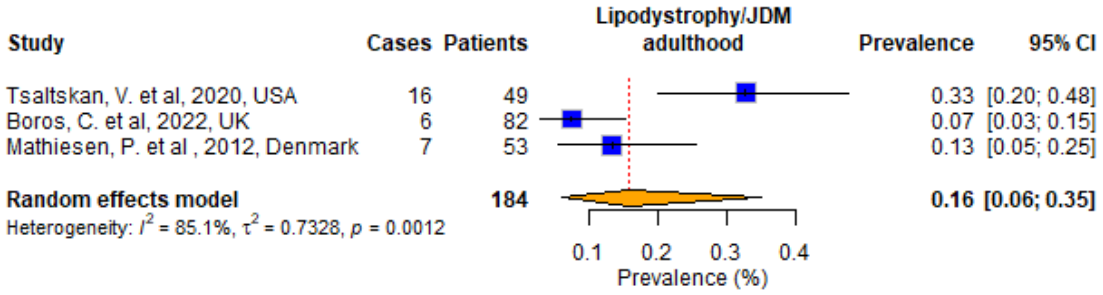
Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tsaltskan, V. et al, 2020, USA        0.5510   [0.4023; 0.6933]       25.1
Sanner, H. et al, 2010, Norway        0.6949   [0.5613; 0.8081]       25.1
Boros, C. et al, 2022, UK             0.2195   [0.1356; 0.3246]       25.3
Mathiesen, P. et al, 2012, Denmark    0.2075   [0.1084; 0.3411]       24.4
Number of studies: k = 4; Number of observations: o = 243; Number of events: e = 97
                     proportion           95%-CI
Random effects model     0.4037 [0.1902; 0.6611]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0797 [0.2878; 16.1291]; tau = 1.0391 [0.5365; 4.0161]
 I^2 = 92.7% [84.6%; 96.6%]; H = 3.71 [2.55; 5.40]
Test of heterogeneity:      Q d.f.  p-value
                        41.34    3 < 0.0001
```

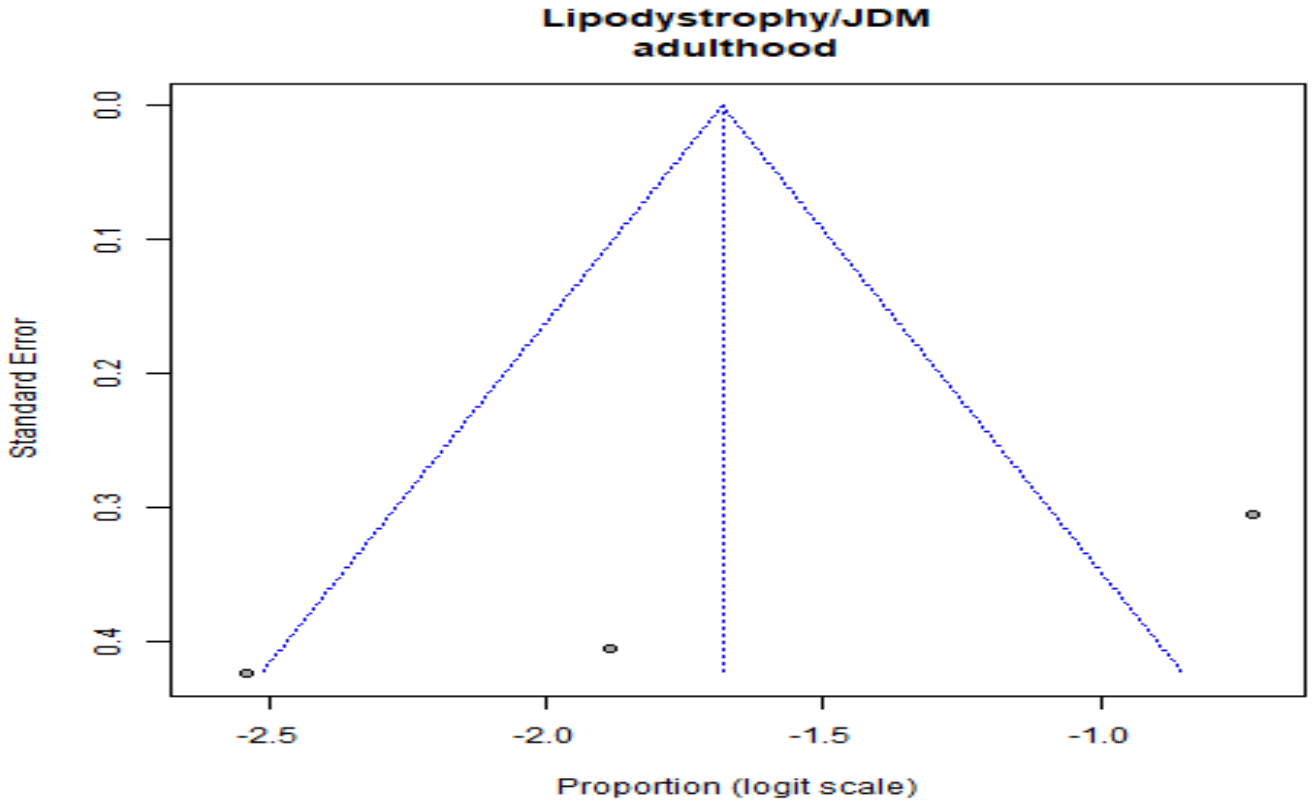
Supplementary Figure S4.75: Forest and Funnel Plots of studies reporting the prevalence of lipodystrophy in adults with Juvenile Dermatomyositis

Lipodystrophy

Forest



Funnel



Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tsaltskan, V. et al, 2020, USA        0.3265  [0.1995; 0.4754]       35.4
Boros, C. et al, 2022, UK             0.0732  [0.0273; 0.1525]       32.0
Mathiesen, P. et al , 2012, Denmark   0.1321  [0.0548; 0.2534]       32.6
Number of studies: k = 3; Number of observations: o = 184; Number of events: e = 29
                     proportion           95%-CI
Random effects model     0.1567 [0.0605; 0.3491]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7328 [0.1017; 33.2331]; tau = 0.8560 [0.3189; 5.7648]
 I^2 = 85.1% [55.9%; 94.9%]; H = 2.59 [1.51; 4.44]
Test of heterogeneity:      Q d.f. p-value
                        13.38    2  0.0012
```